



SCAN FOR THE
ELECTRONIC VERSIONS

KUWAIT PEDIATRIC COUNCIL • BEDSIDE
REFERENCE

Pediatric & Neonatal Common & Emergency Care Guide

A searchable, algorithmic bedside companion for pediatricians and primary care physicians — every medication listed separately with weight-based dosing, first- to third-line therapy, and the can't-miss diagnoses. Tap any topic to open it.

Prepared and proposed to the Kuwait Pediatric Council by

Dr. Athari Almajed — Consultant, General Pediatrics & Pediatric Pulmonology, LTV and Sleep Medicine

Panel of Reviewers

KUWAIT PEDIATRIC COUNCIL • 69 REVIEWERS •
ALPHABETICAL BY FIRST NAME

Dr. Abdulaziz Alosaimi

Specialist, General Pediatrics, Pediatric Cardiology & Critical Care

Dr. Abdulla Alfraj

Consultant, General Pediatrics & Pediatric Critical Care

Dr. Abdullah Abd Alhussain Taqi Ali

Consultant, Pediatric Hematology & Oncology

Dr. Abdullah AlDuwaisan

Assistant Professor of Orthopedic Surgery, Kuwait University

Dr. Abdullah Rajab

Consultant, Adult & Pediatric Surgery

Dr. Aceel Alawadi

Consultant, General Pediatrics & Pediatric Pulmonology, LTV and Sleep Medicine

Dr. Ahlam Mustafa

Consultant in Pediatric Gastroenterology, Hepatology, Endoscopy and Nutrition

Dr. Ahmad Alhaj

Consultant, General Pediatrics & Pediatric Critical Care

Dr. Ahmad Almosawi

Consultant, General Pediatrics & Pediatric Pulmonology and Critical Care

Dr. Ahmad Jaafar

Specialist, General Pediatrics & Pediatric Complex and Palliative Care

Dr. Aisha AlTerkait

Consultant, General Pediatrics & Pediatric Nephrology

Dr. Alaa Shehab

Senior Specialist, General Pediatrics, Pediatric Rheumatology & Systemic Auto-inflammatory Disease

Dr. Alya Hasan Askar Hasan

Consultant, Pediatric Neurosurgery

Dr. Amaar Marefi

Specialist, Pediatric Neurology & Neuroimmunology

Dr. Anfal Taher

Senior Specialist, General Pediatrics & Pediatric Emergency

Dr. Basma Al-Tawari

Specialist, General Pediatrics

Dr. Bibi Alamiri

Consultant Child and Adolescent Psychiatrist

Dr. Dalal Mohammad

Specialist, General Pediatrics & Pediatric Rheumatology

Dr. Dana Marafi

Assistant Professor of Pediatrics, Kuwait University; Pediatric Neurology, Epilepsy & Neurogenetics

Dr. Eiman Al-Hashemi

Specialist, General Pediatrics & Pediatric Critical Care

Dr. Eman Buhamrah

Consultant, General Pediatrics & Pediatric Gastroenterology and Hepatology

Dr. Esmaeel Taqi

Consultant Pediatric Surgery

Dr. Fahed Aljaser

Consultant, General Pediatrics & Pediatric Endocrinology

Dr. Fajer Altammar

Senior Specialist, General Pediatrics, Pediatric Emergency & Pediatric Critical Care

Dr. Fanar Faridoun

Senior Specialist, General Pediatrics

Dr. Fares AlFares

Assistant Professor of Pediatrics, Kuwait University; General Pediatrics & Pediatric Hematology / Oncology

Dr. Fatemah Bukhalaf

Consultant, General Pediatrics & Adolescent Medicine

Dr. Fatoumah Alabdulrazzaq

Consultant, General Pediatrics & Pediatric Emergency; Clinical Pharmacologist & Toxicologist

Dr. Fawaz Albaghli

Consultant, General Pediatrics & Neonatology

Dr. Fawaz AlRefaee

Consultant, General Pediatrics & Pediatric Gastroenterology

Dr. Hanan Alsaleh

Consultant, Pediatric Ophthalmology

Dr. Hashem Alhashemi

Consultant, General Pediatrics & Pediatric Critical Care

Dr. Hessah Alsunaien

Consultant, General Pediatrics

Dr. Hussain Sadeq

Senior Specialist, General Pediatrics

Dr. Ibrahim Alqattan

Senior Registrar, General Pediatrics

Dr. Khaled Kalandar

Consultant, General Pediatrics

Dr. Khalid Alothman

Consultant, General Pediatrics & Pediatric Infectious Diseases

Dr. Lujain Abul

Specialist, General Pediatrics & Pediatric Emergency

Dr. Mariam Ayed

Consultant, General Pediatrics & Neonatology and Neonatal Neuro-Critical Care

Dr. Marwan Alqunae

Senior Specialist, Otorhinolaryngology (ENT)

Dr. Maryam Alhajeri

Specialist, General Pediatrics & Pediatric Nephrology

Dr. Maysoon Al-Musallam

Consultant, General Pediatrics

Dr. Maysoun Al Rushood

Associate Professor of Pediatrics; Consultant, General Pediatrics & Pediatric Allergy and Clinical Immunology

Dr. Meshal Alenezi

Consultant, General Pediatrics & Pediatric Emergency

Dr. Mohammad Alajmi

Specialist, General Pediatrics & Pediatric Emergency

Dr. Mohammad Alghounaim

Consultant, General Pediatrics & Pediatric Infectious Diseases; Medical Microbiologist

Dr. Mohammad AlHumaidan

Consultant, General Pediatrics & Pediatric Emergency

Dr. Moneerah Albesharah

Senior Registrar, General Pediatrics

Dr. Muna Almutairi

Consultant, General Pediatrics & Pediatric Rheumatology

Dr. Munirah AlYaseen

Specialist, General Pediatrics, Pediatric Critical Care & Pediatric Neuro-Critical Care

Dr. Mustafa AlQbandi

Consultant, General Pediatrics & Pediatric Interventional Cardiology

Dr. Osama Shalaby Abdulaziz

Consultant, General Pediatrics & Pediatric Neurology

Dr. Rasha Alsafi

Consultant, General Pediatrics, Clinical Genetics & Metabolic Disorders

Dr. Rashed AlAhmed

Specialist, Orthopedic Surgery; Pediatric & Adult Spine

Dr. Rawan AlFouzan

Senior Specialist, General Pediatrics & Pediatric Pulmonology

Dr. Rehab Bekhit

Practitioner, General Pediatrics & Pediatric Critical Care

Dr. Saadoun Bin-Hasan

Consultant, General Pediatrics & Pediatric Pulmonology and Sleep Medicine

Dr. Saleh Alsaifi

Consultant Adult & Pediatric Orthopedics

Dr. Samir Mohammed Sanad El Mesady

Consultant, Pediatric Emergency Medicine

Dr. Sara Ali

Specialist, General Pediatrics

Dr. Sarah Altamimi

Senior Registrar, General Pediatrics & Pediatric Critical Care

Dr. Suha Atyani

Consultant, General Pediatrics

Dr. Taiba Almousawy

Senior Registrar, General Pediatrics

Dr. Taibah Alansari

Senior Registrar, General Pediatrics

Dr. Talal Alsaleem

Consultant, General Pediatrics

Dr. Tawa Sahar

Senior Specialist, Dermatology

Dr. Wafaa Laymoun

Senior Specialist, General Pediatrics & Pediatric Endocrinology and Diabetes

Dr. Waleed Aldeeweli

Consultant, Pediatric Dentistry

Dr. Yaqoub Jafar

Senior Specialist, Pediatric Surgery and Urology

Dr. Zainab Mustafa

Specialist, Pediatrics, Neonatology & Neonatal Echocardiography and Haemodynamics

Legal & evidence statement. This guide is clinical decision support, not a substitute for clinical judgment, Ministry of Health protocols, or specialist consultation. **This guide is not legally binding in any way** — it is a clinical approach guide intended to support, not to direct or replace, the treating clinician's own judgment, the applicable Ministry of Health regulations, and institutional policies; where it differs from official regulations or institutional policy, those prevail. **It does not substitute for the opinion and decision of the experts responsible for the individual patient in the relevant field.** It reflects the evidence and guidelines current at the date of issue. **If newer high-quality evidence or an updated guideline is published before this guide's scheduled renewal, that newer evidence takes precedence over the recommendations here.** Confirm every weight-based dose against the approved product label and a current pediatric drug reference before administration. Drug-availability notes (marked ♦) flag agents that may be unavailable or not routinely stocked in Kuwait and give an alternative. It is intended for pediatricians and primary care physicians in the governmental and private sectors of the State of Kuwait, working within their scope of practice, and is not patient-facing medical advice. External links (reference DOIs and the document share link) are provided for convenience; the linked content is not under the control or responsibility of this guide or the

Council.

Medical responsibility (State of Kuwait). In Kuwait, the practice of medicine, the standard of care, and professional accountability are governed by **Law No. 70 of 2020** on the practice of the medical and allied health professions, patients' rights, and health facilities, together with the decisions of the **Authority for Medical Responsibility** established under it. Nothing in this guide alters or lessens the treating physician's legal duties under that law — to hold a valid license and practice within their scope, to follow established medical rules and sound diagnostic and therapeutic procedures, to obtain informed consent, and to keep accurate, confidential medical records. Responsibility for every clinical decision, and for verifying it against the law and current Ministry of Health regulations, rests with the treating physician and the treating institution.

Age & scope. This guide covers the newborn through the adolescent. A **neonate** is a baby aged **≤ 28 days**; a **pediatric patient** is any child from birth **up to 18 years** of age. This mirrors Kuwait's **Child Rights Law No. 21 of 2015**, which defines a child as every person who has not completed 18 years. In Kuwait the age of full legal (civil) majority is **21 years**; for any patient under 18, **consent to assessment and treatment is given by a parent or legal guardian**, while the older child/adolescent is involved according to their capacity. Individual

services may apply their own operational age cut-offs, so confirm local policy.

Preparation. This guide was drafted, edited, and formatted with the assistance of an AI-based large language model used under direct human supervision. The tool was not an author and made no clinical judgments. All clinical recommendations, medication names and doses, diagnostic thresholds, and references were written, verified against the primary sources and current guidelines, and approved by the author and the review panel named below, who take full responsibility for the accuracy and integrity of the content. Prepared in line with ICMJE guidance on the use of AI-assisted technologies and with the Ministry of Health **Practical Guide for the Use of Language-based Artificial Intelligence Tools in Healthcare Practice**, adopted and published as a guiding framework for staff of all MOH health facilities by Administrative Decision No. 6189 of 2026 of the Undersecretary of the Ministry of Health (issued under Law No. 70/2020 and Ministerial Decision No. 373/2025 on the code of ethics for health-sector workers).

Author: Dr. Athari Almajed — Consultant, General Pediatrics & Pediatric Pulmonology, LTV and Sleep Medicine.

Reviewers: the review panel named above — reviewed and approved the final version.

The author and reviewers confirm they have read this version, verified the clinical content and references, and approve it for use.

Inquiries, input, or corrections: ayalmajed@moh.gov.kw

Conflicts of interest: none declared. **Funding:** none.

© 2026 Kuwait Pediatric Council. This guide may be reproduced for non-commercial educational use with attribution.

What the colors mean

-  **01 · Resuscitation** — cardiopulmonary arrest & the universal approach
-  **02 · Behavioral & Psychiatric** — agitation, self-harm, drug-induced syndromes
-  **03 · Cardiac & Shock** — arrhythmias, all types of shock, hypercyanotic spell
-  **04 · Common Acute Conditions** — everyday clinic & ED infections and presentations
-  **05 · Complex & Technology-Dependent Care** — tracheostomy, feeding tubes, ECMO, post-ICU, transport
-  **06 · Endocrine & Metabolic** — diabetes, adrenal, inborn errors, electrolytes
-  **07 · Focused Specialty Emergencies** — eye, dental, dermatologic, ENT & neurosurgical
-  **08 · Gastroenterology & Hepatology** — liver failure, GI bleeding, pancreatitis, hepatitis, cholestasis
-  **09 · Hematologic & Oncologic** — neutropenia, tumor lysis, sickle cell, bleeding
-  **10 · Neonatal** — newborn-specific emergencies
-  **11 · Neurologic** — seizures & status epilepticus
-  **12 · Orthopedic & Musculoskeletal** — the limping child, SCFE, compartment syndrome, fractures

-  **13 · Pediatric Trauma & TBI** — primary/secondary survey, traumatic brain injury
-  **14 · Renal & Urology** — AKI, HUS, hypertensive emergency, acute scrotum & testicular torsion
-  **15 · Respiratory, Allergy & Immunology** — airway, breathing, anaphylaxis, allergy & immunology
-  **16 · Rheumatologic** — cytokine storm (HLH/MAS) & Kawasaki
-  **17 · Surgical Emergencies** — recognize, resuscitate, refer
-  **18 · Toxicology & Environmental** — poisoning, burns, heat, drowning, smoke inhalation

Each topic also carries an **urgency tag** showing how fast to act:

MINUTES

immediately life-threatening — act now

HOURS

urgent — stabilize and treat early

ASSESS

evaluate and manage on its own pathway

Call vs refer — in a **hospital**, **call** the in-house specialist team (on site); in a **polyclinic** — or any facility **where the specialty is not available** — **refer** the patient to a facility that has it.

Abbreviations

CPS — Canadian Pediatric Society

GINA — Global Initiative for Asthma

KDIGO — Kidney Disease: Improving Global Outcomes

KNCPP — Kuwait National Child Protection Program

NASPGHAN — North American Society for Pediatric Gastroenterology, Hepatology and Nutrition

PES — Pediatric Endocrine Society

PIDS — Pediatric Infectious Diseases Society

NTSP — National Tracheostomy Safety Project

AAP — American Academy of Pediatrics

NRP — Neonatal Resuscitation Program

AARC — American Association for Respiratory Care

AUA — American Urological Association

BTF — Brain Trauma Foundation

CDC — Centers for Disease Control and Prevention

ELSO — Extracorporeal Life Support Organization

EMA — European Medicines Agency

ESPGHAN — European Society for Pediatric Gastroenterology, Hepatology and Nutrition

EULAR — European Alliance of Associations for Rheumatology

FDA — US Food and Drug Administration

AHA — American Heart Association

IDSA — Infectious Diseases Society of America

ASHP — American Society of Health-System Pharmacists

ANMF — Australasian Neonatal Medicines Formulary

ESC — European Society of Cardiology

ISMP — Institute for Safe Medication Practices

SIDP — Society of Infectious Diseases Pharmacists

AES — American Epilepsy Society

EAN — European Academy of Neurology

PNS — Peripheral Nerve Society

RCPCH — Royal College of Paediatrics and Child Health (UK)

ILAE — International League Against Epilepsy

ISPAD — International Society for Pediatric and Adolescent Diabetes

MOH — Ministry of Health

NICE — National Institute for Health and Care Excellence (UK)

WHO — World Health Organization

DRACMA — Diagnosis and Rationale for Action against Cow's Milk Allergy (WAO)

WAO — World Allergy Organization

AAF — amino acid formula

ABC — airway, breathing, circulation

ABCDE — airway, breathing, circulation, disability, exposure

ABO — ABO blood-group system

ACS — acute chest syndrome

ADAMTS13 — von Willebrand factor-cleaving protease (ADAMTS13)

ADHD — attention-deficit / hyperactivity disorder

AED — automated external defibrillator; also
antiepileptic drug

aEEG — amplitude-integrated
electroencephalography

AEIOU — alcohol, epilepsy, insulin,
opioids/overdose, uremia (coma mnemonic)

AGEP — acute generalized exanthematous
pustulosis

AHI — apnea-hypopnea index

AHO — acute hematogenous osteomyelitis

AHT — abusive head trauma

aHUS — atypical hemolytic uremic syndrome

AKI — acute kidney injury

ALT — alanine aminotransferase

ALTE — apparent life-threatening event

AMPLE — allergies, medications, past history,
last meal, events (history mnemonic)

AMR — antimicrobial resistance

ANC — absolute neutrophil count

AOM — acute otitis media

APGAR — Appearance, Pulse, Grimace, Activity,
Respiration (score)

APML — acute promyelocytic leukemia

aPTT — activated partial thromboplastin time

ARDS — acute respiratory distress syndrome

ASD — atrial septal defect

AST — aspartate aminotransferase

ATOMAC —

Arizona-Texas-Oklahoma-Memphis-Arkansas
Consortium (pediatric trauma guidelines)

ASH — American Society of Hematology

ASO — antistreptolysin O

ATA — American Thyroid Association

ATP — adenosine triphosphate

AUC — area under the (concentration–time) curve

AV — atrioventricular

AVM — arteriovenous malformation

AVPU — alert / voice / pain / unresponsive (conscious-level scale)

AVSD — atrioventricular septal defect

B symptoms — lymphoma constitutional symptoms (Ann Arbor staging): unexplained fever > 38 °C, drenching night sweats, weight loss > 10%

BCG — bacille Calmette–Guérin (vaccine)

BID — twice daily

BMI — body mass index

BNP — B-type natriuretic peptide

BP — blood pressure

BRUE — brief resolved unexplained event

BSPED — British Society for Paediatric Endocrinology and Diabetes

BWPS — Burch–Wartofsky Point Scale (for thyrotoxicosis)

BXO — balanitis xerotica obliterans

C1-INH — C1-esterase inhibitor

CAH — congenital adrenal hyperplasia

CAP — community-acquired pneumonia

CBC — complete blood count

CCHD — critical congenital heart disease

CDH — congenital diaphragmatic hernia

CFU — colony-forming units

CHD — congenital heart disease

CICO — can't intubate, can't oxygenate

CKD — chronic kidney disease

CMC — children with medical complexity

CMPA — cow's milk protein allergy

CMV — cytomegalovirus

CNO — chronic nonbacterial osteomyelitis

CNS — central nervous system

CO — carbon monoxide

COG — Children's Oncology Group

COHb — carboxyhemoglobin

CPAP — continuous positive airway pressure

CPEG — Canadian Pediatric Endocrine Group

CPR — cardiopulmonary resuscitation

CRMO — chronic recurrent multifocal osteomyelitis

CRP — C-reactive protein

CRASH-2 / CRASH-3 — Clinical Randomisation of an Antifibrinolytic in Significant Haemorrhage trials (tranexamic acid in trauma / head injury)

CRRT — continuous renal replacement therapy

CSF — cerebrospinal fluid

CT — computed tomography

CVC — central venous catheter

CVID — common variable immunodeficiency

CVS — cardiovascular system

CXR — chest radiograph (chest X-ray)

DAT — direct antiglobulin (Coombs) test

DDAVP — desmopressin

DEFG — don't ever forget glucose (coma mnemonic)

DIC — disseminated intravascular coagulation

DIVA — difficult intravenous access

DKA — diabetic ketoacidosis

DL — direct laryngoscopy

DMD — Duchenne muscular dystrophy

DNase — deoxyribonuclease (dornase alfa)

DOPE — displacement, obstruction, pneumothorax, equipment (deterioration mnemonic)

DOPES — displacement, obstruction, pneumothorax, equipment, stacked breaths (ventilator mnemonic)

DRESS — drug reaction with eosinophilia and systemic symptoms

DSD — difference / disorder of sex development

DTaP — diphtheria, tetanus, acellular pertussis (pediatric)

EBV — Epstein–Barr virus

ECG — electrocardiogram

ECMO — extracorporeal membrane oxygenation

ECPR — extracorporeal cardiopulmonary resuscitation

ED — emergency department

EEG — electroencephalogram

eGFR — estimated glomerular filtration rate

EHF — extensively hydrolyzed formula

EMLA — eutectic mixture of local anesthetics

ENT — ear, nose and throat (otolaryngology)

ERCP — endoscopic retrograde cholangiopancreatography

ESBL — extended-spectrum β -lactamase

ESETT — Established Status Epilepticus Treatment Trial

ESC-NOW — Eat, Sleep, Console approach for Neonatal Opioid Withdrawal (trial)

ESR — erythrocyte sedimentation rate

ET — endotracheal (route)

ETT — endotracheal tube

EVD — external ventricular drain

eFAST — extended Focused Assessment with Sonography for Trauma (adds lung views for pneumothorax)

FAST — Focused Assessment with Sonography for Trauma (bedside ultrasound for free fluid)

FBC — full blood count

FEIBA — factor eight inhibitor bypassing activity

FHH — familial hypocalciuric hypercalcemia

FiO₂ — fraction of inspired oxygen

FN — febrile neutropenia

FPE — food protein-induced enteropathy

FPIAP — food protein-induced allergic proctocolitis

FPIES — food protein-induced enterocolitis syndrome

FTT — failure to thrive

FVC — forced vital capacity

G6PD — glucose-6-phosphate dehydrogenase

GA²LEN — Global Allergy and Asthma European Network

GAS — group A *Streptococcus*

GBS — group B *Streptococcus*; also Guillain-Barré syndrome

GCS — Glasgow Coma Scale

GER — gastroesophageal reflux

GERD — gastroesophageal reflux disease

GFR — glomerular filtration rate

GGT — gamma-glutamyl transferase

GH — growth hormone

GI — gastrointestinal

GIR — glucose infusion rate

GJ — gastrojejunal (tube)

HA — hypoallergenic (partially hydrolyzed formula label)

HAE — hereditary angioedema

HAST — hypoxia-altitude simulation test

HbA1c — glycated hemoglobin

HBO — hyperbaric oxygen

HCN — hydrogen cyanide

HCT — hematocrit

HDU — high-dependency unit

HEEADSSS — Home, Education, Eating, Activities, Drugs, Sexuality, Suicide, Safety (adolescent interview)

HFNC — high-flow nasal cannula

HHS — hyperosmolar hyperglycemic state

HHV-6 — human herpesvirus 6

HLA — human leukocyte antigen

HLH — hemophagocytic lymphohistiocytosis

HME — heat and moisture exchanger

HR — heart rate

HSP — Henoch–Schönlein purpura (IgA vasculitis)

HSV — herpes simplex virus

HUS — hemolytic uremic syndrome

IABP — intra-aortic balloon pump

IBI — invasive bacterial infection (bacteremia or bacterial meningitis)

ICP — intracranial pressure

ICU — intensive care unit

ID — infectious diseases

IDM — infant of a diabetic mother

IEI — inborn errors of immunity

IgA — immunoglobulin A

IgE — immunoglobulin E

IgG — immunoglobulin G

IgM — immunoglobulin M

IGRA — interferon- γ release assay

IM — intramuscular

iNO — inhaled nitric oxide

INR — international normalized ratio

IO — intraosseous

IPNA — International Pediatric Nephrology Association

ITP — immune thrombocytopenia

IU — international units

IUGR — intrauterine growth restriction

IV — intravenous

IVA — isovaleric acidemia

IVIG — intravenous immunoglobulin

KCl — potassium chloride

KPCC — Kuwait Poison Control Center

LAD — leukocyte adhesion deficiency

LCHADD — long-chain 3-hydroxyacyl-CoA dehydrogenase deficiency

LDH — lactate dehydrogenase

LFTs — liver function tests

LGMD — limb-girdle muscular dystrophy

LGA — large for gestational age

LMA — laryngeal mask airway

LP — lumbar puncture

LOC — loss of consciousness

LRINEC — Laboratory Risk Indicator for Necrotizing Fasciitis (score)

MAS — macrophage activation syndrome

MCAD — medium-chain acyl-CoA dehydrogenase deficiency

MCP — membrane cofactor protein (CD46)

MDRO — multidrug-resistant organism

MEDIF — airline Medical Information Form

MEN — multiple endocrine neoplasia

MenACWY — quadrivalent meningococcal conjugate vaccine (serogroups A, C, W, Y)

MenB — serogroup B meningococcal vaccine

MEP — maximal expiratory pressure

MIC — minimum inhibitory concentration

MIP — maximal inspiratory pressure

min — minutes

MMA — methylmalonic acidemia

MMR — measles, mumps and rubella (vaccine)

MRI — magnetic resonance imaging

MRP — most responsible physician

MRSA — methicillin-resistant *Staphylococcus aureus*

MSSA — methicillin-susceptible *Staphylococcus aureus*

MTP — massive transfusion protocol

MSUD — maple syrup urine disease

NAAT — nucleic acid amplification test

NAC — N-acetylcysteine

NaCl — sodium chloride

NAGS — N-acetylglutamate synthase

NAI — non-accidental injury

NEC — necrotizing enterocolitis

NEJM — The New England Journal of Medicine

NEWS2 — National Early Warning Score 2

NG — nasogastric

NGT — nasogastric tube

NICU — neonatal intensive care unit

NIF — negative inspiratory force

NIV — non-invasive ventilation

NJT — nasojejunal tube

NK — natural killer (cell)

NMS — neuroleptic malignant syndrome

NPA — nasopharyngeal airway

NPO — nil per os (nothing by mouth)

NSAID — non-steroidal anti-inflammatory drug

NT-proBNP — N-terminal pro-B-type natriuretic peptide

NTM — nontuberculous mycobacteria

OGT — orogastric tube

OHP — 17-hydroxyprogesterone (17-OHP)

OI — oxygenation index

OME — otitis media with effusion

OPA — oropharyngeal airway

OR — odds ratio

ORS — oral rehydration solution

ORT — oral rehydration therapy

OSA — obstructive sleep apnea

PA — propionic acidemia; also posteroanterior (radiograph view)

PALF — pediatric acute liver failure

PALS — Pediatric Advanced Life Support

PaO₂ — arterial partial pressure of oxygen

PAT — Pediatric Assessment Triangle

PCR — polymerase chain reaction

PDA — patent ductus arteriosus

PE — phenytoin equivalents

PEA — pulseless electrical activity

PECARN — Pediatric Emergency Care Applied Research Network

PED-TRAX — Pediatric Trauma and Tranexamic Acid study (observational)

PEG — percutaneous endoscopic gastrostomy

PEN-FAST — penicillin-allergy risk-assessment score

PEP — post-exposure prophylaxis

PEWS — Pediatric Early Warning Score

PICC — peripherally inserted central catheter

PICU — pediatric intensive care unit

PIGN — post-infectious glomerulonephritis

PJP — *Pneumocystis jirovecii* pneumonia

PLP — pyridoxal-5-phosphate

PMA — postmenstrual age

PO — by mouth (oral)

POC — portable oxygen concentrator

POCUS — point-of-care ultrasound

PPHN — persistent pulmonary hypertension of the newborn

PPI — proton pump inhibitor

PPV — positive-pressure ventilation

PR — per rectum (rectal route); also PR interval on ECG

PRAM — Pediatric Respiratory Assessment Measure

PRBC — packed red blood cells

PRN — as required (pro re nata)

PROPPR — Pragmatic, Randomized Optimal Platelet and Plasma Ratios trial

PSG — polysomnography

PSGN — post-streptococcal glomerulonephritis

PT — prothrombin time

PTH — parathyroid hormone

PTHrP — parathyroid hormone-related peptide

PTU — propylthiouracil

PUCAI — Pediatric Ulcerative Colitis Activity Index

PVC — premature ventricular contraction

pVT — pulseless ventricular tachycardia

QID — four times daily

QRS — QRS complex (ECG)

QT — QT interval (ECG)

QTc — corrected QT interval

RADT — rapid antigen detection test

RAPD — relative afferent pupillary defect

RBC — red blood cell

RBS — random blood sugar

RBUS — renal and bladder ultrasound

RCT — randomized controlled trial

RDS — respiratory distress syndrome

rFVIIa — recombinant activated factor VII

RIG — rabies immunoglobulin

RLQ — right lower quadrant

ROSC — return of spontaneous circulation

ROTEM — rotational thromboelastometry

RR — respiratory rate

RSI — rapid sequence intubation

RSV — respiratory syncytial virus

RT — respiratory therapist

RV — right ventricle

s — seconds

SA — septic arthritis

SBP — systolic blood pressure

SC — subcutaneous

SCFE — slipped capital femoral epiphysis

SCID — severe combined immunodeficiency

SCIWORA — spinal cord injury without radiographic abnormality

SGA — small for gestational age

SIDS — sudden infant death syndrome

SIP — spontaneous intestinal perforation

SIPA — shock index, pediatric age-adjusted

sJIA — systemic juvenile idiopathic arthritis

SJS — Stevens–Johnson syndrome

SOPA — suction, open mouth, pressure increase, airway alternative (NRP corrective steps)

SpO₂ — peripheral oxygen saturation

SPUR — severe, persistent, unusual, recurrent (immunodeficiency pattern)

SSSS — staphylococcal scalded skin syndrome

SSTI — skin and soft-tissue infection

STAT — immediately (statim)

STEC — Shiga toxin-producing *E. coli*

STI — sexually transmitted infection

SVC — superior vena cava

SVR — systemic vascular resistance

SVT — supraventricular tachycardia

T1DM — type 1 diabetes mellitus

T2DM — type 2 diabetes mellitus

TB — tuberculosis

TBI — traumatic brain injury

TBSA — total body surface area

TBW — total body water

TCA — tricyclic antidepressant

TcB — transcutaneous bilirubin

Tdap — tetanus, diphtheria, acellular pertussis (adolescent/adult)

TEN — toxic epidermal necrolysis

TEN-4-FACESp — bruising clinical decision rule (Torso, Ears, Neck in a child ≤ 4 y or any bruise < 4 months; Frenulum, Angle of jaw, Cheek, Eyelid, Sclera; patterned)

TEG — thromboelastography

TFI — total fluid intake

TID — three times daily

TIF — tracheo-innominate fistula

TLS — tumor lysis syndrome

TM — tympanic membrane

TMA — thrombotic microangiopathy

TMP-SMX — trimethoprim-sulfamethoxazole

TOF — tetralogy of Fallot

tPA — tissue plasminogen activator

TRAb — TSH-receptor antibody

TREC — T-cell receptor excision circle

TREKK — Translating Emergency Knowledge for Kids

TSB — total serum bilirubin

TSH — thyroid-stimulating hormone

TSI — thyroid-stimulating immunoglobulin

TTP — thrombotic thrombocytopenic purpura

TTS — tight-to-shaft (tracheostomy cuff)

TWIST — Testicular Workup for Ischemia and Suspected Torsion (score)

TXA — tranexamic acid

UCD — urea cycle disorder

ULN — upper limit of normal

UTI — urinary tract infection

V_d — volume of distribution

VA — venoarterial (ECMO)

VAD — ventricular assist device

VATS — video-assisted thoracoscopic surgery

VCUG — voiding cystourethrogram

VF — ventricular fibrillation

VL — videolaryngoscopy

VLCADD — very-long-chain acyl-CoA dehydrogenase deficiency

VP — ventriculoperitoneal (shunt)

VRE — vancomycin-resistant enterococci

VSD — ventricular septal defect

VT — ventricular tachycardia

VTE — venous thromboembolism

VUR — vesicoureteral reflux

VV — venovenous (ECMO)

WBC — white blood cell (count)

WOB — work of breathing

WPW — Wolff–Parkinson–White (syndrome)

How to Open & Save This Guide

Which file? The **EPUB** resizes the text and is best for reading on phones and e-readers. The **PDF** keeps the exact page layout. Use whichever you prefer — the steps are the same.



Scan to download any format. Opens kpc-guide.com/downloads — the full guide as Print PDF (A4, A5), Phone PDF, EPUB and an offline HTML copy. The address stays the same when the files are updated.



Scan to install the app. Opens the guide at kpc-guide.com — then **Share** → **Add to Home Screen** (iPhone) or **:** → **Install app** (Android). Works offline after the first open; updates install automatically.

Apple Books

IPHONE • IPAD • MAC

1. Open the EPUB or PDF file.
2. Tap the **Share** icon, then **Copy to Books**.
3. On a Mac, drag the file onto the Books app.
4. Find it any time in your **Library**.

Kindle

APP OR E-READER

1. Install the **Send to Kindle** app, or visit sendtokindle.amazon.com.
2. Add the **EPUB** file — Amazon turns it into a Kindle book.
3. It appears on all your Kindle apps under **Library → Docs**.

Android

PHONES & TABLETS

1. Open the **Google Play Books** app.
2. Go to **Library** and tap **Upload**.
3. Choose the EPUB or PDF — it syncs to all your devices.

Any e-reader app or Adobe Acrobat also opens the files from Downloads.

Install as an App

OFFLINE • HOME-SCREEN ICON

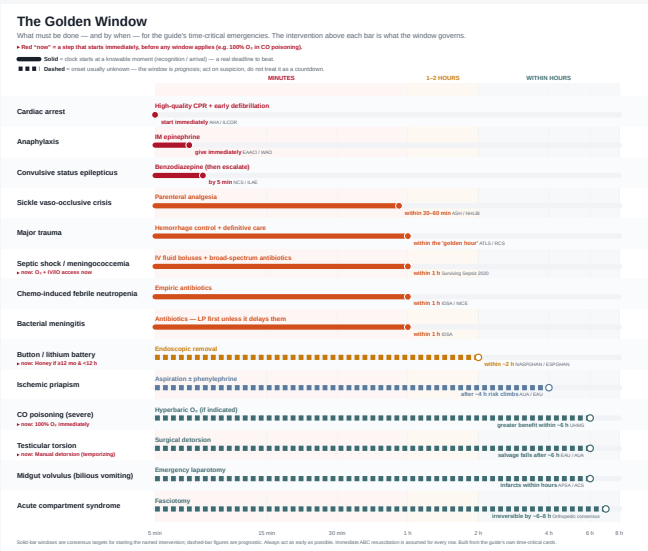
1. Open kpc-guide.com in **Chrome** (Android) or **Safari** (iPhone).
2. Android: menu ☰ → **Install app**. iPhone: **Share** → **Add to Home Screen**.
3. Open it from your home screen — it works **offline**.

Screen looks dark? The pages are white. If they appear black, your reader's **night / dark mode** is on — switch it off in the app to see the guide as designed.

Time-critical emergencies at a glance

Solid bar = clock starts at a knowable moment (a real deadline).

Dashed bar = onset usually unknown – prognosis, act on suspicion.



Contents

Find It by Symptom	38
Presentation · card index	38
01 Resuscitation	40
Cardiopulmonary Arrest	43
02 Behavioral & Psychiatric	51
Acute Severe Agitation	51
Self-Harm / Intentional Ingestion	54
Drug-Induced Hyperthermia Syndromes	56
03 Cardiac & Shock	58
Shock · Recognition & Type	58
Septic Shock	63
Myocarditis / Acute Heart Failure	74
Bradycardia + Poor Perfusion	78
Supraventricular Tachycardia	80
Ventricular Tachycardia (with a pulse)	85
Neonatal & Infant CHD	87
Hypercyanotic (Tet) Spell	92
Blocked Systemic-to-Pulmonary Shunt (BT / Sano)	95
Acute Rheumatic Fever	100
Pericarditis & Cardiac Tamponade	104
Infective Endocarditis (incl. Subacute)	108
04 Common Acute Conditions	115
Fever in a Well-Appearing Infant < 2 Months	115
Gastroenteritis & Dehydration	121
Community-Acquired Pneumonia	128
Complicated Pneumonia · Empyema	133
Acute Otitis Media	138
GAS Pharyngitis	143
Urinary Tract Infection	148
UTI · Treatment & Follow-up	150
Skin & Soft Tissue Infections	154
Cervical Lymphadenopathy	159
Fever · Assessment & Antipyretics	166
The Rash · Approach & Red Flags	168
Preseptal & Orbital Cellulitis	174
Preseptal & Orbital · Management	178
BRUE (Brief Resolved Unexplained Event)	183
Adolescent Emergencies · Kuwait Context	187
The Child with Obesity	191
Pertussis (Whooping Cough)	195
05 Complex & Technology-Dependent Care	198
The Child with Medical Complexity	198
Tracheostomy Emergencies	200

Tracheostomy Tubes & Cuff Types	213
Gastrostomy (Feeding) Tube Problems	216
Enteral Feeding Tubes · Types, Delivery & Complications	221
ECMO · What the General Pediatrician Needs to Know	227
Post-PICU Syndrome (PICS-p)	230
Aeromedical Transport · Pediatric & Neonatal	234
06 Endocrine & Metabolic	245
Diabetic Ketoacidosis	245
Hyperglycemic Hyperosmolar State (HHS)	248
Adrenal Crisis / Salt-Wasting CAH	251
Hypoglycemia (Beyond the Newborn)	253
Acute Electrolyte Emergencies	257
Hypercalcemia	262
Thyroid Storm (Thyrotoxic Crisis)	266
Metabolic Crisis (Inborn Error)	269
07 Focused Specialty Emergencies	274
Pediatric Eye Emergencies	274
Dental & Oral Emergencies	279
Dermatologic Emergencies	287
Pediatric ENT Emergencies	308
Pediatric Neurosurgical Emergencies	312
08 Gastroenterology & Hepatology	316
Upper Gastrointestinal Bleeding	317
Lower Gastrointestinal Bleeding	323
Acute Liver Failure	328
Acute Pancreatitis	333
Acute Hepatitis	337
Neonatal Cholestasis	341
Foreign Body Ingestion	344
Button (Lithium) Battery Ingestion	347
Functional Constipation & Fecal Impaction	350
Reflux & GERD	357
Ingestion Injuries · See Also	360
Acute IBD Flare & Toxic Megacolon	360
09 Hematologic & Oncologic	364
Chemotherapy-induced Febrile Neutropenia	364
Febrile Neutropenia · Otherwise Healthy Child	364
Tumor Lysis Syndrome	375
Sickle Cell Crises	379
Acute Bleeding in Hemophilia	383
Immune Thrombocytopenia (ITP)	385
New-Diagnosis Oncologic Emergencies	389
10 Neonatal	395
Neonatal Resuscitation	395
Critical CHD · Pulse Oximetry Screening	397
Persistent Pulmonary Hypertension (PPHN)	400

Neonatal Hypoglycemia	404
Neonatal Hyperbilirubinemia	410
Neonatal Sepsis · Early-Onset (rapid approach)	418
Omphalitis & Delayed Umbilical Separation	420
Necrotizing Enterocolitis	427
Neonatal Seizures	431
Vitamin K Deficiency Bleeding (VKDB)	436
Neonatal Abstinence Syndrome	439
Neonatal Dosing by Corrected Age (PMA)	439
11 Neurologic	445
Convulsive Status Epilepticus (> 1 month of age)	445
Febrile Seizure	450
Bacterial Meningitis & Encephalitis	451
Brain Tumor · New Presentation	458
Brain Abscess	461
Raised ICP / The Comatose Child	465
Guillain-Barré Syndrome / Acute Flaccid Paralysis	469
The Child with Neuromuscular Disease	473
12 Orthopedic & Musculoskeletal	479
Orthopedic Emergencies · Call Now	479
Osteoarticular Infections	480
Osteoarticular Infections · Management	484
The Limping Child	488
Transient Synovitis vs Septic Arthritis	491
Nursemaid's Elbow	494
Slipped Capital Femoral Epiphysis (SCFE)	497
Fractures · Patterns & Non-Accidental Red Flags	500
Compartment Syndrome	503
Open Fracture	506
Fracture with Pulseless Limb	509
Joint Dislocation	512
Unstable Pelvic Injury	515
Spine Fracture with Neurological Deficit	518
13 Pediatric Trauma & TBI	522
Trauma Team & Roles	522
Major Trauma · Primary Survey & Resuscitation	524
Major Trauma · Secondary Survey & Decisions	529
Traumatic Brain Injury	533
Child Protection & Safeguarding	538
Massive Transfusion & Blood Products	543
14 Renal & Urology	547
Acute Kidney Injury	547
Hemolytic Uremic Syndrome	549
Hypertensive Emergency	557
Nephrotic Syndrome	560
Circumcision · Kuwait Practice	565
Acute Scrotum	568

Testicular Torsion	572
Paraphimosis	574
Balanitis / Balanoposthitis & Phimosis	576
Priapism	579
Gross Hematuria	582
Acute Post-infectious Glomerulonephritis (PIGN)	586
Rhabdomyolysis	589
15 Respiratory, Allergy & Immunology	592
Anaphylaxis	592
Acute Asthma	600
Croup	608
Retropharyngeal Abscess	612
Difficult Airway	615
Bronchiolitis	620
Choking (Foreign-Body Obstruction)	622
Obstructive Sleep Apnea	625
Cow's Milk Protein Allergy	628
Antibiotic Allergy Assessment & Pretesting	633
Hereditary Angioedema · Acute Attack	640
When to Suspect Primary Immunodeficiency (IEI)	642
Tension Pneumothorax & Needle Decompression	650
16 Rheumatologic	652
Cytokine Storm · HLH / MAS	652
Kawasaki Disease	656
IgA Vasculitis (Henoch-Schönlein Purpura)	661
MIS-C (Multisystem Inflammatory Syndrome)	664
17 Surgical Emergencies	667
Malrotation / Midgut Volvulus	667
Intussusception	669
Hypertrophic Pyloric Stenosis	671
Acute Appendicitis	672
Incarcerated Inguinal Hernia	675
18 Toxicology & Environmental	677
Poisoning & Toxic Ingestion	677
Caustic (Corrosive) Ingestion	681
Burns	683
Heat Stroke	686
Drowning	689
Smoke Inhalation & Carbon Monoxide Poisoning	692
Envenomation · Scorpion Sting & Snakebite	697
Quick Reference	700
When to Admit	701
Rabies Post-Exposure Prophylaxis	703
IV · Oral Switch & the DIVA Patient	705
Weight & Airway Sizing	709
Oxygen Delivery Systems & Cylinder Duration	711

Normal Vital Signs by Age	718
Scales, Equations & Synonyms	719
Procedural Sedation, RSI & Analgesia	723
Emergency Drug Doses at a Glance	726
Resuscitation Dose Calculator	730
Resuscitation Doses by Weight	735
Anticoagulation · VTE Prophylaxis & Treatment	737
Maximum Local Anesthetic Doses	739
Kuwait Childhood Immunization Schedule	742
Think Here · Consanguinity & Gulf-Prevalent Conditions	746
High-Alert Drugs & Fluid Safety	747
Decision-Rules Engine · Cutoff · Action	751
How This Guide Is Maintained	754
Shortest Antibiotic Durations	756
Bedside Calculators	761
References	772
Sources	772



Find It by Symptom

The front door — start here: a presentation routes you to the card

Presentation → card index

NAVIGATOR

Presentation → card

You see a presentation, not a diagnosis. Start here and jump to the card.

Presentation	Go to
Bilious vomiting (green)	<u>Malrotation / Midgut Volvulus</u>
The limping / non-weight-bearing child	<u>The Limping Child</u> · <u>Transient Synovitis vs Septic Arthritis</u> · <u>Slipped Capital Femoral Epiphysis (SCFE)</u>
Cyanosis / the blue baby	<u>Hypercyanotic (Tet) Spell</u> · <u>Neonatal & Infant CHD</u> · <u>Critical CHD – Pulse Oximetry Screening</u>
Floppy infant / acute weakness	<u>Guillain-Barré Syndrome / Acute Flaccid Paralysis</u> · <u>The Child with Neuromuscular Disease</u> · <u>Neonatal Sepsis (rapid approach)</u>
Petechiae / purpura	<u>Immune Thrombocytopenia (ITP)</u> · <u>IgA Vasculitis (Henoch-Schönlein Purpura)</u> · <u>Septic Shock</u>

Presentation	Go to
Stridor	<u>Croup</u> • <u>Choking (Foreign-Body Obstruction)</u> • <u>Anaphylaxis</u> • <u>Retropharyngeal Abscess</u>
Wheeze	<u>Acute Asthma</u> • <u>Bronchiolitis</u> • <u>Anaphylaxis</u>
Seizure	<u>Convulsive Status Epilepticus</u> • <u>Febrile Seizure</u> • <u>Neonatal Seizures</u>
Fever + non-blanching rash	<u>Septic Shock</u> • <u>Bacterial Meningitis & Encephalitis</u>
Acute scrotal pain	<u>Acute Scrotum</u> • <u>Testicular Torsion</u>
Jaundiced neonate	<u>Neonatal Hyperbilirubinemia</u> • <u>Neonatal Cholestasis</u>
Bloody diarrhea	<u>Hemolytic Uremic Syndrome</u> • <u>Acute IBD Flare & Toxic Megacolon</u> • <u>Gastroenteritis & Dehydration</u>
Apnea / brief resolved event	<u>BRUE (Brief Resolved Unexplained Event)</u> • <u>Pertussis (Whooping Cough)</u>
Severe abdominal pain	<u>Acute Appendicitis</u> • <u>Intussusception</u> • <u>Malrotation / Midgut Volvulus</u> • <u>Diabetic Ketoacidosis</u>

Presentation	Go to
Bruising / abnormal bleeding	<u>Immune Thrombocytopenia (ITP)</u> · <u>Acute Bleeding in Hemophilia</u> · <u>Vitamin K Deficiency Bleeding (VKDB)</u> · <u>Child Protection & Safeguarding</u>
Fever in the young infant	<u>Fever in the Young Infant</u>
Shock / poor perfusion	<u>Shock – Recognition & Type</u> · <u>Septic Shock</u>
Poisoning / ingestion	<u>Poisoning & Toxic Ingestion</u> · <u>Button (Lithium) Battery Ingestion</u> · <u>Caustic (Corrosive) Ingestion</u>
High glucose / ketones	<u>Diabetic Ketoacidosis</u> · <u>Hyperglycemic Hyperosmolar State (HHS)</u>
Hypoglycemia	<u>Hypoglycemia (Beyond the Newborn)</u> · <u>Metabolic Crisis (Inborn Error)</u> · <u>Neonatal Hypoglycemia</u>
Painful swollen joint + fever	<u>Transient Synovitis vs Septic Arthritis</u> · <u>Osteoarticular Infections</u>
Dark / tea-colored urine	<u>Rhabdomyolysis</u> · <u>Acute Kidney Injury</u> · <u>Gross Hematuria</u>
Agitation / behavioral crisis	<u>Acute Severe Agitation</u> · <u>Self-Harm / Intentional Ingestion</u>
Major hemorrhage / trauma	<u>Massive Transfusion & Blood Products</u> · <u>Major Trauma – Primary Survey & Resuscitation</u>

Presentation	Go to
Anaphylaxis / urticaria + collapse	<u>Anaphylaxis</u>
The unwell child with a device	<u>Tracheostomy Emergencies</u> · <u>Gastrostomy (Feeding) Tube Problems</u> · <u>The Child with Medical Complexity</u>

◆ HOW TO USE

- This routes the commonest bedside **PRESENTATIONS** to the relevant cards — it is a starting point, not a differential.
- Tap a card name to jump straight to it.
- The full alphabetical index and the section colors remain the other two ways in.

Can't-miss patterns – recognition map

See it → name the danger → act. Each node is a sentinel pattern – tap it to jump to the full card (workup + doses). Not exhaustive; the section cards remain the source of truth.



Cardiopulmonary Arrest MINUTES

Unresponsive, apneic, pulseless within 10 seconds → CPR now.

▲ HIGH-QUALITY CPR IS THE PRIORITY

- Rate **100–120 per minute**; depth $\frac{1}{3}$ of chest depth (~4 cm infant, ~5 cm child); full recoil; minimal interruptions¹.
- Ratio **30:2** (single rescuer) · **15:2** (two rescuers). **Adolescent (post-pubertal): 30:2 for both single and two rescuers** (adult BLS)^{1,160}.
- **With an advanced airway (all ages):** continuous chest compressions with **1 breath every 2–3 seconds (20–30 breaths/min)**¹, without pausing compressions.

DURING CPR

- Attach a monitor/defibrillator; establish access — **immediate IO in cardiac arrest if an IV is not already in place** (for other emergencies, go IO if IV fails after 2 attempts or 90 s); consider an **advanced airway** (then continuous compressions with one breath every 2–3 s); give oxygen.¹
- Do not delay epinephrine or defibrillation for access or intubation¹.



DECISION

Rhythm on the monitor?

Defibrillate as early as possible — 2 J/kg → 4 J/kg → ≥ 4 J/kg (subsequent shocks; up to 10 J/kg or the adult dose¹, $\approx$ 200 J biphasic / 360 J monophasic).

- **Resume high-quality CPR** immediately after every shock¹.
- **Reassess** rhythm and a central pulse (brachial in infants; femoral/carotid in children)¹ **every 2 min**, pausing compressions ≤ 10 s. If VF/pVT persists, shock again and resume CPR.
- **Epinephrine** during CPR **after the 2nd shock**¹ (don't pause compressions), then every 3–5 min.
- **Amiodarone** (or lidocaine) **after the 3rd shock**¹ for refractory VF/pVT.

Sequence (a continuous cycle): shock → shock → epinephrine → shock → amiodarone → **↺ back to shock** — keep repeating the 2-min CPR–shock cycle, epinephrine every 3–5 min and a further amiodarone dose as needed¹.

Give epinephrine during CPR as soon as possible, then every 3–5 minutes. Continue high-quality CPR without unnecessary interruptions; check rhythm every 2 minutes. If a shockable rhythm appears, **cross over to the shockable¹ (VF/pVT) pathway** and defibrillate.

**EVERY CYCLE – REVERSIBLE CAUSES**

- **6 H's:** hypoxia, hypovolemia, hydrogen ion (acidosis), hypo-/hyperkalemia, hypoglycemia, hypothermia¹.
- **5 T's:** tension pneumothorax, tamponade (cardiac), toxins, thrombosis (pulmonary), thrombosis (coronary)¹.


ROSC

Post-arrest: start post-arrest care · secure the airway · wean oxygen to target SpO₂ 94–99% (avoid hyperoxic reperfusion injury¹) and normal carbon dioxide · systolic and/or mean arterial BP > 10th centile for age · targeted normothermia (36–37.5 °C) · correct metabolic derangements · treat seizures (EEG if persistent encephalopathy).

MEDICATIONS & ENERGY

Drug / action	Dose	Route & notes
Epinephrine (adrenaline)	0.01 mg/kg (max 1 mg)	IV/IO, 0.1 mg/mL (1:10,000) ¹ concentration; give during CPR without pausing compressions, every 3–5 min
Amiodarone	5 mg/kg (max 300 mg)	IV/IO bolus for refractory VF/pVT; may repeat up to twice (3 doses total; max 15 mg/kg). Avoid in prolonged QT (further prolongs QT) ¹
Lidocaine	1 mg/kg	IV/IO alternative to amiodarone ¹

Drug / action	Dose	Route & notes
Defibrillation	2 → 4 → ≥4 J/kg	Unsynchronized; ¹ up to 10 J/kg or adult dose (≈ 200 J biphasic / 360 J monophasic)

⚠ REMEMBER

- Respiratory failure is the leading cause of cardiac arrest in pediatrics — oxygenate and ventilate early¹.
- Intraosseous access is fast — in **cardiac arrest go straight to IO if an IV is not already in place** (in other emergencies, IO if IV fails after 2 attempts or 90 s); do not delay epinephrine waiting for a vein¹.

◆ ALGORITHMS

- Full AHA PALS flowcharts (cardiac arrest
- bradycardia
- tachycardia
- post-arrest) — cpr.heart.org



▲ ADMIT · DISCHARGE · REFER

- **Admit** — every child with return of circulation → **start post-arrest care** (oxygenation/ventilation targets, hemodynamic support, treat the cause, neuroprotection, avoid fever and hypo/hyperglycemia)¹.
- **Discharge** — not applicable — care after unsuccessful resuscitation follows the death-confirmation and bereavement pathway.
- **Refer** — the specialty for the reversible cause (6 Hs / 5 Ts): cardiology, surgery, toxicology, or metabolic; involve the resuscitation team early.

◆ **SEE ALSO** Situation-specific resuscitation:

Neonatal Resuscitation · **Shock — Recognition & Type** · **Anaphylaxis** · **Difficult Airway** · **Major Trauma — Primary Survey**.

Code Blue – team roles & positions

Assign each role out loud and close the loop: Team Leader (hands-off), Airway Manager, Compressor, Defib Operator / CPR Coach, Medications & Vascular Access (×2), and Recorder / Documenter.



Human-prompted, AI-generated image, reviewed and approved by the panel.

Behavioral & Psychiatric

Agitation · self-harm · drug-induced syndromes

Acute Severe Agitation MINUTES

Keep child and staff safe; look for a treatable cause.

① EXCLUDE AN ORGANIC CAUSE FIRST

Check glucose, oxygenation, head injury, intoxication/withdrawal, pain. Agitation is a symptom — treat the cause where present²²⁵.



② NON-DRUG MEASURES FIRST

Calm low-stimulation environment, verbal de-escalation, familiar caregiver; offer oral medication, including the child's own regular medication, when available.²²⁵

severe / unsafe



③ PHARMACOLOGIC (PER LOCAL PATHWAY)

Second-generation antipsychotics first-line; benzodiazepines where intoxication/anxiety predominates. Physical restraint = brief last resort, continuous monitoring²²⁵.

Drug	Dose (approximate)	Route & notes
Olanzapine	2.5–5 mg (< 40 kg); 5–10 mg (≥ 40 kg)	PO/IM/ODT; the ODT (orally disintegrating — Zyprexa Zydis) is easiest in an agitated child. Do not give within 1 h of a benzodiazepine (respiratory depression) ²²⁵
Risperidone	0.25–1 mg	PO; well tolerated ²²⁵
Lorazepam	0.05–0.1 mg/kg (max 2 mg)	PO/IM/IV; for substance- or anxiety-driven agitation ²²⁵
Haloperidol	0.025–0.05 mg/kg (max 5 mg)	IM; watch for dystonia/QT — second-generation agents preferred ²²⁵

Drug	Dose (approximate)	Route & notes
Diphen- hydramine	1–2 mg/kg/dose (max 50 mg)	IV/IM/PO; antidote for an acute dystonic reaction from an antipsychotic (haloperidol/risperidone) — not a treatment for agitation itself ²²⁵

⚠ CAUTION WITH BENZODIAZEPINES

- Avoid benzodiazepines and diphenhydramine in **delirium** — can worsen confusion (paradoxical disinhibition)²²⁵.
- Also avoid or use with caution in **developmental delay and autism**, where benzodiazepines commonly cause **paradoxical reactions (more agitation)**²²⁵.

▴ ADMIT · DISCHARGE · REFER

- **Admit** — ongoing risk, a medical cause, or need for continued sedation/monitoring.
- **Discharge** — settled, medically cleared, with a safety plan and follow-up.

- **Refer** — psychiatry; medical/toxicology for organic causes.

Self-Harm / Intentional Ingestion

HOURS

Stabilize the patient medically first, then address psychological safety.

① MEDICAL STABILIZATION

ABCDE, glucose, ECG, and a pregnancy test (β -hCG) in all post-menarchal females²²⁹. For any ingestion: identify agent and timing, check acetaminophen and salicylate levels, contact poisons information²²⁹. Treat per the specific antidote pathway.



② SAFETY

Continuous one-to-one observation, safe environment, involve a trusted adult/caregiver. Do not discharge before a mental-health assessment²²⁹.



Refer for psychiatric/psychosocial assessment; arrange follow-up and support before discharge²²⁹.

◆ **CROSS-REFERENCE** General approach, antidotes, and Kuwait Poison Control hotline: see [Poisoning & Toxic Ingestion](#) (Toxicology, Trauma & Environmental).

▲ **ADMIT · DISCHARGE · REFER**

- **Admit** — medical risk from the ingestion/injury, or ongoing suicidality → admit and monitor.
- **Discharge** — medically cleared AND psychiatrically assessed as safe, with follow-up and means restriction²²⁹.
- **Refer** — psychiatry (risk assessment), social work/safeguarding, and toxicology.

Drug-Induced Hyperthermia Syndromes

MINUTES

Three look-alikes with different triggers — and different antidotes.

Syndrome	Clue	Key treatment
Neuroleptic malignant syndrome	Antipsychotic; "lead-pipe" rigidity , slow onset, high creatine kinase ²²⁷	Stop drug; cool; fluids; benzodiazepine; dantrolene; bromocriptine ²²⁷
Serotonin syndrome	Serotonergic drug; clonus & hyperreflexia , rapid onset, agitation ²²⁶	Stop drug; cool; benzodiazepine; cyproheptadine ²²⁶
Malignant hyperthermia	Volatile anesthetic / succinylcholine ²²⁸ rigidity, rising CO ₂	Stop trigger; dantrolene ; aggressive cooling; treat hyperkalemia/rhabdomyolysis; urgent PICU/toxicology/anesthesia — antidotes per local protocol ²²⁸

⚠ ALL THREE

- Active cooling, IV fluids; treat hyperkalemia and rhabdomyolysis²²⁸.

- Secure airway; transfer to PICU.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → PICU (cooling, sedation, dantrolene as indicated).
- **Discharge** — not acutely; after resolution and review of the causative agent.
- **Refer** — toxicology/PICU; psychiatry/pharmacy for a medication review.

Cardiac & Shock

03

Shock recognition & types · septic shock · arrhythmias · myocarditis / heart failure · congenital heart disease · hypercyanotic spell

Shock – Recognition & Type MINUTES

Circulatory failure: tissues not getting enough oxygen. Hypotension is a late sign.

RECOGNIZE EARLY (BEFORE HYPOTENSION)

- Tachycardia, prolonged or flash capillary refill, cool or mottled skin, weak pulses, altered mental status, reduced urine output, rising lactate.⁶

UNIVERSAL FIRST STEPS

Start with ABCDEs · Oxygen · IV/IO access · monitor · check glucose & lactate · identify the type below⁶.



DECISION

Which type of shock?

Type	Cause	Key treatment
Hypovolemic	Vomiting/diarrhea, hemorrhage, burns	Crystalloid or balanced solution 10–20 mL/kg boluses; packed red cells if hemorrhage; stop ongoing losses ⁶
Distributive	Sepsis, anaphylaxis, neurogenic	Fluids + early vasoactive; treat the cause (see <u>Septic shock</u> , <u>Anaphylaxis</u>)
Cardiogenic	Myocarditis, arrhythmia, heart disease	Cautious fluid 5 mL/kg slowly; inotrope/inodilator; treat arrhythmia; early cardiology consult and PICU escalation ²³¹
Obstructive	Tension pneumothorax, tamponade, duct-dependent lesion, pulmonary embolism	Relieve the obstruction (decompress a tension pneumothorax, drain a tamponade, and — for example — prostaglandin for a duct-dependent lesion; massive/high-risk PE → thrombolysis + heparin, specialist-directed) ¹

⚠️ MIMICS SHOCK – NOT CIRCULATORY FAILURE

- **Carbon monoxide, cyanide, hydrogen sulfide, and methemoglobinemia** impair oxygen **delivery or use at the**

hemoglobin / cellular level, not the circulation — so this is **not true shock**, and the usual endpoints (blood pressure, capillary refill) can look normal. **Suspect it** when perfusion and BP appear adequate but there is **unexplained lactic acidosis / tissue hypoxia** with a suggestive history (fire / smoke, enclosed space, oxidizing drugs). **SpO₂ is unreliable** — confirm with **co-oximetry / blood gas¹**.

- It will **not respond to fluids or vasopressors**: give **100% oxygen** and the specific antidote¹ (see **Smoke Inhalation & CO** and **Poisoning**).

RESUSCITATION DRUGS

Drug	Dose	Notes
Balanced crystalloid	10–20 mL/kg (5 mL/kg if cardiogenic)	Ringer's lactate preferred; normal saline if a balanced fluid is unavailable. Bolus; reassess for fluid overload after each
Epinephrine (adrenaline) infusion	0.05–0.3 mcg/kg/min	Titrate to response; first-line vasoactive — start epinephrine or norepinephrine early (PALS) ⁶ . May start peripherally with close observation for extravasation until central access is obtained
Norepinephrine (noradrenaline) infusion	0.05–0.3 mcg/kg/min	Titrate to response; first-line alternative — warm/cold classification no longer directs the choice ⁶ . May start peripherally with close extravasation monitoring until central access is obtained

⚠ FLUID CAUTION

- Reassess after every bolus.
- Vasoactives may start alongside fluids if evidence of

life-threatening end-organ hypoperfusion.

- Slow or stop fluids at first **signs of fluid overload** — new crackles, enlarging liver, gallop, worsening work of breathing, or overload on POCUS⁶ — especially suspected cardiogenic shock.

◆ THERAPEUTIC ENDPOINTS (AIM FOR)

- Heart rate and blood pressure normal for age, capillary refill ≤ 2 s, warm extremities with normal pulses⁶, normal mental status, urine output ≥ 1 mL/kg/h, falling / normalizing lactate (central venous O₂ saturation $\geq 70\%$ where available).
- Reassess after every fluid bolus and every change in vasoactive support.

▴ ADMIT · DISCHARGE · REFER

- **Admit** — any child in shock → **call the PICU team to evaluate** (high-dependency/PICU); begin resuscitation before the type is confirmed⁶.

- **Discharge** — only once the cause is treated and perfusion is normal off vasoactive support.
- **Refer** — by type — septic → PICU/infectious diseases; cardiogenic → cardiology; obstructive → treat the cause (tension pneumothorax, tamponade); hypovolemic/hemorrhagic → surgery/trauma. **Involve PICU in parallel** for any child in shock.

Septic Shock

MINUTES

SSC

Infection + circulatory dysfunction. Defined by the [Phoenix Sepsis Score](#)  (≥ 2 points = sepsis; ≥ 1 cardiovascular point = septic shock).

⚠ THE PHOENIX SCORE IS NOT A SCREENING TOOL

- The Phoenix Sepsis Score identifies **organ dysfunction** — by the time it is positive, sepsis is already established²³⁰.
- It is **not** an early-screening tool.

- Keep using your standard pediatric **early-warning / sepsis-screening** triggers to catch the child **before** organ dysfunction develops²³⁰.

FIRST HOUR — every minute counts



RESUSCITATE

- **Impaired perfusion** = altered mental status; tachycardia with weak or bounding pulses and wide pulse pressure ($\pm$ hypotension); capillary refill > 2 s, mottled cool skin or pallor; oliguria < 1 mL/kg/h (adolescents < 0.5 mL/kg/h).
- 100% oxygen via non-rebreather mask
- two IV/IO
- **antibiotics within 1 hour**⁶
- IV fluid boluses.
- **Labs:** glucose, CBC + differential, PCT or CRP, electrolytes, ionized calcium, renal and liver function, coagulation, blood gas, lactate, cultures (blood $\pm$ urine).
- **Hypotension** (late sign) = systolic BP below, by age:
 - < 60 mmHg — neonate
 - < 70 mmHg — 1–12 mo
 - $< 70 + 2 \times \text{age}(\text{yr})$ — 1–10 yr
 - < 90 mmHg — ≥ 10 yr.⁶



FLUIDS

- **Balanced crystalloid** (Ringer's lactate preferred over 0.9% saline, except with hyponatremia or raised-ICP concern) 10–20 mL/kg boluses; reassess after each, repeat up to ~40 mL/kg⁶.
- **Watch for fluid overload** — new hepatomegaly, crackles, gallop, or B-lines on lung POCUS. Third bolus only if still dehydrated without overload¹⁶⁴.
- Start vasoactives if shock persists after ~40 mL/kg, or earlier if overload develops⁶.

fluid-refractory



VASOACTIVE

- **Call PICU immediately.**
- Start **epinephrine** or **norepinephrine** (either is acceptable first-line; not dopamine)⁶ — the **warm/cold picture** (norepinephrine if warm/vasodilated with bounding pulses, epinephrine if cold/low-output) can **guide but no longer strictly directs**⁶ the choice; a well-sited peripheral IV is acceptable, with close extravasation monitoring while central access is arranged.
- **Catecholamine-resistant shock** → consider **early** hydrocortisone, especially with adrenal-insufficiency risk (purpura fulminans, recent/chronic steroids, hypothalamic–pituitary–adrenal disease): stress dose $\approx 50 \text{ mg/m}^2$ then $50\text{--}100 \text{ mg/m}^2/\text{day}$ ⁶
- OR $1\text{--}2 \text{ mg/kg}$ then $\approx 2\text{--}4 \text{ mg/kg/day}$.
- Give stress-dose hydrocortisone for suspected/documented adrenal insufficiency. In catecholamine-resistant shock, hydrocortisone may restore vascular responsiveness to

catecholamines — adult data suggest a mortality benefit, but pediatric evidence remains limited⁶. Call PICU (SSC).

- **Refractory shock** — persisting despite fluids, ≥ 2 vasoactives, and hydrocortisone, or **low cardiac output on high inotropic support** → **consider VA-ECMO early** and activate an ECMO center⁶; survival falls the longer cannulation is delayed. SSC 2020 suggests VA-ECMO in children with septic shock refractory to all other treatments⁶.

MEDICATIONS

Drug	Dose	Notes
Ceftriaxone (empiric)	100 mg/kg (max 4 g)	IV; empiric regimen by focus (immunocompetent, community-acquired): no obvious focus → ceftriaxone ± vancomycin ± amikacin; meningitis/encephalitis → ceftriaxone + vancomycin ± acyclovir; pneumonia → ceftriaxone ± vancomycin; skin/soft-tissue → cefotaxime + vancomycin ± clindamycin; toxic shock or necrotizing fasciitis → piperacillin-tazobactam + vancomycin + clindamycin; intra-abdominal → piperacillin-tazobactam; neonate → cefotaxime + ampicillin (± acyclovir if HSV suspected, ± vancomycin if MRSA suspected). Immunocompromised or hospital-acquired: add antipseudomonal cover and consider antifungal. Send MRSA screening

Drug	Dose	Notes
		swabs. Meningitic ceftriaxone dose 50 mg/kg/dose every 12 h (max 2 g/dose) ⁶
Balanced crystalloid	10–20 mL/kg	Ringer's lactate preferred; normal saline if unavailable. Bolus, reassess after each
Epinephrine (adrenaline)	Start 0.05–0.1 mcg/kg/min	IV/IO infusion; escalate every 5–10 min to target — escalation and titration must be done by the PICU team. Central line preferred; peripheral is a bridge only with close extravasation monitoring until central access
Norepinephrine (noradrenaline)	Start 0.05–0.1 mcg/kg/min	IV/IO infusion; escalate every 5–10 min to target — escalation and titration must be done by the PICU team. Central line preferred; peripheral is a bridge only with close extravasation monitoring until central access

Drug	Dose	Notes
Hydro-cortisone	1–2 mg/kg (max 100 mg)	IV; give for suspected/documentated adrenal insufficiency. No clear evidence for or against in catecholamine-resistant shock

◆ Vasoactive infusions — how to give them safely

- **ADRENALINE AND NORADRENALINE INFUSIONS SHOULD BE GIVEN VIA A CENTRAL VENOUS CATHETER WHEREVER POSSIBLE. DO NOT DELAY** initiation of vasoactive support to obtain central access (SSC Pediatric 2020)⁶.
- Peripheral use is a **BRIDGE ONLY** — via a large vein, a dedicated dilute infusion, with site review **EVERY 15 MINUTES FOR THE FIRST HOUR THEN HOURLY**, and conversion to central access as soon as feasible.
- **DOSES EXCEEDING THE STATED CENTRAL OR PERIPHERAL MAXIMUM REQUIRE CONSULTANT APPROVAL**, documented in the notes. Escalation beyond **1 MCG/KG/MIN** should prompt reassessment of the shock phenotype and consideration of **VASOPRESSIN AND HYDROCORTISONE**¹⁷⁵,

rather than relying solely on dose escalation.

◆ Preparing the infusion — where you have no local formulary

- **AMPOULES (GULF/ME):** epinephrine (adrenaline) **1 MG/ML (1:1,000)** is the stock ampoule (0.1 mg/mL / 1:10,000 is the pre-filled code syringe); norepinephrine (noradrenaline) as the **BITARTRATE 1 MG/ML** (2 mg/2 mL, 4 mg/4 mL). **DOSE NOREPINEPHRINE AS BASE** — a 4 mg/4 mL bitartrate ampoule = **2 MG BASE**.
- **RULE OF 6 (WEIGHT-BASED):** put **0.3 MG/KG IN 50 ML** (or 0.6 mg/kg in 100 mL) of glucose 5% → **1 ML/H = 0.1 MCG/KG/MIN** (0.05 = 0.5 mL/h; 0.3 = 3 mL/h).
- **STANDARD CONCENTRATION (SAFETY-PREFERRED):** epinephrine **10 / 20 / 40 MCG/ML**; norepinephrine (base) **16 / 32 / 64 MCG/ML**. **RATE (ML/H) = DOSE (MCG/KG/MIN) × WEIGHT (KG) × 60 ÷ CONCENTRATION (MCG/ML)**. Use the most dilute for a peripheral bridge.
- **DILUENT & SAFETY:** dilute in **GLUCOSE 5%** (or 0.9% saline) — **NOT** with bicarbonate/alkaline fluids; **PROTECT FROM LIGHT** and discard if discolored; dedicated line; **NEVER BOLUS OR FLUSH** a vasoactive line; independent

double-check of drug, concentration, weight, and rate.

- **SOURCES:** BNF for Children; ASHP/ISMP Standardize 4 Safety pediatric infusion standards; ANMF adrenaline & noradrenaline standard-concentration monographs (2025–26).

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → PICU/high-dependency; fluid-refractory or vasoactive-requiring shock is always PICU.
- **Discharge** — not from the acute episode — step down once off vasoactives with a controlled source.
- **Refer** — infectious diseases for source control and antimicrobials; surgery for a drainable focus; an ECMO center if refractory.

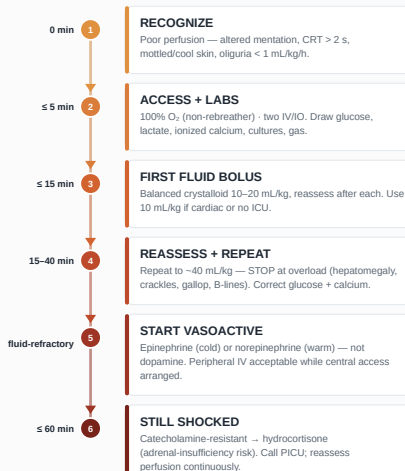
Septic shock – the first hour

The first hour, step by step: recognize → access + labs → fluids → reassess → vasoactive → hydrocortisone/PICU, with antibiotics inside the hour. Details in the card above.



ANTIBIOTICS within 60 minutes

Two agents IV/IO, after cultures — do not delay for the lab.



Myocarditis / Acute Heart Failure

HOURS

The "gastroenteritis" or "asthma" that is really the heart — look for it.

RECOGNIZE

- Viral prodrome, then **tachycardia out of proportion** to the fever, poor feeding, lethargy, grunting or respiratory distress.

- **Hepatomegaly, gallop rhythm, weak pulses, cool peripheries.** Older child: chest pain, syncope, palpitations.
- Commonly mislabeled as gastroenteritis, sepsis, asthma, pneumonia, or bronchiolitis.²³¹

CONFIRM

ECG (sinus tachycardia, low voltages, ST/T changes, arrhythmia, or heart block), chest X-ray (cardiomegaly, pulmonary edema), troponin and BNP raised, echocardiogram for reduced function.²³¹

⚠ **RESUSCITATE WITH CARE**

- Oxygen and monitor with caution; may need positive-pressure ventilation — **call PICU.**
- Do **not** give large fluid boluses.²³¹
- Use 5 mL/kg slowly and reassess for **signs of fluid overload** (worsening hepatomegaly, new crackles, a gallop, or respiratory distress).
- Start inotropes early (cardiology/PICU-directed) — milrinone (preferred if blood pressure allows — an inodilator

that lowers afterload and can drop BP) or dobutamine.

- Tailor to hemodynamics: normotensive with cold peripheries and rising lactate → dobutamine or milrinone; hypotensive with poor perfusion → epinephrine.
- Check and correct electrolytes (calcium, potassium, magnesium, phosphate).
- Treat arrhythmia, give oxygen and ventilatory support.
- Early PICU and pediatric cardiology; refractory shock needs urgent PICU/cardiology transfer — advanced mechanical circulatory support is a specialist/PICU decision.²³¹

INOTROPES / VASOACTIVES

Drug	Dose	Notes
Milrinone	0.25–0.75 mcg/kg/min IV infusion	Inodilator — preferred if the blood pressure allows. Consider omitting the 50 mcg/kg loading dose in acute shock ²³¹ to avoid hypotension.
Dobutamine	2–20 mcg/kg/min IV infusion	Normotensive with cold peripheries and rising lactate.
Epinephrine	0.05–0.3 mcg/kg/min IV infusion	Hypotensive with poor perfusion.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected cases → PICU/cardiology; arrhythmia, low output, or a troponin/BNP rise mandates continuous monitoring.
- **Discharge** — only after cardiology review, recovered function, and a heart-failure plan.
- **Refer** — cardiology urgently (echo, rhythm); a transplant/ECMO/mechanical-support center if low output or refractory arrhythmia.

Bradycardia + Poor Perfusion

MINUTES

Slow heart rate with shock — usually hypoxic.

FIRST

100% oxygen, support airway & breathing, monitor, IV/IO. **Ventilation corrects most pediatric bradycardia.**¹



DECISION

HR < 60 with poor perfusion despite oxygenation & ventilation? Poor perfusion = acutely altered mental status, signs of shock, or low BP.

YES

Start CPR. Give epinephrine. Atropine for a vagal cause or heart block; consider pacing.¹ Treat the reversible cause; call PICU/cardiology early.

NO

Support, observe, treat the cause, seek cardiology input.

LOOK FOR THE CAUSE

- **Hypoxia is the commonest — oxygenate and ventilate first.**¹ Also: hypothermia; drugs/toxins; raised intracranial pressure (Cushing's triad — hypertension, bradycardia, irregular breathing) (see **Raised ICP**); increased vagal tone (e.g., suctioning, intubation); AV / heart block.
- A well, athletic, or sleeping child may be **physiologically** bradycardic.¹

Drug	Dose	Notes
Epinephrine (adrenaline)	0.01 mg/kg (max 1 mg)	IV/IO, 0.1 mg/mL (1:10,000) concentration, every 3–5 min
Atropine	0.02 mg/kg (min 0.1, max 0.5 mg)	IV/IO; for vagal cause, AV block, or cholinergic/organophosphate toxicity ¹

ADMIT · DISCHARGE · REFER

- **Admit** — all → PICU with continuous monitoring, including any child needing atropine, epinephrine, or pacing.
- **Discharge** — only after a reversible cause is corrected, the rhythm is stable, and cardiology has reviewed.

- **Refer** — cardiology for pacing or structural/conduction disease; treat reversible causes (hypoxia, vagal, raised ICP, toxins, electrolytes).

◆ READING THE TRACING

- New to pediatric rhythms?
- See a [stepwise approach to the](#)

[pediatric ECG](#)



and

[common pediatric arrhythmias](#)



(LITFL ECG Library).

Supraventricular Tachycardia

MINUTES

Narrow QRS, no rate variability. HR > 220 infant / > 180 child.

▲ THROUGHOUT

- Run a continuous ECG rhythm strip during every intervention — documents the moment of conversion; can unmask underlying WPW.¹

DECISION

Hemodynamically unstable (shock, poor perfusion, reduced consciousness)?

UNSTABLE

Synchronized cardioversion; sedate only if it will not delay — use an agent that gives **amnesia**, not fentanyl alone (see sedation note). If IV/IO access is already in place, one rapid adenosine dose may be given while the defibrillator charges¹ — never delay the shock.

STABLE

Vagal maneuvers first — ice to the face in infants; modified Valsalva in older children¹ — then adenosine.

✦ MODIFIED VALSALVA

- Sit child semi-recumbent; blow into a 10 mL syringe hard enough to just move the plunger (~15 s); then immediately lay flat and raise legs to ~45° for 15 s.³
- More effective than standard strain in the adult REVERT trial

and a 2025 pediatric randomized trial.^{2,3}

Drug / action	Dose	Notes
Adenosine (1st dose)	0.1 mg/kg (max 6 mg)	⚠ Technique is critical: use the most proximal IV; two-syringe or 3-way-stopcock method — rapid push, then an instant 5–10 mL saline flush while raising the arm ¹ . Often needs two people.
Adenosine (2nd dose)	0.2 mg/kg (max 12 mg)	If no response to the first dose.
If the 2nd dose fails	Consult cardiology	Next steps: synchronized cardioversion , or an antiarrhythmic infusion — amiodarone 5 mg/kg over 20–60 min OR procainamide 15 mg/kg over 30–60 min. Never mix antiarrhythmics. ¹

Drug / action	Dose	Notes
Synchro nized cardioversion	0.5–1 → 2 J/kg	For the unstable child. If it will not delay the shock, sedate with an agent that provides amnesia — do not rely on fentanyl alone (see the sedation note below).

⚠ SEDATION FOR CARDIOVERSION – AMNESIA, NOT JUST ANALGESIA

Fentanyl alone is a pitfall: it gives analgesia but **no amnesia**¹, so the child may be awake and remember the shock; pushed rapidly it can cause **chest-wall rigidity**. **Ketamine**'s myocardial depression matters mainly in **prolonged, catecholamine-depleted** shock; in most children it is a net **sympathomimetic (positive inotrope)**, and ketamine or **etomidate** is standard for brief pediatric procedural sedation — there are no RCTs for this ~60-second window. **Involve a pediatric anesthesiologist** / follow your procedural-sedation protocol.

▶ ADMIT · DISCHARGE · REFER

- **Admit** — first presentation, hemodynamic instability, or need for adenosine/cardioversion → monitored bed/PICU.

- **Discharge** — a stable child who reverts and stays in sinus rhythm, feeding and well, with cardiology follow-up and a clear return plan.
- **Refer** — cardiology for all confirmed SVT (WPW screen, ablation planning) and for recurrent episodes.

◆ READING THE TRACING

- Align terminology with PALS/cardiology: SVT = **narrow-complex tachycardia**, VT = **wide-complex tachycardia** (each with or without a pulse); cross-check the label with cardiology.

- New to pediatric rhythms?
- See a [stepwise approach to the](#)

[pediatric ECG](#)



and

[common pediatric arrhythmias](#)



(LITFL ECG Library).

Ventricular Tachycardia (with a pulse)

MINUTES

Wide-QRS tachycardia. Pulseless VT → use the arrest algorithm.

DECISION

Stable or unstable?

UNSTABLE

Synchronized cardioversion 0.5–1 → 2 J/kg. Sedate if no delay (short-acting agent, e.g., IV fentanyl). Expert input.

STABLE

If **regular and monomorphic**, consider adenosine (diagnostic/therapeutic — may unmask SVT with aberrancy; not for irregular or polymorphic rhythms).

Polymorphic VT / torsades → **magnesium sulfate 25–50 mg/kg IV (max 2 g)** and avoid QT-prolonging antiarrhythmics¹. **Contact cardiology before starting antiarrhythmics** — a single infusion of amiodarone **or** procainamide (not both: additive QT prolongation)¹.

Drug	Dose	Notes
Amiodarone	5 mg/kg (max 300 mg)	IV over 20–60 min if perfusing; monitor BP ¹ & QT
Procainamide	15 mg/kg	IV over 30–60 min; do not combine with amiodarone ¹

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → PICU with continuous monitoring and a defibrillator at the bedside.
- **Discharge** — not from the acute episode — only after cardiology work-up and a management plan.
- **Refer** — cardiology urgently (structural disease, channelopathy, QT/electrolytes); consider genetics and family screening.

◆ READING THE TRACING

- New to pediatric rhythms?
- See a [stepwise approach to the pediatric ECG](#)  and [common pediatric arrhythmias](#)  (LITFL ECG Library).

Neonatal & Infant CHD

MINUTES

RECOGNIZE BY PRESENTATION

Three ways congenital heart disease declares itself in the newborn — cyanosis, shock, or heart failure.

① CYANOSIS – DUCT-DEPENDENT PULMONARY FLOW / MIXING ("5 T'S + 1")

- Transposition of the great arteries (duct-dependent for mixing)
- Truncus arteriosus
- Total anomalous pulmonary venous drainage (obstructed type = emergency)
- Tricuspid atresia
- Tetralogy of Fallot — **only the critical / pulmonary-atresia type is duct-dependent** (ordinary TOF presents with tet spells; see below).
- **+ 1:** critical pulmonary stenosis / atresia.
- **Assess the response to oxygen — a formal hyperoxia test is no longer routinely recommended.** ²³⁹ High FiO_2 can accelerate ductal closure, dangerous in a duct-dependent lesion. Use pre-/post-ductal SpO_2 , chest X-ray + blood gas, and echocardiography instead. In

suspected duct-dependent CHD, **do not delay alprostadil** while waiting for a hyperoxia test²³⁹.

② CARDIOGENIC SHOCK / COLLAPSE – DUCT-DEPENDENT SYSTEMIC FLOW (LEFT-HEART OBSTRUCTION)

- Hypoplastic left heart
- critical coarctation
- interrupted aortic arch
- critical aortic stenosis.
- Gray, mottled shock with **weak/absent femoral pulses**, a pre-/post-ductal saturation²³⁹ & blood-pressure gap, and metabolic acidosis. **Avoid high inspired oxygen** — it lowers pulmonary vascular resistance and steals from systemic flow²³⁹. Start alprostadil.

③ HEART FAILURE – LEFT-TO-RIGHT SHUNTS

- VSD
- AVSD
- PDA
- (large ASD — rarely).
- **Present after the first weeks as pulmonary vascular resistance falls — not as immediate newborn**

emergencies; ASD rarely causes neonatal heart failure.²³⁹

- Tachypnea, poor feeding, sweating with feeds, hepatomegaly, faltering growth.

Suspect a duct-dependent lesion → start Prostaglandin E1 (alprostadil) before the echocardiogram²³⁹



ANTICIPATE

Apnea (be ready to intubate), hypotension, fever. Urgent cardiology + transfer; carry the infusion on transfer.

Drug	Dose	Route & notes
Alprostadil (prostaglandin E1)	start 0.05–0.1 → maintenance 0.01–0.02 mcg/kg/min	IV infusion; two IV lines; never interrupt; once responding, titrate to the lowest effective dose (low dose maintains patency in ~80% with less apnea and fever ²³⁹). Watch for apnea (~12%), hypotension, and fever ²³⁹ — be ready to support ventilation

⚠ DON'T BE CAUGHT OUT

- The collapsed neonate labeled “sepsis,” “bronchiolitis,” or a “cyanotic spell” may be a closing duct.
- If shock or cyanosis does not respond to oxygen and fluids, **start alprostadil empirically** while arranging an urgent echocardiogram²³⁹ — do not wait for imaging.
- In duct-dependent **systemic** lesions keep SpO₂ targets modest.

◆ AVAILABILITY

- Prescribe and label as **ALPROSTADIL** (the single unambiguous name²³⁹ — avoid the ambiguous "prostaglandin E1" and any trade name).
- Stocked in neonatal and cardiac units.

◆ **ARRHYTHMIA**

- Tachycardia — **SVT** (narrow QRS) and **VENTRICULAR TACHYCARDIA** (wide QRS): see the Supraventricular Tachycardia and Ventricular Tachycardia cards; adenosine dosing is given on the SVT card.
- Screen every well newborn before discharge — see **Critical CHD — Pulse Oximetry Screening** (Neonatal).

▴ **ADMIT · DISCHARGE · REFER**

- **Admit** — any duct-dependent lesion or cyanosis/shock → NICU/PICU; keep the duct open with alprostadil.
- **Discharge** — only under cardiology, with a feeding/oxygen/weight plan and clear red-flag advice.

- **Refer** — pediatric cardiology urgently (echo) and a cardiac-surgical center; arrange safe transfer.

◆ **SEE ALSO** Newborn screening & pulmonary hypertension: **Critical CHD — Pulse Oximetry Screening** · **Persistent Pulmonary Hypertension (PPHN)** (Neonatal).

Hypercyanotic (Tet) Spell MINUTES

TETRALOGY OF FALLOT

Sudden deep cyanosis, distress, and a quieter murmur in Tetralogy of Fallot.

① POSITION & CALM

Knee-to-chest position (or let the child squat). Keep calm — minimal handling. Give 100% oxygen.²³⁹



② INCREASE RV PRELOAD

Morphine to settle the cycle; fluid bolus to increase filling.²³⁹



③ INCREASE SYSTEMIC RESISTANCE / SLOW THE HEART

- Phenylephrine; beta-blocker (propranolol or esmolol). **Avoid agents that increase contractility** — they worsen the outflow obstruction.²³⁹
- Phenylephrine, IV beta-blocker, and bicarbonate (for persistent acidosis) are specialist-supported — **call PICU/cardiology early.**

Drug	Dose	Notes
Morphine	0.1–0.2 mg/kg	IV/IM/subcut; settles the spell ²³⁹
Crystalloid bolus	10–20 mL/kg	IV, increases preload
Phenylephrine	5–20 mcg/kg	IV; raises systemic vascular resistance
Propranolol	0.05–0.1 mg/kg	slow IV; alternative to esmolol
Esmolol	500 mcg/kg IV load over 1 min, then 50–300 mcg/kg/min infusion	IV beta-blocker alternative to propranolol ²³⁹
Sodium bicarbonate	1 mmol/kg	IV, if significant metabolic acidosis

▲ ADMIT · DISCHARGE · REFER

- **Admit** — any spell → monitored bed/PICU; recurrent or severe spells need urgent intervention.
- **Discharge** — only after cardiology review, a spell-management plan (knee-chest, hydration, propranolol as advised), and caregiver education.
- **Refer** — pediatric cardiology urgently — a spell often brings surgical repair forward.

◆ **SEE ALSO** Neonatal & Infant CHD for duct-dependent lesions and alprostadil.

Blocked Systemic-to-Pulmonary Shunt (BT / Sano)

MINUTES

Shunt-dependent CHD

Sudden cyanosis with a **lost shunt murmur** — the shunt is occluding; a peripheral-hospital emergency.

RECOGNIZE IT – THE SHUNT IS BLOCKING

- Known **shunt-dependent** cyanotic CHD (modified **Blalock-Taussig** or **Sano** shunt) with **acute, severe cyanosis**.
- **Loss of the continuous shunt murmur** — the single most useful sign (a patent shunt murmurs continuously)²⁴⁰.
- **Sudden desaturation**, then **hypotension and metabolic acidosis**; falling PaO₂, rising lactate, falling cerebral NIRS from baseline²⁴⁰.
- **Echo: reduced or absent flow** across the shunt confirms it — but do **not** delay treatment for imaging.

⚠ CALL FOR HELP NOW

- **Call cardiac ICU/PICU + cardiothoracic surgery + cardiology immediately**, and activate your retrieval service.
- The physiology to fix it: **drive more blood across the shunt — raise systemic vascular resistance (SVR) and lower pulmonary vascular resistance (PVR) — while you anticoagulate.**

① RESUSCITATE (airway & oxygen) —

Secure the airway and ventilate; give **100% oxygen** ($\text{FiO}_2 \approx 10\%$ above the child's baseline) and optimize oxygenation. Urgent echo — but do not delay treatment.



② RAISE SVR (push flow across the shunt) — Fluid bolus **5 mL/kg** aliquots (4% albumin), reassessing; add a vasoconstrictor — **phenylephrine 3–10 mcg/kg slow IV**, or **norepinephrine 0.1 mcg/kg/min**, or **epinephrine up to 0.05 mcg/kg/min**²⁴⁰.



③ LOWER PVR (unload the lungs) —

Deepen **sedation + paralysis**; **hyperventilate** to lower PaCO_2 (mild respiratory alkalosis); keep well oxygenated; **magnesium sulfate 0.4 mmol/kg (50%)**²⁴⁰.



④ ANTICOAGULATE (stop the clot) —

Heparin 100 units/kg IV bolus, then **20 units/kg/h infusion**²⁴⁰; an optional **second 50 units/kg bolus** at 15 min. Restart **prostaglandin (alprostadil)** if the arterial duct has not been surgically ligated²³⁹.



⑤ DEFINITIVE RESCUE — cardiac catheterization (balloon angioplasty, stent, or thrombolysis) or emergency surgical shunt revision; ECMO to bridge a deteriorating child. Confirm with echo and transfer to the cardiac center.

MEDICATIONS – CONFIRM AGAINST YOUR CARDIAC CENTER'S PROTOCOL

Drug / product	Dose	Notes
SVR Phenylephrine	3–10 mcg/kg slow IV	Raises SVR; dilute before use (e.g., 10 mg in 100 mL, then 1 mL made up to 10 mL)
SVR Norepinephrine (noradrenaline)	0.1 mcg/kg/min infusion	Raises SVR; central line preferred, peripheral as a bridge
SVR Epinephrine (adrenaline)	up to 0.05 mcg/kg/min infusion	Inotrope/vasopressor support
SVR Albumin 4%	5 mL/kg aliquots	Volume to support preload/SVR; reassess after each
PVR Magnesium sulfate 50%	0.4 mmol/kg IV	Pulmonary vasodilator — helps lower PVR
CLOT Heparin	100 units/kg IV bolus → 20 units/kg/h; ± 50 units/kg at 15 min	Prevents clot propagation; monitor APTT / anti-Xa

Drug / product	Dose	Notes
I DUCT Prostagl andin E1 (alprosta dil)	per unit protocol	If the duct is not ligated — may reopen ductal pulmonary flow

✦ THE TRAP

The instinct to give a vasodilator, or deep vasodilating sedation without vasopressor support, **drops SVR and worsens shunt flow**. Do the opposite — **raise SVR, lower PVR, heparinize**, and get the child to the cath lab or theatre. **A lost shunt murmur in a cyanotic child is the shunt occluding until proven otherwise.** ²⁴⁰

◆ SOURCES

- NHS Greater Glasgow & Clyde paediatric guideline — immediate management of the modified Blalock–Taussig / central shunt (Paediatrics 386); PIER Network — acute management of the cyanosed infant with a systemic-to-pulmonary (BT/Sano) shunt in the ED; shunt-thrombolysis case & review data (heparin/urokinase; catheter-directed thrombolysis).

- Confirm all doses against your formulary and cardiac center's protocol.

Acute Rheumatic Fever

DIAGNOSE

Jones 2015 • AHA

Prevent the second attack — every recurrence adds valve damage.

▲ DIAGNOSE – JONES 2015 (REVISED)

- Need **evidence of preceding group A strep** (rising/elevated ASO or anti-DNase B, or positive throat swab/RADT, or recent scarlet fever) **plus** either **2 major** or **1 major + 2 minor**²³².
- Recurrence also allows **3 minor**²³². **Chorea** and **indolent carditis** can be diagnostic on their own.
- Get an **echo in every suspected case** (repeat even if first is normal) — subclinical valve regurgitation counts as carditis²³².

⚠️ USE THE MODERATE/HIGH-RISK COLUMN

- Gulf / Middle-East children are a **moderate-to-high-risk** population — applying low-risk thresholds under-diagnoses ARF²³².
- In this group a **single joint (mono-arthritis or even mono-arthralgia)** and a **lower fever/ESR threshold** already count²³².

Jones 2015	Low-risk population	Moderate / high-risk (Gulf)
Major	Carditis (clinical or subclinical on echo) · migratory polyarthritis (flitting, large joints) · chorea · erythema marginatum · subcutaneous nodules	Same, except joints = mono-arthritis, polyarthritis, or polyarthralgia
Minor	Polyarthralgia · fever $\geq 38.5\text{ }^{\circ}\text{C}$ · ESR $\geq 60\text{ mm/h}$ and/or CRP $\geq 3.0\text{ mg/dL}$ · prolonged PR (if carditis not a major)	Mono-arthralgia · fever $\geq 38\text{ }^{\circ}\text{C}$ · ESR $\geq 30\text{ mm/h}$ and/or CRP $\geq 3.0\text{ mg/dL}$ · prolonged PR (if carditis not a major)

A manifestation counts once — polyarthrititis as a major cannot also be counted as a minor²³².

TREAT

Target	What to give
Eradicate GAS (everyone, even if swab negative)	Benzathine penicillin G IM once: < 27 kg → 600,000 units; ≥ 27 kg → 1,200,000 units. Alt: penicillin V PO ×10 d; if allergic, azithromycin 12 mg/kg/day (max 500 mg) ×5 d. ²³³
Arthritis / mild carditis	Aspirin 80–100 mg/kg/day PO divided every 6 h for 2–6 weeks, then taper; watch for tinnitus/hyperpnea (salicylate toxicity) ²³² .
Moderate–severe carditis / heart failure	Add prednisone/prednisolone ~2 mg/kg/day (max ~60 mg) 2–3 weeks then taper, overlapping aspirin. Diuretic + ACE inhibitor for heart failure; bed rest. ²³²
Sydenham chorea	Usually self-limited — quiet environment, protect from injury. If disabling: valproate or carbamazepine (off-label). Salicylates/steroids do not help the chorea itself. ²³²

SECONDARY PROPHYLAXIS – BENZATHINE
PENICILLIN G IM EVERY 3–4 WEEKS (< 27 KG 600,000 U; ≥ 27 KG 1,200,000 U)

Category (from last attack)	Duration
ARF without carditis	5 years or until age 21 (whichever longer)
Carditis, no residual valve disease	10 years or until age 21 (whichever longer)
Carditis with residual valve disease	10 years or until age 40 — often lifelong ²³³

◆ **IF THE IM INJECTION IS DECLINED** Oral **PENICILLIN V 250 MG TWICE DAILY** is an accepted alternative for secondary prophylaxis (sulfadiazine if penicillin-allergic) — but adherence is worse and it is **LESS EFFECTIVE THAN IM BENZATHINE**, which remains preferred²³³.

✦ **THE POINT OF THE CARD**

- Treatment settles the attack; **prophylaxis prevents rheumatic heart disease.**
- Recurrences are cumulative valve damage — document the injection schedule and hand it over so it is never missed.

◆ SOURCES

- AHA 2015 Revised Jones Criteria (Gewitz, Circulation 2015); AHA/AAP prevention statement (Gerber, Circulation 2009); WHO RHD guidance; SPIDS Gulf prophylaxis guidance.
- Confirm doses against your formulary.

Pericarditis & Cardiac Tamponade

MINUTES

ESC 2015

Tamponade is a clinical diagnosis — drain, don't wait for the scan.

▶ PERICARDITIS – 2 OF 4

- Pleuritic chest pain **relieved by sitting forward**
- pericardial **friction rub**
- **widespread ST elevation / PR depression** on ECG (not one coronary territory)²³⁴
- new/worsening **pericardial effusion** on echo.
- Supportive: raised CRP/ESR.

- Causes — viral/idiopathic (commonest), purulent, post-pericardiotomy, autoimmune (SLE, JIA), uremic, TB, malignancy²³⁴.

⚠ **TAMPONADE – RECOGNIZE AND ACT**

- **Beck's triad** (hypotension, muffled heart sounds, and raised JVP) + **tachycardia** and a **narrow pulse pressure; pulsus paradoxus > 10 mmHg** (inspiratory drop in systolic BP)²³⁴.
- Children compensate then crash — tachycardia is the early sign. **A small effusion can tamponade if it accumulates fast** — volume on echo does not equal severity; the clinical picture does²³⁴.
- Beck's triad and pulsus are often incomplete — their absence does not exclude it²³⁴.

DECISION

Hemodynamic compromise (tamponade)?

YES

Emergency echo-guided pericardiocentesis — removing even a small volume rescues output²³⁴. **IV fluid 10–20 mL/kg is a bridge only**, never a substitute. **Maintain systemic vascular resistance** with a vasopressor — **norepinephrine** ($\pm$ **vasopressin**) — as a temporizing bridge to support coronary and systemic perfusion; keep the circulation **full, fast, and tight** (preload, avoid bradycardia, support afterload). **Avoid intubation / positive-pressure ventilation** — it drops preload and can cause arrest²³⁴; if unavoidable, drain first, low tidal volume/PEEP, keep volume-loaded. Avoid diuretics/vasodilators. Call **cardiology + cardiac surgery + PICU** in parallel.

NO

Uncomplicated pericarditis: **NSAID (ibuprofen) + colchicine** — colchicine halves recurrence²³⁴. Treat the cause; restrict strenuous exercise until symptom-free with normal CRP/ECG/echo. Steroids are second-line (increase recurrence)²³⁴.

Drug	Pediatric dose	Notes
Ibuprofen	30–40 mg/kg/day divided every 6–8 h (max ~2.4 g/day)	First-line anti-inflammatory; give with gastroprotection, taper on CRP ²³⁴ .
Colchicine	< 5 yr: 0.5 mg/day · > 5 yr: 1–1.5 mg/day divided	First-line adjunct — halves recurrence; reduce in renal impairment.
Purulent pericarditis	Urgent drainage + IV vancomycin + 3rd-gen cephalosporin, ~3–4 weeks	Surgical emergency — NSAIDs alone are inadequate; narrow to culture ²³⁴ .

✦ THE TRAP

Don't reach for the ventilator in a crashing tamponade — positive-pressure ventilation can be the last straw. **Drain first.** And treat **purulent pericarditis** as the surgical emergency it is.

◆ SOURCES

- ESC 2015 pericardial diseases guidelines (Adler, Eur Heart J); pediatric echo/tamponade reviews (Frontiers Pediatr 2017); pediatric

colchicine data (Pediatr Cardiol 2024).

- Confirm doses against your formulary.

Infective Endocarditis (incl. Subacute)

HOURS

Modified Duke / AHA

Fever + a new murmur + positive cultures — think endocarditis; diagnose by the modified Duke criteria.

▲ SUSPECT IT – SUBACUTE CLUES

- Persistent/unexplained fever with a **new or changed regurgitant murmur** — the classic pairing.
- Subacute course over weeks: malaise, weight loss, night sweats, anorexia. **Viridans-group streptococci** are the commonest cause of subacute native-valve disease; **S. aureus** dominates acute disease²³⁷.
- Peripheral stigmata (late, immune-mediated): **splinter hemorrhages**, tender **Osler nodes**,

painless **Janeway lesions**, **Roth spots**²³⁷, splenomegaly, embolic phenomena.

- At-risk: **congenital heart disease**, prosthetic valve/material, previous IE, indwelling central lines, IV drug use.

▶ **MODIFIED DUKE CRITERIA – HOW TO CLASSIFY**

- **DEFINITE** (clinical) = **2 major**, or **1 major + 3 minor**, or **5 minor**.
- **POSSIBLE** = **1 major + 1 minor**, or **3 minor**.
- **REJECTED** = firm alternative diagnosis, or resolution with ≤ 4 days of antibiotics²³⁷.

DECISION

The two **MAJOR** criteria

MICROBIOLOGY

Typical organism from ≥ 2 **separate blood cultures** (viridans streptococci, *S. gallolyticus*/bovis, HACEK group, *S. aureus*, or community-acquired enterococci with no primary focus), **or persistently positive** cultures of a compatible organism²³⁷.

ENDOCARDIAL

Echo evidence of **endocardial involvement: vegetation, abscess, new partial dehiscence of a prosthetic valve, or new valvular regurgitation** (worsening of an existing murmur is not enough)²³⁷.

▶ THE FIVE MINOR CRITERIA

- **Predisposition** — predisposing heart condition or IV drug use.
- **Fever** $\geq 38^{\circ}\text{C}$.
- **Vascular phenomena** — arterial emboli, septic pulmonary infarcts, mycotic aneurysm, intracranial or conjunctival hemorrhage, **Janeway lesions**.
- **Immunologic phenomena** — glomerulonephritis, **Osler nodes**, **Roth spots**, rheumatoid factor.
- **Microbiologic** evidence not meeting a major criterion²³⁸.

⚠ WORKUP – GET IT RIGHT THE FIRST TIME

- **Draw ≥ 3 sets of blood cultures** from separate venipunctures **before** starting antibiotics — the

single most important step for diagnosis and tailoring therapy²³⁸.

- Echo: **TTE first**; proceed to **TEE** if TTE is non-diagnostic²³⁵, a prosthetic valve is present, or suspicion stays high.
- Add ESR/CRP, FBC, and urinalysis (microscopic hematuria supports it). Do **not** delay antibiotics in a septic/unstable child once cultures are drawn.

◆ Endocarditis prophylaxis (AHA / ESC 2023) — who

Prophylaxis needs **BOTH** a highest-risk cardiac condition **AND** a high-risk procedure. The **HIGHEST-RISK CARDIAC CONDITIONS**:

1. **PROSTHETIC HEART VALVE** (surgical or transcatheter) or prosthetic material used for valve repair.
2. **PREVIOUS INFECTIVE ENDOCARDITIS**.
3. **UNREPAIRED CYANOTIC CHD** (including palliative shunts/conduits).
4. Completely **REPAIRED CHD WITH PROSTHETIC MATERIAL/DEVICE** — during the **FIRST 6 MONTHS** after the procedure.
5. **REPAIRED CHD WITH A RESIDUAL DEFECT** at or adjacent to the prosthetic patch/device.
6. **CARDIAC TRANSPLANT** recipients who develop **VALVULOPATHY**.

7. **VENTRICULAR ASSIST DEVICE (VAD)** — added by **ESC 2023**²³⁵.

◆ **ESC 2023 EMPHASIS:** for all **INTERMEDIATE- AND HIGH-RISK** patients, **GOOD ORAL HYGIENE AND REGULAR DENTAL REVIEW** prevent more endocarditis than any single antibiotic dose²³⁵.

◆ Endocarditis prophylaxis (AHA) — which procedures

- **DENTAL** — any procedure that manipulates **GINGIVAL TISSUE** or the **PERIAPICAL REGION** of teeth, or **PERFORATES THE ORAL MUCOSA** (extractions, scaling, periodontal work, implants). Routine fillings, radiographs, and shedding of deciduous teeth do **NOT** count²³⁶.
- **RESPIRATORY TRACT** — procedures with **INCISION OR BIOPSY OF THE RESPIRATORY MUCOSA** (e.g., **TONSILLECTOMY, ADENOIDECTOMY**, bronchoscopy **WITH BIOPSY**). Cover staphylococci if the airway procedure treats an established infection (add anti-staphylococcal agent).
- **INFECTED SKIN, SKIN STRUCTURES, OR MUSCULOSKELETAL TISSUE** — regimen should cover **STAPHYLOCOCCI AND B-HEMOLYTIC STREPTOCOCCI**.

- **NOT RECOMMENDED** for **GI OR GU PROCEDURES**, or for bronchoscopy/endoscopy **WITHOUT** incision/biopsy, solely to prevent endocarditis.
- **◆ ESC 2023** is more restrictive for non-dental procedures — it advises prophylaxis for respiratory, GI, GU, skin, or musculoskeletal procedures **ONLY WHEN PERFORMED AT AN INFECTED OR COLONIZED SITE**²³⁵. Where the two guidelines differ, follow your local protocol.

PROPHYLAXIS REGIMEN – SINGLE DOSE 30-60 MIN BEFORE THE PROCEDURE

Situation	Pediatric dose (single)	Max
Able to take oral	Amoxicillin 50 mg/kg PO ²³⁶	2 g
Unable to take oral	Ampicillin 50 mg/kg IM/IV, or cefazolin / ceftriaxone 50 mg/kg IM/IV	Ampicillin 2 g · cefazolin/ceftriaxone 1 g
Penicillin allergy — oral	Cephalexin 50 mg/kg, or azithromycin / clarithromycin 15 mg/kg, or doxycycline (< 45 kg 2.2 mg/kg; ≥ 45 kg 100 mg)	Cephalexin 2 g · azithromycin/clarithromycin 500 mg · doxycycline 100 mg

Situation	Pediatric dose (single)	Max
Penicillin allergy — no oral	Cefazolin or ceftriaxone 50 mg/kg IM/IV	1 g

✦ THE TRAP

Clindamycin is no longer recommended (dropped in the 2021 AHA update²³⁶ — *C. difficile* risk and adverse effects). Prophylaxis is **not** indicated for most CHD, nor for routine GI/GU procedures. Avoid cephalosporins if there is a history of anaphylaxis, angioedema, or urticaria to penicillin²³⁶. **Good oral hygiene and regular dental care prevent more endocarditis than any single antibiotic dose.**

◆ SOURCES

- AHA prevention of IE statement (Wilson, *Circulation* 2007) and 2021 update (Wilson, *Circulation* 2021), and AHA IE wallet card; **2023 ESC ENDOCARDITIS GUIDELINES** (Delgado & Ajmone Marsan, *Eur Heart J* 2023;44:3948); Modified Duke criteria (Li, *Clin Infect Dis* 2000); 2023 Duke-ISCVID criteria (Fowler, *Clin Infect Dis* 2023).

- Confirm doses against your formulary.

Common Acute

04

Conditions

Everyday clinic & ED infections and presentations

Fever in a Well-Appearing Infant < 2 Months

HOURS

≤ 60 DAYS • CPS 2026 / AAP 2021

Most are viral — but ~8–13% have a serious bacterial infection¹⁶. Define fever as rectal $\geq 38.0^{\circ}\text{C}$; stratify risk and spare the low-risk from over-testing.

► SCOPE & FIRST MOVE

- This pathway is for the **well-appearing, full-term (≥ 37 wk), previously healthy** infant with a documented **rectal temperature $\geq 38.0^{\circ}\text{C}$. Hypothermia $< 36.0^{\circ}\text{C}$** carries the same risk — manage identically¹⁶.

- **This pathway does not apply to the ill-appearing at any age.**
- **Risk factors that are not applicable to this pathway:** prematurity < 37 wk, prior or prolonged hospitalization, a chronic / genetic / technology-dependent condition, immunodeficiency, recent antibiotics, or any focal bacterial infection (e.g., omphalitis, cellulitis).

⚠ KEEP A HIGH INDEX OF SUSPICION

- Even a well-appearing infant can deteriorate quickly.
- The youngest (**≤ 21–28 days**) have the highest risk of **invasive bacterial infection (IBI)** — bacteremia or bacterial meningitis) and the **lowest threshold** for full work-up, LP, empiric antibiotics, and admission¹⁶.

▶ EVERYONE GETS

- **All ages:** history + exam, **urinalysis + urine culture** (catheter or suprapubic; clean-catch acceptable), **blood culture, CBC + differential**, and an **inflammatory marker** — procalcitonin (PCT) or CRP.

- **Risk-stratify with a validated rule:** **PECARN** or **Step-by-Step** (both use PCT), or **Aronson** when PCT is unavailable — keep a **lower threshold for LP** with Aronson¹⁶.
- **CSF pleocytosis** = $> 15 \text{ cells/mm}^3$ ($\leq 28 \text{ d}$) or $> 9 \text{ cells/mm}^3$ ($> 28 \text{ d}$) → treat as bacterial meningitis. Don't delay the LP if there are neurological signs.¹⁶

RISK STRATIFICATION – VALIDATED LOW-RISK RULES

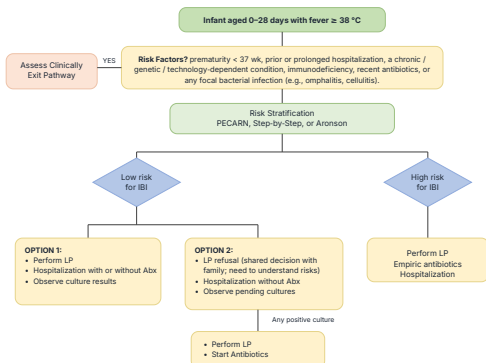
Rule	Low risk if ALL criteria are met
PECARN (uses PCT)	1) Urinalysis negative (leukocyte esterase, nitrite, $\leq 5 \text{ WBC/hpf}$); 2) $\text{ANC} \leq 4000/\mu\text{L}$; 3) $\text{PCT} \leq 0.5 \text{ ng/mL}$ ¹⁶
Step-by-Step (uses PCT)	1) Well-appearing; 2) 22–90 days; 3) urinalysis negative for leukocytes; 4) $\text{PCT} < 0.5 \text{ ng/mL}$; 5) $\text{CRP} \leq 20 \text{ mg/L}$ and $\text{ANC} \leq 10,000/\mu\text{L}$ ¹⁶
Aronson (no PCT — keep a lower LP threshold)	Low risk if ≤ 1 point: age $< 21 \text{ d}$ (1) · ED temp $38.0\text{--}38.4^\circ\text{C}$ (2) or $\geq 38.5^\circ\text{C}$ (4) · $\text{ANC} \geq 5185/\mu\text{L}$ (2) · urinalysis positive (3) ¹⁶

◆ APPLYING THE RULES – PCT TIMING & ALTERNATIVES

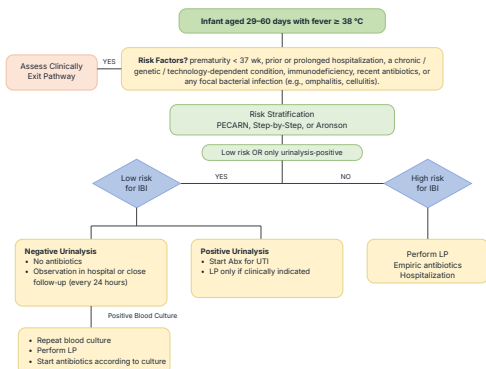
- Where **PROCALCITONIN IS NOT READILY AVAILABLE**, PECARN and Step-by-Step (both require PCT) **CANNOT BE FULLY APPLIED**. Use an alternative: the **ARONSON** rule (no PCT needed) and keep a **LOWER THRESHOLD FOR LP; CRP $\leq$ 20 MG/L** with a normal ANC and urinalysis still support lower risk.
- When PCT **IS** obtained, draw it **$\geq$ 2 H AFTER FEVER ONSET** — it rises ~3–4 h after a bacterial trigger, so very early samples can read falsely low¹⁶. It typically **STAYS LOW (< 0.5 NG/ML)** in isolated viral illness.
- Missed invasive infection with Step-by-Step clusters in the **22–28-DAY** group and in infants tested very early — keep a low threshold in these.¹⁶

FEVER PATHWAY ALGORITHMS (WELL-APPEARING, ≥ 37 WEEKS)

Algorithm 1 – infant 0–28 days with fever ≥ 38°C



Algorithm 2 – infant 29–60 days with fever ≥ 38°C



EMPIRIC ANTIBIOTICS – WELL-APPEARING, ≥ 37 WEEKS

Age	First-line IV	Modifiers
0–7 days	Ampicillin 100 mg/kg/dose q8h (reduced neonatal renal clearance; Lexicomp) ¹⁶ + gentamicin 4 mg/kg/dose q24h	Meningitis: add — or replace the aminoglycoside with — cefotaxime 50 mg/kg/dose q8h. HSV risk: add acyclovir 20 mg/kg/dose q8h.
8–28 days	Ampicillin 75 mg/kg/dose q6h + gentamicin 5 mg/kg/dose q24h	Meningitis: add — or replace the aminoglycoside with — cefotaxime 50 mg/kg/dose q6h (200 mg/kg/day for meningitis at this age). HSV risk: add acyclovir 20 mg/kg/dose q8h.
29–60 days	Ceftriaxone 100 mg/kg/dose q24h	Meningitis: add vancomycin 15 mg/kg/dose q6h. HSV risk: add acyclovir. Low-risk UTI: cefixime 8 mg/kg/dose q24h PO. ¹⁶

SOURCE

Canadian Pediatric Society — Management of well-appearing febrile young infants aged ≤ 90 days (updated May 2026); AAP — Pantell et al.,

Evaluation and Management of Well-Appearing Febrile Infants 8–60 Days Old, *Pediatrics* 2021.

◆ **BEYOND 60 DAYS** The well-appearing infant **61–90 DAYS** falls outside this pathway: risk of invasive infection is lower and routine blood culture / full septic work-up is **NOT REQUIRED**¹⁶. Test **SELECTIVELY** (urinalysis ± inflammatory markers) and use clinical judgment; escalate only the ill-appearing or those with a positive screen.

◆ **SEE ALSO** For the neonate (≤ 28 days), see **Neonatal Sepsis — rapid approach** for neonatal-specific empiric therapy (ampicillin + gentamicin/cefotaxime ± acyclovir).

Gastroenteritis & Dehydration

HOURS

Assess hydration: alertness, eyes, mucosa, skin turgor, capillary refill, urine output.

▲ CLINICAL SIGNS OF DEHYDRATION

- Altered mental status
- sunken eyes
- dry oral mucosa

- poor skin turgor
- delayed capillary refill
- decreased urine output
- tachycardia
- abnormal peripheral perfusion.
- The more signs present, the greater the fluid deficit; no single sign is reliable alone.⁹⁰

DECISION

Degree of dehydration?

NONE-MODERATE

Oral rehydration with ORS; a single ondansetron dose can reduce vomiting and the need for IV.⁹⁰

SEVERE / SHOCK

IV/IO crystalloid bolus, repeat to reverse shock; then deficit + maintenance. Check glucose & electrolytes.

Drug / fluid	Dose	Route & notes
Oral rehydration solution	~50–100 mL/kg over 4 h	Oral/NG + replace ongoing losses. First-line for minimal ⁹⁰ –moderate dehydration.
0.9% sodium chloride (bolus)	20 mL/kg	IV/IO rapid bolus for hypovolemia/shock; reassess during & after each bolus and repeat until perfusion is restored. If shock seems out of proportion to the diarrhea/vomiting, consider sepsis, DKA, a surgical abdomen, or myocarditis. Balanced crystalloid (e.g., Ringer's lactate) is an acceptable alternative. ⁹⁰
0.9% sodium chloride (moderate)	10 mL/kg over 30–60 min	Moderate dehydration not tolerating oral — then reassess for a further bolus or transition back to ORT if tolerated.

Drug / fluid	Dose	Route & notes
Ondansetron	0.15 mg/kg/dose – single dose, max 8 mg (either route). Weight-based: 2 mg (8–15 kg) • 4 mg (15–30 kg) • 8 mg (> 30 kg)	PO preferred; age ≥ 6 months only. Reduces vomiting and the need for IV. Restart ORT 15–30 min after the dose. Ondansetron prolongs the QT interval dose-dependently — do not exceed 8 mg in a single pediatric dose (the 16 mg cap is an adult chemotherapy figure, not for gastroenteritis)⁹⁰.

Drug / fluid	Dose	Route & notes
Zinc (adjunct to ORS)	10 mg/day (< 6 mo) 20 mg/day (≥ 6 mo)	Oral, 10–14 days . A WHO/UNICEF measure for low- and middle-income settings where zinc deficiency is common — shortens diarrhea by ~½ day (~1 day if malnourished) concentrates at ≥ 6 months in zinc-deficient or malnourished children. Not routine in high-income, well-nourished populations including Kuwait ; reserve for malnutrition, documented zinc deficiency, or persistent / chronic diarrhea. Little benefit under 6 months. (ESPGHAN/ESPID acute-gastroenteritis guideline: zinc not recommended for well-nourished European children.) ⁹⁰

Drug / fluid	Dose	Route & notes
Racecadotril — optional adjunct	1.5 mg/kg/dose PO TID with ORS; sachets 10 mg ($\leq$ ~12 kg) / 30 mg ($>$ ~13 kg). Age $>$ 3 months. Continue until diarrhea stops (2 normal stools), max 7 days.	Antisecretory enkephalinase inhibitor — reduces stool water loss without slowing gut motility (unlike loperamide, which is contraindicated). Adjunct to ORS, not a substitute. Licensed and used in Kuwait/Europe/Asia; not FDA/NICE/AAP/CPS/WHO-ESPGHAN/ESPID 2014: “may be considered” (II-B, weak recommendation, moderate-quality evidence; responder NNT $\approx$ 4). Cochrane 2019 found it safe with limited benefit (low-certainty). ⁹⁰

PEARLS

- Resume normal feeding (including breastfeeding) early. Antibiotics only for specific pathogens.⁹⁰
- Red flags: bilious or bloody vomit, severe pain/distension (think surgical), inability to tolerate ORS.

- Zinc dosing & evidence: [WHO eLENA](#)



▲ ADMIT · DISCHARGE · REFER

- **Admit** — shock or severe dehydration, failed oral/NG rehydration, intractable vomiting, dangerous electrolyte derangement, or diagnostic uncertainty.
- **Discharge** — minimal or mild dehydration tolerating oral rehydration, with carers confident on fluids and red-flag advice.
- **Refer** — surgical review if a surgical abdomen is suspected; PICU for refractory shock or severe electrolyte disturbance.

Community-Acquired Pneumonia

HOURS

CPS 2024

The healthy, immunized child with fever, cough, and tachypnea — treat *S. pneumoniae*, mostly orally, and keep it short.

RECOGNIZE & ASSESS SEVERITY

- Acute fever + cough + **tachypnea** (the most useful sign) ± increased work of breathing, chest or abdominal pain, poor feeding. Wheeze + hypoxia points to bronchiolitis or asthma, not bacterial CAP.⁶⁰
- **Tachypnea cut-offs (breaths/min):**
 - < 2 mo > 60
 - 2–12 mo > 50
 - 1–5 yr > 40
 - > 5 yr > 30.⁶⁰
- Check SpO₂ in everyone — hypoxemia may not be clinically obvious; **give oxygen for SpO₂ < 92%.**
- **Likely cause by age:**
infants/preschool — mostly **viral** (RSV, influenza, human metapneumovirus); *Streptococcus pneumoniae* is the leading bacterial

cause at all ages; school-age — add atypicals (⁶⁰*Mycoplasma*, *Chlamydophila*).

◆ KEEP INVESTIGATIONS TARGETED

- A chest radiograph is not essential for a non-severe outpatient with a typical picture; obtain posteroanterior + lateral when the diagnosis is uncertain or the child is admitted (extent, effusion, abscess).¹⁷⁷
- Do not image typical bronchiolitis or asthma.
- In hospitalized children: complete blood count (CBC)/C-reactive protein (CRP) + blood culture before antibiotics; consider respiratory viral testing in influenza season.

DECISION

Outpatient or admit?

OUTPATIENT • NON-SEVERE

Well-hydrated, tolerating oral intake, no respiratory compromise. **Oral amoxicillin** is the mainstay — high-dose covers *S. pneumoniae*. Most do not need a chest radiograph.

Inadequate oral intake or intolerance, **hypoxemia / grunting / nasal flaring / apnea**, severe or complicated pneumonia, or **age < 6 months**. Start **IV ampicillin**.

Amoxicillin (oral) or ampicillin (IV) is first-line — cover *S. pneumoniae*.

EMPIRIC THERAPY

Setting / drug	Dose	Notes
Amoxicillin — outpatient, first-line	25–30 mg/kg/dose PO q8h (max 4 g/day)	Drug of choice for non-severe CAP. TID dosing (not BID). 5 days is enough if responding. ¹⁷⁷
Ampicillin — admitted, first-line	50 mg/kg/dose IV q6h (max 12 g/day)	IV first-line for hospitalized, non-life-threatening CAP. Step down to oral amoxicillin on improvement. ⁶⁰
Ceftriaxone — severe / respiratory failure	50 mg/kg/dose IV q12–24h (max 2 g/dose)	For respiratory failure or septic shock (or cefotaxime 50 mg/kg/dose q6–8h). Broader cover incl. β -lactamase <i>H. influenzae</i> .

Setting / drug	Dose	Notes
+ Vancomycin — if MRSA suspected	15 mg/kg/dose IV q6h	Add for rapidly progressing multilobar disease or pneumatocoles until cultures return; send nasal MRSA screen first; stop if MRSA excluded.
Azithromycin — atypical (<i>Mycoplasma</i>)	10 mg/kg day 1, then 5 mg/kg × 4 d (max 500 mg)	School-age with atypical features. Macrolide resistance rising; doxycycline (any age, short courses). ¹⁷⁷

⚠ DURATION, COURSE & FOLLOW-UP

- **5 days** for non-severe CAP (ambulatory, or admitted and responding within 24–36 h); **7–10 days** for more significant disease.¹⁷⁷
- Expect improvement (appetite, fever, tachypnea, O₂ need) within **48 h**.
- If **not improving by 48–72 h** → re-image (radiograph/ultrasound) for empyema or abscess and reconsider the diagnosis (foreign body, tuberculosis, atypical).⁶⁰
- Routine end-of-therapy radiograph is not needed — radiographic

clearance lags⁶⁰ 4–6 weeks.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — hypoxia ($\text{SpO}_2 < 90\%$; $< 92\%$ in high-risk infants), respiratory distress, dehydration or feeding failure, an effusion/empyema, or failure of outpatient therapy.
- **Discharge** — well child maintaining saturations in air, feeding, on oral antibiotics with follow-up arranged.
- **Refer** — PICU for respiratory failure or sepsis; surgical or respiratory review for empyema or non-resolving disease.

◆ **DIFFICULT IV ACCESS?** Reassess for an early IV→oral switch at every line failure rather than re-sticking — see [IV → Oral Switch & the DIVA Patient](#) (Quick Reference).

Complicated Pneumonia – Empyema

HOURS

IDSA/PIDS 2026 • CPS 2024

Pneumonia + pus or a moderate-to-large pleural effusion — antibiotics plus, when needed, drainage with a fibrinolytic.

◆ DEFINITION & STAGES

- Empyema = intrapleural pus, or a moderate-to-large exudative parapneumonic effusion.
- It evolves: **STAGE 1** exudative (free-flowing) → **STAGE 2** fibrinopurulent (loculated) → **STAGE 3** organized (thick fibrinous peel)⁵⁹.
- Small parapneumonic effusions are common in uncomplicated pneumonia and do **NOT** need drainage.⁵⁹
- Lung abscess and necrotizing pneumonia are separate complications.

▲ RECOGNIZE & DIAGNOSE

- Suspect it when an initially uncomplicated pneumonia is **poorly responsive** — persistent fever after 48–72 h, worsening distress or

hypoxia, or new effusion signs (decreased breath sounds, dullness, reduced expansion).

- **Imaging:** chest radiograph first, then **chest ultrasound** (confirms effusion, estimates size, distinguishes free-flowing from loculated). Computed tomography (CT) is not routine — reserve for a suspected alternative diagnosis (e.g., malignancy).⁵⁹
- **Pathogens:** *S. pneumoniae* (most common), *S. aureus* (incl. MRSA), *S. pyogenes* (group A streptococcus). Send drained fluid for **culture and Gram stain** ($\pm$ pneumococcal PCR where available); blood cultures positive in only $\sim 10\%$ ⁵⁹. Labs often show high white-cell count/ESR/CRP, thrombocytosis, hypoalbuminemia — these normalize with recovery.

DECISION

How big is the effusion, and how sick is the child?

SMALL / MILD – ANTIBIOTICS ALONE

Small effusion, mild distress, no mediastinal shift → a trial of **antibiotics alone** is reasonable with close monitoring.

MODERATE-LARGE / COMPROMISE – DRAIN

Moderate-to-large effusion, significant respiratory distress, or mediastinal shift → **drainage plus antibiotics**, early. Consult thoracic surgery / interventional radiology promptly.

▴ DRAIN WHEN

Moderate-to-large effusion or empyema causing respiratory compromise, or occupying much of the hemithorax ± **mediastinal shift; loculated (stage 2) or organized** collection on ultrasound, or frank **pus** on aspiration; or **failure of antibiotics alone** (not improving, or progressing, after an adequate trial).

◆ FIBRINOLYTIC (ALTEPLASE / TPA)

- First-line, most cost-effective drainage = **SMALL-BORE (≤ 12 FR) PERCUTANEOUS CHEST TUBE + INTRAPLEURAL FIBRINOLYTIC**, preferred over surgical debridement (VATS) in

most cases — reserve VATS for refractory disease or extensive loculation. **ALTEPLASE 0.1 MG/KG (MAX 4 MG)** in normal saline instilled via the chest tube, **CLAMP/DWELL 1 HOUR, THEN DRAIN – ONCE DAILY FOR UP TO 3 DAYS** (a fixed 4 mg dose daily × 3 is an accepted alternative). **TPA ALONE IS NOW PREFERRED OVER TPA + DNASE** (no added benefit, higher cost).⁵⁹

- Prescribe and label as **ALTEPLASE**.

ANTIBIOTICS FOR EMPYEMA

Agent	Dose	Notes
Cefotaxime / ceftriaxone — first-line	Cefotaxime 50 mg/kg/dose q6–8h; ceftriaxone 50 mg/kg/dose q12–24h	Empiric IV cover for the usual pathogens; choose per local antibiogram. ¹⁷⁷
+ Vancomycin — MRSA	15 mg/kg/dose IV q6h	Reserve for culture-proven or strongly suspected MRSA / severe disease.
Oral step-down — culture-negative	Amoxicillin (high-dose) or amoxicillin-clavulanate	Switch once drainage complete, improving, and off oxygen.

⚠ DURATION, COURSE & FOLLOW-UP

- Total **3–4 weeks** is reasonable with adequate drainage and no further complication; transition to **oral** at or just before discharge once drained, improving, and off oxygen. **Persistent fever > 72 h is common** and — if the child is otherwise improving — usually **not** treatment failure; serial CRP can confirm improvement.⁷⁸
- Outcome is excellent: lung function and radiograph normalize over weeks–months (a repeat film at 2–3 months is reasonable).
- Watch necrotizing pneumonia for **bronchopleural fistula** → involve surgery / respirology / infectious diseases.⁵⁹

▶ ADMIT · DISCHARGE · REFER

- **Admit** — clinically significant empyema or a moderate-to-large effusion, for IV antibiotics ± drainage. A **small parapneumonic effusion in a well child** can be managed as community-acquired pneumonia with close follow-up rather than admitted.

- **Discharge** — not acutely; complete therapy with follow-up imaging.
- **Refer** — respiratory/surgery for drainage (chest drain ± fibrinolytics or VATS).

Acute Otitis Media ASSESS

INFANT-CHILD • CPS 2016 / AAP 2013

Diagnose by the drum, not the symptoms — then choose between watchful waiting and antibiotics.

▲ CLINICAL DIAGNOSIS – EFFUSION + ACUTE INFLAMMATION

- Acute onset (otalgia, ear-tugging, or non-specific signs in the non-verbal child) **plus** a middle-ear effusion **plus** signs of acute inflammation.
- **A bulging tympanic membrane is the key sign** — highest predictive value for bacterial AOM, especially if yellow, cloudy, or hemorrhagic.⁴⁴
Acute perforation with purulent otorrhea also confirms a bacterial cause — distinguish from otitis externa, chronic drainage, or tube discharge.

- **Not AOM:** a non-bulging drum, or erythema / air-fluid level alone without bulging. Effusion without inflammation is OME⁴⁴ — do not give antibiotics.
- **Scope:** the **watchful-waiting** option applies only to healthy children ≥ 6 **months** (CPS 2016 / AAP 2013). **Infants < 6 months are excluded from observation**⁴⁴ — treat all with antibiotics. Excludes tympanostomy tubes, craniofacial anomalies, immunocompromise, or recurrent AOM.

DECISION BY AGE & SEVERITY

DECISION

How old is the child, and how severe?

< 6 MONTHS

Antibiotics for all — no watchful waiting (below the CPS/AAP observation range; the exam is less reliable and complications are more likely). In a **febrile infant ≤ 90 days**, do **not** attribute the fever to AOM alone — evaluate for invasive bacterial infection (see [Fever in the Well-Appearing Infant](#)).

≥ 6 MO – TREAT NOW

Bulging, erythematous TM **and** any of:
highly febrile ($\geq 39^{\circ}\text{C}$),
moderately-to-severely ill, severe
otalgia, symptoms ≥ 48 h, perforation
with otorrhea, or complicated disease
(mastoiditis, facial palsy).⁴⁴

≥ 6 MO – WATCHFUL WAITING (24–48 H)

Otherwise-healthy child, mildly ill, alert,
responding to antipyretics, low-grade
fever ($< 39^{\circ}\text{C}$), mild otalgia — or when
the diagnosis is uncertain. Needs
assured follow-up.

◆ HOW TO WATCHFUL-WAIT (CHOOSE ONE)

- Reassess within 24–48 h to document the course; **OR** have the caregiver return if no improvement or worsening by 48 h; **OR** give a “safety-net” prescription to fill only if not improved by 48 h.
- Always give analgesia advice; ensure ready access to care.

Drug	Dose	Route & notes
Amoxicillin (first choice)	40–45 mg/kg/dose BID	PO. Drug of choice — covers <i>S. pneumoniae</i> ; excellent middle-ear penetration. ⁴⁴
Amoxicillin-clavulanate	40–45 mg/kg/dose BID (amoxicillin component)	PO. If AOM + purulent conjunctivitis, amoxicillin in the last 30 days, or failure at 48–72 h. Specify the 14:1 suspension. ⁴⁴
Cefuroxime-axetil	15 mg/kg/dose BID (30 mg/kg/day; max 1 g/day)	PO, twice daily. Non-anaphylactic penicillin allergy (cefprozil is an alternative).
Ceftriaxone	50 mg/kg IM/IV once daily × 3 days	If oral not tolerated or amoxicillin-clavulanate failed. Reserve.

◆ DURATION – BY AGE

- < 2 years: 10 days. ≥ 2 years, uncomplicated: 5 days.⁴⁴
- Also 10 days for a perforated TM or recurrent AOM.
- Symptoms should improve within 24 h and resolve in 2–3 days — reassess if they persist or worsen.

⚠ **ANALGESIA IS PART OF EVERY PLAN**

- Pain control (acetaminophen or ibuprofen) is essential whether or not antibiotics are given — it is often the child's main need.
- Persistent effusion after treatment does **not** warrant a change of antibiotic; a middle-ear effusion can linger for months.⁴⁴

▴ **ADMIT · DISCHARGE · REFER**

- **Admit** — a complication (mastoiditis, intracranial spread, facial palsy), a toxic child, or failure of second-line therapy.
- **Discharge** — uncomplicated disease — analgesia, with observation or antibiotics per age and severity, and safety-netting.
- **Refer** — ENT for suspected mastoiditis, recurrent or complicated disease, or a non-resolving effusion.

GAS Pharyngitis

ASSESS

≥ 3 YR • IDSA 2025

Most sore throats are viral. Score first to decide who needs a test — then treat the confirmed case.

▶ WHEN TO TEST – SCORE FIRST

- Sore throat is usually viral, and clinician judgment alone is unreliable. GAS causes up to ~30% of childhood (and ~15% of adult) pharyngitis.⁴⁵
- IDSA 2025 suggests a **clinical scoring system to decide who to test**. A score is most useful for spotting **low-probability** patients in whom testing is unlikely to help — prefer a score that needs no lab (Centor / McIsaac).⁴⁵
- **Does not apply under 3 years** — GAS in this age may not show the typical features; test only for an outbreak, scarlet fever, or a household case.⁴⁶

- Score criteria point to GAS: fever, tonsillar swelling or exudate, tender anterior cervical nodes, and **absence of cough**. Cough, coryza, or hoarseness point to a virus.

Centor / McIsaac – one point each	Point
Tonsillar exudate or swelling	+1
Tender / swollen anterior cervical nodes	+1
History of fever	+1
No cough	+1
Age 3–14 years (McIsaac)	+1

◆ HOW TO TEST

- Read the score as a probability of GAS: **≤1 LOW** (~1–10%) → testing can be deferred; **2–3 INTERMEDIATE** (~11–35%) → test; **≥4 HIGH** (~50%) → test, and still confirm before treating.⁴⁵
- Test with a rapid antigen test (RADT), a molecular test (NAAT), or throat culture.
- In children, confirm a **NEGATIVE** rapid antigen test with culture, or use a molecular test.⁴⁶

- The score selects *who to test* — never a treatment threshold.
- Do **NOT** use ASO titers to diagnose acute pharyngitis.⁴⁵
- Start treatment within 9 days of onset.

⚠ TEST EVEN IF THE SCORE IS LOW

Test regardless of a low score when the patient is high-risk: a **household GAS case** (shared indoor living or sleeping space), prior **rheumatic fever**, or features of complicated or systemic GAS — peritonsillar or retropharyngeal abscess, scarlet fever, or toxic shock.⁴⁵

Lab-confirmed GAS	Drug & dose	Duration
First choice	Amoxicillin 50 mg/kg once daily (max 1 g), or penicillin V ⁴⁶	10 days
Poor adherence expected	Benzathine penicillin G IM – < 27 kg: 600,000 U; ≥ 27 kg: 1,200,000 U	1 dose

Lab-confirmed GAS	Drug & dose	Duration
Non-immediate (delayed / non-IgE) penicillin allergy	Cephalexin 20 mg/kg/dose BID (max 500 mg/dose) ♦ if unavailable: cefuroxime-axetil 20 mg/kg/day ÷ BID (max 500 mg/day) PO	10 days
Documented penicillin anaphylaxis / IgE-mediated (immediate) reaction	Azithromycin 12 mg/kg on day 1 (max 500 mg) then 6 mg/kg (max 250 mg) once daily on days 2–5 (IDSA, as corrected 2014) ⁴⁶ – the FDA-labeled 12 mg/kg once daily × 5 days (max 500 mg/day) is an accepted alternative ²³³ ; or clarithromycin 7.5 mg/kg/dose BID (max 250 mg/dose) × 10 days; or clindamycin 7 mg/kg/dose TID (max 300 mg/dose) × 10 days	5–10 days

⚠️ WHY WE TREAT

- Antibiotics prevent suppurative complications and acute rheumatic fever (but not post-streptococcal glomerulonephritis). **All GAS is penicillin-susceptible**, but macrolide/clindamycin resistance reaches ~20% and varies locally — reserve those for true penicillin anaphylaxis and confirm local susceptibility.⁴⁶
- Acetaminophen or ibuprofen for pain and fever; corticosteroids are not recommended as analgesic adjuncts.⁴⁵

▶ ADMIT · DISCHARGE · REFER

- **Admit** — rarely — complications (peritonsillar abscess, dehydration, systemic illness).
- **Discharge** — most, with antibiotics and safety-net advice.
- **Refer** — ENT for abscess or recurrent tonsillitis.

Urinary Tract Infection

HOURS

> 2 MONTHS

A common occult source of fever in infants — get a reliable sample before antibiotics.

◀ WHEN TO SUSPECT & TEST

- **Preverbal child** with unexplained fever ($> 39^{\circ}\text{C}$, no source) — especially **fever $> 48\text{ h}$** without another focus. A clear viral source (coryza, cough, diarrhea, rash) makes UTI unlikely.⁴⁷
- **Verbal child ($\geq 3\text{ yr}$):** dysuria, frequency, hematuria, abdominal/back pain, or new daytime incontinence. Beware bubble-bath / hygiene vulvovaginitis — sterile cultures, often mis-treated as UTI.
- **Usual pathogens:** *E. coli* (most), then *Klebsiella*, *Enterobacter*, *Citrobacter*, *Serratia*; *S. saprophyticus* in adolescent girls.⁴⁷
- Do not over-diagnose: low colony counts, mixed growth, or **no pyuria** usually mean contamination or asymptomatic bacteriuria.

SAMPLING – BEFORE ANTIBIOTICS

- **Toilet-trained:** clean-catch midstream. **Not toilet-trained:** catheter or suprapubic aspirate for culture.
- A bag specimen may screen the urinalysis but its culture is unreliable (contamination up to 63%) — never use it to diagnose.⁴⁷
- A single antibiotic dose sterilizes urine; always collect first.⁴⁷

◆ **INTERPRETING THE URINALYSIS** UTI is unlikely if the urinalysis is completely normal.

LEUKOCYTE ESTERASE OR NITRITE POSITIVE ≈ 93% sensitive; **NITRITE ALONE** is highly specific (~98%) but often falsely negative (frequent voiding, Gram-positive / non-nitrate organisms)⁴⁷. **PYURIA** (> 5 WBC/HPF, or ≥ 10/μL) supports infection.

DIAGNOSTIC COLONY-COUNT THRESHOLDS

Collection method	Minimum count (CFU/mL)	Note
Clean catch / midstream	≥ 10 ⁵	Mixed growth usually = contamination.
In-and-out catheter	≥ 5 × 10 ⁴	Indwelling-catheter specimens less reliable.

Collection method	Minimum count (CFU/mL)	Note
Suprapubic aspirate	Any growth	Most reliable method. ⁴⁷

ADMIT · DISCHARGE · REFER

- **Admit** — young infants, ill/septic, unable to tolerate oral, or complicated/obstructive UTI.
- **Discharge** — well child on oral antibiotics with follow-up.
- **Refer** — nephrology/urology for recurrent, atypical, or obstructive UTI, or abnormal imaging.

UTI – Treatment & Follow-up HOURS

LOCAL SUSCEPTIBILITY

Oral is fine for most non-toxic children — pick the empiric agent from your antibiogram; plan for ESBL.

DECISION

Cystitis or febrile UTI / pyelonephritis?

CYSTITIS • AFEBRILE LOWER TRACT

Dysuria/frequency, no fever (mainly post-pubertal girls). **2–4 days** of an oral agent chosen by local *E. coli* susceptibility. No imaging needed.⁴⁹

FEBRILE UTI / PYELONEPHRITIS

Treat **7–10 days**. **Oral is as effective as IV** in non-toxic children likely to tolerate every dose.⁴⁷ Use IV initially if toxic, < 2–3 months, vomiting, or unable to tolerate oral — then step down.

◆ ESBL RISK – PLAN FOR IT

- ESBL-producing Enterobacterales are increasingly isolated even without prior antibiotics — often resistant to cephalosporins and many oral agents.
- Where local ESBL rates are high or the child is high-risk: **AMIKACIN (IV)** is the most reliable aminoglycoside — active against the large majority of pediatric ESBL isolates, whereas gentamicin is variable (roughly half)⁹⁵. **TMP-SMX (PO)** is a useful oral step-down once susceptibility is confirmed, though ESBL isolates are often co-resistant.

- Fluoroquinolones are not routine in prepubertal children but may be used if the isolate is resistant to other oral options.

ANTIMICROBIAL OPTIONS – IF THE ISOLATE IS SUSCEPTIBLE

Drug	Dose	Notes
Amikacin (IV)	15 mg/kg once daily	ESBL-active empiric choice; monitor levels ⁹⁵ & renal function if > 48 h.
Gentamicin (IV/IM)	5–7.5 mg/kg once daily	ESBL activity variable (~half of isolates) — confirm susceptibility. Monitor levels / renal function > 48 h.
Ceftriaxone (IV/IM)	50–75 mg/kg once daily (max 2 g)	Less nephrotoxic but broader; not reliable if ESBL.
Ampicillin (IV)	50 mg/kg/dose q6h (max 2 g/dose)	± aminoglycoside; many <i>E. coli</i> now resistant. (Lexicomp)
TMP-SMX (PO)	4–5 mg/kg/dose (TMP) BID	Preferred oral step-down once susceptible — covers many ESBL isolates if tested S. ⁴⁷
Cefixime (PO)	8 mg/kg/dose once daily	Common first oral choice pending susceptibilities (if low ESBL).

Drug	Dose	Notes
Cefdinir (PO)	7 mg/kg/dose BID or 14 mg/kg once daily (max 600 mg/day)	Palatable oral step-down; not always on the MOH formulary . Like cefixime, not reliable against ESBL .
Cephalexin (PO)	12.5 mg/kg/dose QID	Oral cephalosporin option.
Amoxicillin-clavulanate (PO)	13 mg/kg/dose (7:1) TID	If susceptible; amoxicillin alone often resisted.
Ciprofloxacin (PO) *	15 mg/kg/dose BID	* Not routine in prepubertal children — reserve for resistant isolates.

◁ COMPLICATED UTI – ESCALATE & IMAGE EARLY

- Hemodynamic instability, **rising creatinine**, abdominal/bladder mass, poor urine flow, or **no improvement by 24 h / fever not settling by 48 h** on appropriate antibiotics.
- Use **IV until clearly improving**; start with a **renal & bladder ultrasound (RBUS)** to look for obstruction or abscess.⁴⁷

⚠️ IMAGING & FOLLOW-UP

- **RBUS** after a first febrile UTI in a child **< 2 yr** (during, or within ~2 weeks of, the illness)⁴⁷.
- **VCUG is not routine** — obtain it only if RBUS shows **marked hydronephrosis, a dilated ureter, a duplex system / ureterocele, or obstruction**, or after a **second febrile UTI**. A normal RBUS does not exclude VUR (**VCUG** grades reflux; **DMSA** for scarring)⁴⁹.
- Routine antibiotic **prophylaxis is not recommended** while awaiting RBUS.⁴⁷
- Repeat urinalysis / culture to confirm cure is **not recommended** unless symptoms persist.
- Counsel families to test urine early in any future unexplained fever.
- Recurrent afebrile UTI is often driven by **bladder–bowel dysfunction** — assess and treat constipation (a major, modifiable risk factor).⁴⁷

Skin & Soft Tissue Infections

ASSESS

Match the antibiotic to the layer and the pathogen — most are oral, short-course, and cephalexin-first.

▴ TYPES – BY DEPTH & MORPHOLOGY

- **Impetigo** (epidermis): non-bullous (honey-crusted) or bullous; highly contagious. Mostly group A streptococcus (GAS) / *S. aureus*.
- **Erysipelas / cellulitis** (dermis → subcutis): spreading erythema, warmth, tenderness. Non-purulent cellulitis is more often **GAS**; purulent points to *S. aureus*.¹⁹¹
- **Folliculitis / furuncle / carbuncle / abscess**: discrete pus collections — *S. aureus* (methicillin-susceptible, MSSA, or methicillin-resistant, MRSA). Source control is the priority.
- **Necrotizing infection** (fascia/muscle): pain out of proportion, rapid spread, necrosis, crepitus¹⁹¹ — surgical emergency. GAS ± polymicrobial.

DECISION

How severe, and is there pus?

MILD / NON-PURULENT

Impetigo or small localized lesion, systemically well — often resolves with skin/wound care alone. Limited: topical

mupirocin 2% TID × 5 days.

Widespread/multiple lesions or systemic features: oral **cephalexin**.

PURULENT – ABSCESS / FURUNCLE

Incision & drainage is the treatment.

For a small, drained abscess in a healthy host without surrounding cellulitis, antibiotics are usually **not** needed; stop them once source control is achieved.¹⁹¹

SEVERE / DEEP / NECROTIZING

Extensive or rapidly progressive, systemic toxicity, necrosis or crepitus

→ **admit, IV antibiotics, urgent surgery.** Add an anti-toxin agent (clindamycin) and consult infectious diseases.

Oral cephalexin is the antimicrobial of choice when antibiotics are warranted.

BY PATHOGEN / SETTING

Target	Drug & dose	Notes
MSSA / GAS — oral, first-line	Cephalexin 25 mg/kg/dose (max 1 g) PO QID; high-dose TID for moderate cellulitis	Drug of choice. Covers MSSA and GAS. High-dose enables outpatient moderate cellulitis. ¹⁹¹
MSSA / GAS — IV (admitted)	Cefazolin 33 mg/kg/dose IV q8h (max 2 g/dose)	IV first-line for non-toxic inpatients. Step down to oral cephalexin on improvement.
GAS — streptococcal	Amoxicillin 25 mg/kg/dose PO BID; penicillin G IV if severe	For confirmed/likely pure GAS (erysipelas, non-purulent cellulitis).
MRSA — oral, first-line	Trimethoprim-sulfamethoxazole 4–6 mg/kg/dose (trimethoprim component) PO BID	Preferred oral MRSA agent ¹⁹¹ — also covers GAS. Use when colonized or MRSA strongly suspected.
MRSA — alternative	Clindamycin 10 mg/kg/dose PO/IV q8h	Only where local clindamycin susceptibility is high. Anti-toxin role in severe disease.

Target	Drug & dose	Notes
MRSA — IV (severe)	Vancomycin 15 mg/kg/dose IV q6h	Add when MRSA suspected in toxic/deep infection; stop once MRSA excluded. <i>K. kingae</i> is resistant to vancomycin and clindamycin and shows reduced susceptibility (high MICs) to cloxacillin — so cefazolin/cephalosporins, which cover it, are preferred empirically in under-4s. ¹⁹¹

◆ DURATION & STEP-DOWN

- **5-7 DAYS IS USUALLY ENOUGH**, even for many severe SSTIs (rarely > 7 days).¹⁹¹
- Failure to improve → re-evaluate diagnosis and source control, not simply broaden.
- Avoid routine combination therapy (e.g., cephalexin + TMP-SMX) and broad outpatient IV (ceftriaxone).
- If a child improves on a non-MRSA agent but is later found MRSA-colonized/positive, **DO NOT SWITCH** — clinical response governs.

⚠️ PREVENTING RECURRENCE

- Protect the skin barrier: optimize **eczema** control, diligent wound care, cover open lesions, treat scabies/lice, and encourage varicella vaccination.
- Reserve **decolonization** (5-day nasal mupirocin + chlorhexidine washes, or dilute bleach baths) for recurrent moderate–severe SSTIs or household clusters — success is variable and re-colonization common.¹⁹¹

▀ ADMIT · DISCHARGE · REFER

- **Admit** — systemic illness, rapid spread, an abscess needing drainage, immunocompromise, or failed oral therapy; suspected necrotizing infection → emergency.
- **Discharge** — localized, well child on oral antibiotics with review.
- **Refer** — surgery for drainage or necrotizing infection.

Cervical Lymphadenopathy

ASSESS

COMMON

Very common and usually reactive — your job is to separate the benign node from the few that need antibiotics, imaging, or a malignancy work-up.

▀ WHAT COUNTS AS ABNORMAL

- A small, mobile, palpable node is **normal** in most healthy children. Treat a **cervical node > 1 cm** as enlarged (inguinal > 1.5 cm; epitrochlear / post-auricular > 0.5 cm).²⁰⁴
- A **supraclavicular** node is **never normal** — it is the single highest-risk site for malignancy (up to ~75%). Image and investigate every one.²⁰³
- Sort by **tempo & laterality** — it points to the cause.

PATTERN → LIKELY CAUSE

Pattern	Think	Action
Acute, bilateral, tender	Reactive viral URTI (commonest); EBV / CMV if prolonged with splenomegaly	Supportive, no antibiotics, safety-net. EBV serology if a glandular-fever picture.
Acute, unilateral, tender, warm ± fluctuant	Bacterial lymphadenitis – <i>S. aureus</i> (incl. MRSA) & group A strep	Empiric oral antibiotics 10 d; ultrasound if fluctuant → drain an abscess. Review at 48 h.
Subacute (2–6 wks) / chronic (> 6 wks), often non-tender	Nontuberculous (atypical) mycobacteria (violaceous, "cold", young child), cat-scratch (Bartonella), TB, toxoplasma	Don't just re-course antibiotics — investigate. NTM → complete surgical excision; cat-scratch → usually self-limited (azithromycin if bulky/symptomatic). ²⁰⁴

△ CAN'T-MISS – RED FLAGS FOR MALIGNANCY (LYMPHOMA / LEUKEMIA / METASTASIS)

- **Supraclavicular** node; **hard**, **matted**, **fixed**, or **painless / rubbery**; **> 2 cm and enlarging**; persisting or progressive **beyond 4–6 weeks**; **generalized** nodes (> 2 non-contiguous regions) or **hepatosplenomegaly**; **B symptoms** — unexplained fever >

38°C, drenching night sweats, weight loss > 10%; a **mediastinal mass** on chest X-ray.

- Any one → urgent bloods + imaging and refer. **Strongest independent predictors (2024 cohort):** a larger node — each extra centimeter roughly **doubles** the odds, a node > **3 cm** ≈ **12% malignant** — and **older age** (> 12 yr ≈ 20% vs ≤ 2.5 yr < 1%); firm consistency, weight loss, and an abnormal chest X-ray also raise the odds.²⁰³

▶ **WORK-UP (WHEN IT ISN'T SIMPLE REACTIVE)**

- **Bloods:** CBC + differential + **blood film** (blasts, cytopenias), **ESR / CRP, LDH**; EBV / CMV serology; a **TB test (IGRA or tuberculin)**; Bartonella / toxoplasma serology if exposed.
- **Imaging: ultrasound first-line** (size, abscess, architecture); chest X-ray if any red flag (mediastinal mass).
- **Biopsy** (excisional preferred) for a red-flag or supraclavicular node, **failure to regress by 4–6 weeks**, or progression despite treatment.²⁰¹

EMPIRIC ANTIBIOTICS – ACUTE BACTERIAL LYMPHADENITIS (10 DAYS)

Setting	Oral	Notes
Cover <i>S. aureus</i> + GAS	Cephalexin 25–50 mg/kg/day ÷ 3–4 OR amoxicillin-clavulanate	Review at 48 h; most settle within days. Add source control if an abscess forms. ²⁰⁴
MRSA risk / penicillin allergy	Clindamycin 20–40 mg/kg/day ÷ 3–4	Also covers GAS; use per local MRSA rates.
Fluctuant / abscess	Incision & drainage + culture	Ultrasound-guided if deep; send pus for routine and mycobacterial culture.

✦ DON'T FORGET

- A cervical node **> 1.5 cm** is a clinical criterion of **Kawasaki disease** — reassess if fever has lasted ≥ 5 days.³⁵
- Tender anterior nodes with exudative pharyngitis suggest **GAS pharyngitis**.
- A firm supraclavicular or matted node in a child with pallor or bruising → treat as possible malignancy and refer **same day**.

✦ YOUR ROLE – THE GENERAL PEDIATRICIAN

- **Most nodes are yours to manage.** Do the first-line assessment and reassurance; classify by duration (**acute ≤ 2 wks** · **subacute 2–6 wks** · **chronic > 6 wks**) and actively look for a red flag.
- Treat acute bacterial lymphadenitis empirically and **review at 48 h. Safety-net every child and set a definite review point** — a node that has not begun to shrink by **~4 weeks**, or is growing, needs work-up.
- You can start the first-line bloods and request the ultrasound yourself. **Refer early rather than re-course antibiotics** for a persistent or red-flag node: **same-day to hematology-oncology** for a hard/matted/supraclavicular node or systemic features; **surgery/ENT** for excision biopsy (persisting > 6 weeks or progressive); **infectious diseases** for suspected TB or atypical mycobacterial disease.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — systemically unwell, a large / fluctuant node needing drainage, airway or deep-space concern, or a strong malignancy suspicion.
- **Discharge** — the well child with a small, mobile, reactive node → safety-net and review in 2–4 weeks.
- **Refer** — surgery / ENT for excision (NTM, abscess, biopsy); hematology-oncology for a red-flag node; infectious diseases for TB or atypical infection.

SOURCE

Pediatric cervical lymphadenopathy: an evidence-based approach for GPs (*Br J Gen Pract* 2026;76(764):137); Refining indications for pediatric cervical lymph-node biopsy (*Eur J Pediatr* 2026); Risk factors for malignancy in subacute/chronic focal craniocervical lymphadenopathy (*Front Pediatr* 2024); AAFP — Management of Cervical Lymphadenitis in Children (*Am Fam Physician* 2008); AAP Red Book 2024 (nontuberculous mycobacteria, cat-scratch disease).

Fever – Assessment & Antipyretics

ASSESS

NICE NG143

Fever $\geq 38^{\circ}\text{C}$. Treat the child's discomfort and risk — not the number.

▲ RISK-ASSESS BY TRAFFIC LIGHT (AND BY AGE)

- Any infant **under 3 months** with fever is high-risk → use the **Fever in the Young Infant** pathway.
- **Red (high-risk):**
pale/mottled/ashen/blue, no response to social cues, unrousable, weak/high-pitched cry, grunting, respiratory rate > 60 , reduced skin turgor, bulging fontanelle, neck stiffness, non-blanching rash, focal seizures/neurology.²⁴¹
- **Amber (intermediate):** pallor, decreased activity, poor feeding, dry mucous membranes, reduced urine, nasal flaring, tachycardia, fever ≥ 5 days, rigors, non-weight-bearing/limb swelling.²⁴¹
- **Green (low-risk):** normal color, responsive, content, moist membranes — manage at home with safety-netting.

ANTIPYRETIC DOSING

Drug · route	Dose	Max · notes
Acetaminophen — PO	10–15 mg/kg/dose every 4–6 h PRN	≤ 1 g/dose, ≤ 5 doses/24 h; max 75 mg/kg/day (not > 4 g/day). Onset 30 ²⁴² –60 min.
Acetaminophen — IV	15 mg/kg/dose every 6 h	< 50 kg: max 15 mg/kg (750 mg/dose); ≥ 50 kg: 1 g/dose. ≤ 75 mg/kg/day.
Ibuprofen — PO	5–10 mg/kg/dose every 6–8 h	Age ≥ 6 months. Max 400 mg/dose, 40 mg/kg/day. Give with food + ensure hydration. ²⁴²
Ibuprofen — IV	10 mg/kg/dose every 6–8 h	Age ≥ 6 months. Max 400 mg/dose, 40 mg/kg/day.

◆ AVOID

- Acetaminophen — severe liver failure.
- Ibuprofen — dehydration, renal impairment, active GI bleeding / ulcer disease, or any poorly hydrated child.

⚠ USE ANTIPYRETICS FOR COMFORT, NOT THE

THERMOMETER

- Give only while the child is uncomfortable.²⁴²
- Combining or alternating the two clears fever faster and gives more fever-free time (PITCH trial), but does **not** improve comfort — so for comfort one agent is usually enough.¹⁶⁶
- Switch agents if discomfort is not relieved; alternate only if discomfort recurs before the next dose; counsel carefully to avoid dosing errors.
- Tepid sponging is not recommended; don't over- or under-wrap. **Antipyretics do not prevent febrile seizures.**²⁴²
Encourage fluids; safety-net for red/amber signs.

The Rash – Approach & Red Flags

MINUTES

Two questions first: does the child look unwell, and does the rash blanch?

▲ STEP 1 – DOES THE CHILD LOOK UNWELL?

(BEFORE FOCUSING ON THE RASH)

- Rapid ABCDE-style screen: work of breathing; color and perfusion (capillary refill > 2–3 s, mottled, cold peripheries); heart rate and respiratory rate for age; mental status (irritable, lethargic); fever; neck stiffness, photophobia, bulging fontanelle in infants.
- Any red flag → urgent assessment + empiric IV antibiotics (see **Septic Shock**), **in parallel** with the rash exam. Looks well → Step 2.
- **A blanching rash does not rule out sepsis** — meningococcal disease often begins as a blanching maculopapular rash before turning petechial or purpuric, and the classic non-blanching rash can be a late sign.¹²² Escalate an unwell child whatever the rash looks like.
- **Darkly pigmented skin:** erythema, petechiae and purpura may appear dark brown or violaceous and be masked — check palms, soles, conjunctivae and mucosa, and lower the threshold to escalate.¹²²

STEP 2 – DOES THE RASH BLANCH? (PRESS A CLEAR GLASS AGAINST IT)

DECISION

Does the rash blanch under glass?

NON - BLANCHING

Petechiae (< 2 mm) or purpura (> 2 mm) — **purpura carries higher risk.**¹²²

Unwell or febrile → treat as invasive meningococcal disease / sepsis until excluded. A **well** child still needs a diagnosis and follow-up (see below) — do not dismiss just because they look well.

BLANCHING

Usually benign — viral exanthem, urticaria, or eczema. Treat symptomatically; safety-net with clear advice to "come back if the child becomes unwell, even if the rash looks the same or fades."²⁴¹

NON-BLANCHING – SORT THE CAUSE

Cause	Key features
Meningococcal / invasive bacterial	Ill, febrile — emergency (identified at Step 1)

IgA vasculitis (Henoch– Schönlein)	Palpable purpura on buttocks and legs in a <i>well</i> child; abdominal pain, arthralgia/arthritis; check urine for blood and protein (renal involvement)
Immune thrombocytopenia (ITP)	Well child, isolated low platelets, bruising; often post-viral; watch for mucosal bleeding (epistaxis, gums)
Mechanical purpura	Forceful cough, vomiting or breath-holding; classically above the nipple line (head and neck, SVC territory) — benign
Leukemia	Pallor, hepatosplenomegaly, fatigue, bone pain, unexplained bruising

BLANCHING – SORT THE CAUSE

Cause	Key features
Viral exanthems	Roseola, measles, rubella, slapped-cheek (parvovirus B19); hand-foot-and-mouth and varicella are vesicular, not purely maculopapular
Scarlet fever	Sandpaper rash + strawberry tongue → penicillin

Urticaria

Transient wheals → antihistamine;
airway or systemic features → see
[Anaphylaxis](#)

Kawasaki disease

Fever ≥ 5 days + rash +
non-purulent conjunctivitis +
mucositis/strawberry tongue +
extremity changes + cervical node
— easily mistaken early for scarlet
fever or a viral exanthem (see
[Kawasaki Disease](#))

⚠ ANY UNWELL CHILD IS THE EMERGENCY

- Escalate immediately — with or without a rash, and whatever the rash looks like.
- This alone is sufficient.

▶ ESCALATE URGENTLY – REGARDLESS OF BLANCHING – ALSO IF

- Non-blanching rash with fever, ill appearance, or rapid spread.
- **Retiform purpura** (branching, angulated, stellate) rather than round petechiae/purpura — suggests vascular occlusion (DIC, severe sepsis, vasculitis) and is higher-acuity than ordinary purpura.¹²²

- **Mucosal involvement** (eyes, mouth, genitals) with blistering or skin peeling → Stevens–Johnson syndrome / TEN or toxic shock syndrome. Superficial peeling with **perioral crusting and radial fissures but mucous membranes spared**, especially in young children → **staphylococcal scalded skin syndrome (SSSS)** — mucosal sparing is the classic clue that separates SSSS from SJS/TEN. The **Nikolsky sign** (gentle shearing pressure separates the epidermis) is positive in SJS/TEN, SSSS, and pemphigus.
- **Vesicles in a neonate** → herpes simplex. Skin vesicles are **absent in roughly 40–50% of neonatal HSV with CNS or disseminated disease** — keep the threshold low for a full sepsis + HSV workup (and empiric acyclovir) in any unwell neonate, even with no rash.¹²³
- Fever $\geq$ 5 days with rash plus any Kawasaki features.³⁵

◆ BEDSIDE TIPS

Faded or changed rash? Ask whether a parent photographed it — useful for evanescent eruptions like urticaria.

Stroke test: a wheal along a tongue-blade stroke = dermatographism (common, benign); a wheal raised specifically over a pigmented lesion = **Darier's sign** (mastocytosis — worth recognizing in recurrent unexplained wheals). **Herpes zoster** in a dermatomal band is usually benign and immunocompetent in children, especially after early-life varicella or vaccination; consider underlying immunodeficiency only if it is multidermatomal, sacral, recurrent, prolonged or treatment-refractory, or has invasive features.

Preseptal & Orbital Cellulitis HOURS

CPS 2026

The orbital septum is the firewall — eye signs separate sight-threatening orbital disease from benign preseptal swelling.

► THE DISTINCTION THAT DRIVES EVERYTHING

- The **orbital septum** is the anatomic firewall. **Preseptal (periorbital)** = infection anterior to it — usually from skin trauma or sinusitis; mostly benign. **Orbital** = posterior to it — a

complication of **ethmoid sinusitis**
eroding the lamina papyracea;
threatens vision and brain.⁹¹

- The distinction drives etiology, imaging, antibiotics, and disposition.⁹¹

DIFFERENTIATE AT THE BEDSIDE

Clinical feature	Preseptal	Orbital
Eyelid swelling & erythema	Present	Present
Pain with eye movement	Absent	Present
Extraocular movements	Normal	Impaired / restricted
Proptosis	Absent	Often present
Chemosis / diplopia / ↓ acuity / RAPD	Absent	May be present
Fever / toxic appearance	May be present	Usually present

◁ ETIOLOGY & MICROBIOLOGY

- **Preseptal — traumatic:** skin flora after a bite/scratch → *S. pyogenes*, *S. aureus*.
- **Preseptal — atraumatic:** a complication of bacterial rhinosinusitis.
- **Orbital:** mirrors acute sinusitis — *Streptococcus (anginosus, pyogenes, pneumoniae* groups); *S. aureus* and oral anaerobes in chronic sinusitis / older children; *H. influenzae* if under-immunized.⁹¹
- **MRSA:** generally low but rising — risk factors do not reliably predict it⁹¹; a **negative nasal MRSA swab reliably excludes** it and guides the oral step-down.

◆ IMAGING – CONTRAST CT ORBITS + SINUSES

- Well children with uncomplicated **PRESEPTAL** disease need **NO IMAGING**.⁹¹
- Contrast CT is first-line; MRI preferred if intracranial extension or cavernous sinus thrombosis is suspected.
- Its job: find who needs surgery or has a complication.

- Image **IF** toxic appearance, optic-nerve signs (abnormal vision, RAPD), intracranial signs (focal deficit, seizure), prominent forehead swelling, or high-risk labs (CRP > 120 mg/L, WBC > 20,000/ μ L)⁹¹.
- Otherwise reasonable to **DELAY CT 24-36 H** to judge response when ENT / ophthalmology are available.⁹¹

▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — orbital signs, systemic illness, age < 1, and all true orbital cellulitis → IV antibiotics.
- **Discharge** — mild preseptal in a well child on oral antibiotics with 24 h review.
- **Refer** — ophthalmology and ENT for orbital involvement; surgery for an abscess.

Preseptal & Orbital – Management

HOURS

RISK-BASED

Preseptal is mostly oral outpatient; orbital is admitted, IV, and shared with ENT + ophthalmology.

DECISION

Preseptal or orbital?

PRESEPTAL – MOSTLY OUTPATIENT

Well, immunized child, eyelid erythema/edema with **normal painless eye movements, normal vision, no proptosis** → oral antibiotics + review. Admit the very young, severe, or not-improving.

ORBITAL – ADMIT, EMERGENCY

Pain/restriction on eye movement, **proptosis, chemosis, diplopia, reduced acuity, or RAPD** → admit: CBC/CRP, blood cultures, IV antibiotics, **same-day ENT + ophthalmology**, CT per criteria. Do not swab the conjunctiva (contamination).

WHAT TO SEND

- Uncomplicated preseptal, well child: no work-up.
- Severe / uncertain / young child:
CBC + CRP. Severe disease: **blood cultures before antibiotics**. Drained pus / abscess → culture ± PCR.
MRSA nasal swab where available — a negative result reliably excludes MRSA and guides the oral step-down.

EMPIRIC ANTIMICROBIAL THERAPY

Scenario	First-line	Dose (max) · MRSA / notes
Traumatic preseptal (bite / broken skin)	PO cephalexin IV cefazolin	Cephalexin 25 mg/kg/dose q6h (100 mg/kg/day, max 4 g/day); cefazolin 33 mg/kg/dose q8h (100 mg/kg/day). MRSA suspected → TMP-SMX 4–6 mg/kg/dose (TMP) q12h or clindamycin. ⁹¹
Atraumatic preseptal (from sinusitis)	PO cephalexin IV cefazolin	Doses as above. Escalate to / add amoxicillin-clavulanate (co-amoxiclav 30 mg/kg [amoxicillin] q8h) if no response by 24–48 h or the child is under-immunized. ⁹¹

Scenario	First-line	Dose (max) · MRSA / notes
Orbital cellulitis (admitted)	IV ceftriaxone + vancomycin	Ceftriaxone 50 mg/kg q12h (max 4 g/day) + vancomycin 15 mg/kg q6h (target levels). Alternative: ceftriaxone + clindamycin (adds anaerobe/MSSA cover) or amoxicillin-clavulanate. PO step-down by culture + MRSA swab. ⁹¹
Severe · toxic / intracranial / optic-nerve	IV ceftriaxone (or cefotaxime) + vancomycin + metronidazole	Ceftriaxone 50 mg/kg q12h + vancomycin 15 mg/kg q6h + metronidazole 10 mg/kg q8h. CNS-penetrating triple cover; consult ID; assess for surgery.

◆ WHY THESE CHOICES – ADAPTED TO LOCAL EPIDEMIOLOGY

- In Kuwait, near-universal Hib and pneumococcal conjugate vaccination reduces the *H. influenzae* / *S. pneumoniae* contribution, so a first-generation cephalosporin (cephalexin/cefazolin — strong streptococcal + MSSA cover) is a reasonable narrow empiric choice for skin- and

sinus-origin preseptal disease; escalate to amoxicillin-clavulanate on non-response or under-immunization.

- For orbital disease, **VANCOMYCIN IS INCLUDED EMPIRICALLY** given community-associated MRSA prevalence higher than North America; ceftriaxone covers sinusitis streptococci with good tissue/CNS penetration; metronidazole covers anaerobes / intracranial spread.⁹¹
- Always tailor to immunization status, prior antibiotics, and the **LOCAL ANTIBIOGRAM**.

Condition	Duration
Preseptal cellulitis	5–7 days
Orbital cellulitis (uncomplicated) ⁹¹	10–14 days
Orbital + subperiosteal / orbital abscess (no intracranial)	2–3 weeks

✦ SURGERY, ADJUNCTS & STEP-DOWN

- **Surgery:** subperiosteal or orbital abscess → prompt ENT + ophthalmology for possible drainage (most small

subperiosteal collections settle medically)⁹¹; intracranial complications → neurosurgery.

- **IV → oral** once clearly improving, narrowed by cultures / MRSA swab.
- **Adjuncts:** intranasal saline/steroids only with subspecialty input; systemic corticosteroids are **not** routinely recommended.⁹³ Escalate to ID for systemic illness, non-response, large abscess, or intracranial extension.

⚠ **SIGHT & BRAIN AT RISK – RED FLAGS**

- Falling visual acuity, a relative afferent pupillary defect, severe or bilateral proptosis, ophthalmoplegia, cranial-nerve palsies, or altered consciousness → cavernous sinus thrombosis or intracranial spread.
- Image urgently and escalate immediately.
- Bilateral involvement or rapid progression despite IV therapy is an emergency.

◆ **DIFFICULT IV ACCESS?** Reassess for an early IV→oral switch at every line failure rather

than re-sticking — see [IV → Oral Switch & the DIVA Patient](#) (Quick Reference).

BRUE (Brief Resolved Unexplained Event)

ASSESS

< 1 YR • AAP 2016

A frightening but now-resolved episode in a well-appearing infant — a diagnosis of exclusion.

DEFINE

- An infant < 1 year, when an observer reports a **sudden, brief, now-resolved** episode of one or more of: cyanosis or pallor; absent, decreased, or irregular breathing; a marked change in tone (hyper- or hypotonia); or altered responsiveness.³²
- Diagnose a BRUE **only when no explanation is found** after appropriate history and examination.³²

▀ LOWER-RISK – ALL OF

- Age > 60 days; gestation $\geq$ 32 wk and corrected (postmenstrual) age³² $\geq$ 45 wk.
- First event (no prior BRUE, not in clusters); duration < 1 min; no CPR by a trained provider; no concerning history or examination findings.³²

▀ LOWER-RISK – DO

- **Should** — educate caregivers about BRUEs and use shared decision-making to guide evaluation, disposition, and follow-up; offer resources for caregiver CPR training.
- **May** — obtain pertussis testing and a 12-lead ECG; briefly monitor with continuous pulse oximetry and serial observations.³²

✗ LOWER-RISK – DO NOT / NEED NOT

Should not: blood counts or cultures, CSF, electrolytes/renal panel/calcium, ammonia, blood gases, or metabolic screen (urine organic acids, plasma amino acids, acylcarnitine); chest X-ray, echocardiogram, EEG, or studies for reflux; home cardiorespiratory monitoring; acid-suppression or

antiepileptic drugs. **Need not:** viral respiratory testing, urinalysis, glucose, bicarbonate, lactate, anemia work-up, or neuroimaging; admission **solely** for cardiorespiratory monitoring.³²

⚠ CONCERNING FEATURES → NOT LOWER-RISK, EVALUATE

- Toxic appearance, lethargy, vomiting, or respiratory distress at assessment; significant compromise / CPR during the event; recurrent or prior events; family history of sudden cardiac death, long-QT syndrome, unexplained drowning or motor-vehicle death, SIDS, or sibling BRUE; history suggesting seizure; paroxysmal cough or pertussis exposure; any trauma or concern for maltreatment; unexplained sibling death or dysmorphic features.
- These mandate **targeted** evaluation — the testing restrictions above apply to lower-risk infants only.

✦ PEARL – “LOWER-RISK” IS NOT “NO-RISK”

- In the multicenter BRUE network, about **4%** of infants had a serious underlying condition (most often a seizure disorder or an airway abnormality) and events recurred in roughly **7%** discharged from the ED and **11%** discharged from hospital.³³
- The AAP criteria reliably identify low-risk infants but over-classify many as higher-risk.³³
- Safety-net every family, offer CPR training, and consider asking carers to video any recurrence for diagnostic clarity.

▴ ADMIT · DISCHARGE · REFER

- **Admit** — higher-risk BRUE (age < 2 months, prematurity, repeated events, or concerning features) for monitoring.
- **Discharge** — lower-risk BRUE with education, CPR resources, and follow-up.
- **Refer** — by cause (cardiology, neurology, safeguarding).

Adolescent Emergencies – Kuwait

Context

HOURS

ADOLESCENT

Mental-health crises and menstrual emergencies — within the Kuwait consent and confidentiality framework.

▶ MENTAL HEALTH & SELF-HARM

- Screen the psychosocial context in a calm, private setting — Home, Education, Eating, Activities, Drugs, Sexuality, Suicide/safety (HEEADSSS).²²⁹
- **Acute self-harm, suicidal ideation, or intentional ingestion → manage per the Self-Harm / Intentional Ingestion card** (continuous one-to-one observation, remove the means, treat the ingestion, mental-health assessment before discharge). This card adds the **adolescent consent, confidentiality, and safeguarding** framework below.

▶ MENSTRUAL EMERGENCIES

- **Heavy menstrual bleeding:** assess for hemodynamic compromise and anemia. Check CBC, ferritin,

coagulation screen and **von Willebrand studies** (especially at menarche)²⁴³, thyroid function, and a pregnancy test — performed within the local legal/ethical and safeguarding framework, and discussed with a senior clinician where culturally sensitive.

- **Severe dysmenorrhea:** first-line is a scheduled NSAID; escalate to hormonal therapy for a menstrual indication if refractory.²⁴³
- **Amenorrhea:** investigate primary amenorrhea if no menarche by age 15 with normal secondary sexual characteristics (or by 13 with none); in secondary amenorrhea, exclude pregnancy first.

Drug	Dose	Route & notes
Tranexamic acid	15–25 mg/kg/dose (max 1.5 g) every 8 h	Acute heavy menstrual bleeding; PO or IV ²⁴³
Combined oral contraceptive / oral progestogen	per gynecology protocol	Hormonal control of acute bleeding or refractory dysmenorrhea

Drug	Dose	Route & notes
NSAID (ibuprofen / mefenamic acid)	ibuprofen 10 mg/kg/dose; mefenamic acid per protocol	Dysmenorrhea; also reduces menstrual blood loss
Iron	3–6 mg/kg/day elemental iron	Iron-deficiency anemia from chronic loss

⚠ SAFEGUARDING OVERRIDES CONFIDENTIALITY

- Confidentiality with an adolescent is real but **not absolute**.
- When there is imminent risk of serious harm — self-harm, suicide, abuse, or exploitation — override it: involve the guardian and escalate through the hospital safeguarding pathway.
- Treat life- or limb-threatening conditions immediately, with or without consent.

◆ KUWAIT LEGAL FRAMEWORK

- Under **LAW NO. 70/2020** (Medical Profession & Patients' Rights), informed consent is required before any intervention and, for a minor, is given by the **LEGAL GUARDIAN (WALI)**.

- **CONFIDENTIALITY (ARTICLE 13):** the practitioner must not disclose a patient's secret — whether it came to their knowledge, was discovered through practice, or was entrusted to them — except in defined circumstances: a written order from a court, the public prosecution, or a competent investigating authority; disclosure to prevent or report a crime; notification of a communicable disease to the authority designated by the Ministry; the patient's written consent; information exchange required between the care team for the patient's management; or reporting of child abuse (Law 21/2015) or elder abuse/neglect (Law 18/2016).
- Under the **MENTAL HEALTH LAW NO. 14/2019**, a physician may hold a patient for **UP TO 72 HOURS** for psychiatric assessment when there is danger to self or others or an inability to self-care/consent (Art. 11); involuntary referral is also permitted by judicial order or at the request of a relative (to second kinship), the treating physician, or an investigator. Disclosing a patient's psychological information is a **CRIMINAL OFFENCE**.
- Confirm exact article wording against the official gazette text and

The Child with Obesity ASSESS

AAP 2023

Obesity reshapes acute care — weight-based dosing, a harder airway, and comorbidities that present as emergencies.

▀ DEFINE

- **Obesity** = BMI $\geq$ 95th percentile for age and sex; **severe obesity** $\geq$ 120% of the 95th percentile (or BMI $\geq$ 35). A chronic disease with genetic, physiologic, and environmental contributors — not a failure of willpower.¹²⁹
- Focuses on how obesity **modifies acute care** — drug dosing, the airway, resuscitation, and the comorbidities that present as emergencies.

▀ APPROACH & WORK-UP

- Confirm the **BMI percentile**; targeted history (diet, activity, sleep/snoring, family history, medications) and exam for comorbidity signs —

acanthosis nigricans, hypertension (correct-size cuff), hepatomegaly, hip/knee findings, papilledema.

- **Screen ≥ 10 years:** fasting glucose/HbA1c (T2DM), ALT (fatty liver), and a lipid panel; blood pressure with an appropriately sized cuff.¹²⁹

MANAGEMENT – DRUG DOSING IS THE KEY BEDSIDE TRAP

Drug	Dose	Route & notes
Resuscitation drugs · defibrillation	Actual (total) body weight – but cap at the adult dose	Length-based tapes (Broselow) mis-estimate weight in obesity ¹³⁰ — verify actual weight.
General rule	No universal scalar – dosing is drug-specific	There is no valid one-size default ; the correct weight basis depends on each drug's pharmacokinetics. Confirm with clinical pharmacy / the formulary. Never exceed the adult dose; scaling a hydrophilic drug to total body weight can cause overdose. ¹³⁰

Drug	Dose	Route & notes
Lipophilic drugs, large V_d	Loading dose nearer total body weight	Then maintain on lean/adjusted weight (e.g., some sedatives, induction agents).
Cefazolin succinylcholine antineoplastics	Total body weight (TBW)	TBW is the correct scalar for these. ¹³⁰
Aminoglycosides hydrophilic drugs	Adjusted or lean body weight	Volume of distribution altered; use adjusted body weight. ¹³⁰

◆ **RULE OF THUMB** When total body weight exceeds ~120% of ideal body weight, choose the weight basis deliberately — **PER DRUG, NOT BY A BLANKET RULE. NEVER EXCEED THE RECOMMENDED ADULT DOSE** for the indication, regardless of weight. For any drug not listed above, or with a narrow therapeutic index, **INVOLVE CLINICAL PHARMACY**.

⚠ **AIRWAY & VENTILATION**

- Anticipate **difficult bag-mask ventilation and intubation** and **rapid desaturation** (reduced functional residual capacity).

- Pre-oxygenate well, position **ramped/HELP** (ear-to-sternal-notch), and have adjuncts ready.
- Higher risk of OSA and obesity-hypoventilation.

▲ **RED FLAGS – DON'T-MISS ACUTE COMORBIDITIES**

- **T2DM → DKA or HHS** — hyperglycemic hyperosmolar state is disproportionately common in adolescents with obesity and T2DM.
- **SCFE** — hip, thigh, or knee pain with a limp in an adolescent¹³⁸ (see the [SCFE](#) card).
- **Idiopathic intracranial hypertension** — headache, papilledema, visual change.
- **Cholelithiasis / pancreatitis**, VTE in adolescents, worsening asthma control.

✦ **PEARL – WEIGH THE CHOICE, NOT JUST THE CHILD**

Document both actual weight and ideal body weight, pick the scalar per drug

class, and avoid codeine — the same CYP2D6/respiratory risk applies.¹²⁹

Pertussis (Whooping Cough)

MINUTES

AAP Red Book 2024 • CDC

In the young infant, the danger is apnea and cyanosis — the “whoop” is often absent.

RECOGNIZE

- Paroxysmal coughing fits ± inspiratory whoop or post-tussive vomiting, worse at night, with well periods in between; often marked lymphocytosis.
- **Infants < 6 months / unimmunized:** may present with **apnea, cyanosis or bradycardia** and little cough — a low threshold to admit and monitor.¹⁶¹

SEVERE / MALIGNANT PERTUSSIS

Extreme leukocytosis ($WBC > 50 \times 10^9/L$), pneumonia, pulmonary hypertension and refractory hypoxemia carry high mortality in young infants — escalate early to critical care; exchange transfusion/leukoreduction is considered in specialist units.¹⁶¹

WORKUP

- **Nasopharyngeal PCR** — test of choice in the first 3–4 weeks of cough; culture (< 2 weeks) for confirmation/typing; serology in later/adolescent disease.¹⁶¹
- CBC — marked **lymphocytosis**; **extreme leukocytosis** predicts pulmonary hypertension and death in young infants.
- Chest X-ray if respiratory distress.

Age	Azithromycin	Notes
< 6 months	10 mg/kg once daily for 5 days	Macrolide of choice — lowest risk of infantile pyloric stenosis ¹⁶¹
≥ 6 months	10 mg/kg (max 500 mg) day 1, then 5 mg/kg (max 250 mg) days 2–5	PO

PUBLIC-HEALTH & CONTACTS

- Pertussis is **NOTIFIABLE** — inform public health.
- Droplet precautions until 5 days of macrolide.¹⁶¹
- Offer **POST-EXPOSURE PROPHYLAXIS** (same macrolide dosing) to close contacts, prioritizing households

with infants < 12 months or pregnant women¹⁶¹, and check the family's immunization status.

- Reinforce **MATERNAL TDAP** each pregnancy (27–36 weeks) as the main protection for young infants.¹⁶¹

▲ ADMIT · REFER

- **Admit** — any infant < 6 months, apnea/cyanosis, feeding difficulty, pneumonia or extreme leukocytosis.
- **Refer/Call** — critical care for severe pertussis.

Complex & Technology-Dependent Care

Technology-dependent & medically complex children — tracheostomy, feeding tubes, ECMO, post-ICU, transport

The Child with Medical Complexity

ASSESS

When a Child with Medical Complexity presents unwell — start from their baseline, and treat the carer as the expert.

▲ WHO – AND WHY THEY DECOMPENSATE FAST

- **Children with Medical Complexity (CMC)** have chronic conditions with **medical fragility**, multiple coexisting problems needing **multidisciplinary care**, and often **medical technology dependence** — a tracheostomy, long-term ventilation, enteral feeding tubes and pumps, a central venous catheter, a ventriculoperitoneal shunt, or an intrathecal baclofen pump. They usually have

polypharmacy, coordinated care needs, and frequent healthcare use.¹⁰³

- They have little reserve, present **atypically**, and the **carer knows the baseline** — when they say “this is not normal for him,” believe it and look hard.¹⁰⁴

▲ **THE UNWELL CHILD – A STRUCTURED APPROACH**

- **Start from baseline:** their usual saturations, tone, secretions, feeds, alertness, and seizure pattern.
Deviation from baseline is the alarm, not the absolute number.
- **Read the emergency care plan first** — diagnosis, baseline, devices and sizes, drugs, allergies, and the agreed goals of care.
- **Think the common six:**
 - infection
 - **device malfunction**
 - constipation
 - pain / dystonia
 - seizures
 - medication problem (including **withdrawal**).

- These cause most acute presentations.¹⁰⁴

TECHNOLOGY PROBLEM – WHERE TO LOOK

Device	Red flag → action
Tracheostomy / home ventilator	Desaturation, distress, or alarm → blocked or displaced tube, tracheitis → see Tracheostomy Emergencies . Ventilator alarm → think DOPES (displacement, obstruction, pneumothorax, equipment, stacked breaths)
Gastrostomy / jejunostomy	Dislodged or blocked tube, leak, or peritonism → see Gastrostomy (Feeding) Tube Problems ; a fresh stoma can close within hours
Central venous catheter	Fever = line infection until proven otherwise – paired cultures (line + peripheral) before antibiotics; do not pull a tunnelled line lightly. Occlusion or breakage → specialist
VP shunt	Vomiting, headache, drowsiness, irritability, or a bulging fontanelle = shunt malfunction / raised ICP until proven otherwise → urgent imaging + neurosurgery

Device	Red flag → action
Baclofen pump	Itch, rigidity, fever, agitation, autonomic instability = acute withdrawal (can be life-threatening) → urgent specialist; bridge with benzodiazepines ± oral baclofen

⚠ POLYPHARMACY & DO-NOT-STOP DRUGS

- Do **not** abruptly stop chronic medications — **most should be continued** — including anticonvulsants, baclofen, and long-term steroids, or long-term oxygen / ventilation.¹⁰⁴
- If on long-term steroids and acutely unwell, give **stress-dose steroids** (see [Adrenal Crisis](#)).
- Reconcile every drug, check interactions, and bring the child's own equipment and ventilator settings with them.

◆ **PLAN BEFORE THE CRISIS** Agree and document the **BASELINE, THE EMERGENCY CARE PLAN, AND THE GOALS OF CARE** with the family and the medical home in advance — and make sure it travels with the child.¹⁰⁴

Tracheostomy Emergencies MINUTES

The deteriorating child with a tracheostomy — oxygenate both ways, then work the algorithm: patent, obstructed, or displaced?

▲ FIRST MOVES – CALL FOR HELP, OXYGENATE BOTH WAYS

- **Call for help.** Apply **high-flow oxygen to both the stoma and the face** as soon as it is available — use **two sources if you have them**; with a **single oxygen source, the stoma is the priority** (it is the airway in most tracheostomy patients), moving to the face if the stoma is not the effective route⁹⁶. Attach **pulse oximetry ± waveform capnography** (exhaled CO₂ = a patent or partly patent airway). Bring the **tracheostomy emergency kit**: same-size and one-size-smaller tubes, a suction machine + catheters, water-based lubricant, a cuff syringe, and a bag-valve-mask. Use the **bedside emergency card** (tube type and size, cuff status, **suction depth**); suction with a **premarked catheter to a set depth** — just past the tube tip, no deeper.⁹⁶

Routine **humidification** (HME / “Swedish nose”) prevents the mucus plugs behind most obstructions.

- **Rapid airway assessment:** look, listen, and feel at **BOTH the mouth and the stoma** — assess **chest rise, breath sounds (air entry), airflow, and exhaled CO₂** (if available).

DECISION

① **Is the tracheostomy tube in place?**

NO

Accidental decannulation. If trained, **reinsert the tracheostomy tube.** If not trained or reinsertion fails, **ventilate via whichever route achieves chest rise** (upper airway or stoma). Continue ventilation and await expert airway support.⁹⁶

YES

Tube is in place — proceed to the steps below.

② REMOVE WHAT MAY BE CAUSING THE PROBLEM

- Remove any **speaking valve or cap**, remove the **inner cannula** (if present), and **check the ventilator / circuit** (if ventilated) — this alone often relieves the obstruction.
- (Some inner cannulas must be reinserted to connect a breathing circuit.)

DECISION

③ Can you pass a suction catheter?

PASSES

Tube is **patent** — suction to clear secretions (a **mucus plug** is the commonest cause; if it will not clear, **change the tube** — do not persist)⁹⁶.
Ventilate via the tracheostomy if not breathing. Continue ABCDE.

WON'T PASS

Tube is **obstructed or displaced**.
Deflate the cuff (if present) — if this improves things, air is moving around the tube⁹⁶ → **remove the tracheostomy tube**.

after tube removal / still not ventilating



④ VENTILATE – VIA WHICHEVER ROUTE ACHIEVES CHEST RISE

- **Upper airway** (if patent): cover the stoma (gauze / finger) and bag-mask via the mouth/nose ($\pm$ oral or nasal airway), or an LMA.
- **Stoma** (if the upper airway is not patent or oral/nasal ventilation is ineffective — e.g., laryngectomy, subglottic stenosis): ventilate over the stoma with an appropriately sized pediatric face-mask, or an LMA.

when trained help arrives

⑤ RE-ESTABLISH THE AIRWAY

- Re-cannulate with the **same-size tube**; if it will not pass, use **one size smaller**. If it still will not pass, **do not force — stop and keep ventilating** until expert airway support arrives, or **orally intubate** (an uncut / longer ETT, cuff sited **beyond the stoma**)
- Confirm with **capnography**. A **neck roll** (shoulder roll, neck extended) increases the likelihood of successful insertion.⁹⁶

▶ DOPE – CAUSES OF DETERIORATION

- **D — Displacement** (accidental decannulation)
- **O — Obstruction** (mucus plug, blood, granulation tissue)
- **P — Pneumothorax**
- **E — Equipment failure** (ventilator / circuit, speaking valve, cap, inner cannula, oxygen failure)

✦ KEEP THE OBTURATOR AT THE BEDSIDE

- Keep the tracheostomy **obturator at the bedside at all times.**
- In an emergency it may be needed to **reinsert the child's existing tube** when a new tube is unavailable or there is no time to open one — reinsertion is **difficult or impossible without it.** ²⁰⁷

⚠ NO UPPER AIRWAY? (LARYNGECTOMY / FIXED OBSTRUCTION)

- If the child **cannot be ventilated from above** (laryngectomy, severe subglottic stenosis), the **stoma is the only airway** — do not waste time on face-mask ventilation;

oxygenate and re-cannulate or intubate the stoma.

- Know this from the care plan **before** the emergency.

⚠ BLEEDING – EXCLUDE TRACHEO-INNOMINATE FISTULA (TIF)

- **Massive bleeding — or a brief “sentinel” bleed — 3 days to 6 weeks after tracheostomy is TIF until proven otherwise** (high risk in neurologically impaired children)²⁰⁷.
- This is an airway/ENT/surgical emergency — call expert help (ENT, surgery, anesthesia/ICU) immediately.
- As an expert-performed bridge while help arrives: **overinflate the cuff** to tamponade (~85% effective); if it fails, **orally intubate past the stoma** (ETT cuff distal to the bleed) and apply the **Utley maneuver** — a finger through the stoma compressing the artery against the back of the sternum — then **activate massive transfusion, call ENT + surgery, and go to OR.**²⁰⁷

- Any tracheostomy bleeding needs ENT review before discharge.

▲ TRACHEITIS

- Increasingly **purulent / copious secretions, fever, plugging, rising work of breathing** — can obstruct. Send a **tracheal aspirate**; cover *S. aureus* (including MRSA) and *Pseudomonas* per the local antibiogram; step up humidification and suction.²⁰⁷

⚠ WHEN NOT TO CHANGE THE TUBE YOURSELF – CALL ENT / THE TRACHEOSTOMY TEAM

As a general pediatrician, **do not attempt an elective tube change** when: it is the **first change** (the surgical / ENT team performs or directly supervises the first change, usually at **day 3–7** post-formation, to confirm a patent tract)²⁰⁸; the stoma is **fresh / immature (< 7 days**, especially percutaneous) — high risk of a **false passage** or an unrecoverable airway; you are **not trained in that specific tube**, or the correct-size tubes and suction are not at hand; the child is **unstable, on high ventilator / oxygen support, or has a known difficult airway** — change only

where **ENT / anesthesia and airway rescue are immediately available**; or there is **active bleeding or suspected tracheo-innominate fistula** (an emergency, not a routine change); the tracheostomy was placed for **critical upper airway obstruction** — the upper airway is **not a reliable rescue route**, so a lost or displaced tube can mean a **lost airway** (treat the stoma as the only airway and change only with ENT / anesthesia and airway rescue at the bedside); or the child has had **major airway / tracheal reconstruction** (laryngotracheal reconstruction, slide tracheoplasty, tracheal resection & anastomosis) — the tube may cross or protect a **fresh repair**, so changes are done by the **surgical team only** (risk of disrupting the anastomosis or a false passage). Otherwise **keep the current tube, oxygenate, and get expert help**.

ELECTIVE TUBE CHANGE (ROUTINE)

- Two people.
- Supine with a **shoulder roll, neck extended**; pre-oxygenate and suction.
- Load the obturator, lubricate, deflate the cuff.
- Remove the old tube along its curve; insert the new one on the **obturator** (tip toward the posterior wall, then rotate downward) on expiration — **never force. Remove the obturator immediately**, insert the inner cannula, inflate the cuff, and confirm (chest rise, breath sounds, capnography, suction catheter passes)²⁰⁸
- Secure ties to a one-finger gap.

⚠ CALL ENT / ANESTHESIA EARLY

- Any bleeding · suspected TIF (emergency) · failed or difficult reinsertion · obstructing granulation · **accidental decannulation with a fresh stoma (< 7 days)** — prefer oral intubation and avoid blind reinsertion.
- An established stoma can still narrow > 50% within ~12 h of decannulation, so if decannulated

> ~8 h and the child is stable, call ENT before repeated attempts (dilation may be needed).²⁰⁷

SOURCE

National Tracheostomy Safety Project (NTSP) pediatric emergency algorithm; Care of Infants and Children with Tracheostomies — American Thoracic Society Clinical Practice Guideline (*Am J Respir Crit Care Med* 2025;211(11):2001–2020).

▲ ADMIT · DISCHARGE · REFER

- **Admit** — any airway compromise → ED/PICU.
- **Discharge** — after the tube is secure and patent and the caregiver is confident.
- **Refer** — ENT and the home-ventilation/tracheostomy team.

◎ LEARN MORE – OPEN-ACCESS PEDIATRIC RESOURCES (no login required)

- **CINCINNATI CHILDREN'S – TRACHEOTOMY CARE** — caregiver handbook (Arabic available) plus short videos

including an **EMERGENCY TUBE CHANGE**:

cincinnatichildrens.org



- **ABOUTKIDSHEALTH (SICKKIDS) – TRACHEOSTOMY & VENTILATION** — home care, suctioning and **EMERGENCY MANAGEMENT**:

aboutkidshealth.ca/trachvent



Tracheostomy Tubes & Cuff Types

REFERENCE

Know the exact tube before you change or ventilate — same stated size from different manufacturers is not interchangeable.

TUBE – WHAT IT IS AND WHEN

Tube	Material & features	Notes
PVC tube	PVC, firm, thermosensitive radiopaque line; neonatal / pediatric / pediatric-long; cuffed (soft air cuff) or uncuffed. Pediatric sizes are single-lumen (no inner cannula).	Single-patient, disposable. Sized by inner diameter (ID, mm) . ²⁰⁷
Silicone tube	Silicone, soft and flexible, reusable; cuff options tight-to-shaft (TTS) , air, or foam; extended-proximal and adjustable-length	Silicone is MRI-conditional — wire-reinforced / metal variants may not be ; check. Sized by ID. ²⁰⁷

Tube	Material & features	Notes
	(wire-reinforced) variants exist.	

CUFF TYPES – INFLATE WITH WHAT, WATCH WHAT

Cuff	Inflate with	Use / caution
Uncuffed	–	Most children; airflow around the tube (speech), least mucosal injury.
Air cuff	Air	Positive-pressure ventilation or aspiration protection; keep cuff pressure < 20–25 cmH₂O (minimal-leak) — over-inflation causes mucosal ischemia. ²⁰⁸
Tight-to-shaft (TTS)	Sterile water (never air)	Deflated, it sits flush like an uncuffed tube (speech / airflow); inflate only when needed. Air diffuses out of silicone ²⁰⁸ — use water.
Foam cuff	Self-inflating (pilot open to air)	For a difficult seal or tracheomalacia; actively deflate and cap the pilot to insert / remove, then release to self-inflate.

⚠ CUFF SAFETY

- Never use a **speaking valve or cap with the cuff inflated** — no route to exhale (can be fatal)²⁰⁸; deflate cuff first.
- Deflate cuff whenever a seal is not needed, to protect mucosa and allow speech.

▲ BEFORE ANY TUBE CHANGE

- Same-size tubes from **different manufacturers differ in outer diameter and length — not interchangeable**; if you must substitute in an emergency, be ready to **downsize**.²⁰⁷
- Keep a **same-size and one-size-smaller** spare at the bedside; know the exact tube from the **bedside emergency card**.
- After insertion, confirm airway — chest rise, **capnography**, suction catheter passes.

🕒 LEARN MORE – OPEN-ACCESS PEDIATRIC RESOURCES (no login required)

- **CINCINNATI CHILDREN'S – TRACHEOTOMY CARE** — caregiver handbook (Arabic

available) plus short videos including an **EMERGENCY TUBE CHANGE**:

cincinnatichildrens.org



- **ABOUTKIDSHEALTH (SICKKIDS) – TRACHEOSTOMY & VENTILATION** — home care, suctioning and **EMERGENCY MANAGEMENT**:

aboutkidshealth.ca/trachvent



Gastrostomy (Feeding) Tube Problems

HOURS

A dislodged tube is time-critical — the stoma can close within hours. Tract age decides who replaces it.

▶ **DISLODGED TUBE – THE CLOCK IS THE STOMA**

- Empty tract narrows within hours — per **ESPGHAN / NASPGHAN** it can **close within ~6 h**, usually lost by **24 h**.¹⁰⁵ Keep it open **now**, then replace.
- **Tract age decides the route** — ask when the tube was first placed.

DECISION

How old is the stoma tract?

IMMATURE < 6 WEEKS

High risk of a **false tract / intraperitoneal placement** → **do NOT insert a tube into the stoma**. Cover stoma with gauze, keep nil-by-stoma, get to **hospital** for guided replacement by **pediatric surgery / gastroenterology / interventional radiology**.¹⁰⁵

MATURE ≥ 6 WEEKS

Epithelialized tract → bedside replacement usually safe. Replace **promptly** with a **same-size balloon tube**; lubricate, advance gently, inflate the balloon with water. **Caveat: a first PEG, or poor wound healing** (malnutrition, chemotherapy), may mature more slowly — up to **~3 months**; if unsure, **treat as immature**.¹⁰⁵

IF THE SAME SIZE WILL NOT PASS

- Place the largest **Foley catheter** that passes atraumatically to hold the stoma open, then **serially dilate in 2-Fr steps** to the target size.
- **Do not inflate a Foley balloon inside the tract to dilate.**¹⁰⁵
- If > 24 h out, or any resistance / false-passage concern → stop and refer.

◆ CONFIRM BEFORE FEEDING

- Verify intragastric position before use: **ASPIRATE GASTRIC CONTENTS (PH ≤ 5.5) – BEWARE A FALSE PH ON PROTON-PUMP INHIBITORS.**
- After any **DIFFICULT OR UNCERTAIN** replacement, confirm with a **WATER-SOLUBLE CONTRAST STUDY** — the ESPGHAN standard before first use of a replaced balloon tube.¹⁰⁵ **NEVER USE AIR INSUFFLATION** to confirm.
- Document tube type, French size, and balloon volume.

▴ BLOCKED TUBE & STOMA ISSUES

- **Blocked:** flush gently with **warm water**; do not force. Hold feeds until patency is confirmed.

- **Buried bumper:** suspect if the tube **will not rotate or advance freely**, or pain, leakage, or resistance on feeding — prevented by avoiding traction and rotating + advancing the tube regularly. Buried bumper or **suspected peritonitis** (pain, feed intolerance, fever, rigidity) → stop feeds, urgent **surgical review**.¹⁰⁵
- **Granulation, leak, redness, excoriation:** see [Enteral Feeding Tubes — Types & Complications](#).

⚠ ADMIT / ESCALATE

- Peritonitis or systemic infection
- **buried bumper syndrome**
- uncertain or **difficult placement**
- concern of a **false passage**
- any immature-tract dislodgement
- **tube out > 24 h** → see surgery / gastroenterology.

▶ ADMIT · DISCHARGE · REFER

- **Admit** — peritonitis, a dislodged tube with a lost tract, or inability to replace it.
- **Discharge** — a replaced, patent tube with feeds tolerated.

- **Refer** — surgery/gastroenterology for tract problems or early dislodgement (< 4 weeks).

© **LEARN MORE – OPEN-ACCESS PEDIATRIC RESOURCES** (no login required)

- **CINCINNATI CHILDREN'S – G-TUBE HOME CARE** — caregiver handbook + procedure videos (Arabic available):

cincinnatichildrens.org



- **TOGETHER BY ST. JUDE – TYPES OF TUBES & DEVICES** — illustrated visual guide to NG / ND / NJ and G / GJ / J tubes and low-profile buttons:

together.stjude.org



Enteral Feeding Tubes – Types, Delivery & Complications

REFERENCE

Know the route before you feed or give a drug — the stomach tolerates boluses, the jejunum does not.

TUBE TYPES – ROUTE, HOW TO FEED, AND WHAT NOT TO GIVE

Tube	Route & how to feed	Do NOT
NGT / OGT	Stomach; short-term; bolus or continuous. Nasogastric vs orogastric route is a unit / clinical choice – evidence favors neither. ¹⁹⁶ Choose by expected duration, gag strength, and respiratory status: prefer OGT when nasal airflow matters (preterm, CPAP,	Never feed with position unconfirmed. Use OGT (not NGT) in suspected base-of-skull injury.

Tube	Route & how to feed	Do NOT
	respiratory distress) or the gag is weak; NGT is more secure for a stronger gag or longer use. Confirm gastric aspirate pH $\leq$ 5.5 before each use (x-ray if in doubt; local policy: confirm by x-ray). ¹⁰⁵	
NJT	Nose → jejunum (post-pyloric); for gastroparesis or high aspiration risk. Continuous feeds only.	No bolus (dumping). pH is unreliable — confirm position by imaging. ¹⁰⁵
Gastrostomy (PEG / high-profile / button)	Abdominal wall → stomach; long-term. Bolus or continuous.	Force or over-rotate a PEG bumper (buried bumper). A low-profile button needs an extension set to feed.

Tube	Route & how to feed	Do NOT
Gastro-jejunosomy (GJ)	Through the gastric stoma, tip in jejunum; gastric + jejunal ports. Jejunal port = continuous only ; gastric port = venting / medicines.	Blind replacement — needs fluoroscopy. No bolus down the jejunal port.
Jejunostomy (J)	Abdominal wall → jejunum. Continuous feeds only, small volumes.	Bolus feed. Attempt a bedside change — refer (surgical tract).

⚠ MEDICINES THROUGH A FEEDING TUBE

- Use **liquid formulations** where possible; **never crush enteric-coated or modified / sustained-release** tablets (blocks the tube, alters the drug).¹⁰⁵
- Give each drug **separately** and **flush with water** before, between, and after. **The route matters:** down a **jejunal (post-pyloric)** tube avoid hypertonic or large-volume drugs (dumping, perforation), and **some medications cannot be**

given jejunally at all — check with pharmacy.

- Some **parenteral (injectable) preparations** can be given enterally when no oral liquid exists (e.g., **clonidine, glycopyrrolate**) — confirm suitability and dose with pharmacy first.

▲ DIFFERENTIATE THE TUBE AT THE BEDSIDE

- **Where it enters** — nose / mouth (NGT/OGT) vs the abdomen (gastrostomy / GJ / J).
- **Profile** — an external bumper with a **balloon port** (water valve) = balloon gastrostomy; **flush to the skin** needing an extension = low-profile **button**; a firm internal flange that resists gentle traction = **PEG**.
- **Ports** — **two labelled ports (GASTRIC + JEJUNAL)** = GJ; a single feed port plus a balloon port = gastrostomy.
- Read the **French size, length, and brand** off the tube and the care plan. Unsure of the route? Confirm with **contrast** — do not feed to find out.

MANAGE THE COMMON STOMA COMPLICATIONS

Problem	What & why	Management
Granulation on (granuloma)	Red, moist, friable overgrowth from friction and moisture.	Stop the movement (secure the tube, correct balloon volume); saline soaks ; topical silver nitrate (protect surrounding skin with a barrier) or a short topical steroid ; silver / foam / hydrocolloid dressing. ¹⁰⁵
Leak	Gastric contents track around the tube.	Do not just upsize the tube (enlarges the stoma). Correct balloon volume; treat constipation, retching, delayed emptying (vent, prokinetic, slow the feed); treat granulation / infection; barrier cream. Consider buried bumper. ¹⁰⁵
Redness	Irritation vs infection.	< 5 mm erythema is usually irritation — barrier protection. Spreading redness, pain, swelling, or pus = infection → swab, then topical or systemic antibiotics (cover skin flora).

Problem	What & why	Management
Excoriation	Enzymatic burn from leaked feed and acid.	Stop the leak at source; barrier products (zinc or dimethicone cream, pectin barrier ring / powder); keep dry; involve the stoma nurse.

◆ GOLDEN RULES

- Confirm route before feeding or giving a drug; **GASTRIC TOLERATES BOLUSES, JEJUNAL IS CONTINUOUS ONLY**; flush to keep patent.
- A fresh gastrostomy stoma can close within hours if the tube falls out — see **Gastrostomy (Feeding) Tube Problems.**

◎ LEARN MORE – OPEN-ACCESS PEDIATRIC RESOURCES (no login required)

- **TOGETHER BY ST. JUDE – TYPES OF TUBES & DEVICES** — illustrated visual guide to NG / ND / NJ and G / GJ / J tubes and low-profile buttons:

together.stjude.org



- **CINCINNATI CHILDREN'S – G-TUBE HOME CARE** — caregiver handbook +

procedure videos (Arabic available):

cincinnatichildrens.org



ECMO – What the General Pediatrician Needs to Know

ASSESS

KNOW WHEN TO CALL • ELSO

Temporary heart–lung support for severe but **reversible** failure — your job is to recognize the candidate and call **early**.

▴ WHAT IT IS

- Temporary machine support that **oxygenates the blood and removes CO₂** (± supports the circulation) to buy time for **potentially reversible** heart or lung failure. It is a **bridge** — to recovery, a decision, or transplant — **not a cure**, and is run only in a specialized PICU / ECMO center.
- **VV (veno-venous) = lung support** (the heart still works); **VA (veno-arterial) = heart ± lung support**, and covers **ECPR** (ECMO during a refractory but reversible cardiac arrest).²⁴⁴

⚠ **WHEN TO THINK ECMO – CALL EARLY**

Severe **cardiac or respiratory failure not responding to maximal conventional therapy**, in a child with a **reversible cause** (e.g., fulminant myocarditis, ARDS / severe pneumonia, PPHN; neonatal MAS or CDH). **Call the PICU / ECMO center early** — before catastrophic deterioration or irreversible brain injury; timely referral improves outcomes, so do not wait for arrest. **The center decides candidacy, thresholds, and timing.**²⁴⁴

⚠ **USUALLY NOT A CANDIDATE**

- An **irreversible** condition with no exit strategy · **severe irreversible brain injury** · a **lethal chromosomal / congenital anomaly** (e.g., trisomy 13 or 18 — not 21) · **uncontrolled bleeding**.
- Extreme prematurity, very low birth weight, and many days of prior ventilation lower the chance of benefit.
- The final decision rests with the ECMO team.²⁴⁴

✦ YOUR ROLE – BEFORE AND AFTER

- You are **not** expected to run ECMO — you **recognize the failing-yet-reversible child and escalate**.
- Resuscitate and support maximally, **protect the brain and other organs**, avoid irreversible injury, and **call the PICU / ECMO center early**.
- Afterwards, **survivors need long-term multidisciplinary follow-up**: up to about $\frac{1}{3}$ have some neurodevelopmental impairment and new problems can appear later — track development and arrange **hearing tests** (sensorineural loss can be progressive / late).²⁴⁴
- Support the family through a **high-risk, uncertain course** (major complications include bleeding, clots, and stroke).

► ADMIT · DISCHARGE · REFER

- **Escalate** — refractory cardiac or respiratory failure with a reversible cause → **PICU / ECMO center now**.

- **Discuss early** — call **before** the child is in extremis; deterioration is often rapid.

SOURCE

- Extracorporeal Life Support Organization (ELSO) general and pediatric / neonatal guidelines.
- Candidacy, thresholds, and timing are set by the PICU / ECMO center.

Post-PICU Syndrome (PICS-p)

ASSESS

AFTER CRITICAL ILLNESS

Surviving the PICU is only the start — new physical, cognitive, emotional, and social problems can follow the child and family for years.

WHAT IT IS

- **Post-Intensive Care Syndrome — pediatric (PICS-p):** new or worsening impairment across **four domains** after critical illness — **physical, cognitive, emotional, and social** — affecting the **child and the**

family. It emerges over the months after discharge and can persist for years.²¹⁰

- You, the general pediatrician, are the one who **sees these children back** — so you screen for it, coordinate care, and reassure.

THE FOUR DOMAINS – WHAT TO LOOK FOR

Domain	Watch for
Physical	ICU-acquired weakness, fatigue, pain, poor feeding / growth, reduced exercise tolerance, poor sleep; residual respiratory or organ problems.
Cognitive	Attention, memory, and executive difficulties; developmental regression in the young; declining school performance.
Emotional	PTSD, anxiety, depression, nightmares / delusional memories, fear, irritability or aggression, behavioral change.
Social	Trouble reintegrating at school, absenteeism, withdrawal from peers and activities, identity and participation problems.

▲ HIGHER RISK AFTER...

- **Longer PICU stay, mechanical ventilation, sedation and delirium, renal replacement, ECMO, cardiac arrest, sepsis, ARDS, or major trauma; oncology / neurology diagnoses; an unscheduled admission; younger age and pre-existing conditions.**²¹⁰

⚠ DON'T FORGET THE FAMILY (PICS-F)

- Parents and siblings are affected too: at PICU discharge up to **~60% of parents are at risk of PTSD and ~75% of depression**, alongside anxiety, exhaustion, financial strain, and impaired family functioning.²¹⁰
- Ask how the **family** is coping, and support them.

✦ YOUR ROLE

- At the first post-discharge visits and beyond: **screen every domain** (development, school, mood / sleep, physical function) and the family's wellbeing; **coordinate multidisciplinary**

follow-up — physiotherapy / occupational therapy / speech-language therapy, psychology, neurodevelopmental assessment, and **school liaison**; reconcile medications and immunizations; and treat the family as part of the patient.

- Normalize and follow up rather than dismiss — recovery is a trajectory, not an event.

▲ ADMIT · DISCHARGE · REFER

- **Screen** — every child at follow-up after a significant PICU admission.
- **Refer** — neurodevelopment / rehabilitation, psychology / psychiatry, and the relevant specialty for persistent domain-specific problems; social work for family support.
- **Escalate** — new severe mental-health symptoms (e.g., self-harm risk) or acute physical deterioration.

SOURCE

- Manning JC, Pinto NP, Rennick JE, Colville G, Curley MAQ.

- Conceptualizing Post Intensive Care Syndrome in Children — the PICS-p Framework. *Pediatr Crit Care Med*. 2018;19(4):298–300; pediatric PICS reviews.

Aeromedical Transport – Pediatric & Neonatal

ASSESS

PRE-FLIGHT ASSESSMENT • MOH A-ADM-014

Match the transport tier to clinical risk — two predictable cabin stresses (hypoxia, gas expansion) drive the pre-flight assessment.

▲ TWO CABIN STRESSES – ASSESS BEFORE EVERY FLIGHT

- **Hypobaric hypoxia** — cabins are pressurized to a maximum **8,000 ft** equivalent ($\approx$ breathing **15% oxygen** at sea level); a healthy PaO_2 falls to **$\sim 60\text{--}65$ mmHg** ($\text{SpO}_2 \sim 89\text{--}94\%$). Limited cardiopulmonary reserve desaturates further. **Resting SpO_2 alone is insensitive. If available, a HAST (hypoxia-altitude simulation test) best predicts in-flight oxygen need in a child with chronic lung or**

cardiac disease.²⁴⁵ **If HAST is not available**, use the practical **alternative**: base the decision on **resting oximetry, the diagnosis and trajectory, and any prior flight tolerance**, and — when in doubt — **give empiric in-flight oxygen** (double an existing flow) or choose a **sea-level cabin / air ambulance** for higher-risk children.

- **Trapped-gas expansion** — gas in a closed space expands ~**35%** from sea level to 8,000 ft. Pneumothorax, recent surgery with retained gas, and intracranial or intraocular gas need a **sea-level cabin (air ambulance)** when the gas cannot be excluded. A **current pneumothorax requires a chest drain before any urgent flight**²⁴⁵ — even if it is minimal.

⚠ NEONATAL & INFANT AIR TRANSPORT – STABILITY, NOT AGE, GOVERNS

- **Term neonate**: defer non-urgent flying for roughly the **first 7 days** (ideally to ~40 weeks corrected) to confirm feeding, thermoregulation, and no undiagnosed cardiorespiratory

problem or silent pneumothorax.²⁴⁵

- **Preterm without chronic lung disease:** the “~3 months corrected age” figure is a **soft guide, not a fixed cutoff** — fitness depends on **gestational maturity + demonstrated cardiorespiratory stability + being apnea-free**, judged individually. Avoid flying during a respiratory infection until ~6 months corrected age (apnea risk). Former preterms with **BPD / neonatal chronic lung disease:** discuss with a respiratory pediatrician and, where available, a **HAST** before flying (~70% desaturate on hypoxic challenge).²⁴⁵
- **An incubator is not mandatory for a stable infant** — a physiologically stable infant may travel in an aircraft **bassinet / in-arms**. A **transport incubator** is needed for thermal instability, continuous monitoring, controlled oxygen/humidity, or mechanical ventilation.
- **Respiratory support changes the route, not by age:** a ventilated or monitoring-dependent infant goes by **dedicated air ambulance**

(incubator/transport platform, continuous SpO₂ + ECG, capnography if ventilated, altitude-appropriate settings, sea-level-cabin option). A **stable, monitored, ventilated infant can still be flown** for urgent surgery with a specialist team.

- **Tier by stability:** stable, self-ventilating, thermally stable → **bassinet/in-arms commercial** (after the term deferral / individualized preterm assessment); oxygen-dependent or apnea-risk → **commercial + medical clearance + in-flight O₂** (roughly double the flow at cruise); unstable/ventilated/monitoring-dependent → **dedicated air ambulance**.

IN-FLIGHT OXYGEN – INFANTS & YOUNG CHILDREN

Current support / status	Action
Already on oxygen	Double the flow rate in flight as a starting point; confirm it is deliverable in the cabin.
Needs > 4 L/min at baseline	Commercial flight not advised → escalate to air ambulance . ²⁴⁵

© OXYGEN LOGISTICS

- Carriers supply oxygen at fixed flows (often 2–4 L/min, set rates or none).
- A need for precise or higher flow, or a corrected requirement > ~4 L/min, cannot be reliably managed on a scheduled flight.
- Carry an airline-approved **PORTABLE OXYGEN CONCENTRATOR (POC)** with battery for $\geq 150\%$ of journey time; the US prohibits passenger oxygen cylinders (use a POC or carrier oxygen).²⁴⁵

TRANSPORT TIER – ASSIGN THE LOWEST TIER THAT SAFELY COVERS THE WHOLE JOURNEY

Tier	Mode & escort	Typical pediatric patient
0	Commercial seat, no escort	Stable, self-caring; PEWS low.
1	Commercial seat + non-medical attendant	Minor / vulnerable; needs help with mobility, not clinical care.

Tier	Mode & escort	Typical pediatric patient
2	Commercial seat + a competent trained caregiver; a nurse/RT is required only if oxygen- or ventilation-dependent, or other risk is expected	Stable tracheostomy on room air (with a trained caregiver competent in the device), stable oxygen, or pump feeding.
3	Commercial seat or stretcher + physician + nurse (add RT for home-vent)	Stable established home ventilation, or a fragile child with deterioration potential.
4	Dedicated air ambulance + critical-care team	Critically ill or unstable (meets air-ambulance criteria below). ²⁴⁶

© SCORING & PERSONNEL (PER POLICY)

- The policy stratifies patients by **LEVEL OF CARE 0-4** and a **NEWS-BASED RISK LEVEL** (Low = 1-4 · Medium = 5-6 · High = ≥ 7). In children, substitute an age-appropriate **PEWS** (NEWS2 is validated only ≥ 16 yr).²⁴⁶
- If findings fall between tiers, choose the **HIGHER TIER**. A low score with **HIGH DETERIORATION POTENTIAL** (end-stage lung disease, potential

airway/hemodynamic compromise) is classified up by the accompanying physician.

- **AT LEAST TWO COMPETENT PERSONNEL** escort any patient; for a child, include a **PARENT/LEGAL GUARDIAN** where possible.

⊙ **LOCAL AVAILABILITY (KUWAIT)** A **DEDICATED PEDIATRIC / NEONATAL AEROMEDICAL TRANSPORT TEAM IS NOT YET ESTABLISHED** — transfers are currently staffed by the **ACCOMPANYING PEDIATRIC / TREATING TEAM**. Confirm current MOH capability early and coordinate through the A-ADM-014 pathway (Directorate of Treatment Abroad).²⁴⁶

⚠ DEDICATED AIR AMBULANCE (SEA-LEVEL CABIN) – WHEN COMMERCIAL CARRIAGE WILL NOT DO

Any one absolute criterion mandates it: **acute / unstable ventilation** or a credible need to intubate or escalate in transit; **hemodynamic instability** needing vasoactive / inotropic infusion; a **trapped-gas condition** needing sea-level cabin (unresolved pneumothorax, recent intracranial / intraocular gas, significant bowel obstruction); need for **invasive monitoring or mechanical support** (arterial-line titration, ICP monitoring, IABP, **ECMO**, VAD, external pacing);

refractory hypoxemia or oxygen need beyond carrier capability; **status epilepticus** / uncontrolled seizures; or a communicable disease needing **isolation**. Under the policy, **Medevac (air ambulance)** also applies to **priority 1–2 cases**, expedited transfer of an unstable patient, brain-death repatriation, or when the commercial route/port is not cleared. **Absolute contraindications to flight until treated:** untreated pneumothorax, hemothorax or a compromising effusion, cardiac tamponade, recent abdominal surgery / immediate post-op, or a penetrated eye injury.²⁴⁶

✦ **AIRWAY & CUFFS AT ALTITUDE**

- The cabin is a poor place to rescue a failing airway — **secure a difficult or unstable airway definitively before departure**, not aloft; do not defer a foreseeable intubation to the aircraft.
- Endotracheal / tracheostomy **cuff gas expands $\sim\frac{1}{3}$** as cabin pressure falls — **check for a leak and adjust the cuff volume as needed**; monitor cuff pressure during ascent and descent.²⁴⁵
- Confirm the ventilator and oxygen are airline-approved with battery

for $\geq 150\%$ of journey time.

▶ TEAM, CLEARANCE & HANDOVER

- **Escort competence:** current **pediatric / neonatal resuscitation** (PALS / NRP) and competence with the child's devices; the accompanying physician is the **MRP** en route.
- **Clearance:** pre-flight medical report + **risk assessment** reviewed and approved by the **MRP** (senior registrar or above) **7–14 days and again 48–72 h** before departure; airline **MEDIF** for commercial carriage. A **return/repatriation reassessment is mandatory** before flying home.²⁴⁶
- **Handover:** document status before, during and on arrival, and give a structured handover to the receiving team using the policy's **handover tools**.

▶ ADMIT · DISCHARGE · REFER

- **Escalate** — any air-ambulance criterion, or a deteriorating trajectory → critical-care transport team +

treatment-abroad / transport committee.

- **Proceed commercial** — stable child at Tier 0–3 with a completed pre-flight clinical assessment (HAST where available) and confirmed oxygen/device support.
- **Refer** — the MOH Directorate of Treatment Abroad (DTA) decides transport type; involve the relevant clinical council for team and equipment.

SOURCE

- State of Kuwait — MOH Aeromedical Evacuation & Transport Policy **A-ADM-014** (effective Oct 2025; review 2027) — the transport forms and checklists are in the policy on the MOH website.
- Pre-flight assessment thresholds align with the British Thoracic Society clinical statement and pediatric air-travel evidence. Neonatal/infant guidance: BTS air-travel recommendations (*Thorax* 2011); Fitness to fly in the paediatric population (*Eur J Pediatr* 2018); Khanna et al.,

in-flight hypoxia in healthy infants
(*Thorax* 2013).

06

Endocrine & Metabolic

Diabetic Ketoacidosis

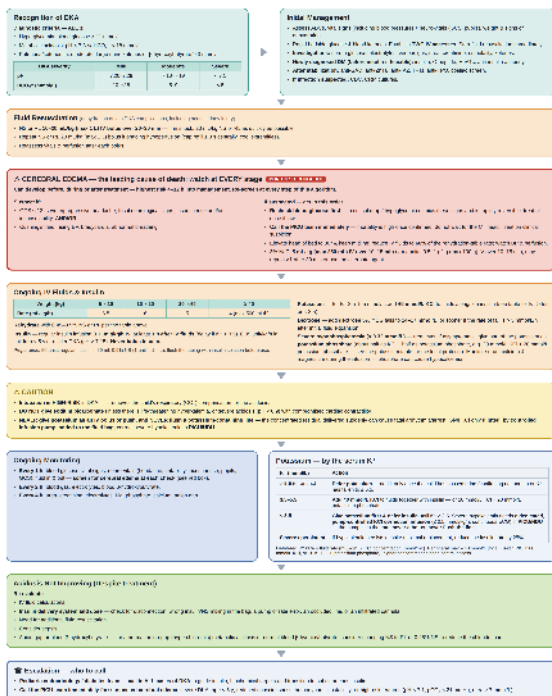
HOURS

ISPAD 2022 • TREKK/CPEG 2023

Glucose > 11 mmol/L (200 mg/dL) + pH < 7.3
or bicarbonate < 18 + ketonemia (β -hydroxybutyrate $\geq$ 3 mmol/L).

Diabetic Ketoacidosis – bedside algorithm

Adapted from TREKK; aligned with the Kuwait PICU Task Force · JSPAD 2022.



◆ SEVERITY & PICU

- **For DKA managed in the PICU, follow the latest Kuwait PICU Task Force DKA guideline** — the local task-force protocol governs fluid, insulin and electrolyte targets in the unit.¹⁷⁶
- Consider PICU for severe DKA, young children (age < 5 y), reduced consciousness, long symptom duration, compromised circulation, or high-risk features (pH < 7.1, pCO₂ < 21 mmHg, urea > 7 mmol/L).⁷

△ CEREBRAL INJURY – FURTHER DETAIL

- If focal neurological deficits develop, obtain an urgent CT to exclude cerebral venous thrombosis or intracranial hemorrhage. **Do not postpone treatment to obtain a CT.** Perform hourly GCS so cerebral injury is caught early — it is the leading cause of death in pediatric DKA.⁷

◆ AVOID WHERE POSSIBLE

Intubation (ventilation removes the child's respiratory compensation for the acidosis), central venous catheters (thrombosis risk; a large dead space can make insulin delivery erratic if insulin runs through the CVC), and routine urinary catheterization.⁷

✦ PEARL – THE MASQUERADES OF NEW-ONSET DKA

- First-presentation diabetes is missed when DKA wears a disguise: **abdominal pain and vomiting** taken for gastroenteritis or appendicitis, **Kussmaul breathing** taken for pneumonia or an asthma attack, and **polyuria** taken for a urinary tract infection or, in a younger child, secondary enuresis.
- A bedside glucose and ketones settle it in seconds.

▴ ADMIT · DISCHARGE · REFER

- **Admit** — all children with DKA are admitted, but the **level of care follows severity**⁷: **mild DKA** can be managed on a general pediatric ward with hourly monitoring;

moderate/severe, young age (< 5 y), reduced consciousness, or high-risk features → high-dependency/PICU.

- **Discharge** — not from DKA; transition to subcutaneous insulin with diabetes-team education.
- **Refer** — pediatric endocrinology/diabetes team for all; PICU for cerebral-injury risk.

Hyperglycemic Hyperosmolar State (HHS)

MINUTES

ISPAD 2022

Not "severe DKA" — very high glucose, very high osmolality, little ketosis, and higher mortality.

▴ DEFINE (DIFFERS FROM DKA)

- Glucose > **33 mmol/L (600 mg/dL)**, serum osmolality > **320 mOsm/kg**, **pH > 7.3** and **bicarbonate > 15**, only mild ketosis, with altered consciousness/profound dehydration. Increasingly seen in adolescents with obesity/type-2 diabetes.⁷

⚠️ **MANAGE IT DIFFERENTLY FROM DKA**

- Fluid deficits are larger (~12–15%) and **fluid — not insulin — is the priority first.**⁷
- Start insulin **later and lower** (only once glucose stops falling with fluids alone⁷, ~0.025–0.05 unit/kg/h).
- Watch for **rhabdomyolysis, malignant-hyperthermia-like fever, thrombosis and arrhythmia from potassium/phosphate shifts.**⁷
- Mortality is higher than DKA.⁷

✦ **MIXED PICTURES EXIST**

- Some children have features of both HHS and DKA.
- When ketoacidosis coexists, follow DKA fluid/insulin caution but expect a **bigger fluid requirement** and monitor osmolality and creatine kinase closely.

▴ **WORKUP**

- Glucose, **venous blood gas**, beta-hydroxybutyrate/ketones.

- Electrolytes with **corrected sodium**, potassium, **phosphate**, **magnesium**, **calcium**; BUN/creatinine.
- **Calculated effective osmolality**; ECG + continuous cardiac monitoring.
- **Creatine kinase** — screen for rhabdomyolysis.

Step	Approach	Notes
Fluids first	0.9% saline ; restore circulation, then replace deficit steadily	Aim to lower glucose gradually with fluid alone initially
Insulin (delayed)	0.025–0.05 unit/kg/h once glucose plateaus on fluids	No bolus; avoid rapid osmolar shifts
Monitor	Osmolality, K^+ , PO_4 , Ca^{2+} , creatine kinase, neuro obs	Anticipate rhabdomyolysis and hyperthermia

ADMIT · REFER

- **Admit** to high-dependency/PICU.
- **Call** pediatric endocrinology and critical care.

Adrenal Crisis / Salt-Wasting CAH

MINUTES

BSPED 2023

Shock + low sodium + high potassium + low glucose; collapse with illness.

RECOGNIZE

- Known adrenal insufficiency/CAH under stress, or a neonate (1–4 weeks) with vomiting, poor feeding, weight loss, hyperpigmentation, ambiguous genitalia (in females).¹⁵
- Labs: low sodium, high potassium, metabolic acidosis, low glucose.

① STEROID – DON'T WAIT

Hydrocortisone immediately (draw 17-OHP/cortisol first only if no delay).²⁴⁷



② FLUIDS + GLUCOSE

Saline bolus, repeat; treat low glucose; maintenance with dextrose-saline.



③ ELECTROLYTES & CAUSE

High potassium usually corrects with fluids + steroid (ECG for peaked T waves). Treat the trigger (often infection); cover empirically for sepsis if suspected. Endocrine + PICU.

Drug / fluid	Dose	Route & notes
Hydro-cortisone	< 1 yr 25 mg 1–5 yr 50 mg > 5 yr 100 mg	IV/IM, then 50–100 mg/m ² /day divided every 6 h ²⁴⁷
Sodium chloride 0.9%	20 mL/kg	IV bolus, repeat
Dextrose 10%	5 mL/kg	IV, for low glucose
Fludro-cortisone	0.1 mg once daily	Oral — maintenance only, not for the acute crisis

⚠ DO NOT USE DEXAMETHASONE FOR SALT-WASTING CRISIS

- It has **no mineralocorticoid effect**.¹⁵
- Give hydrocortisone immediately — do not delay it for lab results.
- High-dose hydrocortisone (> 50 mg) covers the mineralocorticoid

need acutely; fludrocortisone is added later for maintenance.¹⁵

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → monitored bed; hydrocortisone before labs, treat hypoglycemia/hyponatremia.
- **Discharge** — only after stabilization, endocrine review, and stress-dosing/emergency-injection education.
- **Refer** — pediatric endocrinology.

Hypoglycemia (Beyond the Newborn)

MINUTES

PES 2015

Plasma glucose < 3.9 mmol/L (70 mg/dL) — treat now; draw the critical sample first only when it will not delay treatment.⁸³

▲ RECOGNIZE

- **Autonomic:** sweating, tremor, tachycardia, hunger, pallor, anxiety.
Neuroglycopenic: lethargy, irritability, poor feeding, behavior change, confusion, seizures, coma.

- **Whipple triad** confirms it: low plasma glucose + consistent symptoms + resolution with glucose. Confirm any bedside reading with a **laboratory plasma glucose** (glucometers are unreliable when low).⁸³
- Red-flag context: hepatomegaly (glycogen storage / fatty-acid oxidation defect), midline defects or micropenis (hypopituitarism), hyperpigmentation (adrenal insufficiency), fasting- or illness-triggered episodes, or a high glucose-infusion requirement.

⚠ DRAW THE CRITICAL SAMPLE – BUT NEVER DELAY TREATING SEVERE OR SYMPTOMATIC HYPOGLYCEMIA

Obtain the sample before glucose **whenever it can be drawn immediately**. If the child is **symptomatic or severely hypoglycemic, treat first** and take the sample as soon as access allows — do not delay treatment for the sample. At the moment of hypoglycemia (ideally glucose < 2.8 mmol/L): **lab glucose, venous gas/bicarbonate, insulin + C-peptide, β -hydroxybutyrate + free fatty acids** (the pivotal branch point), **lactate, ammonia, cortisol + growth**

hormone, acylcarnitine profile + carnitine, plasma amino acids; and the **first-void urine** for ketones, organic acids ± toxin/sulfonylurea screen. One missed sample can cost the diagnosis.⁸³

① **CORRECT NOW**

IV **D10 2 mL/kg** (200 mg/kg) slow push — avoid large hyperosmolar boluses. If alert with a safe airway: oral rapid carbohydrate ~0.3 g/kg.⁸³



② **MAINTAIN**

- Continue **D10 at maintenance** and titrate the **glucose-infusion rate** to keep glucose > 3.9 mmol/L (central line for > D12.5)
- A requirement > **8–10 mg/kg/min** **strongly suggests hyperinsulinism** (a hallmark⁸³ — confirm on the critical sample); maintain euglycemia with escalating dextrose as an initial measure until the endocrinology team advises.



Glucagon 0.5–1 mg IM/IV/SC (or 0.03 mg/kg, max 1 mg) when there is no IV access or hyperinsulinism is suspected — a rise > 1.7 mmol/L (30 mg/dL) suggests hyperinsulinism. It fails in glycogen-depleted states (ketotic hypoglycemia, fatty-acid oxidation defects).⁸³

▴ **READ THE KETONES/FREE-FATTY-ACIDS BRANCH**

- **Low ketones + low free fatty acids** (non-ketotic) with detectable insulin → **hyperinsulinism** (congenital K_{ATP} , Beckwith-Wiedemann, exogenous insulin/sulfonylurea, dumping).⁸³
- **Low ketones + high free fatty acids** (hypoketotic) → **fatty-acid oxidation defect** (e.g. VLCADD or LCHADD) — abnormal acylcarnitines ± hyperammonemia/cardiomyopathy.
- **High ketones + high free fatty acids** (appropriate) → **ketotic hypoglycemia of childhood** (commonest, 1–5 yr, illness-triggered, diagnosis of exclusion), **GH/cortisol deficiency**, or **glycogen storage disease** (hepatomegaly, high lactate).⁸³

▲ ADMIT · DISCHARGE · REFER

- **Admit** — unexplained symptomatic or persistent hypoglycemia → until fasting safety is documented and a cause established.
- **Discharge** — only after a documented safe fast and a sick-day/emergency plan (home glucagon where indicated).
- **Refer** — pediatric endocrinology/metabolic; hyperinsulinism or a glucose infusion rate > 8–10 mg/kg/min → **consult endocrinology with congenital hyperinsulinism (CHI) expertise.**

Acute Electrolyte Emergencies

MINUTES

Stabilize the heart, stop the seizures, then correct at a safe rate.

▲ HYPERKALEMIA – GET AN ECG FIRST

- **Hyperkalemia** = serum K^+ > **5.5 mmol/L** (mild 5.5–6.0 · moderate 6.0–6.5 · severe > **6.5** or ECG changes);
hypokalemia = K^+ < **3.5 mmol/L** (severe < 2.5).²⁴⁸

- ECG progression: peaked T waves → flattened P / prolonged PR → widened QRS → sine wave → arrest. Treat urgently if there are **ECG changes or potassium > 6.5 mmol/L.**²⁴⁸
- If the child is well with a normal ECG, repeat a **free-flowing venous sample** first — hemolysis causes pseudohyperkalemia.²⁴⁸ **Stop all potassium intake** and potassium-raising drugs.

HYPERKALEMIA – STABILIZE, SHIFT, REMOVE

Step / drug	Dose	Route & notes
STABILIZE • Calcium gluconate 10%	0.5 mL/kg (max 20 mL) IV over 5–10 min	Cardiac monitor; protects the heart, does not lower potassium. Repeat after 15 min if ECG still abnormal. Not in the same line as bicarbonate (precipitates); caution in digoxin toxicity ²⁴⁸
SHIFT • Insulin (soluble) + glucose	0.1 unit/kg (max 10 units) + glucose 0.5 g/kg IV (= 5 mL/kg of dextrose 10%)	Onset 10–20 min; monitor glucose closely — early and late hypoglycemia ²⁴⁸

Step / drug	Dose	Route & notes
SHIFT · Salbutamol (nebulized)	2.5–5 mg (high dose 10–20 mg)	Additive to insulin; may cause tachycardia ²⁴⁸
SHIFT · Sodium bicarbonate	1 mmol/kg IV	Only if acidotic; modest, adjunctive effect
REMOVE · Potassium binder	0.5–1 g/kg (max 30 g) oral / rectal	Sodium/calcium polystyrene sulfonate; slow onset — not for immediately life-threatening hyperkalemia; avoid with ileus, bowel obstruction, or NEC risk ²⁴⁸
REMOVE · Furosemide	1 mg/kg IV	If making urine
REMOVE · Dialysis	–	Refractory or severe; involve renal / PICU early

◆ THE LOGIC

- Calcium and the shifters (insulin, salbutamol, bicarbonate) only **BUY TIME**; only binders, loop diuretics, and dialysis actually remove potassium from the body.²⁴⁸

- Recheck potassium and glucose after treating.

SODIUM – CORRECT AT A SAFE RATE

⊙ **DEFINITIONS** Hyponatremia = $\text{Na}^+ < 135 \text{ MMOL/L}$ (severe < 125); hypernatremia = $\text{Na}^+ > 145 \text{ MMOL/L}$ (> 150 in hypernatremic dehydration).¹⁰¹

Problem	Treatment	Notes
Hyponatremia with seizures / encephalopathy	3% saline 2–5 mL/kg ($\approx 1\text{--}2.5 \text{ mmol/kg}$) IV over ~ 20 min; may repeat	Aim to raise sodium by 4–6 mmol/L to stop the seizures — that is usually enough ¹⁰¹
Hyponatremia (no severe symptoms)	Treat the cause; correct slowly	Do not exceed 8 mmol/L per 24 h (osmotic demyelination). Volume depletion $\rightarrow$ 0.9% saline; SIADH $\rightarrow$ fluid restriction ¹⁰¹
Hypernatremic dehydration	Resuscitate shock with isotonic fluid first, then rehydrate over $\sim 48 \text{ h}$	Lower sodium $\leq$ 0.5 mmol/L per hour ($\leq 10\text{--}12$ per 24 h) — faster risks cerebral edema ¹⁰¹

CALCIUM & MAGNESIUM

⊙ **DEFINITIONS** Hypocalcemia = total calcium $< 2.1 \text{ MMOL/L}$ or ionized $< 1.1 \text{ MMOL/L}$ (neonate $< 2.0 \text{ mmol/L}$; correct total calcium for albumin). Hypomagnesemia = $\text{Mg}^{2+} < 0.7 \text{ MMOL/L}$ (blocks correction of low K^+ and Ca^{2+}).

Problem	Treatment	Notes
Symptomatic hypocalcemia	Calcium gluconate 10% 0.5–1 mL/kg slow IV, cardiac monitor; then infusion	Tetany, seizures, prolonged QT, Chvostek / Trousseau signs. Avoid extravasation (tissue necrosis) ¹
Refractory hypocalcemia or hypokalemia	Check and replace magnesium	Low magnesium blocks correction of both — replace it first ²⁴⁸

⚠ THE SAFE-RATE RULE

Move fast to **stabilize the heart** (calcium) and **stop seizures** (3% saline), but correct chronic sodium **slowly**: overcorrecting low sodium causes osmotic demyelination, and dropping high sodium too fast causes cerebral edema.¹⁰¹

▴ ADMIT · DISCHARGE · REFER

- **Admit** — symptomatic or severe derangements, or those needing infusions/monitoring → high-dependency/PICU.
- **Discharge** — mild, asymptomatic, correctable causes, with a follow-up recheck.
- **Refer** — nephrology/endocrinology by cause; PICU for arrhythmia risk or rapid shifts.

Hypercalcemia HOURS

Corrected calcium > 2.6 mmol/L (10.5 mg/dL); > 3.5 mmol/L (14 mg/dL) is a crisis.²⁴⁹

▴ RECOGNIZE

- **Stones, bones, groans & psychiatric moans:** nephrocalcinosis/stones; bone pain; constipation, nausea, anorexia, pancreatitis; lethargy, confusion, weakness, hypotonia, coma. Infants: poor feeding, failure to thrive, irritability.²⁴⁹
- **Polyuria → dehydration** (nephrogenic diabetes insipidus) — vicious cycle worsening the calcium.

Cardiac: **short QT**, bradyarrhythmia;
monitor ECG when
severe/symptomatic.²⁴⁹

- **Correct for albumin:** add 0.8 mg/dL per 1 g/dL albumin below 4.0; confirm with **ionized calcium**.²⁴⁹
- Mild 2.6–3.0
- moderate 3.0–3.5
- severe > 3.5 mmol/L.

▴ FIND THE CAUSE – PTH IS THE BRANCH POINT

- **High or inappropriately normal PTH**
→ primary hyperparathyroidism (think MEN) or **familial hypocalciuric hypercalcemia** (low urine Ca/Cr — benign, do **not** operate).²⁴⁹
- **Suppressed PTH** → malignancy (PTHrP/osteolysis), vitamin D intoxication, granulomatous disease, immobilization, thiazides, milk-alkali. Send PTHrP, 25-OH & 1,25-(OH)₂²⁴⁹ vitamin D, phosphate, magnesium, urine Ca/Cr.
- **Infants:** subcutaneous fat necrosis, Williams syndrome, idiopathic infantile hypercalcemia, neonatal severe hyperparathyroidism.

LOWER THE CALCIUM – HYDRATE, THEN REMOVE

Step / drug	Dose	Route & notes
Stop intake	–	Stop all calcium & vitamin D (diet, fluids, supplements); stop thiazides, lithium, vitamin A/D
0.9% saline (cornerstone)	10–20 mL/kg bolus, then ~2–3× maintenance (~3000 mL/m ² /day)	IV; restores volume then drives calciuresis; target brisk urine output ²⁴⁹
Furosemitide	1 mg/kg IV	Only after volume repletion ; for fluid overload; monitor K ⁺ /Mg ²⁺
Calcitonin (salmon)	4 IU/kg SC/IM every 6–12 h (up to 8 IU/kg q6h)	Rapid onset (2–6 h), modest; tachyphylaxis by 48 h ²⁴⁹ — a bridge. <i>Salmon</i> is the potent form used clinically (≈40× human calcitonin).

Step / drug	Dose	Route & notes
Zoledronic acid (preferred)	0.025–0.05 mg/kg/day IV (max 4 mg); 4 mg/5 mL vial, dilute 1:20 in 0.9% saline, infuse over 15 min	Preferred bisphosphonate for high bone turnover (malignancy, immobilization) — higher response rate (complete response ~88% vs ~70%) and a much shorter infusion (15 min vs 2–4 h) than pamidronate (Major, <i>J Clin Oncol</i> 2001). ²¹¹ Onset 24–48 h; ensure volume replete; watch for hypocalcemia; reduce/withhold in renal impairment
Pamidronate (alternative)	0.5–1 mg/kg IV over 2–4 h	Alternative where zoledronic acid is unavailable; same indications ²¹¹ ; onset 24–48 h; ensure volume replete; watch for hypocalcemia
Glucocorticoid	Prednisone 1–2 mg/kg/day	For vitamin D-mediated, granulomatous disease, or lymphoma/leukemia ²⁴⁹
Dialysis	–	Severe/refractory, renal failure, or cannot tolerate a saline load

▲ ADMIT · DISCHARGE · REFER

- **Admit** — moderate–severe, symptomatic, or > 3.5 mmol/L → monitored bed/PICU with ECG.
- **Discharge** — mild & asymptomatic with an outpatient workup and offending agents stopped.
- **Refer** — pediatric endocrinology ± oncology/nephrology; parathyroidectomy for primary hyperparathyroidism (not for FHH).

Thyroid Storm (Thyrotoxic Crisis)

MINUTES

ATA 2016 · BWPS

Decompensated hyperthyroidism: fever, tachyarrhythmia and agitation, often with a trigger.

▲ RECOGNIZE

- Known/undiagnosed Graves' disease with **hyperthermia**, marked **tachycardia or tachyarrhythmia** (atrial fibrillation/heart failure), agitation or reduced consciousness,

and GI upset — usually with a trigger (infection, surgery, DKA, medication non-adherence).

⚠ **TREAT ON CLINICAL SUSPICION**

- Do not wait for thyroid function results.
- Provide supportive care (cooling, fluids, treat the trigger) and block the thyroid axis in the right **order** — β -blockade and a thionamide first, and give **iodine only after** the thionamide to avoid fuelling hormone synthesis.²⁵⁰

▴ **WORKUP**

- **TSH (suppressed), free T4, free T3, TRAb/TSI.**
- CBC, LFTs, glucose, electrolytes, calcium; **ECG.**
- Diagnosis is **clinical** (Burch-Wartofsky²⁵⁰ ≥ 45 highly suggestive) — do not delay treatment for labs.

Step / drug	Dose	Route & notes
1 · Propranolol (β-block)	0.25–0.5 mg/kg/dose PO/NG q6h or 0.01–0.05 mg/kg IV (max 1 mg/dose)	Controls heart rate, tremor and agitation; slows peripheral T4→T3. Caution in heart failure — treat failure alongside ²⁵⁰
2 · Methimazole (thionamide)	0.5–1 mg/kg/day PO/NG/PR divided q8h (max 30 mg/day)	Blocks new hormone synthesis; give the thionamide before iodine. Carbimazole is the equivalent prodrug ²⁵⁰
2 · PTU (thionamide — alternative)	5–10 mg/kg/day PO/NG divided q8h	Also blocks peripheral T4→T3. Either thionamide is acceptable — no difference in mortality (Lee, <i>JAMA Netw Open</i> 2023); ATA 2016 has historically listed PTU first in storm ²¹²
3 · Potassium iodide (strong iodine solution)	1–5 drops PO/NG q8h	Blocks hormone release ²⁵⁰ ; give ≥ 1 h after the thionamide
4 · Hydrocortisone	1–2 mg/kg/dose IV q6h (or 50–100 mg/m ² /day)	Reduces peripheral T4→T3 conversion and covers relative adrenal insufficiency ²⁵⁰

▲ ADMIT · REFER

- **Admit** to PICU/HDU.
- **Call** pediatric endocrinology and critical care.

Metabolic Crisis (Inborn Error)

MINUTES

ANY AGE

Suspected inborn error of metabolism —
"sepsis that doesn't fit."

▲ RECOGNIZE (THINK INBORN ERROR)

- Encephalopathy, vomiting, seizures, or apnea after a symptom-free interval or a catabolic trigger (fasting, intercurrent illness, surgery); poor feeding; tachypnea. Family history or parental consanguinity.²⁵¹
- Mimics sepsis — "sepsis that doesn't fit." Presents at **any age**, not only the newborn. Not every metabolic crisis is hyperammonemic — the pattern points to the group:²⁵¹

Pattern	Acid-base · ammonia	Pointers
Urea-cycle (hyperammonemia)	Respiratory alkalosis; very high ammonia; low urea ²⁵¹	Little or no acidosis; encephalopathy ± seizures
Organic acidemia (MMA / PA / IVA)	High-anion-gap metabolic acidosis + ketosis; ammonia high (secondary)	Hypoglycemia; offending acid on urine organic acids ^{251,13}
MSUD	Ketosis with normal / near-normal ammonia; little or no acidosis	Maple-syrup odor (urine/cerumen); high plasma leucine ²⁵¹

① STOP CATABOLISM

Stop all protein — a temporary emergency measure (first ~24–48 h) until the metabolic team advises; prolonged protein restriction itself worsens catabolism, so protein is reintroduced on their guidance. Run high-dextrose fluids; avoid hypotonic fluids and lactated Ringer's.²⁵¹ Treat low glucose. **This card is initial stabilization pending specialist consultation.**



② IMMEDIATE LABS + CALL METABOLIC TEAM

- Ammonia (free-flowing sample, **on ice, run STAT and process immediately**)²⁵¹, blood gas, glucose, lactate, ketones, electrolytes/anion gap.
- Send as **confirmatory** (not routine/STAT): plasma amino acids, urine organic acids, acylcarnitines.

high ammonia

③ NITROGEN SCAVENGERS + COFACTORS

See drug table — coordinate with the metabolic service.

ammonia > ~500 / rising / coma

Urgent hemodialysis or CRRT. Transfer to a metabolic center.²⁵¹

Drug / measure	Dose	Route & notes
Dextrose 10%	1.5× maintenance; glucose infusion rate 8–10 (up to 15–20) mg/kg/min	IV — reverses catabolism
Sodium benzoate + sodium phenylacetate	250 mg/kg load over 90–120 min, then 250 mg/kg/day	IV; central line preferred, peripheral acceptable when life-saving — ask pharmacy for an appropriately diluted preparation. ²⁵¹ In MMA / PA , phenylacetate/phenylbutyrate is not the primary scavenger (depletes glutamine) — benzoate ± carglumic acid per the metabolic team. ¹³
L-arginine hydrochloride	200–600 mg/kg	IV load then per day; omit in arginase deficiency ²⁵¹

Drug / measure	Dose	Route & notes
Carglumic acid	Acute 100–250 mg/kg/day (2–4 divided doses), then maintenance ~50 mg/kg/day (range 10–100) in 2 divided doses – titrate to plasma ammonia	Oral/NG; for NAGS deficiency & some organic acidemias ^{251,13}
L-carnitine	100 mg/kg	IV / oral; for suspected organic acidemia

◆ AVAILABILITY

- Nitrogen scavengers, carglumic acid and L-carnitine are **SPECIALIST STOCK** held in tertiary metabolic / PICU units, not general wards.
- Phone the metabolic on-call **EARLY** — high-dextrose fluids, protein cessation and dialysis access can start anywhere while drugs are sourced. **AVOID VALPROATE.**²⁵¹

▴ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected → PICU; stop protein, high-dose dextrose, treat hyperammonemia.

- **Discharge** — not acutely; under the metabolic team with an emergency regimen.
- **Refer** — metabolic/genetics urgently and a center capable of ammonia scavengers/hemofiltration.

Focused Specialty

07

Emergencies

Focused eye, dental, dermatologic, ENT & neurosurgical emergencies

Pediatric Eye Emergencies

MINUTES

SIGHT-THREAT

Chemical burn — irrigate before anything else. Leukocoria, orbital cellulitis, and a ruptured globe are the sight- and life-threats not to miss.

⚠️ CHEMICAL INJURY — IRRIGATE FIRST

- Immediate, copious irrigation with **saline (or clean water)** for **≥ 15–30 min BEFORE** examination or imaging; evert the lids, sweep out particulate matter, recheck pH

until neutral ($\approx 7.0-7.5$). **Alkali** (drain/oven cleaner, cement, wet button batteries) penetrates deeper than acid.¹⁹²

- Then urgent ophthalmology.

▶ THE CAN'T-MISS DIAGNOSES

- **Orbital (post-septal) cellulitis** — **proptosis, pain on or limited eye movement, reduced acuity or color vision, a relative afferent pupillary defect**, chemosis → emergency. **CT orbits + sinuses, IV antibiotics, ophthalmology + ENT**. Distinguish from **preseptal** (lid swelling only, with **normal** movement, vision, and pupil)¹⁹² (see [Preseptal & Orbital Cellulitis](#)).
- **Ruptured / penetrating globe** — after trauma with a **distorted (teardrop) pupil**, prolapsed dark uveal tissue, or marked subconjunctival hemorrhage → cover with a **rigid shield (never a pad or any pressure)**, keep **NPO**, give analgesia + antiemetic, update tetanus, and refer urgently. **No drops, no ointment.**¹⁹²

- **Leukocoria (white pupil reflex)**, a new squint, or an **absent red reflex** → **retinoblastoma** until proven otherwise → urgent ophthalmology (a life-threatening tumor).¹⁹²
- **Ophthalmia neonatorum** — hyperacute purulent conjunctivitis in the **first days of life** → gonococcal risk (corneal perforation) → emergency: **systemic ceftriaxone or cefotaxime is the main treatment** (adequate systemic therapy makes topical antibiotics unnecessary; prefer cefotaxime in the jaundiced neonate or with calcium-containing IV fluids), frequent saline irrigation of discharge, admit and swab; **test for both *N. gonorrhoeae* and *C. trachomatis***.¹⁶¹
- **Acute red painful eye** with photophobia/reduced vision, or **hyphema** (blood layering in the anterior chamber after trauma) → same-day ophthalmology.
- **Corneal ulcer (microbial keratitis)** — red, painful, photophobic eye with a **white/opaque corneal infiltrate or haze** ± discharge; high risk in **contact-lens** wearers and after

trauma → sight-threatening,
same-day ophthalmology. **Do not
patch**; stop contact-lens wear.¹⁹²

- **Congenital / infantile glaucoma** —
the infant triad of **watering
(epiphora), photophobia, and
blepharospasm** with an **enlarged or
cloudy cornea (buphthalmos)** →
urgent ophthalmology (needs
surgery); do not dismiss as a
blocked tear duct.¹⁹²
- **New anisocoria (unequal pupils)** —
may signal a **third-nerve palsy,
Horner syndrome, or raised ICP /
herniation** (see [Raised ICP](#)); with
reduced consciousness, ptosis, or
trauma it is a red flag → urgent
assessment. Check pupils in **both
light and dark** to localize.¹⁹²

⚠ SAFEGUARDING – RETINAL HEMORRHAGES

Multilayered retinal hemorrhages
extending to the periphery in an infant
with encephalopathy suggest **abusive
head trauma** — involve safeguarding and
ophthalmology (see [Fractures —
Non-Accidental Red Flags](#)).

✦ **PEARL – CHECK THE RED REFLEX IN EVERY**

CHILD

A symmetrical red reflex helps exclude retinoblastoma, cataract, and vitreous pathology; any **white, asymmetric, or absent** reflex needs urgent ophthalmology.¹⁹²

✦ YOUR ROLE – THE GENERAL PEDIATRICIAN

- You are the first responder, not the eye surgeon.
- Three time-critical moves are yours: **(1) irrigate immediately** for any chemical splash — don't wait for ophthalmology; **(2) shield** (never pad or press) a suspected ruptured globe, keep **NPO**, give analgesia + antiemetic; **(3) check the red reflex in every child** and treat leukocoria / an absent reflex as retinoblastoma.
- Document **visual acuity, pupils, and eye movements** before referral. **Avoid drops, ointment, or any pressure** on a possibly open globe, and never patch a suspected corneal ulcer.
- Then refer to ophthalmology (add ENT for orbital cellulitis) within the timeframe the diagnosis demands.

REFER

- **Ophthalmology urgently** — chemical injury, suspected globe rupture, leukocoria/absent red reflex, orbital cellulitis (add **ENT**), acute vision loss, or hyphema.

Dental & Oral Emergencies

MINUTES

TIME-CRITICAL

An avulsed permanent tooth is a true emergency — reimplant within minutes. Never reimplant a primary tooth. Watch the swelling that threatens the airway.

DENTAL TRAUMA

- **Luxation** (displaced or mobile tooth) — **do not attempt to reposition it**; analgesia and same-day dental referral.¹⁸²
- **Crown fracture** — preserve any fragment and keep it moist (milk or saline); analgesia; prompt dental review — pulp exposure needs same-day care.
- **Primary tooth avulsion** — **do NOT reimplant** (risks of damaging the permanent successor); reassure and

arrange dental review.¹⁸²

- **Post-extraction bleeding** — firm gauze pressure 15–20 min ± tranexamic acid; if it persists, consider an undiagnosed **bleeding disorder**.

⚠ AVULSED PERMANENT TOOTH – MINUTES MATTER

- **Reimplant as soon as possible.** Hold the tooth by the **crown** — **never touch or scrub the root** (protect the periodontal ligament).
- If dirty, rinse ~10 s in **cold milk or saline**, reinsert, have the child bite on gauze.
- If reimplantation isn't possible, transport the tooth in **Hank's solution or cold milk** — **never water**.¹⁸²
- Then urgent dental care for splinting; **assess tetanus prophylaxis** — indicated for a contaminated avulsion and guided by both the child's immunization status and the nature of the wound; consider antibiotics.
- Prognosis falls sharply once **extra-oral dry time exceeds ~60 min**, and periodontal-ligament

cells are then considered non-viable — **reimplant anyway.**¹⁸²

✦ PEARL – MILK BEATS WATER

If Hank's solution isn't to hand, **cold milk** is the best readily-available storage medium; tap water is hypotonic and kills periodontal-ligament cells, worsening the outcome.¹⁸² **Time is tooth.**

▴ PRIMARY HERPETIC GINGIVOSTOMATITIS (HSV-1)

- The commonest cause of a painful mouth with fever in the **1–5-year-old: fever** (usually **3–5 days**, occasionally up to a week), **swollen, friable, bleeding gingiva**, and crops of **painful vesicles → ulcers** on the gingiva, tongue, lips, and buccal mucosa (often with perioral spread), drooling, and **refusal to eat or drink**. Tender **cervical lymphadenopathy** is usual and, being reactive, resolves as the illness settles. Self-limited — the illness runs **~10–14 days.**¹⁹⁹
- **Management is supportive** — the priority is **hydration and analgesia** (acetaminophen ± ibuprofen); offer cold, bland fluids. **Dehydration from**

poor intake is the main reason to admit. Avoid topical lidocaine/anesthetic gels in young children (toxicity).¹⁹⁸

- **Oral acyclovir — 15–20 mg/kg/dose 4×/day for 5–7 days** (max single dose 800 mg); immunocompromised children **20 mg/kg/dose 4×/day for 7–10 days**; immunocompromised adolescents 400 mg 3×/day for 10 days or until resolution.¹⁹⁸

Significant or disseminated disease in an immunocompromised child needs IV acyclovir (5–10 mg/kg/dose q8h) rather than a prolonged oral course. Shortens fever, oral lesions, and viral shedding **only if started within 72 h of symptom onset**; not routine for the well child presenting late.

- Distinguish from **herpangina** (coxsackievirus — ulcers on the soft palate/tonsillar pillars, **gingiva spared**) and hand-foot-and-mouth disease.¹⁹⁹ **Admit/refer** the dehydrated, the immunocompromised, or suspected **eczema herpeticum** / disseminated HSV.

▶ ORAL SOFT-TISSUE INJURIES

- **Tongue lacerations** — most heal without sutures (rich blood supply). **Repair** if the wound gapes at rest, is large ($> \sim 1\text{--}2\text{ cm}$), bleeds persistently, is through-and-through, or splits the free edge/tip (a bisecting flap) — these risk a notched tongue.
- **Lip lacerations** — a wound that **crosses the vermilion border** needs precise realignment of the border first (a 1 mm step is visible); through-and-through wounds need layered closure. Palpate and image for an **embedded tooth fragment** or foreign body after a fall onto the teeth.
- **Buccal & gingival mucosa** — most intraoral tears heal spontaneously; suture only large, gaping, or food-trapping wounds. A deep cheek wound can injure the **parotid duct or facial nerve** — refer.
- **Embedded foreign bodies** — account for **every tooth fragment**: a missing piece may be buried in the lip/tongue (soft-tissue radiograph or ultrasound) or aspirated (**chest**

X-ray if unaccounted for). Remove grit/glass, irrigate, and assess tetanus status.

- **Intruded tooth** (driven into the socket) — **refer to a pediatric dentist.** A **primary** intruded tooth is usually left to **re-erupt spontaneously** (do not reposition — protect the permanent tooth bud); a **permanent** intruded tooth needs dental assessment for repositioning.¹⁸²
- Repair intraoral and lip wounds within ~24 h with analgesia; give antibiotics for through-and-through lip wounds, bites, or grossly contaminated wounds.

▀ **DENTOALVEOLAR ABSCESS / FACIAL SWELLING**

- Analgesia and **dental source control** — **incision and drainage, root canal (pulp) treatment, or extraction of the causative tooth** — with a **mandatory dental referral.** For a **localized** abscess without systemic involvement or spreading, definitive dental treatment is the primary management and **systemic antibiotics are generally not required; reserve antibiotics for**

facial cellulitis, systemic signs, significant spreading, or suspected deep-space infection (per clinical judgment and MOH protocols). These commonly present **directly to dental clinics**, where the dentist identifies the source and delivers timely definitive treatment.¹⁹⁸ **Watch for spread to the deep neck, floor of mouth, or orbit** — a spreading facial-space infection can progress to Ludwig's angina (next).

⚠ **AIRWAY – LUDWIG'S ANGINA**

Firm submandibular swelling — in children often **unilateral at onset** before becoming bilateral — with a **raised (elevated) tongue**, drooling, or trismus is a deep-space infection that can obstruct the airway. **Call for airway help early — anesthesia, oral & maxillofacial surgery, ENT, or pediatric surgery** (all are equipped to manage these cases), give IV antibiotics, and do **not** lay the child flat.²⁰⁰ **Time-critical — urgently call or refer.**

✦ **YOUR ROLE – THE GENERAL PEDIATRICIAN**

- Most of what reaches you is yours to start.

- For an **avulsed permanent tooth**, reimplant or store it in **cold milk** and send urgently — minutes decide the tooth; **never reimplant a primary tooth**.
- Give analgesia, control post-extraction bleeding with pressure ± tranexamic acid, **assess tetanus** for contaminated wounds, and suture only the soft-tissue lacerations that need it — account for **every tooth fragment**.
- Treat herpetic gingivostomatitis supportively, watching hydration.
- Escalate the two true emergencies fast: a spreading facial-space infection or **Ludwig's angina** (airway — call/refer anesthesia, oral & maxillofacial surgery, ENT, or pediatric surgery), and refer to dental / maxillofacial for avulsion, pulp exposure, or abscess.

REFER

- **Dental / maxillofacial urgently** — avulsion, fractures with pulp exposure, facial fractures, or a facial-space infection; refer an **intruded tooth to a pediatric dentist**;

for any airway threat call **anesthesia, oral & maxillofacial surgery, ENT, or pediatric surgery** — all are equipped to manage these cases.

Dermatologic Emergencies HOURS

CAN'T-MISS

Fever + rash — find the killers:
meningococcemia / purpura fulminans,
SJS/TEN, staph scalded skin, and eczema
herpeticum.

⚠ **NON-BLANCHING RASH + FEVER**

- Petechiae or purpura that **do not blanch** in an unwell/febrile child = **invasive meningococcal disease / sepsis / purpura fulminans** until proven otherwise — take blood cultures, give **empiric IV antibiotics immediately**, follow the **Septic Shock** pathway.¹²²
- Do not wait for investigations.

▶ THE CAN'T-MISS RASHES

- **SJS / TEN** — dusky/targetoid macules with skin detachment and **erosions of ≥ 2 mucosal sites** (eyes, mouth, genitals), 1–3 weeks after a **drug** (antibiotics, anticonvulsants).¹⁸⁵
Stop the culprit; fluids, skin + eye care, urgent dermatology + ophthalmology, transfer to a **burns/PICU** unit. **Avoid the culprit drug and relevant cross-reacting agents** — broad beta-lactam avoidance applies mainly when a beta-lactam is the suspected culprit (see [Antibiotic Allergy](#)).
- **Staphylococcal scalded skin syndrome (SSSS)** — young child, fever, **tender diffuse erythema with superficial peeling (Nikolsky +)** and perioral crusting; **mucosae spared**.
Toxin-mediated: IV anti-staphylococcal antibiotic (**cefazolin** first-line in Kuwait, as flucloxacillin is not available locally) + analgesia + fluids.¹⁸⁸
- **Eczema herpeticum** — clusters of **monomorphic punched-out vesicles/erosions on eczematous skin** in an unwell child → start **acyclovir** urgently; ophthalmology if periocular. (HSV superinfection.)¹⁸⁹

- **DRESS** — fever, widespread morbilliform rash, **facial edema**, lymphadenopathy, **eosinophilia**, and organ (liver) involvement 2–8 weeks after a drug → stop the drug, supportive care ± steroids.¹⁹⁰
- **Necrotizing fasciitis** — **pain out of proportion**, rapidly spreading dusky/edematous skin, systemic toxicity → surgical emergency + broad-spectrum antibiotics.¹⁹¹

◆ SJS / TEN – FOCUSED WORKUP & MANAGEMENT

- **Confirm the signs** — dusky/targetoid macules with **skin detachment** and **erosions of ≥ 2 mucosal sites**; **Nikolsky +** (lateral shearing pressure detaches epidermis on non-blistered skin), **Asboe–Hansen +** (a blister extends sideways with slight tangential pressure), and **widespread skin tenderness** — typically 1–3 weeks after a culprit drug.¹⁸⁵
- **Prognosis** — calculate **SCORTEN** (best at day 1 and day 3). **In children it over-estimates mortality** (pediatric death rates are far lower), so use it for guidance, not as a ceiling.¹⁸⁵
- **Investigations** — CBC with differential, urea & electrolytes, LFTs, glucose, bicarbonate, coagulation profile, CRP/ESR, magnesium, phosphate, lactate, urinalysis. **Sepsis is the leading cause of death** — infective screen (blood cultures; skin swabs for **HSV/VZV and bacteria**) and treat superadded infection. **Chest X-ray**;

***Mycoplasma* testing guided by the clinical picture** (in children, **reactive infectious mucocutaneous eruption [RIME]** — the current term, which subsumes the former *Mycoplasma*-induced rash and mucositis [MIRM] — is a distinct, milder mimic with **prominent mucositis and little skin loss**; RIME can also follow other respiratory pathogens, so test by presentation rather than routinely).²¹³ **Skin biopsy** if the diagnosis is unclear.

- **Act now — stop the culprit drug immediately** (earlier withdrawal improves survival); **transfer to a burns unit or PICU**.¹⁸⁶
- **Supportive care is the mainstay** — IV fluids & electrolytes, **thermoregulation**, nutrition, **analgesia; gentle handling, minimize friction/shear; non-adherent dressings** (petrolatum-impregnated gauze or silicone/advanced dressings); avoid **prophylactic** systemic antibiotics — treat proven infection. Early multidisciplinary team: **dermatology**,

ophthalmology (prevent ocular scarring; amniotic-membrane grafting in severe eye disease), **urology/gynecology** (prevent genital adhesions), wound care, pain, psychology.¹⁸⁶

- **Systemic immunomodulation — specialist decision, low-certainty evidence.** Drug withdrawal + supportive care remain the priority. If an agent is used, **ciclosporin** and **etanercept (anti-TNF)** carry the most promising (still low-certainty) mortality signals; **corticosteroids** are controversial; **IVIG monotherapy is not supported** (used, if at all, combined with corticosteroids). Individualized by the specialist team.¹⁸⁷
- **After recovery** — screen for **ocular, cutaneous, urogenital and psychological** sequelae; document the culprit, give a clear **allergy alert** (avoid re-exposure and cross-reacting agents), and consider **HLA-guided avoidance** where

relevant (e.g. HLA-B*15:02 & carbamazepine, HLA-B*58:01 & allopurinol). ¹⁸⁵

Evidence: BAD children & young people
SJS/TEN guideline 2018; European S3 guideline 2024; RIME/MIRM terminology per Damari et al., *J Hosp Med* 2025. Culprit-drug withdrawal + supportive care have the clearest benefit; all systemic immunomodulators remain low-certainty.

◆ SSSS (STAPHYLOCOCCAL SCALDED SKIN)

– FOCUSED WORKUP & MANAGEMENT

- **Recognize** — young child, fever, tender diffuse erythema with **superficial peeling (Nikolsky +); desquamation of the flexures and radial fissuring/crusting around the eyes, mouth and nose; mucosae are spared.**

Toxin-mediated (exfoliative toxin from a distant staphylococcal focus).¹⁸⁸

- **Cultures** — **swab the focus, not the blisters.**

Nasopharynx/nares, conjunctivae, throat, umbilicus, urine, blood, and any suspected primary site; in a **neonate** include the **umbilical stump and circumcision site. Bulla fluid is sterile** (unlike bullous impetigo) — do not culture it. **Skin biopsy is usually unnecessary** — reserve it (or a frozen section) for when SJS/TEN cannot be excluded (SSSS splits high in the granular layer; TEN is full-thickness).¹⁸⁸

- **Antibiotics** — start an **IV anti-staphylococcal** agent: **cefazolin** is first-line in Kuwait

(the penicillinase-resistant penicillins — flucloxacillin, nafcillin, oxacillin — are the international first-line but are not available locally), switching **IV** → **oral** once improving; **clindamycin** only as an **anti-toxin** adjunct where susceptible (added benefit uncertain in children, resistance rising); **vancomycin** if MRSA is suspected/prevalent or first-line fails. Doses below — de-escalate once afebrile and per sensitivities.¹⁸⁸

- **Supportive care** — treat like a **superficial burn**: fluid & electrolyte balance, thermoregulation, nutrition, **analgesia**, and **minimal handling**. **Wound care** with **petrolatum/Vaseline-impregnated gauze** (apply to the dressing — do not smear emollient directly onto raw skin); **warm compresses** to facial crusting. Watch for fluid loss and secondary sepsis.
- **Prognosis** — good in children with prompt treatment (low mortality); higher risk in

neonates and with extensive involvement.¹⁸⁸

Antibiotic	Dose (pediatric)	Route & notes
Cefazolin (first-line IV)	25–33 mg/kg/dose q8h (max 1 g/dose)	IV; first-generation cephalosporin — anti-staphylococcal first-line in Kuwait (flucloxacillin not available locally)
Flucloxacillin / nafcillin / oxacillin (where available)	25–50 mg/kg/dose q6h (max 2 g/dose)	IV; penicillinase-resistant anti-staphylococcal — the international first-line, not currently available in Kuwait
Clindamycin (anti-toxin adjunct)	10 mg/kg/dose q8h (max 900 mg/dose)	IV/PO; optional — suppresses toxin, only where susceptible; benefit uncertain in children
Vanco- mycin (if MRSA)	15 mg/kg/dose q6h (max 1 g/dose)	IV; if MRSA suspected/prevalent or first-line fails
Oral step-down	Cephalexin 25 mg/kg/dose q6–8h (max 1 g/dose)	PO once afebrile & improving; complete ~7-day course (dicloxacillin/flucloxacillin where available)

Evidence: pediatric SSSS systematic review (Pediatr Dermatol 2025); Merck Manual (professional). Bullae are sterile — culture the primary focus; anti-staphylococcal therapy + supportive care is the mainstay.

◆ ECZEMA HERPETICUM — FOCUSED WORKUP
& MANAGEMENT

- **Recognize** — clusters of **monomorphic punched-out vesicles/erosions** on eczematous skin in an unwell child; **honey-colored crusting** means the lesions are **impetiginized** (secondary *S. aureus*).¹⁸⁹
- **Isolate** — contact precautions: **gloves and gown** until lesions are dry and crusted.
- **Confirm (don't delay treatment)** — **deroof an intact vesicle** with a sterile needle and swab the clear fluid for **HSV-1/2 & VZV PCR** ($\pm$ Tzanck smear as a rapid bedside adjunct); take a **bacterial swab (C&S)** for secondary infection. Diagnosis is clinical — start antivirals empirically.¹⁸⁹
- **Baseline labs** — CBC and a metabolic/renal panel, and check hydration, **before IV acyclovir** (renal safety).
- **Antivirals early — they save lives** — **IV acyclovir 5–10 mg/kg/dose q8h** (dose by ideal

body weight; ensure hydration) if **systemically unwell, severe/disseminated, periocular, or poor oral intake; oral acyclovir per local protocol** for milder disease. Continue ~10–14 days / until lesions crust; step IV → oral on improvement.¹⁸⁹

- **Cover secondary infection** — if impetiginized, add an **oral anti-staphylococcal antibiotic** for *S. aureus*.
- **Eyes** — facial/periocular involvement warrants **urgent ophthalmology** to exclude **corneal disease (herpetic keratitis)**; start topical ophthalmic antiviral if corneal lesions are suspected.
- **Skin / eczema care** — once antivirals are started (many **pause topical steroids for ~48 h, then restart the appropriate-potency topical corticosteroid** as the child improves — steroids must never delay antivirals).¹⁸⁹ **Bland ointment emollients** (avoid creams — they sting); **non-soap cleansers**; lukewarm **5-min**

baths, pat (never rub) dry, no soap; warm compresses to crusted areas; loose cotton clothing; avoid friction.

- **Supportive & referral** — fluids/electrolytes, analgesia, nutrition; admit if extensive/systemic/dehydrated. **Neurology** if CNS involvement is suspected; **dermatology at discharge** for ongoing eczema management.

Evidence: StatPearls (Eczema Herpeticum); pediatric emergency guidance. Early antivirals are the priority; topical steroids treat the underlying eczema but must not delay antivirals.

◆ DRESS / DIHS — FOCUSED WORKUP & MANAGEMENT

- **Recognize** — **morbilliform rash with facial edema** starting on the **face** → **upper trunk** → **extremities** (can **worsen even after the drug is stopped**), plus fever, lymphadenopathy, **eosinophilia** and atypical lymphocytes. Onset is **late: 2–8 weeks** after starting the culprit — but within **~24 h on re-challenge**. Commonest culprits: **anticonvulsants** (carbamazepine, phenytoin, lamotrigine, phenobarbital), **allopurinol**, and **antibiotics** (sulfonamides, vancomycin, minocycline). First-degree relatives carry up to **~25% risk** with the same drug (HLA-linked).¹⁹⁰
- **Organ involvement** — **liver first** (hepatitis), then lung, kidney, **heart (myocarditis)**, spleen, GI and neurologic; **HHV-6/7, EBV, CMV reactivation** is characteristic. **Late endocrine autoimmunity** (hypothyroidism, type 1 diabetes) can appear months afterwards.¹⁹⁰

- **Diagnose** — clinical, scored with the **RegiSCAR** or **J-SCAR** criteria. **Labs:** CBC + differential (eosinophilia, atypical lymphocytes), **LFTs** (may stay abnormal for months), renal panel + urinalysis (proteinuria/hematuria), CK, lipase, CRP/ESR/**ferritin**, baseline **TFTs + glucose**, viral PCR (HHV-6/7, EBV, CMV), **ECG + troponin + baseline echo**, and CXR if respiratory symptoms; extend the work-up by clinical suspicion. **Skin biopsy** to exclude mimics.¹⁹⁰
- **Manage** — **stop the culprit immediately** and **label the patient hypersensitive**; supportive care; **high-potency topical corticosteroids**. For organ involvement, **systemic corticosteroids** — prednisolone **0.5–1 mg/kg/day (max 60 mg/day)** with a **slow taper over weeks–months** to avoid relapse; adjuncts/alternatives (specialist): **ciclosporin**, IVIG (limited evidence), JAK inhibitors (anecdotal).¹⁹⁰

- **Monitor & avoid** — screen for **autoimmune sequelae** (thyroiditis, type 1 diabetes): **TSH, free T4 and glucose for up to 2 years.**¹⁹⁰ **Strict lifelong avoidance** of the culprit and cross-reacting drugs; document the allergy clearly.

Evidence: EAACI position paper on DRESS in children 2025; RegiSCAR / J-SCAR criteria. Prompt drug withdrawal + staged (skin-only vs organ) therapy; steroids taper slowly to prevent relapse.

◆ NECROTIZING FASCIITIS – FOCUSED
WORKUP & MANAGEMENT

- **Recognize — pain out of proportion and edema extending beyond the visible erythema;** rapidly spreading dusky skin, fever, **crepitus**, bullae, necrosis or ecchymosis, and a **hard "wooden" feel** of the subcutaneous tissue; **cutaneous anesthesia is a late sign.** Lower > upper extremities. Special forms: **perineum (Fournier gangrene)**, head & neck, and neonatal infection (e.g. from omphalitis).¹⁹¹
- **It is a surgical emergency — call the surgeon now.** Early aggressive debridement is the single biggest determinant of survival; **do not let the LRINEC score or imaging delay surgery.**¹⁹¹
- **Scoring / labs — the LRINEC score** can help distinguish necrotizing fasciitis from other soft-tissue infection, **but a low score does not exclude it** (poor sensitivity in children).¹⁹¹ Send CBC + differential, basic metabolic panel, glucose, CRP,

serum lactate, coagulation profile, and **blood cultures (2 sets)** — then start antibiotics without waiting.

- **Imaging** — CT (or MRI) may support the diagnosis but **must never delay surgical exploration**.
- **Empiric antibiotics** (after cultures) — broad-spectrum covering **gram-positive incl. MRSA + gram-negative + anaerobes**, plus **clindamycin for its antitoxin effect**: e.g. **vancomycin 15 mg/kg/dose q6h (max 1 g/dose) + piperacillin-tazobactam 100 mg/kg/dose (piperacillin) q6–8h (or meropenem 20 mg/kg/dose q8h, max 1 g/dose) + clindamycin 10–13 mg/kg/dose q8h (max 900 mg/dose)**; de-escalate on sensitivities.¹⁹¹
- **Surgery = the gold standard** — surgical exploration is both the **definitive diagnosis and the treatment** (expect repeat looks/debridement); send intra-operative specimens for **Gram stain, culture, and histology** — alert microbiology

in advance: processing is **time-critical once tissue reaches the lab** (expedite the Gram stain and keep direct communication).

- **Who leads — shared surgical responsibility.** The operative lead follows the **anatomical site: orthopedics** (limbs), **ENT or maxillofacial** (head & neck), **plastics** (soft-tissue coverage), **urology** (perineum / Fournier), **general/pediatric surgery** (trunk / abdomen). In complex or multi-region cases, **several surgical subspecialties may need to collaborate together in the ED.**
- **Support & referral —** aggressive hemodynamic resuscitation / PICU; **IVIG** as an adjuvant in selected cases (streptococcal toxic shock).¹⁹¹ Refer to **infectious diseases and plastic/reconstructive surgery** (with pediatric/general surgery).

Evidence: IDSA skin & soft-tissue infection guidelines 2014. Surgery is time-critical — neither the LRINEC score nor imaging should delay operative exploration.

⚠ **NEONATE**

Vesicles, or an unwell neonate with a rash, may be **neonatal HSV** — start **high-dose IV acyclovir 20 mg/kg/dose every 8 h** and a full sepsis work-up without delay. Treat for **14 days** (skin-eye-mouth disease) or **21 days** (CNS or disseminated disease); for **CNS disease**, repeat CSF HSV PCR near the end of therapy and continue until it is negative, then give **oral acyclovir suppression — 300 mg/m²/dose 3×/day for 6 months.** ¹⁶¹

✦ **PEARL – READ THE PATTERN**

Non-blanching + unwell = sepsis;
monomorphic vesicles on eczema = eczema herpeticum; widespread peeling with **mucosal erosions** = SJS/TEN, versus **mucosa spared / superficial** = SSSS.

▴ **REFER**

- **Dermatology + the relevant team urgently; PICU/burns** for extensive TEN; **ophthalmology** for any

mucosal/ocular involvement;
urology/gynecology for genital involvement.

Pediatric ENT Emergencies

MINUTES

AIRWAY / ENT

The airway threats, bleeds, and batteries that can't wait — stabilize first, then call ENT.

⚠ **BUTTON BATTERY IN THE NOSE OR EAR – HOURS MATTER**

- A disc battery lodged in the nose or ear causes **liquefactive necrosis and septal/ostial perforation within hours; unilateral foul or bloody discharge** is the clue.
- ENT emergency — **urgent removal**; do **not irrigate** a nasal battery.¹⁹⁴ (Swallowed batteries → **Button Battery Ingestion.**)

▶ **RECOGNIZE & ACT**

- **Acute upper-airway obstruction** — stridor, drooling, tripod posture → epiglottitis / bacterial tracheitis /

severe croup: keep the child **calm and upright, minimal handling**, call **anesthesia + ENT**, and **do not examine the throat**¹⁶¹ (see [Croup](#), [Difficult Airway](#), [Retropharyngeal Abscess](#)).

- **Post-tonsillectomy bleeding** — any bleed (classically **day 5–10**) is an emergency: sit up, suction, **IV access + fluids/blood**, tranexamic acid, call ENT + anesthesia, prepare for the operating room. Swallowed blood **hides the true loss**.
- **Deep-neck / peritonsillar / retropharyngeal abscess** — trismus, muffled “hot-potato” voice, neck stiffness or torticollis → IV antibiotics, imaging, ENT; watch the airway.¹⁶⁷
- **Acute mastoiditis** — post-auricular swelling/erythema with a **protruding pinna** after otitis media → IV antibiotics + ENT (risk of intracranial spread / venous sinus thrombosis).¹⁶¹
- **Severe epistaxis** — sit forward, **pinch the soft part 10–15 min**, topical vasoconstrictor; if uncontrolled or recurrent, ENT — consider a **bleeding disorder**, or juvenile angiofibroma in an adolescent boy.

- **Ear/nose foreign body** — refer for removal; for a live insect, instill lidocaine or oil first.

⚠ **AIRWAY**

Do **not** try to visualize the airway or lay the child flat when epiglottitis or a deep-neck abscess is suspected — summon **anesthesia/ENT**, keep the child calm and upright.

✦ **PEARL – UNILATERAL DISCHARGE = FOREIGN BODY / BATTERY**

Unilateral foul or bloody nasal discharge in a toddler is a **foreign body — and a battery until excluded**; examine, image if needed, and remove urgently.¹⁹⁴

✦ **YOUR ROLE – THE GENERAL PEDIATRICIAN**

- Secure the child first, not the diagnosis.
- In suspected epiglottitis or a deep-neck abscess, keep the child **calm and upright**, give oxygen, do **not** examine the throat or lay them flat, and summon **anesthesia + ENT**.
- For post-tonsillectomy bleeding, get **IV access, fluids/blood, and tranexamic acid** while ENT is

called — assume swallowed blood hides the true loss.

- Recognize the emergencies that can't wait: a **nasal/aural button battery** and **unilateral bloody discharge** need same-hour ENT.
- Control epistaxis with **10–15 min of firm pressure** before escalating.
- Your job is early recognition, first stabilization, and a timely ENT / anesthesia call.

▲ REFER

- **ENT urgently** — nasal/aural battery, post-tonsillectomy bleed, deep-neck abscess, mastoiditis, or uncontrolled epistaxis; **anesthesia** for any airway threat.

Pediatric Neurosurgical Emergencies

MINUTES

TIME-CRITICAL

Shunt failure, rising ICP, and cord compression — recognize early, control the pressure, and call neurosurgery.

⚠ VP SHUNT MALFUNCTION – TREAT AS RAISED ICP

- In a child with a shunt, **headache, vomiting, drowsiness, irritability, a bulging fontanelle, or any change from baseline = a blocked shunt until proven otherwise.**¹⁹³
- Assess, image (**CT or rapid-sequence MRI + shunt series**), call **neurosurgery urgently**.
- Shunt **infection** (fever, redness/swelling along the tract, or a new malfunction) carries the same urgency — manage on a defined pathway: **sample CSF from the shunt/reservoir plus blood cultures**; start empiric antimicrobials covering **staphylococci (including MRSA) and Gram-negative organisms** — e.g. **vancomycin + an**

anti-pseudomonal beta-lactam (cefepime, ceftazidime, or meropenem); plan removal or externalization of the infected hardware; and involve neurosurgery and infectious diseases early.²¹⁴

- Controlling raised ICP is the priority while awaiting neurosurgery.

■ **RECOGNIZE & ACT**

- **Raised ICP / impending herniation** — falling GCS, an unequal or fixed pupil, **Cushing's triad** (hypertension, bradycardia, irregular breathing), or abnormal posturing → **head up 30°, midline neck, normocapnia, treat seizures, hyperosmolar therapy** (hypertonic saline or mannitol), urgent neurosurgery (see¹ [Raised ICP](#)).
- **Non-traumatic intracranial hemorrhage** — sudden severe headache, focal deficit, or coma (AVM, aneurysm, tumor) → urgent CT + neurosurgery.
- **Depressed / open skull fracture or penetrating head injury** → neurosurgery + antibiotics + tetanus

(see [Traumatic Brain Injury](#)).

- **Acute spinal cord compression — neck or back pain with arm or leg weakness**, a sensory level, or bladder/bowel dysfunction (tumor, abscess, trauma) → **urgent MRI**, dexamethasone if tumor, and **urgent spine-surgical referral — neurosurgery or pediatric orthopedic spine surgery** (either may be consulted for extradural pathology: epidural abscess, vertebral/extradural tumor, or bony trauma) ± oncology — time-critical for recovery. In **traumatic spinal cord injury**, also involve **PICU for hemodynamic (MAP) support** — maintain mean arterial pressure and avoid hypotension to preserve cord perfusion; watch for **neurogenic shock** (hypotension with bradycardia).
- **CSF leak** — clear rhinorrhea/otorrhea or a tense pseudomeningocele after surgery/trauma → neurosurgery (meningitis risk).

⚠ DON'T DELAY IMAGING

A shunted child who is drowsy or vomiting needs **urgent imaging and a**

neurosurgical call — deterioration can be rapid; do not attribute it to a viral illness.

✦ **PEARL – ASK THE PARENTS**

Parents of a shunted or hydrocephalic child usually recognize their own child's **“shunt-failure pattern”** — take that history seriously and image.

✦ **PEARL – PROTECT THE PUPIL EXAM**

Have **ophthalmology** look for **papilledema**, but **avoid dilating the pupils in a drowsy or deteriorating child** — mydriatic drops abolish the pupillary response, a key sign of herniation.

✦ **YOUR ROLE – THE GENERAL PEDIATRICIAN**

- Recognize, control the pressure, and call — you are not expected to operate.
- In any shunted child, treat headache / vomiting / drowsiness / any change from baseline as a **blocked shunt until proven otherwise**: image (CT or rapid-sequence MRI + shunt series) and call neurosurgery.
- While waiting, deliver the **ICP first-aid that is yours** — head up

30°, midline neck, normocapnia, treat seizures and fever, and give hyperosmolar therapy for impending herniation.

- Take a parent's **"this is his shunt-failure pattern"** seriously.
- Don't anchor on a viral illness, and refer **time-critically** for cord compression (urgent MRI) and open / depressed skull fractures.

REFER

- **Neurosurgery urgently** — shunt malfunction/infection, raised ICP, intracranial hemorrhage, or a depressed/open skull fracture.
Acute spinal cord compression — neurosurgery **or** pediatric orthopedic spine surgery (either for extradural pathology); add **PICU** for traumatic cord injury (MAP support).

Gastroenterology & Hepatology

08

GI bleeding · liver failure · pancreatitis · hepatitis · neonatal cholestasis · ingestion

Upper Gastrointestinal Bleeding

MINUTES

Hematemesis or melena — resuscitate first, then find and treat the source.

▶ RECOGNIZE & STRATIFY

- Bleeding proximal to the ligament of Treitz: **hematemesis** (fresh or coffee-ground) or **melena**. Confirm true blood — swallowed maternal or epistaxis blood and red foods mislead.⁵⁶
- **Clinically significant** = hemodynamic instability, a hemoglobin fall ≥ 2 g/dL, or a transfusion requirement — most need PICU.⁵⁶
- **Age-based causes:** neonate — swallowed maternal blood, vitamin K deficiency bleeding, stress gastritis; infant/child — reflux esophagitis, gastritis, peptic ulcer, Mallory-Weiss tear; any age with liver disease — **varices**. Consider an **ingested button battery** (urgent chest X-ray; see **Button Battery Ingestion**).²⁶¹

○ **Investigate — order → look for → what it means**

Order	Look for	What it tells you
CBC + serial hemoglobin	Hb fall ≥ 2 g/dL; thrombocytopenia	Severity & transfusion need ⁵⁶
Blood group + crossmatch	—	Prepare blood early in any significant bleed
Coagulation — PT/INR, aPTT, fibrinogen	Coagulopathy	Correct before/with endoscopy; a prolonged INR also flags underlying liver disease ²⁶¹
LFTs + albumin	High bilirubin, low albumin	Chronic liver disease → suspect a variceal source ²⁶¹
Urea & creatinine	Urea raised out of proportion to creatinine	Supports an upper (vs lower) GI source — absorbed blood-protein load ²⁶¹

Venous
blood gas +
lactate

Metabolic
acidosis, rising
lactate

Hypoperfusion
→ the bleed is
severe;
escalate²⁶¹

① RESUSCITATE FIRST

ABC, two large-bore IVs, crystalloid;
send crossmatch; correct
coagulopathy/thrombocytopenia.
Protect the airway in brisk
hematemesis.



DECISION

Hemodynamically unstable?

UNSTABLE / MASSIVE

Resuscitate to **hemodynamics** and
activate the **massive transfusion
protocol** — **not** a fixed Hb target. PICU;
urgent endoscopy once stabilized.²²

STABLE

Restrictive transfusion — transfuse at
Hb < 7 g/dL, target 7–9 g/dL.
Over-transfusion raises portal pressure
and can worsen variceal bleeding.²²



DECISION

Variceal suspected?

(known/suspected liver disease, splenomegaly, umbilical-catheter history)

VARICEAL

Octreotide infusion (start early, after senior/GI discussion). In **cirrhotic** portal hypertension, add **broad-spectrum IV antibiotic prophylaxis** (per local susceptibility) for 5–7 days (e.g., ceftriaxone); role in non-cirrhotic PH unclear.²²⁰ Urgent endoscopy — band ligation/sclerotherapy.

NON-VARICEAL

High-dose IV **proton pump inhibitor**. Endoscopy for hemostasis — injection + cautery/clips ($\pm$ hemostatic powder)²³ — **early (≤ 24 h)** once resuscitated; emergent/urgent (≤ 6 –12 h) endoscopy only if still unstable despite resuscitation.²⁹²



Uncontrolled bleeding → balloon tamponade (Sengstaken-Blakemore) as a bridge in massive variceal bleeding; interventional radiology (TIPS) or surgery — keep the massive transfusion protocol running. ²⁶¹

Drug	Dose	Route & notes
Octreotide	bolus 1–2 mcg/kg (max 50 mcg), then 1–2 mcg/kg/h infusion (titrate up to ~5 mcg/kg/h if bleeding continues)	IV; variceal bleeding. Once bleeding stops, wean gradually (rebound risk); monitor glucose ²¹
Pantoprazole / omeprazole	1–2 mg/kg (max 40 mg) IV bolus, then 0.1–0.2 mg/kg/h infusion (max 8 mg/h) for severe bleeding, or intermittent high-dose IV	Non-variceal; also stress-ulcer prophylaxis ²³

Drug	Dose	Route & notes
Ceftriaxone	50 mg/kg/dose once daily (max 2 g)	Broad-spectrum IV prophylaxis per local susceptibility, 5–7 days , in cirrhotic portal hypertension with acute variceal bleeding; benefit in non-cirrhotic portal hypertension unclear ²²⁰
PRBC transfusion	restrictive: transfuse at Hb < 7 g/dL, target 7–9 g/dL	Especially variceal/cirrhosis; transfuse promptly for hemodynamic instability or ongoing massive bleeding regardless of Hb ²²

⚠ AT THE BEDSIDE

Do not routinely place a nasogastric tube — it does not improve outcomes, is complicated in about a third of patients, and endoscopy is the diagnostic and therapeutic tool.^{283,284} **If varices are suspected** (known chronic liver disease or portal hypertension, splenomegaly, prior varices), **avoid blind nasogastric tube insertion** — it can traumatize varices and provoke bleeding, and gains nothing before endoscopy; leave gastric access to the endoscopy/anesthesia team.²⁸³ **Protect the airway** in brisk

hematemesis; do not over-transfuse a stable variceal bleed.²²

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all significant bleeds → resuscitation and endoscopy.
- **Discharge** — a minor self-limited bleed after observation, with follow-up.
- **Refer** — gastroenterology (endoscopy); surgery/interventional radiology for uncontrolled bleeding.

Lower Gastrointestinal Bleeding

HOURS

AGE-STRATIFIED

Hematochezia — most are benign, but resuscitate first and never miss the surgical causes.

▲ RECOGNIZE & STRATIFY

- Bleeding distal to the ligament of Treitz: **hematochezia** (fresh or maroon blood/clots per rectum), or melena from a brisk right-colonic/small-bowel source. Confirm it is blood — red foods, drugs, and bismuth mislead.²⁶¹

- **Separate the sick from the well first:**
a pale, tachycardic child, or blood with bilious vomiting or severe pain, points to a **surgical emergency** (intussusception, malrotation with volvulus, or NEC in a neonate) — see the red lines below.²⁶¹
- **Age-based commonest causes:**
 - **Neonate** — swallowed maternal blood, anal fissure, cow's-milk protein allergy, **NEC**, vitamin K deficiency bleeding, malrotation/volvulus.
 - **Infant / toddler** — anal fissure, cow's-milk protein colitis, **intussusception, Meckel diverticulum** (painless, brisk), infectious colitis.
 - **Child / adolescent** — infectious colitis, **juvenile polyp** (painless), anal fissure, inflammatory bowel disease, Henoch-Schönlein purpura, hemorrhoids.

261

○ **Investigate — order → look for → what it means**

Order

Look for

What

CBC + serial hemoglobin	Anemia, Hb fall, thrombocytopenia; microcytosis	Severe chronic (microcytic) anemia
Coagulation — PT/INR, aPTT	Coagulopathy	Bleeding diathesis; vitamin K deficiency; bleeding neonates
Blood group + crossmatch	—	Preparation for an anticipated significant bleed
Urea & creatinine	A raised urea:creatinine ratio	Suggests upper GI bleed instead of lower
Stool — culture, <i>C. difficile</i> , ova/parasites; fecal calprotectin	Pathogen; raised calprotectin	Infection vs inflammatory bowel disease
Abdominal ± Doppler ultrasound	Target / pseudo-kidney sign; free air	Intussusception / NEC / perforation
^{99m} Tc-pertechnetate (Meckel) scan	Ectopic gastric mucosa	Meckel's diverticulum; painless bleeding

① RESUSCITATE FIRST

ABC, IV access, crystalloid, crossmatch; transfuse by the **restrictive** strategy (Hb < 7 g/dL) unless unstable or massively bleeding.²²



DECISION

Sick child, or surgical features?

SURGICAL / UNSTABLE

Bilious vomiting, peritonism, a palpable mass or **currant-jelly** stool, or a distended tender neonatal abdomen → **urgent surgical review + imaging** (ultrasound for intussusception; contrast study for malrotation). Do not delay.²⁶¹

WELL / STABLE

Treat by the likely cause — fissure care and address constipation; trial a cow's-milk-protein-free diet in a well infant with colitis; supportive care for infectious colitis.²⁶¹



Persistent, significant, or unexplained bleeding → endoscopy (colonoscopy ± upper endoscopy) by pediatric gastroenterology; a Meckel scan for painless brisk bleeding. ²⁶¹

⚠ **RED LINES – DO NOT MISS**

- **Intussusception** — colicky pain, drawing up the legs, currant-jelly stool, a sausage-shaped mass — ultrasound first, then air/contrast enema reduction.
- **Malrotation with volvulus** — bilious vomiting with rectal bleeding is a surgical emergency until excluded.
- **Meckel diverticulum** — massive **painless** rectal bleeding in a young child.
- **NEC** in a preterm/neonate — distension, bloody stools, systemic upset (see **Necrotizing Enterocolitis**).

²⁶¹

▴ **ADMIT · DISCHARGE · REFER**

- **Admit** — any significant bleed, hemodynamic compromise, or a suspected surgical cause.

- **Discharge** — a well child with a minor self-limited cause (fissure, small polyp bleed) and follow-up.
- **Refer** — pediatric surgery for intussusception/volvulus/Meckel/NEC; gastroenterology for endoscopy, suspected IBD, or recurrent/unexplained bleeding.

Acute Liver Failure HOURS

Coagulopathy not corrected by vitamin K — transfer to a transplant center early.

▴ DEFINE (PALF)

- Biochemical liver injury with a **coagulopathy not corrected by vitamin K**: $\text{INR} \geq 1.5$ **with** hepatic encephalopathy, or $\text{INR} \geq 2.0$ **without** it — and no known chronic liver disease.⁵⁷
- Encephalopathy is hard to read in infants (irritability, poor feeding, reversed sleep). Look for hypoglycemia, jaundice, hyperammonemia.
- Seek a treatable cause: acetaminophen, herpes simplex virus, autoimmune hepatitis, Wilson

disease, metabolic (galactosemia, tyrosinemia), gestational alloimmune liver disease.⁵⁷

○ Investigate — order → look for → what it means (NASPGHAN)

Order	Look for	What it means
PT / INR	INR $\geq$ 1.5 with encephalopathy, or $\geq$ 2.0 without it — not corrected by vitamin K	Meets criteria for severe liver disease; the t... prog...
Ammonia, glucose, venous gas/bicarbonate, lactate ($\pm$ pyruvate)	Hyperammonemia, hypoglycemia, metabolic acidosis, high lactate	Complicates severe liver disease; PICU escalation
Bilirubin, albumin/total protein, AST/ALT/GGT	Rising bilirubin, low albumin	Degrading liver function; hepatic failure & syndrome
CBC, electrolytes, Ca/Mg/phosphate	Cytopenias, electrolyte derangement	Supports severe liver disease; target organ risk ²
Etiology panel: acetaminophen level, HSV PCR + viral serology, autoimmune markers, ceruloplasmin/Wilson studies, metabolic	A treatable cause	Directly impacts therapy; acyclovir, immunomodulation, chelation, exchange

screen
(galactosemia,
tyrosinemia),
gestational
alloimmune liver
disease workup

**Suspect it → contact a pediatric liver
transplant center / PICU immediately**



SUPPORT

- Treat hypoglycemia (continuous dextrose, glucose infusion rate 6–8 mg/kg/min)
- Give one dose of parenteral vitamin K. **Do not routinely correct the INR without bleeding or a pre-invasive procedure** — it is the prognostic marker the transplant team needs.⁵⁷
- Manage hyperammonemia/encephalopathy with transplant-hepatology/PICU guidance (lactulose/ammonia-lowering, nurse head-up, avoid sedatives, transfer early); cover empirically for sepsis if unwell.
- Consider adding acyclovir in a neonate with suspected HSV infection, and N-acetylcysteine if acetaminophen toxicity is suspected.

Drug	Dose	Route & notes
10% dextrose infusion	glucose infusion rate 6–8 mg/kg/min, titrate	Keep glucose normal; check frequently
Vitamin K	single parenteral dose	To exclude vitamin K-responsive coagulopathy

Drug	Dose	Route & notes
N-acetylcysteine	per acetaminophen protocol	Acetaminophen toxicity; not routine in non-acetaminophen disease — the pediatric RCT showed no survival benefit, so any use is a transplant-center/hepatology decision ²⁰
Acyclovir	consider in a neonate/infant with suspected HSV	HSV is a treatable cause ¹⁶¹

⚠️ AVOID HARM

- Avoid sedatives/benzodiazepines (not cleared by a failing liver; they mask encephalopathy) and hepatotoxic drugs. **Do not chase a normal INR with plasma** in a non-bleeding child — it erases the key trend.
- Watch for cerebral edema (rising intracranial pressure) as encephalopathy advances.

▀ ADMIT · DISCHARGE · REFER

- **Admit** — all → PICU/transplant center.
- **Discharge** — not acutely.

- **Refer** — hepatology/transplant urgently; PICU for encephalopathy/cerebral edema.

Acute Pancreatitis HOURS

NASPGHAN 2018

Diagnose on two of three — then fluids, analgesia, and early feeding.

▴ **DIAGNOSE (≥ 2 OF 3)**

- Abdominal pain compatible with acute pancreatitis (epigastric, may radiate to the back).
- Amylase and/or **lipase $\geq 3\times$ the upper limit of normal** (lipase is more specific).⁵⁴
- Imaging consistent with acute pancreatitis (ultrasound first; CT/MRI if needed).
- Look for a cause in every case: structural/biliary, trauma (including inflicted injury), drugs, infection, metabolic, systemic illness, genetic.

○ **Investigate — order → look for → what it means**

Order	Look for	What it means
Lipase (± amylase)	≥ 3× the upper limit of normal	Diagnosis of pancreatitis; more than 3 times the upper limit of normal for amylase or lipase
CBC, CRP, hematocrit	Leukocytosis, rising CRP, hemoconcentration	Inflammation; burden of disease; severe
BUN, creatinine	Rising BUN, acute kidney injury	Key predictor of escalation of fluid resuscitation
Glucose, calcium	Hyperglycemia, hypocalcemia	Severe disease; markers of severity
ALT, bilirubin, GGT ± triglycerides	Raised ALT/bilirubin; high triglycerides or calcium	Biliary obstruction (gallstones, metastasis, pancreatitis) causing obstruction; suspicion of obstruction; MRCP then needed to determine whether there is obstruction; procedure needed
Abdominal ultrasound first-line; MRCP for suspected biliary or pancreatic-duct obstruction/anomaly;	Gallstones, necrosis, ductal anomaly, choledocholithiasis	Etiology of obstruction; common

CT for suspected complications

① FLUIDS

Crystalloid **1.5–2× maintenance** first 24–48 h (10–20 mL/kg bolus for hypovolemia). Target urine output > 0.5–1 mL/kg/h; monitor cardiac, pulmonary, and renal status closely for 48 h.⁵⁴



② ANALGESIA & FEEDING

- Adequate analgesia, including opioids when needed. **Start enteral nutrition early** (within 48–72 h, as tolerated — oral/gastric/jejunal); enteral is preferred over parenteral.
- Prolonged gut rest is no longer recommended.⁵⁴



No routine prophylactic antibiotics — reserve them for proven infection (e.g., infected necrosis).⁵⁴

Therapy	Dose / setting	Notes
Crystalloid (balanced, or saline + 5% dextrose)	1.5–2× maintenance, first 24–48 h; avoid overly aggressive fluids – reassess for pulmonary edema, AKI, and shock, especially in severe disease	Most important early intervention
Morphine	0.05–0.1 mg/kg IV, titrate	Opioid analgesia when needed
Early enteral nutrition	within 48–72 h	Oral/gastric/jejunal; enteral preferred over parenteral

⚠ WATCH FOR SEVERE DISEASE

- Most complications declare within 48 h.
- Persistent organ dysfunction (respiratory, cardiovascular, renal) defines severe acute pancreatitis — escalate to PICU.
- Gallstone pancreatitis with suspected obstruction: **MRCP first** (first-line and prerequisite), then **ERCP** for stone clearance if

obstruction is confirmed; involve surgery and GI.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — essentially all confirmed cases — for IV fluids, analgesia, and monitoring; ICU for organ dysfunction or predicted severe disease.
- **Discharge** — mild disease once pain is controlled, feeding is re-established, and the cause is being addressed.
- **Refer** — surgical or gastroenterology review for biliary/structural causes, necrosis, or complications.

Acute Hepatitis HOURS

Most recover — the job is to catch the few progressing to liver failure.

▲ ASSESS

- Raised transaminases ± jaundice, nausea, anorexia, right-upper-quadrant pain, dark urine, pale stools.

- Causes: viral (hepatitis A/B/C/E, EBV, CMV, adenovirus, enterovirus, herpes simplex virus, SARS-CoV-2), drug-induced (including acetaminophen), autoimmune hepatitis, Wilson disease, metabolic, ischemic.¹⁶¹
- **Check synthetic function and a coagulation profile (INR) at presentation** — it separates uncomplicated hepatitis from evolving liver failure. In a sick or deteriorating child, also send **ammonia, lactate, glucose**, and a **toxicology (acetaminophen) screen**.⁵⁷

○ **Investigate — order → look for → what it means**

Order	Look for	What it tells
AST/ALT, bilirubin	Markedly raised transaminases ± jaundice	Confirms hepatocellular injury
PT/INR, albumin	Prolonged INR, low albumin	Synthetic → evolving liver failure separates uncomplicated hepatitis ⁵⁷

Ammonia, glucose, lactate, venous gas	Hyperammonemia, hypoglycemia, acidosis	Complicated severity in sick/deteriorating child ⁵⁷
Acetaminophen level; viral panel — hepatitis A/B/C/E, EBV, CMV, HSV PCR	A treatable/infectious cause	Directs specific therapy (e.g., acyclovir) and public-health notification
Autoimmune markers, ceruloplasmin, ferritin	Autoimmune hepatitis, Wilson disease, HLH	Specific etiologies missed ⁵⁷

MANAGE

- Mostly supportive: hydration, antiemetics, avoid hepatotoxins, review every drug.
- Test for **viral hepatitis** as indicated; send autoimmune markers, ceruloplasmin, and **ferritin**. Notify per public-health requirements for notifiable viral hepatitis.



Red flags → treat as acute liver failure: a rising INR not corrected by vitamin K, hypoglycemia, or any encephalopathy. Refer urgently.

⚠ DON'T MISS THE TURN

- A child with hepatitis who develops coagulopathy (INR $\geq$ 1.5–2.0), hypoglycemia, or altered behavior is no longer “just hepatitis” — escalate to the Acute Liver Failure pathway (see Acute Liver Failure) and contact a transplant center.
- Consider acetaminophen toxicity and Wilson disease in any severe presentation.⁵⁷

▀ ADMIT · DISCHARGE · REFER

- **Admit** — any coagulopathy, hypoglycemia, encephalopathy, severe vomiting or dehydration, or diagnostic uncertainty.
- **Discharge** — well child with stable or improving enzymes, normal INR, tolerating oral intake, with follow-up.
- **Refer** — urgent transplant-center/hepatology referral if it evolves to acute liver failure (rising INR, encephalopathy).

Neonatal Cholestasis

ASSESS

NASPGHAN/ESPGHAN 2025

Conjugated jaundice is never physiological — the biliary-atresia clock is ticking.

THE RULE

- **Any infant still jaundiced at 2 weeks** (3 weeks if breastfed and well) needs a **fractionated (total and direct) bilirubin** — do not assume breast-milk or physiological jaundice.¹⁹⁵
- **Cholestasis** = conjugated/direct bilirubin **> 1.0 mg/dL (> 17 μ mol/L), regardless of the total bilirubin** — always pathological (NASPGHAN/ESPGHAN 2017; the older “> 20% of total” rule is no longer used).^{312,195}
- Pale (acholic) stools and dark urine point to biliary obstruction — use a stool-color reference card.
- **Prioritize the treatable causes** — exclude biliary atresia (time-critical), sepsis, urinary tract infection, hypothyroidism, hypopituitarism, galactosemia, and tyrosinemia.¹⁹⁵

○ **Investigate — order → look for → what it means**

Order	Look for	What it tells you
Fractionated (total + direct) bilirubin	Direct > 1.0 mg/dL (> 17 μmol/L), regardless of the total	= cholestasis — always pathological ^{312,195}
GGT, ALT/AST, albumin	Pattern of injury (high- vs low-GGT)	Narrows the differential ¹⁹⁵
Coagulation (PT/INR)	Coagulopathy	Fat-soluble vitamin deficiency → give vitamin K ¹⁹⁵
Abdominal ultrasound (fasting), first-line	Absent/small gallbladder, triangular-cord sign	Clue to biliary atresia (time-critical) ¹⁹⁵
Septic + metabolic screen — glucose, ammonia, lactate, blood gas, urine reducing substances, thyroid, cultures	A treatable cause	Exclude sepsis/UTI, hypothyroidism, galactosemia, tyrosinemia ¹⁹⁵

Neonatal conjugated hyperbilirubinemia = a medical emergency → urgent pediatric gastroenterology consultation



ACT NOW

Start the workup above the **same day**; give **vitamin K** (cholestasis impairs fat-soluble vitamin absorption and risks bleeding); do not wait for an outpatient appointment.¹⁹⁵

⚠ BILIARY ATRESIA IS TIME-CRITICAL

- Kasai portoenterostomy works best **within the first 60 days of life** — outcomes fall sharply after.¹⁹⁵
- A delayed referral for “prolonged jaundice” can cost the native liver.
- When in doubt, fractionate the bilirubin and refer the same day.

▴ ADMIT · DISCHARGE · REFER

- **Admit** — for workup and any decompensation — a medical emergency for evaluation.
- **Discharge** — a stable infant with an expedited outpatient workup.

- **Refer** — pediatric hepatology urgently (exclude biliary atresia before 6 weeks).

Foreign Body Ingestion HOURS

ESPGHAN 2026

X-ray to locate it first; then the object and where it sits decide the timing.

▲ FIRST STEP

- Ask **what, when, and how many**.
- **Plain radiographs** — a **chest X-ray (PA or AP frontal + lateral**, including the neck / upper esophagus) and an **abdominal X-ray**; together these cover neck-to-pelvis. The **lateral CXR** locates and characterizes an esophageal object and shows the button-battery **double-ring / halo** sign (vs a coin); the **abdominal film** tracks an object that has passed into the stomach or bowel.
- **Radiolucent objects** (most plastic, wood, glass, thin aluminum, fish/food bones) may not show — **a**

normal film does not exclude them;
consider further imaging or
endoscopy if suspicion is high.²⁴

- Keep child fasting (nothing by mouth) until disposition decided.²⁴

Timing	Which swallowed objects
Now endoscopy ≤ 2 h	Button battery in the esophagus; two or more magnets , or a magnet plus a metal object (any level) ²⁴ ; sharp or pointed object in the esophagus; esophageal blockage — drooling, unable to swallow saliva; airway distress or signs of perforation; any symptomatic child .
Within 24 h next day at the latest	Any other object lodged in the esophagus — never leave one there beyond 24 h. Sharp object, or magnets, sitting in the stomach or duodenum. Object ≥ 2.5 cm (25 mm, ~1 inch) wide or ≥ 5–6 cm (~2 inches) long — too big to pass the pylorus or duodenum.
Observe at home, elective	A single blunt object (e.g., a coin) already past the esophagus into the stomach, < 2.5 cm wide and < 5 cm long, with no battery or magnet. 80–90% pass within 1–2 weeks. Re-image only if symptoms appear; remove if it is still in the stomach at ~ 2–4 weeks. ²⁴

Water beads

admit & observe; surgery, not endoscopy

Superabsorbent (water) beads are radiolucent — plain films miss them; use **abdominal ultrasound**. Small ingestions (a few beads) usually pass: admit, serial abdominal examination, watch for bilious vomiting, distension or pain. **Endoscopy rarely retrieves them** and is not the pathway. **Involve pediatric surgery early** for large ingestions or any sign of obstruction — most symptomatic children need an operation.^{314,315}

⚠ RED LINES

- Batteries and magnets are never simply observed — an esophageal battery or two or more magnets is an emergency at any level (see **Button Battery Ingestion**).
- An esophageal foreign body left beyond 24 h risks pressure necrosis, perforation, and aorto-/tracheo-esophageal fistula.²⁴
- Return for pain, vomiting, fever, drooling, bleeding, or black stool.

◆ **WHO REMOVES IT** A foreign body lodged in the **UPPER (CERVICAL) ESOPHAGUS, AT THE LEVEL OF THE CLAVICLES (T2)**, typically removed by ENT with a **RIGID ESOPHAGOSCOPE**; objects

lower (esophagus/stomach) usually managed by GI with flexible endoscopy.

◆ **SEE ALSO** Button (Lithium) Battery Ingestion.

▲ **ADMIT · DISCHARGE · REFER**

- **Admit** — symptomatic, an esophageal object, or high-risk (magnets, sharp, long).
- **Discharge** — asymptomatic gastric blunt object with return advice.
- **Refer** — GI/surgery for endoscopy (esophageal, magnets, sharp objects).

Button (Lithium) Battery Ingestion

MINUTES

ESPGHAN 2026

An esophageal battery burns within 2 hours — move fast.

LOCALIZE

Urgent **chest X-ray (PA or AP frontal + lateral)** covering neck to upper esophagus (add an **abdominal X-ray** if the battery is not seen). A "double-ring / halo" sign distinguishes battery from coin. Esophageal location = emergency.²⁵



DECISION

Battery in the esophagus?

YES — ESOPHAGEAL

Endoscopic removal within 2 hours.

See mitigation doses below — do not let them delay removal.²⁵

PAST ESOPHAGUS, asymptomatic

Often expectant with follow-up imaging; remove if symptomatic, co-ingested magnet, or fails to progress.

Mitigation	Dose	Route & notes
Honey	10 mL every 10 min, up to 6 doses	Oral; within 12 h. Pre-hospital. Only if $\geq$ 12 months — never give honey under 12 months. ²⁵
Sucralfat e	1 g every 10 min, up to 3 doses	Oral; in-hospital, before removal ²⁵

⚠ AFTER REMOVAL

Injury continues — watch for delayed strictures and tracheo-/aorto-esophageal fistula.²⁵

▴ ADMIT · DISCHARGE · REFER

- **Admit** — all esophageal batteries → emergency.
- **Discharge** — after confirmed passage past the esophagus in an asymptomatic child, per protocol.
- **Refer** — GI/ENT/surgery IMMEDIATELY for esophageal lodgement — do not delay.

Functional Constipation & Fecal Impaction

ASSESS

ESPGHAN/NASPGHAN • NICE

Common, benign, and clinical — but disimpact before maintenance and screen for red flags.

RECOGNIZE (ROME IV)

- **< 4 years** — ≥ 2 of the following for ≥ 1 month; **≥ 4 years** — ≥ 2 at least once a week for ≥ 1 month (with insufficient criteria for IBS):
 - ≤ 2 defecations/week
 - painful or hard stools
 - large-diameter stools
 - a large fecal mass in the rectum
 - excessive stool retention
 - (if toilet-trained) ≥ 1 episode of fecal incontinence/week or a history of large stools that may obstruct the toilet.¹⁵³
- Retentive posturing / withholding — often presents as **overflow soiling** mistaken for diarrhea.
- Acute presentations: abdominal pain, a palpable fecal mass, urinary symptoms, or a distressed withholding child.

▀ DIAGNOSE – CLINICAL

- History + examination make the diagnosis. **Without a red flag, no routine investigations** — no bloods (celiac, thyroid, calcium, lead), imaging, allergy testing, or transit studies.¹⁵³
- **A digital rectal exam is not needed when ≥ 2 Rome IV criteria are met** — reserve it for a single criterion, a red flag, or to confirm impaction when the abdominal exam is unclear. A plain abdominal film is not used to diagnose constipation.¹⁵³

⚠ RED FLAGS – THINK ORGANIC DISEASE, DO NOT JUST TREAT

- Onset from birth or delayed meconium > 48 h (Hirschsprung)
- ribbon stools
- failure to thrive
- unintentional weight loss or slow growth
- bilious vomiting
- severe distension
- blood without a fissure
- abnormal anal position or tone, absent anal/cremasteric reflex,

sacral dimple, hair tuft or lower-limb weakness (spinal)

- perianal scarring or extreme fear on exam (safeguarding)
- family history of autoimmune disease or cystic fibrosis.
- Examine the perineum and back.¹⁵²

TREAT

- Disimpact first if a fecal mass is present — **maintenance alone will worsen overflow soiling**. Then start maintenance the same day, add behavioral measures, and review at 2 weeks.¹⁵²
- **Goal — one soft, painless stool daily with no withholding or soiling**. Continue maintenance at goal for ≥ 2 months and until **symptom-free for ≥ 1 month** — in a toilet-training child, continue until trained — then **wean gradually rather than stopping abruptly**; do not stop while symptoms persist.¹⁵³
- **Break the pain-withholding cycle** — an anal fissure or painful defecation drives retention; keeping stool soft (with topical fissure care) is what lets maintenance work.²⁶²

POLYETHYLENE GLYCOL (PEG 3350/4000) – FIRST-LINE FOR BOTH STEPS

Step	Dose	Notes
Disimpaction	PEG 1–1.5 g/kg once daily (oral) for 3–6 consecutive days	Oral is as effective as enemas and less invasive; expect a few messy days. A rectal enema is an alternative if oral is refused or fails. ¹⁵²
Maintenance	PEG 0.4–1.5 g/kg once daily (start ~0.4 g/kg, titrate up to ~1.5 g/kg/day as needed)	First-line and superior to lactulose . Titrate to the goal above; reassess at 2 weeks if inadequate. ¹⁵³
Add-on / alternative	Stimulant (senna or bisacodyl, age-appropriate), lactulose if PEG unavailable	Second-line adjunct when PEG alone is insufficient — not a substitute for disimpaction.

◆ PEG – WITH OR WITHOUT ELECTROLYTES

- Same drug (macrogol); the only difference is added balancing salts. **With electrolytes** for high-volume **disimpaction**; **without** (more palatable) is an equal option for **maintenance**. The electrolyte-containing macrogol stocked locally covers both; improve palatability by

chilling and mixing it into a flavored drink.^{281,282}

◆ SIDE-EFFECTS & PRECAUTIONS

- **PEG (osmotic)** — dose-related loose stools, bloating, flatulence and nausea; over-titration can cause diarrhea or overflow soiling — **reduce the dose, do not stop**. Maintain adequate fluid intake. Clinically significant electrolyte disturbance is rare (the electrolyte form guards against it further), and there is **no tolerance or dependence** with long-term use.¹⁵³
- **Stimulants (senna, bisacodyl)** — abdominal cramps and diarrhea; in nappy-wearing or incontinent children, prolonged skin contact with senna can cause a blistering perianal ("senna") dermatitis, so keep the skin clean and barrier-protected. No good evidence for a "lazy bowel" at therapeutic doses.¹⁵³
- **Lactulose** — flatulence, bloating and cramps, with more adverse events than PEG and inferior efficacy; reserve it for where PEG is unavailable or unlicensed.²⁸⁰

- **General** — titrate to one soft, painless stool daily; do not stop abruptly (relapse); recheck adherence and dose before escalating.

◆ INFANTS < 1 YEAR

- PEG is effective and first-line; **LACTULOSE IS THE ALTERNATIVE** where PEG is unavailable or unlicensed.
- In a **REFRACTORY** infant, consider a **2-4 WEEK TRIAL OF A COW'S-MILK-PROTEIN-FREE DIET**.
- **INFANT DYSCHIZIA** — straining then passing a *soft* stool, < 9 months — is **NOT** constipation and needs no laxative.¹⁵³

✦ WHEN MAINTENANCE ISN'T WORKING

- **Check adherence and dose first** — under-dosing and stopping too early are the commonest reasons for relapse.
- Re-check for a **retained impaction**; re-disimpact if present.
- Add a **stimulant** (senna or bisacodyl); then reconsider the diagnosis / red flags.

- Prucalopride, lubiprostone and linaclotide are **not established in children** — specialist use only.¹⁵²

✦ DIET, BEHAVIOR & COUNSELING

- Normal fiber and fluid; **fiber supplements, extra fluid, and probiotics are not proven** and do not replace a laxative.¹⁵³
- Demystify and remove blame; scheduled toilet sits ~5 min after meals with foot support; a stool diary and reward system.
- **Long-term PEG is safe and not habit-forming** — reassure families to keep the dose adequate.¹⁵³
- Relapse is common and treatment often runs months; about half are off laxatives by 6–12 months — set expectations and follow up.¹⁵²

▴ ADMIT · DISCHARGE · REFER

- **Admit** — rarely — failed disimpaction or severe symptoms.
- **Discharge** — most, with disimpaction, maintenance laxatives, and education.

- **Refer** — pediatric **gastroenterology** for red flags / suspected organic disease, failure of optimized maintenance after ~3 months, diagnostic uncertainty, or a child needing ongoing second-line therapy; **surgery for suspected Hirschsprung** — rectal suction biopsy is the gold standard (contrast enema and anorectal manometry are adjuncts).¹⁵³

Reflux & GERD ASSESS

INFANT-FOCUSED • 2018

Distinguish the thriving "happy spitter" from true reflux disease.

▶ REFLUX VS DISEASE

- **GER (physiological):** effortless regurgitation in a comfortable, thriving infant; peaks ~4 months, resolves by 12–18 months. Not a disease²⁸ — reassurance.
- **GERD:** reflux causing troublesome symptoms or complications — feeding refusal, faltering growth, esophagitis, recurrent aspiration.²⁸

① REASSURE + CONSERVATIVE

- Parental education; avoid overfeeding; smaller, more frequent feeds; thickened feeds; hold upright after feeds; continue breastfeeding.
- Do **not** use prone/side positioning for sleep (SIDS risk).²⁸



② TRIAL IF NOT IMPROVING (INFANT)

2–4 weeks of extensively hydrolyzed (or amino acid) formula, or maternal dairy exclusion if breastfed — overlaps with cow's milk protein allergy.²⁸



INFANT • still not responding

Refer to gastroenterology — do not start acid suppression in an infant yourself. The gastroenterologist assesses and decides the next step: a PPI / H2 trial, or upper-GI endoscopy or a contrast study.²⁸

OLDER CHILD • confirmed GERD

Time-limited **4–8 week** trial of acid suppression (proton-pump inhibitor or H2-antagonist), then **wean and reassess.**²⁸

⚠ **DO NOT ACID-SUPPRESS SIMPLE REFLUX**

- Proton-pump inhibitors are **not** recommended for unexplained crying, distress, or visible regurgitation in an otherwise well infant — no benefit and real harms.²⁸
- Avoid acid suppression for isolated cough/wheeze. **Red flags** (investigate / refer): bilious or projectile vomiting, blood in vomit or stool, onset after 6 months or persisting beyond 12–18 months, faltering growth, fever, lethargy, or a bulging fontanelle.

▴ **ADMIT · DISCHARGE · REFER**

- **Admit** — rarely — only for complications (apnea, faltering growth, aspiration) or diagnostic uncertainty.
- **Discharge** — uncomplicated reflux with reassurance and feeding advice.
- **Refer** — gastroenterology for red flags, faltering growth, or treatment failure.

Ingestion Injuries – See Also

Corrosive ingestion is covered in the Toxicology section.

◆ CROSS-REFERENCE CAUSTIC (CORROSIVE)

INGESTION (Toxicology & Environmental) — acid/alkali esophageal injury; no charcoal, no emesis, airway first, endoscopy at 12–24 h.

Acute IBD Flare & Toxic Megacolon

MINUTES

ESPGHAN-ECCO 2025

Bloody diarrhea with systemic upset — and the colon that dilates and can perforate.

▲ RECOGNIZE

- Known or new IBD with **frequent bloody stools**, abdominal pain, fever, tachycardia and raised inflammatory markers; grade severity (e.g. **PUCAI** in ulcerative colitis).
- **Toxic megacolon** — colonic dilatation with systemic toxicity (fever, tachycardia, tenderness/distension); a surgical emergency.²⁵²

⚠ WHAT MAKES IT WORSE

- **Avoid antimotility agents, opioids and anticholinergics** — they can precipitate toxic megacolon.²⁵²
- Exclude infection (including **C. difficile** and enteric pathogens) before/while escalating immunosuppression.
- Rising distension, worsening toxicity, or free air means **surgery**.

▀ WORKUP

- **PUCAI at admission and daily** (score it below, or use the **PUCAI calculator**) — acute severe colitis = $\text{PUCAI} \geq 65$. **Day 3 > 45** → plan second-line therapy (screen, discuss); **day 5 $\geq$ 65** → start second-line; **35–65 on day 5** → continue IV steroids 2–5 more days with daily scoring; **< 35** → responding, continue and transition to oral.²⁵²
- CBC, CRP/ESR, albumin, electrolytes, renal and liver function.
- **Stool infection panel including C. difficile** (consider CMV on biopsy if steroid-refractory).

- **Abdominal X-ray** for colonic dilatation / toxic megacolon; limited unprepped sigmoidoscopy where needed.

◆ PUCAI – SIX CLINICAL ITEMS, NO LABS (0-85)

Item	Score
Abdominal pain	none 0 · can be ignored 5 · cannot be ignored 10
Rectal bleeding	none 0 · small amount, in < 50% of stools 10 · small amount with most stools 20 · large amount (> 50% of stool content) 30
Stool consistency (most stools)	formed 0 · partially formed 5 · completely unformed 10
Stools per 24 h	0–2 0 · 3–5 5 · 6–8 10 · > 8 15
Nocturnal stools (waking the child)	no 0 · yes 10
Activity level	no limitation 0 · occasional limitation 5 · severely restricted 10

- **< 10** remission · **10–34** mild · **35–64** moderate · **≥ 65** severe (acute severe colitis — admit for IV steroids). ^{313,252}

Step	Approach	Notes
Resuscitate	IV fluids, correct electrolytes (esp. K ⁺), transfuse if needed	Monitor for toxic megacolon
Treat the flare	IV corticosteroids for severe colitis	Gastroenterology-directed; escalate to infliximab or cyclosporine by day 5 if inadequate response; pharmacologic VTE prophylaxis for all hospitalized ²⁵²
Toxic megacolon	Bowel rest, decompression, urgent surgical review	Colectomy if deteriorating/perforation

ADMIT · REFER

- **Admit** all acute severe colitis.
- **Call/Refer** — pediatric gastroenterology and surgery (early joint care for toxic megacolon).

Chemotherapy-induced Febrile Neutropenia

MINUTES

Fever in a child on chemotherapy — a medical emergency until proven otherwise.

◆ NOT ON CHEMOTHERAPY?

- This card is for the **oncology / chemotherapy (or post-transplant) patient**. For a **well, previously healthy, immunocompetent child** with an incidental fever + neutropenia, use [Febrile Neutropenia — Otherwise Healthy Child](#) instead — empiric antibiotics are usually **not** required.^{157,62}

▴ DEFINE

- Fever — single oral temp ≥ 38.3 °C, or ≥ 38.0 °C sustained over 1 h between checks, **without antipyretic**

use (even if first reported at home)
— **plus** absolute neutrophil count < 500 cells/ μ L (or expected to fall below).³⁶

Empiric broad-spectrum antibiotics within 1 hour — do not wait for the count³⁶



WORKUP

- Blood cultures (peripheral + each central line lumen — at least one must be drawn, central preferred; do not delay antibiotics waiting for both), full blood count with differential, renal panel (electrolytes, creatinine, glucose), lactate, urinalysis $\pm$ culture, focused exam (skin, perianal, line site)
- Start antibiotics immediately if unstable — even without labs collected.³⁶

Drug	Dose	Route & notes
Piperacillin-tazobactam	75–80 mg/kg/dose (piperacillin component) every 6 h (max 4000 mg/dose)	IV; first-line anti-pseudomonal monotherapy ^{36,286}
Cefepime	50 mg/kg/dose every 8 h (max 2000 mg/dose)	IV; alternative monotherapy if piperacillin-tazobactam is unavailable ²⁸⁶
Meropenem	20 mg/kg/dose every 8 h (max 1 g/dose; 2 g/dose if CNS/severe)	IV; for penicillin / β -lactam allergy; also for resistant organisms, CNS infection, or instability ²⁸⁶
Ceftazidime (+ vancomycin)	50 mg/kg/dose every 8 h (max 2000 mg/dose)	IV; if piperacillin-tazobactam is unavailable — give with vancomycin for gram-positive cover ²⁸⁶
Vancomycin (add only if indicated)	15 mg/kg/dose every 6 h if < 12 yr; every 8 h if $\geq$ 12 yr (max 1 g/dose; level-guided)	IV; for instability, line/skin infection, known resistant organism — not routine; requires drug-level monitoring

◆ LOCAL CHOICE

- Follow your local antibiogram — the empiric agent should match local resistance.

- Add an **AMINOGLYCOSIDE** (amikacin or gentamicin) **ONLY** for hemodynamic instability or known/suspected multidrug-resistant gram-negative rods, and until culture results are available.²⁸⁶
- **SELECTED LOW-RISK** (including but not limited to: anticipated neutropenia $\leq$ 7 days, no comorbidity, stable, tolerating oral intake, reliable caregiver & transport) may be managed as an outpatient as per local protocol: day-1 IV **CEFTRIAXONE** (100 mg/kg, max 4 g) + gentamicin, stepping down to oral ciprofloxacin + amoxicillin-clavulanate; review daily.²⁸⁶
- Do not routinely add vancomycin (other than as the gram-positive companion to ceftazidime above); granulocyte colony-stimulating factor only in consultation with hematology/oncology. **AGE < 2 MONTHS** (rare): ampicillin + ceftazidime, in consultation with infectious diseases / pediatric hematology-oncology.³⁶

ADMIT · DISCHARGE · REFER

- **Admit** — high-risk febrile neutropenia (oncology/chemotherapy) →

immediate empiric antibiotics.

- **Discharge** — selected low-risk after risk assessment with close follow-up (the well-appearing non-oncologic child).
- **Refer** — oncology/hematology.

Febrile Neutropenia – Otherwise Healthy Child

ASSESS

IMMUNOCOMPETENT CHILD

The well child with fever, commonly a viral illness, found to have an incidental low ANC — usually transient and benign, unlike chemotherapy-induced neutropenia from myelosuppression. Some malignancies (e.g., a localized brain tumor) do not suppress the marrow and so do not cause neutropenia.

◆ THE KEY QUESTION – WHICH CAME FIRST?

- **INFECTION → NEUTROPENIA** (a virus transiently suppresses or consumes neutrophils in a healthy child) behaves very differently from **NEUTROPENIA → INFECTION** (marrow failure, malignancy, post-transplant).¹⁵⁷

- Most well, febrile children with incidental neutropenia are the former and do **NOT** need empiric febrile-neutropenia antibiotics.¹⁵⁷

▴ DEFINITIONS & MECHANISM

- **Neutropenia = absolute neutrophil count (ANC) < 1500/μL.**
- Mild 1000–1500
- moderate 500–1000
- **severe < 500**; very severe < 200.
- Normal ANC varies by age and ethnicity (benign ethnic neutropenia).¹⁵⁷
- **Fever (neutropenic definition):** single oral temp ≥ 38.3 °C, or ≥ 38 °C sustained > 1 h, or twice > 38 °C in 12 h. Avoid rectal temperatures (mucosal trauma).¹⁵⁷
- **Common viral triggers:** RSV, influenza A/B, parvovirus B19, Epstein-Barr virus (EBV), cytomegalovirus (CMV), human herpesvirus 6 (HHV-6), varicella-zoster. Other unidentified viruses may also cause neutropenia or other transient cytopenias. Mechanisms: transient marrow

suppression, peripheral destruction/margination, or an autoimmune process.¹⁵⁷

▀ WHAT MAKES IT LOW-RISK (USUALLY PRESUMED VIRAL)

- **Well-appearing**, immunocompetent, clear viral context, no features of sepsis or primary immunodeficiency.
- **Short anticipated duration** — viral neutropenia typically resolves within ~2–4 weeks, rarely beyond 6 weeks; risk rises mainly when neutropenia is prolonged¹⁵⁷ (> 30 days).
- No prior neutropenia, recurrent/severe infections, failure to thrive, or family history of a congenital syndrome; not on a neutropenia-inducing drug; not immunosuppressed.

DECISION

Ill-appearing or any high-risk feature?

LOW-RISK • WELL CHILD

Well-appearing, presumed viral, short-duration, no clue to an underlying cause. **Low risk of serious infection** — manage supportively with close follow-up. A history of a previously normal ANC is reassuring here.

HIGH-RISK

Ill-appearing (hypothermia, hypotension, lethargy), or known/suspected **severe congenital neutropenia (no prior normal ANC on record; e.g., Kostmann), aplastic anemia, or other marrow-failure/immunodeficiency** → medical emergency.

Most transient neutropenia is viral — it does not require antimicrobial therapy.¹⁵⁷

LOW-RISK – EVALUATION & DISPOSITION

Step	What to do	Notes
Cultures	Blood culture stratified by severity: not required in mild (ANC 1000–1500), consider in moderate (500–1000), obtain in severe (< 500); consider urine culture (esp. young / female)	Canadian guidance leans toward routine culture. Don't rely on urinalysis — pyuria is often absent in neutropenia. ⁶²
Blood film	Peripheral blood film in all newly discovered neutropenia (if not already done)	Look for blasts, dysplasia or a left shift — helps exclude malignancy / marrow failure.
Identify the virus	Consider a respiratory multiplex PCR (not just influenza/RSV)	Confirming a viral cause supports a benign, transient course; may not always identify a specific viral species.
Targeted tests	As indicated: chest radiograph, abdominal imaging, lumbar puncture, stool testing	A normal radiograph can mislead — infiltrates may appear only as the ANC recovers.

Step	What to do	Notes
Disposition	Discharge home with close follow-up if well and follow-up assured	Recheck CBC to confirm recovery ¹⁵⁷ ; most normalize within ~2–4 weeks.

HIGH-RISK – MANAGEMENT

Empiric therapy	Agent / approach	Notes
Act promptly	Full infectious work-up, hospitalize, start empiric antimicrobials without waiting for the ANC trend	Involve hematology ± infectious diseases early.
First-line	Piperacillin-tazobactam (broad-spectrum anti-pseudomonal)	Bedside use of high-risk empiric cover, as in chemotherapy-related FN. ³⁶
Tailor by risk	Adjust for known colonization (MDRO, MRSA), prior prophylaxis, local antibiogram	Add MRSA cover (vancomycin) for hemodynamic instability, line, or skin/soft-tissue infection.

⚠️ DON'T OVER-TREAT – OR UNDER-TREAT

- The well child with an incidental low ANC during a viral illness is **not** chemotherapy-induced myelosuppression — reflexive broad-spectrum antibiotics add resistance and harm with little benefit.
- Conversely, never anchor on “just a virus”: an **ill-appearing** child, **prolonged** neutropenia (> 30 days), recurrent/severe or unusual infections, oral ulcers, or failure to thrive warrant a congenital/marrow-failure work-up and prompt treatment.¹⁵⁷

▶ ADMIT · DISCHARGE · REFER

- **Admit** — only if febrile with concerning features or a suspected/confirmed bacterial infection.
- **Discharge** — well child — expectant management with a scheduled recheck of counts.

- **Refer** — hematology for persistent (> 6 weeks), profound (< 500 consistently), or recurrent neutropenia.

Tumor Lysis Syndrome MINUTES

Massive cell breakdown — after starting chemotherapy, or spontaneously in high-burden leukemia/lymphoma.

THE BIOCHEMICAL QUARTET

- **High** potassium, **high** phosphate, **high** uric acid; **low** calcium → acute kidney injury (see [Acute Kidney Injury](#)), arrhythmia, seizures.³⁷

① HYDRATION IS THE BACKBONE OF CARE

- Vigorous IV fluids (up to ~3–4 L/m²/day per renal status; if body surface area unknown or unmeasurable, use weight-based hydration — 4-2-1 rule at 1.5× maintenance). **No potassium** in fluids.
- Routine urinary alkalinization not recommended.³⁷



DECISION

Risk level?

INTERMEDIATE

Allopurinol + hydration.³⁷

HIGH / ESTABLISHED

Rasburicase + hydration. Do not give allopurinol on a rasburicase day. High risk includes (not limited to) white cell count > 50,000, elevated or rapidly rising uric acid, or suspected lymphoma with bulky disease (mediastinal mass, multiple significantly enlarged nodes).³⁷



② TREAT THE ELECTROLYTES

- Manage hyperkalemia (see [Electrolyte Emergencies](#)).
- Treat symptomatic hypocalcemia with appropriate replacement.
- For refractory derangement or oliguric kidney injury, escalate urgently to nephrology/PICU — dialysis is a specialist decision (AEIOU indications).

Drug	Dose	Route & notes
Allopurinol	10 mg/kg/day PO divided every 8 h (max 800 mg/day)	PO; start before chemotherapy; prevents new urate ³⁷
Rasburicase	Fixed by weight, 1 dose/24 h: < 15 kg → 1.5 mg 15–25 kg → 3 mg 26–35 kg → 4.5 mg > 35 kg → 6 mg (max)	IV over 30 min; breaks down existing urate. Reassess daily by uric acid — often 1 dose is adequate. ⁶⁹ Do NOT give in known G6PD deficiency.

⚠ **RASBURICASE + G6PD DEFICIENCY**

- Rasburicase is **contraindicated** in glucose-6-phosphate dehydrogenase deficiency (causes hemolysis and methemoglobinemia) — screen before giving — **quantitative G6PD level** if status unknown (e.g., new diagnosis not yet screened); especially relevant here given high prevalence.⁶⁹
- Send urate samples on ice.³⁷

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all at-risk or established TLS → hyper-hydration and monitoring (allopurinol for intermediate risk; rasburicase for high risk or rapidly rising uric acid); severe/refractory → PICU with the dialysis team (nephrology).
- **Discharge** — after resolution of TLS and confirmed stability; not acutely; after resolution.
- **Refer** — oncology/nephrology (dialysis for refractory).

Sickle Cell Crises

MINUTES

Pain crisis, acute chest syndrome, splenic sequestration, and stroke — plus every patient is functionally asplenic.

VASO-OCCLUSIVE (PAIN) CRISIS

Step	Action	Notes
Analgesia — fast	Opioid within 30–60 min; add NSAID/acetaminophen	Use a pain protocol; reassess every 15–30 min. Do not delay analgesia — splinting (shallow, guarded breathing from pain) can progress to acute chest syndrome. ^{38,71}
Hydration	Maintenance fluids (0.75× maintenance if concern for acute chest syndrome)	Avoid over-hydration (acute chest risk) ³⁸
Oxygen + warmth	If SpO ₂ < 95% or below baseline	Keep warm; treat the trigger ³⁸

⚠️ ACUTE CHEST SYNDROME

- **Definition:** new pulmonary infiltrate + respiratory symptoms / fever / chest pain — the leading cause of death in patients with sickle cell disease.

- **Treatment:** oxygen (keep $\text{SpO}_2 \geq 95\%$), broad-spectrum antibiotics covering atypicals (cefotaxime + azithromycin/clarithromycin), bronchodilator if wheezing, incentive spirometry (10 breaths every 2 h while awake).
- **Escalation:** early hematology/PICU discussion for **simple vs exchange transfusion** if severe hypoxia or falling hemoglobin.³⁸

⚠ **SPLenic SEQUESTRATION**

- Rapidly enlarging spleen + falling hemoglobin **and/or platelet count** → hypovolemic shock. Give fluids for shock and consult hematology immediately.
- Transfuse **cautiously** (~5 mL/kg toward partial correction) — over-transfusion causes rebound hyperviscosity as the spleen releases trapped cells.³⁸

⚠ **ACUTE STROKE**

- Sudden focal deficit, seizure, or new altered consciousness → **urgent neuroimaging** with

immediate hematology and neurology involvement.

- **Ischemic stroke** — transfuse urgently, do not delay. **Exchange transfusion is the definitive treatment** (goal HbS < 30%; keep total Hb $\leq$ 10 g/dL to avoid hyperviscosity), under hematology and neurology. If the child is significantly anemic (**Hb < 9 g/dL**) or exchange is not immediately available, give a **simple (top-up) transfusion as an immediate bridge** — but do not raise Hb above ~10 g/dL. Aim to start exchange **within 6 h.**²⁸⁵
- **Hemorrhagic stroke** is managed case by case, in tandem with neurosurgery.

◆ **FEVER IN SICKLE CELL DISEASE IS AN EMERGENCY**

All patients functionally asplenic, at risk of overwhelming infection by encapsulated organisms — give prompt empiric antibiotics — **CEFTRIAXONE** is the agent of choice (once-daily; enables outpatient management of low-risk cases). Use **CEFOTAXIME IN INFANTS < 1 MONTH**, in whom ceftriaxone is contraindicated (bilirubin displacement); another broad-spectrum agent covering encapsulated bacteria may be used if allergic.^{72,38} Assess for the

crises above. **CONSIDER ORAL ANTIBIOTICS ONLY IF STRICTLY LOW-RISK (PER LOCAL PROTOCOL):**

age > 1 year and clinically well with normal vitals ($\text{SpO}_2 \geq 95\%$); no focal source, severe pain, acute chest, or sequestration; reassuring labs ($\text{WBC } 5\text{--}30 \times 10^9/\text{L}$, reticulocytes $> 100 \times 10^9/\text{L}$, platelets $> 100 \times 10^9/\text{L}$); a first parenteral antibiotic dose given in the ED with 2–4 h observation; and a reliable home situation with guaranteed 24-h follow-up.

▲ **ADMIT · DISCHARGE · REFER**

- **Admit** — severe uncontrolled pain, acute chest syndrome, sequestration, stroke, priapism, or sepsis.
- **Discharge** — pain controlled on oral analgesia, well and hydrated, with follow-up.
- **Refer** — hematology; PICU for acute chest/stroke; exchange-transfusion decisions.

Acute Bleeding in Hemophilia

MINUTES

"Factor First" — replacing clotting factor is always the priority; give factor before imaging or detailed assessment so there is no delay to life/limb-saving treatment.

PRINCIPLE

- In any serious bleed — especially head injury — give factor **immediately**, before the CT scan or full work-up. Each unit/kg of factor VIII (factor 8) raises the level by ~2%; each unit/kg of factor IX (factor 9) by ~1%.³⁹

Scenario / agent	Dose	Target & notes
Hemophilia A — severe / head injury	Factor VIII ~50 units/kg	IV; aim factor level 80 ³⁹ –100%
Hemophilia A — minor bleed	Factor VIII ~25 units/kg	IV; aim ~50% ³⁹
Hemophilia B — severe	Factor IX 100–140 units/kg	IV; aim 80 ³⁹ –100%

Scenario / agent	Dose	Target & notes
Mild hemophilia A	Desmopressin (DDAVP)	Only if a documented prior response; not for major bleeds ³⁹
Known inhibitor — life/limb threat	rFVIIa 90 mcg/kg, or FEIBA 75–100 units/kg	IV; bypassing agents — call hematology urgently for use and guidance ³⁹
Tranexamic acid (mucosal / oral bleeds)	25 mg/kg/dose PO every 8 h, or 10 mg/kg/dose IV every 8 h	Excellent adjunct for oral/mucosal bleeds. Avoid with FEIBA (thrombosis risk) and in hematuria (ureteric clot obstruction). ³⁹

⚠ HEAD INJURY

- Treat to 100% factor level immediately and **before** the CT.
- If weight unknown, give roughly three times the usual joint-bleed dose.
- Contact the on-call hematologist.
- Do not hesitate to give factor; use the Breslow tape to estimate weight if unknown.⁷⁴

▲ ADMIT · DISCHARGE · REFER

- **Admit** — major or life-threatening bleeds (head, neck, gut, compartment).
- **Discharge** — minor bleeds controlled after factor, with a home plan.
- **Refer** — hematology for all; neurosurgery/surgery as needed — give factor before imaging.

Immune Thrombocytopenia (ITP)

ASSESS

ASH 2019

Isolated thrombocytopenia in a well child — treat the bleeding, not the number.

▲ RECOGNIZE

- Sudden bruising, petechiae or mucosal bleeding in an otherwise **well** child (often 1–4 weeks after a viral illness), **isolated** platelets $< 100 \times 10^9/L$ with normal hemoglobin, white count and film.²⁵³
- Grade the bleeding, not just the count: **“dry” purpura** (skin only) vs **“wet” purpura** (mucosal — oral

blood blisters, epistaxis, GI/urinary)
— wet bleeding drives treatment.²⁵³

⚠ **RED FLAGS – THIS IS NOT SIMPLE ITP**

- Hepatosplenomegaly, lymphadenopathy, bone/joint pain, pallor or fatigue, **any other cytopenia**, or an atypical film → suspect **leukemia, aplastic anemia, HUS or NAI** and involve hematology before steroids.
- Steroids can partially treat and mask leukemia — **never start empiric steroids without a film reviewed by hematology**²⁵³.

✦ **FIRST DECISION – OBSERVE OR TREAT**

- For a child with **no or minor (dry) bleeding**, observation is preferred **regardless of platelet count** — most childhood ITP is self-limiting.²⁵³
- Drug therapy is for **significant mucosal (wet) bleeding** or bleeding that affects daily life, not for a low number alone.²⁵³
Actively bleeding toddlers with counts $< 10 \times 10^9/L$ may need a

short oral-steroid course —
decided case by case.

▲ WORKUP

- **CBC + peripheral blood film** — confirm **isolated** thrombocytopenia and look for blasts, schistocytes or other cytopenias.
- **Blood group**, and a **direct antiglobulin test** (Coombs test) before anti-D.²⁵³
- **No routine bone marrow** in typical ITP — reserve for atypical features (other cytopenias, organomegaly, constitutional symptoms) or before splenectomy.²⁵³

Option	Dose	Route & notes
Prednisolone / prednisolone	2–4 mg/kg/day (max 120 mg/day) for 5–7 days, then stop	PO; short courses need no taper (taper only if prolonged/recurrent); ideally the film is seen first; cost-effective with tolerable short-course side-effects ²⁵³
IV immunoglobulin (IVIG)	0.8–1 g/kg as a single dose (may repeat once)	IV; fastest platelet rise — first-line when a rapid rise is needed; expensive and requires special access ²⁵³
Anti-D immunoglobulin	50–75 µg/kg once	IV; only if Rh(D)-positive, non-splenectomized, no autoimmune hemolysis ²⁵³ — watch for hemolysis
Tranexamic acid	10–15 mg/kg/dose (max 1000 mg) every 8 h	PO/IV; adjunct for mucosal bleeding — avoid in hematuria ²⁵³
Platelet transfusion	Only for life-threatening bleeding	Give with IVIG + steroid; consumed rapidly — not for the count alone ²⁵³

◆ BRIDGE TO DEFINITIVE CARE

- For life-threatening bleeding (intracranial, massive GI): platelets + IVIG 1 g/kg + IV methylprednisolone

+ tranexamic acid together, and call hematology/critical care.²⁵³

- Avoid IM injections, aspirin/NSAIDs and contact sport while counts are low.

▲ ADMIT · REFER

- **Admit** — wet purpura, severe bleeding, or any diagnostic doubt.
- **Refer/Call** — hematology for all new severe thrombocytopenia; urgently if red flags for malignancy — do not hesitate to reach out for guidance.

New-Diagnosis Oncologic Emergencies

MINUTES

COG · AAP 2026

The child who arrives before anyone knows they have cancer — new malignancies can present with emergencies that are lethal in hours.

▲ PATTERNS TO RECOGNIZE

- **Hematologic malignancy (5):** febrile neutropenia / sepsis (functionally neutropenic despite a high WBC),

hyperleukocytosis, mediastinal mass, tumor lysis syndrome, and severe coagulopathy (a subset of leukemia — APL).

- **Solid tumors (3):** tumor lysis syndrome, mediastinal mass (by origin/location), and local effects (spinal cord compression, raised ICP).

▶ **MEDIASTINAL MASS (SVC COMPRESSION SYNDROME)**

- Orthopnea, stridor, facial/neck swelling and plethora, distended neck veins, cough worse when supine — often T-cell lymphoma/leukemia.²⁵⁴
- **Do NOT sedate or lie the child flat** and avoid general anesthesia — the airway and great vessels can collapse irreversibly. Keep **upright**, give oxygen, get an **erect** chest film / echo, and call oncology + anesthesia/critical care.²⁵⁴ **Always consult hematology-oncology once a mediastinal mass is identified.**
- **Getting tissue safely:** obtain the diagnosis by the least-invasive route that avoids general anesthesia — peripheral node, bone marrow, or

pleural/pericardial fluid. If a procedure is unavoidable in a cooperative child, use light sedation with local/regional anesthesia and **preserve spontaneous ventilation** (avoid muscle relaxants and positive-pressure ventilation). When general anesthesia, rigid bronchoscopy, or ECMO might be needed, give **steroids** and — if the child is stable — postpone to let the mass shrink; if general anesthesia is unavoidable in moderate-to-high-risk disease, plan to escalate fast: rigid bronchoscopy, airway stent, and ECMO on standby.²⁸⁸

▲ **HYPERLEUKOCYTOSIS / LEUKOSTASIS**

- WBC > 100 ×10⁹/L with headache, hypoxia, confusion, priapism or visual change.²⁵⁴
- Hydrate (2–3× maintenance), start **tumor-lysis** prophylaxis (see [Tumor Lysis Syndrome](#)), and **avoid red-cell transfusion** unless critically anemic — it raises viscosity. Keep platelets ≥ 20–30 ×10⁹/L. Urgent oncology ± leukapheresis.²⁵⁴

▶ COAGULOPATHY (SUSPECT APML)

- Deranged coagulation studies in suspected leukemia are a medical emergency. Inform hematology/oncology and the hematopathologist promptly to identify **acute promyelocytic leukemia (APML)** on morphology.²⁸⁷
- If APML is identified, **All-Trans Retinoic Acid (ATRA)** is life-saving and given without delay — under hematology/oncology guidance after hematopathology confirmation.²⁸⁷

▶ MALIGNANT SPINAL CORD COMPRESSION

- Back pain (esp. night/waking) — or **neck pain with arm weakness or numbness** when the compression is cervical — leg weakness, a sensory level, urinary retention or constipation — a neurosurgical-oncological emergency.²⁵⁴
- Give **dexamethasone** immediately, arrange **whole-spine MRI**, and call oncology plus the **pediatric spine service — orthopedics or**

neurosurgery, depending on local coverage — outcome depends on how quickly power is preserved.²⁵⁴

■ WORKUP

- **All:** CBC + differential, **peripheral blood smear**, electrolytes, renal & liver profile, **uric acid, LDH, phosphate, potassium, calcium**, and coagulation screen (tumor-lysis / DIC).²⁵⁴
- **Mediastinal mass / SVC:** **erect + supine chest X-ray**, CT chest and echocardiogram; obtain the **least-invasive** diagnostic tissue (peripheral blood, effusion, marrow).
- **Leukostasis:** WBC, coagulation, type & crossmatch; neuroimaging if neurological signs.
- **Cord compression: whole-spine MRI with contrast** — do not wait for it to start dexamethasone if the deficit is progressing.²⁵⁴

Drug	Dose	Route & notes
Dexamethasone (cord compression)	1–2 mg/kg IV once (max 10 mg), then per oncology	IV; do not delay for imaging if deficit is progressing — but confirm no undiagnosed leukemia first where possible ²⁵⁴
Methylprednisolone (anterior mediastinal mass)	16 mg/m ² /dose IV every 12 h; if IV access is difficult, PO prednisone 20 mg/m ² /dose every 12 h	IV/PO; obtain a biopsy first if the patient's condition allows — steroids can rapidly shrink lymphoma and obscure the diagnosis. If a malignant effusion is present, it may be aspirated and sent for flow cytometry. Assess airway security before any sedation or positioning. ²⁵⁴
IV fluids (leukostasis / TLS)	2–3× maintenance, no potassium initially	IV; with rasburicase/allopurinol per Tumor Lysis Syndrome

⚠ TWO TRAPS

- Sedating a child with an anterior mediastinal mass for a “quick CT” or line has caused fatal airway collapse — imaging is **erect and unsedated**.
- And giving steroids before a firm in possible hematological malignancy

can obscure the diagnosis —
coordinate with oncology.²⁵⁴

REFER

- **Call** pediatric oncology urgently for all of the above; involve critical care, anesthesia, neurosurgery and hematopathology as required.

10

Neonatal

NRP 2025 · AAP hyperbilirubinemia 2022

Neonatal Resuscitation

MINUTES

NRP 2025

Effective ventilation is the single most important step.

DECISION

Term? Good tone? Breathing or crying?

YES

Delayed cord clamping ≥ 60 s, dry, skin-to-skin, keep warm, monitor.⁴

NO

Warm, dry, stimulate, position airway, clear secretions if needed. SpO₂ (right hand) + ECG.

apnea / gasping / HR < 100

GOLDEN MINUTE

Start PPV within 60 s. Term 21% oxygen; preterm 21–30%. Assess for chest rise after 15 s; if none, perform MR SOPA corrective steps.⁴

HR < 60 after 30 s effective PPV

COMPRESSIONS

Coordinate 3:1 (90 compressions + 30 breaths/min), increase to 100% oxygen, ideally via a secured airway. Reassess after 60 s. **Compressions are effective only after establishing good chest rise.**⁴

HR still < 60

Give epinephrine; treat hypovolemia; reconsider airway, pneumothorax, dose, technique.⁴

Drug / fluid	Dose	Route & notes
Epinephrine (adrenaline)	IV/IO 0.02 mg/kg (0.2 mL/kg) ET 0.1 mg/kg (1 mL/kg)	0.1 mg/mL (1:10,000). IV/IO preferred; ET only while establishing access. ⁴ Repeat every 3–5 min.
Volume (saline or blood)	10 mL/kg	IV/IO, for hypovolemia ⁴

◆ **ALGORITHM** Full AHA/AAP neonatal resuscitation algorithm — cpr.heart.org



Critical CHD – Pulse Oximetry Screening

ASSESS

≥ 24 H • AAP 2025

Screen every well newborn before discharge — catches duct-dependent lesions before they collapse.

WHO & HOW TO SCREEN

- **Who:** asymptomatic infants at **≥ 24 hours** of age (or shortly before discharge if earlier), in the well-baby nursery, **not on supplemental oxygen.**⁵

- **How:** measure SpO₂ in the **right hand (pre-ductal)** and **either foot (post-ductal)**.⁵

DECISION

Result of the screen?

PASS

SpO₂ $\geq$ **95%** in **BOTH** limbs **and** $\leq$ 3% difference between them. Routine newborn care.⁵

IMMEDIATE FAIL

Any reading **< 90%** in either limb → straight to echocardiography.⁵

borderline

RETEST ONCE

90–94% in either/both limbs, **or** > 3% hand–foot difference → repeat **once after 1 hour** (one retest, not two).⁵ Still not passing → fail.

any fail

Urgent echocardiography — and rule out sepsis, PPHN, and respiratory causes (4–5× more common than CCHD)⁵

⚠ **SCREENING DOES NOT REPLACE EXAMINATION**

- It can **miss left-sided obstruction**, especially coarctation of the aorta.
- Check **femoral pulses** and four-limb saturations/blood pressure on every newborn.
- Some CCHD is not hypoxemic in the first hours — a normal screen does not exclude it.⁵
- For the lesions, presentation, and emergency management, see **Neonatal & Infant CHD** (Cardiac & Shock).

▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — a failed screen → keep for echo; cyanosis/shock → NICU + alprostadil.
- **Discharge** — a passed screen with a normal exam.
- **Refer** — pediatric cardiology for any failed screen (echo before discharge).

◆ **SEE ALSO** Lesion recognition & management: **Neonatal & Infant CHD** and

the **Hypercyanotic (Tet) Spell** (Cardiac & Shock).

Persistent Pulmonary Hypertension (PPHN)

MINUTES

TERM / LATE PRETERM

Labile, severe hypoxemia out of proportion to the lung disease — the circulation never made the switch.

RECOGNIZE

- Severe cyanosis/desaturation **poorly responsive to oxygen**, often labile (large swings with handling).⁵¹
- **Differential cyanosis:** pre-ductal SpO₂ (right hand) higher than post-ductal (foot) by **> 5–10%** — right-to-left shunt across the duct.⁵¹
- Common settings: meconium aspiration, sepsis/pneumonia, birth asphyxia, congenital diaphragmatic hernia, or idiopathic.⁵¹
- Confirm by **echocardiography** — and exclude duct-dependent congenital heart disease before committing to therapy.⁵¹

① OPTIMIZE LUNGS, AVOID SWINGS

- **Intubate and ventilate** — recruit the lung (gentle ventilation; surfactant if parenchymal disease — meconium aspiration / RDS). **Minimal handling**, cluster cares, sedation/analgesia.
- Keep temperature, glucose, calcium, and pH normal.⁵¹



② TARGETS

Pre-ductal SpO₂ low-to-mid 90s, PaO₂ 55–80 mmHg (7.3–10.6 kPa). Support blood pressure and cardiac output with fluids/inotropes⁵¹ so systemic pressure $\geq$ pulmonary pressure.



③ PULMONARY VASODILATOR

- Suspected PPHN needs urgent neonatology/cardiology/PICU care; iNO, milrinone, noradrenaline, and ECMO referral are specialist-directed.⁵¹
- Start **inhaled nitric oxide** once the oxygenation index is ≥ 20 –25 (after lung recruitment)
- Response is a clear rise in oxygenation within ~30 min.¹⁵⁵



No response with OI rising toward ~40, or refractory shock → ECMO — refer to an ECMO center early.⁵¹

⚠ ECHOCARDIOGRAM – ALWAYS

- A neonatal echocardiogram should **always** be performed when PPHN is suspected — to confirm the diagnosis and exclude underlying congenital heart disease; iNO can be detrimental in some CHDs.⁵¹

Therapy	Dose / setting	Notes
Inhaled nitric oxide	start 20 ppm	Above 20 ppm rarely adds benefit and raises methemoglobin/NO ₂ — monitor. Assess response after 20–30 min. Do not stop abruptly (rebound) ¹⁵⁵
Milrinone / inotropes	specialist-directed	Support cardiac output; milrinone for left-ventricular dysfunction, in addition to its pulmonary vasodilator properties

Therapy	Dose / setting	Notes
Noradrenaline	specialist-directed	Add-on to milrinone when blood pressure drops — raises systemic vascular resistance and further lowers the pulmonary/systemic pressure ratio ²⁹⁰

◆ NOTE

- The approach above is suggested when the cause of pulmonary hypertension is **INCREASED PULMONARY VASCULAR RESISTANCE (PVR) AND** evidence of an ischemic hit is absent. Other scenarios of pulmonary hypertension are approached differently.
- A trial of **ORAL SILDENAFIL** can be attempted where iNO is not available, after specialist consultation.²⁸⁹

⚠ OXYGENATION INDEX & A KEY CONTRAINDICATION

- $OI = (\text{mean airway pressure} \times FiO_2 \times 100) \div PaO_2 \text{ in mmHg}$. **Do not give inhaled nitric oxide in duct-dependent congenital heart**

disease — confirm the heart on echo first.⁵¹

- Wean FiO₂ and settings slowly; sudden small drops can trigger prolonged desaturation.⁵¹

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → NICU/PICU.
- **Discharge** — not acutely; after weaning support, with follow-up.
- **Refer** — neonatology/cardiology; an ECMO center if refractory.

◆ **SEE ALSO** Neonatal & Infant CHD — exclude duct-dependent CHD before attributing cyanosis to PPHN.

Neonatal Hypoglycemia HOURS

CPS 2019

Often asymptomatic — screen the at-risk, treat early, find the persistent few.

▲ RECOGNIZE & WHO TO SCREEN

- Signs are non-specific: jitteriness, poor feeding, lethargy, hypotonia, temperature instability, apnea — and,

when severe, seizures. **Many affected newborns are asymptomatic, so at-risk infants must be screened.**⁸³

- Screen at-risk infants from ~2 h of age, pre-feed: infant of a diabetic mother (IDM), large- or small-for-gestational-age (LGA / SGA-IUGR), late-preterm / preterm, perinatal stress or asphyxia, hypothermia or sepsis, maternal beta-blocker, and congenital syndromes (e.g. Beckwith-Wiedemann) or midline / microphallus features (suggesting hypopituitarism).⁸³
- Confirm a low point-of-care (POC) reading with a **laboratory plasma glucose** — whole-blood / POC reads ~15% lower and falls further with processing delay — but **treat first; do not wait for the lab**⁸³.

Phase	Threshold	Meaning
Transitional — first 72 h	< 2.6 mmol/L (47 mg/dL)	The physiologic nadir is normal — catch the at-risk or persistently low infant, don't over-treat healthy transitional dips. ⁸³
Persistent — beyond 72 h	< 3.3 mmol/L (60 mg/dL)	Also applies to hypoglycemia first presenting after 72 h — below this warrants investigation for a hypoglycemia disorder. ⁸³
On IV dextrose	hold > 2.6 mmol/L (47 mg/dL)	Suspect a disorder if glucose cannot be held ⁸³ > 2.6 mmol/L on GIR 5–8 mg/kg/min.

DECISION

First 72 h — symptomatic (any level), or asymptomatic glucose < 2.6 mmol/L (47 mg/dL)?

SYMPTOMATIC, or < 1.8 mmol/L (32 mg/dL)

IV D10 mini-bolus 2 mL/kg, then infusion at GIR 5–8 mg/kg/min. Recheck in 20–30 min; escalate GIR in 2 mg/kg/min steps.⁸³ No IV access → glucagon IM.

ASYMPTOMATIC 1.8–2.5 mmol/L (32–45 mg/dL)

Feed + buccal 40% dextrose gel 0.5 mL/kg; recheck in 30 min.⁸⁵

still 1.8–2.5 mmol/L

REPEAT ONCE

Feed ± a second buccal gel dose; recheck in 30 min.

second repeat, or symptoms appear

Start IV D10 — mini-bolus 2 mL/kg then GIR 5–8 mg/kg/min; escalate as needed.

Drug	Dose	Route & notes
Dextrose 10%	2 mL/kg (200 mg/kg) bolus	IV/IO; then infusion at GIR 5–8 mg/kg/min, escalate by 2 mg/kg/min as needed. Concentrations > 12.5% need central access.
Dextrose gel 40%	0.5 mL/kg (200 mg/kg) buccal	First-line for asymptomatic; massage into buccal mucosa, then feed. May repeat once. ⁸⁵

Drug	Dose	Route & notes
Glucagon	0.2 mg/kg (200 mcg/kg) – max 1 mg	IM/IV as a bridge when IV access is not yet available; less effective in hyperinsulinism; watch for hyponatremia. ⁸³
Diazoxide	1.7–5 mg/kg/dose PO q8h	Endocrinology-led, for confirmed hyperinsulinism; usually with chlorothiazide. Pre-treatment echo (fluid retention, pulmonary hypertension risk). ⁸³

⚠ PERSISTENT / REFRACTORY – THINK HYPERINSULINISM

Persisting beyond 48–72 h, recurring, or needing **GIR > 8–10 mg/kg/min** to stay euglycemic warrants a **critical (hypoglycemia) sample**, drawn at glucose < 2.6–2.8 mmol/L (best < 50 mg/dL) *before* correcting: plasma glucose, insulin, C-peptide, beta-hydroxybutyrate (ketones), free fatty acids, cortisol, growth hormone — plus ammonia, lactate, and an acylcarnitine profile / urine organic acids. Hyperinsulinism = inappropriately low ketones and free fatty acids with detectable insulin, a glycemic response

to glucagon, and a high glucose requirement. Refer to endocrinology / metabolic.⁸³

✦ PEARLS

- Glucose is normally lowest in the first 24–72 h (transitional); the goal is to catch the at-risk or persistently low infant, not to over-treat healthy transitional dips.
- Do not discharge an at-risk infant until persistent hypoglycemia is excluded — i.e. pre-feed glucose stays above target⁸³ through a normal feed–fast cycle.

▀ ADMIT · DISCHARGE · REFER

- **Admit** — symptomatic, persistent, or needing IV dextrose/high glucose-infusion rate.
- **Discharge** — transient, resolved, feeding well with stable pre-feed glucose.
- **Refer** — endocrinology/metabolic for persistent or recurrent hypoglycemia (hyperinsulinism workup).

Neonatal Hyperbilirubinemia

HOURS

≥ 35 WK • AAP 2022

Use total serum bilirubin against the hour- and gestation-specific nomogram.

ASSESS & DECIDE

- Any jaundice under 24 h, or visible jaundice → measure bilirubin.
Transcutaneous bilirubin (TcB) is a valid screen; **confirm with a total serum bilirubin (TSB)** when the TcB is within 3 mg/dL (50 μmol/L) of the phototherapy threshold, is ≥ 15 mg/dL (250 μmol/L), or once phototherapy has started.³⁰⁵ Use the **total** value against the threshold (do not subtract the direct fraction).¹²
Severe hyperbilirubinemia = peak TSB > 425 μmol/L (25 mg/dL) or need for exchange transfusion.³⁰⁵
- Thresholds for phototherapy and exchange depend on **gestational age (35–40 wk), age in hours, and neurotoxicity risk factors** — read the AAP nomogram or app each time, never a single fixed number.¹²
- Neurotoxicity risk factors:** gestation < 38 weeks, albumin < 3.0 g/dL (30 g/L), isoimmune hemolysis (positive

DAT), G6PD deficiency or other hemolysis, sepsis, clinical instability in the prior 24 h — these **lower** the threshold.¹²

- **Suspect hemolysis** when the TSB rises $\geq 5 \mu\text{mol/L/h}$ (0.3 mg/dL/h) in the first 24 h or $\geq 3.5 \mu\text{mol/L/h}$ (0.2 mg/dL/h) thereafter, or fails to respond to intensive phototherapy → CBC, reticulocytes, smear, DAT, G6PD/pyruvate-kinase assay; sepsis screen if clinically indicated.³⁰⁵
- **Find the cause** when the level is high or rising — hemolytic: ABO/Rh isoimmunization (DAT), G6PD deficiency, hereditary spherocytosis; non-hemolytic: cephalohematoma or bruising, polycythemia, sepsis, hypothyroidism, and breast-milk vs inadequate-breastfeeding jaundice.¹²
- **Conjugated (direct) hyperbilirubinemia is a different problem** — for neonatal cholestasis see [Neonatal Cholestasis](#) in the GI & Hepatology section.

TREAT

At/above the phototherapy line → phototherapy + feeding/hydration + find the cause.¹²

h 2 mg/dL (30 μ mol/L) of the exchange line – pre-exchange

ESCALATION OF CARE

Intensive phototherapy + IV fluids without delay; withhold enteral feeds; urgent TSB, direct bilirubin, CBC, albumin, chemistries, type & crossmatch, hemolysis work-up; **repeat TSB every 2–3 h**; consult neonatology and prepare transfer to a unit that can perform exchange. IVIG is not routine: give 0.5–1 g/kg over 2 h (may repeat once at 12 h) for **isoimmune** hemolytic disease at the pre-exchange threshold when exchange cannot be performed promptly.^{12,305}

at/above exchange line, or encephalopathy

Immediate exchange transfusion for any sign of acute bilirubin encephalopathy (high tone, arching, high-pitched cry, lethargy). NICU.¹²

◆ NOMOGRAM & FOLLOW-UP

- Kuwait laboratories report bilirubin in **MMOL/L**. Plot the hour-specific total serum bilirubin on the **AAP 2022 THRESHOLD CURVES** — in μ mol/L using the **CPS 2026 FIGURES** (Figures 2–7,

AAP data reproduced with permission; cps.ca)^{305,12} or the online calculators below, which accept either unit (± 17.1 converts $\mu\text{mol/L}$ to mg/dL). Decision values on this card are given in both units.

- Calculators: peditools.org/bili2022



or bilitoool.org



- Before discharge, set the follow-up interval by the **GAP BETWEEN THE TSB AND THE PHOTOTHERAPY THRESHOLD**: the smaller the gap, the sooner the recheck (often within 24 hours).¹²

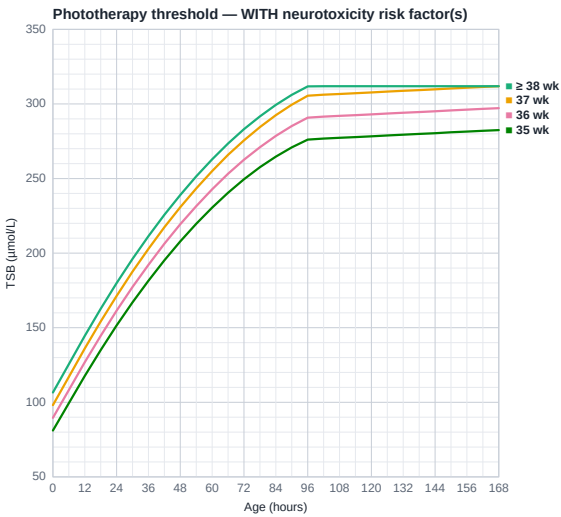
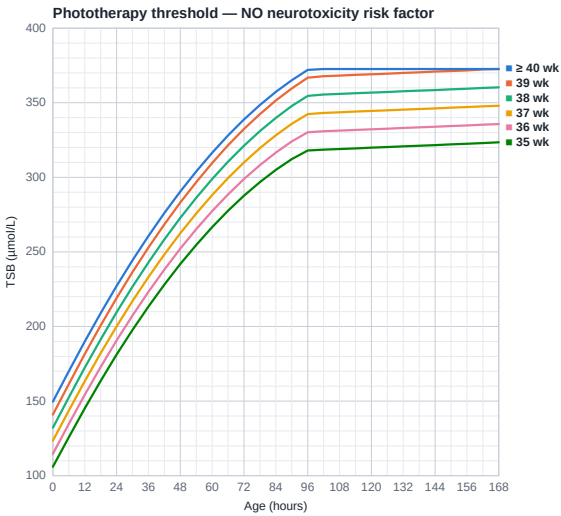
Treatment thresholds in $\mu\text{mol/L}$ – AAP 2022 curves, 35 to ≥ 40 weeks

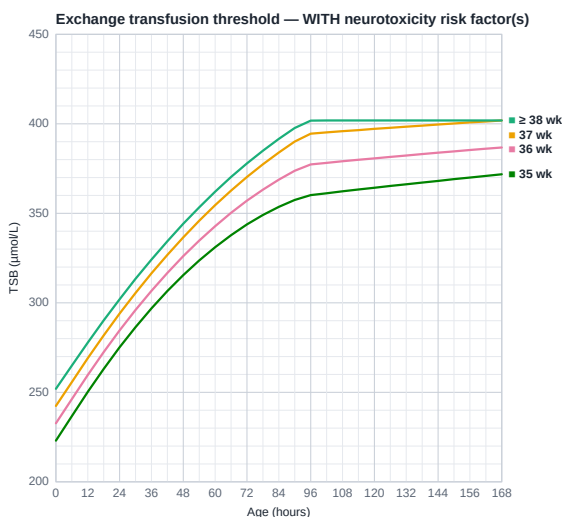
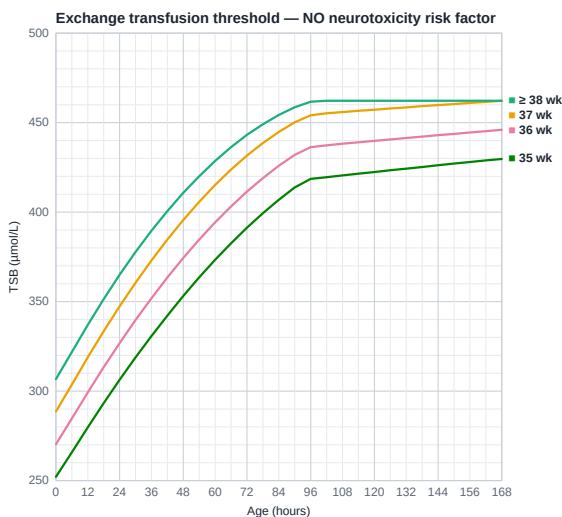
Redrawn from the AAP 2022 hour-specific thresholds as published in $\mu\text{mol/L}$ by the Canadian Paediatric Society (2026), gridlines every 10 $\mu\text{mol/L}$ and 6 h, 0–168 h; beyond 168 h the 35–37-week lines rise by $\leq 20 \mu\text{mol/L}$ and the ≥ 38 -week lines are flat – use the calculator for days 8–14.

Full text: [CPS 2026 guideline \(doi:10.1093/pch/pxaf034\)](https://doi.org/10.1093/pch/pxaf034) ·

[AAP 2022 guideline \(doi:10.1542/peds.2022-058859\)](https://doi.org/10.1542/peds.2022-058859).

Neurotoxicity risk factors: albumin < 30 g/L, isoimmune or other hemolytic disease, sepsis, significant clinical instability in the preceding 24 h. ^{305,12}





◆ **ΔTSB – THE “RULES OF 30” (CPS 2026)**

- **ΔTSB** = hour-specific phototherapy threshold – measured TSB or TcB (screen at ≥ 12 h of age, before discharge;

TcB valid only if not on phototherapy).³⁰⁵

- **Δ TSB $\leq$ 30 $\mu\text{mol/L}$** $\rightarrow$ delay discharge: < 24 h old, consider starting phototherapy; ≥ 24 h old, repeat TSB in 4–12 h and start phototherapy if rising. **31–60** $\rightarrow$ repeat TSB in 12–24 h. **61–90** $\rightarrow$ repeat in 24–48 h. **> 90** $\rightarrow$ routine follow-up with the primary provider.³⁰⁵
- With neurotoxicity risk factors, phototherapy may be started when the TSB is within 30 $\mu\text{mol/L}$ of the threshold.³⁰⁵
- **Stop phototherapy** once the TSB is > 30 $\mu\text{mol/L}$ below the threshold (≥ 38 wk) or > 60 $\mu\text{mol/L}$ below (35–37 wk). Re-check TSB 12–24 h after stopping for rebound (re-crossing the threshold within 72–96 h); with hemolytic disease re-check at 6–12 h, then every 24 h until the TSB is falling and $\Delta\text{TSB} > 60$. TcB may be used ≥ 18 h after phototherapy stops.³⁰⁵

⚠ DON'T RELY ON A SINGLE VALUE

- Plot against time.

- A rapidly rising level, early jaundice, or hemolysis can cross thresholds fast — recheck on the interval the nomogram specifies.¹²
- TcB should **not** be used for follow-up when the baby is on phototherapy — use serum bilirubin.¹²

▲ ADMIT · DISCHARGE · REFER

- **Admit** — levels at/above phototherapy or exchange thresholds, or a rapid rise.
- **Discharge** — below threshold, feeding well, with a bilirubin-recheck plan.
- **Refer** — hematology for hemolysis; an exchange-capable unit near exchange thresholds; **audiology** (formal hearing assessment) if peak TSB > 400 $\mu\text{mol/L}$, the pre-exchange threshold was reached, or exchange was performed;
neurodevelopmental follow-up if the pre-exchange threshold was reached, exchange was performed, or neurological signs occurred; after

immune hemolytic disease, check hemoglobin at 4–10 weeks for late anemia.³⁰⁵

Neonatal Sepsis – Early-Onset (rapid approach)

MINUTES

Non-specific: temperature instability, poor feeding, lethargy, apnea, distress.

▲ ACT ON SUSPICION

- Septic screen (blood culture ± CSF/urine, blood count, CRP), check glucose. Start empiric antibiotics promptly — do not wait for results. Support airway/breathing/circulation; NICU.²⁵⁵

◆ EARLY- VS LATE-ONSET

Early-onset (≤ 72 h) is vertically acquired — group B *Streptococcus*, *E. coli*, *Listeria*; empiric **ampicillin + gentamicin** (below). **Late-onset (> 72 h)** is usually nosocomial (coagulase-negative staphylococci, *S. aureus*, Gram-negatives) and needs

different empiric cover per your unit protocol.

Drug	Dose	Route & notes
Ampicillin	50 mg/kg/dose (meningitis 100 mg/kg/dose)	IV; interval by postnatal age. Use the meningitic dose if CSF not obtained or CNS involvement. ²⁵⁵
Gentamicin	4–5 mg/kg	IV (interval by gestation/age) ²⁵⁵
Cefotaxime	50 mg/kg	IV, add if meningitis suspected ²⁵⁵
Acyclovir	20 mg/kg every 8 h	IV, add if herpes simplex suspected ²⁵⁵

⚠ BILIOUS VOMITING IN A NEONATE

Green vomiting is **malrotation with volvulus until proven otherwise** — see Surgical.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all with risk factors or clinical signs → empiric antibiotics after cultures.
- **Discharge** — not until sepsis is excluded/treated; a reassuring 36–48 h course with negative cultures may stop antibiotics.

- **Refer** — neonatology/PICU for instability.

◆ **SEE ALSO** The neonate (≤ 28 days) is always high-risk in the febrile-infant pathway — use [Fever in the Young Infant](#) for the full age-stratified workup; this card is the neonatal-specific rapid approach.

Omphalitis & Delayed Umbilical Separation

HOURS

NEONATE

Red or discharging umbilical stump — treat as neonatal sepsis; watch for necrotizing fasciitis.

RECOGNIZE

- Periumbilical **erythema, induration, tenderness**; purulent or malodorous discharge from the stump.¹⁶²
- Spreading abdominal-wall cellulitis, or systemic signs (poor feeding, lethargy, temperature instability, tachycardia) = **invasive infection / sepsis**.¹¹¹

- Uncommon with dry cord care (~1 in 1000) but can progress rapidly¹¹¹ — treat as a neonatal emergency.
- **Risk factors:** low birth weight or prematurity, prolonged rupture of membranes or maternal infection, non-sterile or home delivery, umbilical catheterization, and unclean cord practices (e.g., cow-dung application — also a tetanus risk). Onset is typically around day 3.¹⁶²

⚠ **RED FLAGS → NECROTIZING FASCIITIS / MYONECROSIS (SURGICAL EMERGENCY)**

- Pain or tenderness out of proportion, rapidly spreading or **dusky / bullous** skin, crepitus, peau-d'orange edema, systemic toxicity.
- It can deteriorate fast despite mild initial findings — reassess often and mark the margins.
- Any of these → **urgent pediatric surgery for debridement**; do not delay for imaging.¹¹¹
- Debridement resects the umbilical vein, both arteries, and any urachal remnant (even if they look normal); reported mortality is high

(~60–85%), so err toward early surgical exploration in the critically ill infant.¹¹¹

INVESTIGATE & TREAT

- Blood culture (+ discharge swab), CBC, CRP; full neonatal sepsis workup.
- Start IV antibiotics **immediately** — cover staphylococci (incl. MRSA), group A Streptococcus, Gram-negatives (E. coli, Klebsiella, Proteus) and anaerobes (Bacteroides, Clostridium); the exact regimen follows the local antibiogram and neonatal formulary.¹⁶²
- If systemic: keep nil by mouth (septic ileus), IV fluids / parenteral nutrition; manage as neonatal sepsis.

EMPIRIC IV ANTIBIOTICS – DOSES SHOWN ARE GENERAL PEDIATRIC (PER DOSE · INTERVAL); ADJUST FOR NEONATES BY GESTATIONAL/POSTNATAL AGE AND FOR HEPATIC/RENAL IMPAIRMENT; CONFIRM AGAINST THE LOCAL FORMULARY AND ANTIBIOGRAM

Drug	Dose (per dose · interval)	Role
Cefotaxime	50 mg/kg/dose q6h	Gram-negative cover
+ Cefazolin	33–50 mg/kg/dose q8h	Gram-positive (MSSA) — use vancomycin where MRSA is prevalent
± Metronidazole	7.5–10 mg/kg/dose q6–8h	Add for anaerobes — foul discharge or maternal chorioamnionitis
Meropenem	40 mg/kg/dose q8h	OR (severe cases) — alternative for necrotizing fasciitis (with vancomycin + clindamycin)
± Vancomycin	15 mg/kg/dose q6h (max 1 g/dose)	Where MRSA is prevalent; monitor levels
± Clindamycin	5–10 mg/kg/dose q6–8h	Necrotizing fasciitis (antitoxin effect)

◆ **DURATION**

- About 7 days for uncomplicated periumbilical cellulitis; 10–14 days

IV with bacteremia or sepsis; 14–21 days if meningitis; prolonged and surgically-guided in necrotizing fasciitis.

- Tailor to cultures, susceptibilities, and clinical response — older infants may step down to oral once clearly improving.¹¹¹

◆ DELAYED UMBILICAL SEPARATION

- Normal separation is by ~2 weeks (range 3–45 days, mean ~10 days); antiseptic cord care delays it versus dry care, so a **well** infant is usually benign.
- Not off by **3–4 weeks** = delayed (up to ~10% of newborns). ⚠ **Suspect leukocyte adhesion deficiency (LAD)** if delayed separation + omphalitis or soft-tissue infection that forms **no pus** + persistent marked leukocytosis without infection (often $40\text{--}60 \times 10^9/\text{L}$) + poor wound healing → **consult immunology whenever LAD is suspected**. There is a growing group of LAD syndromes with variable defects; normal expression of CD18 alone is not sufficient to exclude LAD-1 —

functional assays must be performed if clinical suspicion is high.¹⁶²

- Most isolated delay in a well infant with a normal white count is **not** LAD.¹⁶²
- Other causes: a patent urachus or omphalomesenteric duct (see below), factor XIII deficiency (stump bleeding).

UMBILICAL GRANULOMA

- Moist pink granulation after separation → silver-nitrate cautery (protect surrounding skin) or topical salt.
- Refer if persistent discharge or a polyp suggests a duct remnant.¹⁶²

⚠ DON'T MISS – PATENT DUCT REMNANTS

- **Patent omphalomesenteric (vitelline) duct** = persistent ileum-to-umbilicus connection;
patent urachus = bladder-to-umbilicus. Both present after cord separation with intermittent umbilical drainage:
small-bowel contents (omphalomesenteric duct) or **clear**

urine-like fluid, more with crying or voiding (urachus). ^{306,307}

- A sinus or small opening may be visible at the umbilical base; **ultrasound** may show a tubular structure extending deep to the umbilicus. If an opening is present it can be cannulated for a **sinogram/fistulogram** to demonstrate communication with the ileum or bladder. ^{307,308}
- Other remnants that mimic granuloma: **umbilical polyp** (bright-red, mucosa-covered, does not respond to silver nitrate — ectopic ileal/gastric mucosa) and urachal or omphalomesenteric **cyst/sinus** (persistent wetness, infected periumbilical mass). ³⁰⁹
- ► **Refer all to pediatric surgery** — prolapsed bowel through a patent omphalomesenteric duct is a surgical emergency; small urachal remnants often close spontaneously in infancy, so excision is decided by the surgeon. ^{307,308}

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → IV antibiotics (neonatal-sepsis pathway).

- **Discharge** — not acutely; complete the IV course — older infants may step down to oral.
- **Refer** — pediatric surgery for necrotizing fasciitis or a suspected duct remnant; immunology for suspected leukocyte adhesion deficiency.

Necrotizing Enterocolitis

HOURS

Modified Bell

Preterm or stressed neonate: feeding intolerance, distension, bloody stools.

RECOGNIZE

- Feeding intolerance, bilious or rising gastric residuals, **abdominal distension and tenderness**, bloody stools.⁸⁶
- Systemic: lethargy, apnea, temperature instability, shock. Mostly preterm (onset often the 2nd–4th week, later at lower gestation); term infants with congenital heart disease or asphyxia are also at risk.⁸⁶

- Risk rises with prematurity and formula feeding; human (breast) milk and standardized feeding protocols are protective.⁸⁶

CONFIRM

- Abdominal X-ray (supine + left-lateral decubitus): **pneumatosis intestinalis** (intramural gas — the radiologic hallmark), portal venous gas, a persistent fixed dilated loop, or free air.⁸⁷
- Bloods: CBC, CRP, blood gas, lactate, blood culture, coagulation, electrolytes.
- Bowel ultrasound (wall thickness/perfusion, free fluid) is a useful adjunct.⁸⁷
- Repeat X-ray every 6 h while acute.⁸⁷

Modified Bell stage	Findings	Bowel rest + antibiotics
I — Suspected	Nonspecific signs (apnea, temperature instability), distension, occult or gross blood; X-ray normal or ileus	NPO + NG drainage ~3 days; stop antibiotics if NEC is excluded ⁸⁷
II — Definite (proven)	Above + absent bowel sounds ± tenderness; pneumatosis ± portal venous gas (IIB: metabolic acidosis, thrombocytopenia) ⁸⁷	NPO + NG ~7–10 days; antibiotics 7–10 days
III — Advanced	Critically ill: shock, acidosis, DIC, peritonitis. IIIA bowel intact; IIIB perforated (pneumoperitoneum) ⁸⁷	NPO + NG ~14 days; antibiotics 14 days; surgery

TREAT

- Stop enteral feeds (NPO), nasogastric tube on free drainage, IV fluids and parenteral nutrition.
- Broad-spectrum antibiotics covering Gram-negatives and anaerobes — e.g. ampicillin + gentamicin + metronidazole (or piperacillin-tazobactam); add vancomycin if a central line or a resistant organism is suspected.⁸⁷
- Correct acidosis, thrombocytopenia, and coagulopathy; support blood pressure.
- Serial exams and X-rays.
- **Surgical consultation** if no improvement or signs of perforation are present.

⚠ SURGERY NOW

Pneumoperitoneum (perforation) is an absolute indication. Also: deterioration despite maximal medical therapy, a fixed dilated loop, abdominal-wall erythema, a palpable mass, or a positive paracentesis (stool / bile / pus). Options: laparotomy with resection, or a primary peritoneal drain in the unstable extremely-low-birth-weight infant.⁸⁷

✦ PEARLS

- **Pneumatosis intestinalis is the radiologic hallmark**; portal venous gas signals more extensive disease and a worse prognosis.⁸⁶
- **Spontaneous intestinal perforation (SIP) is a distinct entity** — earlier onset (first week), often after indomethacin or steroids, a focal perforation with *no* pneumatosis; treat with anaerobe-covering antibiotics (~7 days) plus drainage or laparotomy.⁸⁶

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → NICU, nil by mouth, gut decompression, antibiotics.
- **Discharge** — not acutely; after recovery, with feeding advice.
- **Refer** — pediatric surgery (perforation or deterioration).

Neonatal Seizures

MINUTES

ILAE 2023

Often subtle — and often a treatable cause behind them.

▲ RECOGNIZE & CONFIRM

- Subtle signs: lip-smacking, cycling or pedaling, eye deviation, apnea, focal clonic or tonic movements.³⁰

- **Only focal clonic and focal tonic seizures can be diagnosed clinically** — all other types need EEG or amplitude-integrated EEG (aEEG) to confirm. Electrographic-only seizures are common, especially after the first drug (electroclinical dissociation), so monitor with EEG and **treat electrographic seizures.**³⁰
- Common causes: hypoxic-ischemic encephalopathy (most common), stroke, intracranial hemorrhage, infection / meningitis, metabolic (low glucose, calcium, sodium, magnesium), inborn errors, and genetic or structural epilepsies.³⁰

FIND THE CAUSE

- Glucose, calcium, magnesium, sodium; blood gas, lactate, ammonia.
- Sepsis screen including lumbar puncture, with **CSF HSV PCR** (start empiric IV acyclovir if neonatal HSV is possible)³⁰
- Cranial ultrasound, then MRI. If unexplained: metabolic work-up (plasma amino acids, urine organic acids, CSF glucose / lactate / glycine) and genetic testing for early-onset epilepsy (e.g. KCNQ2, SCN2A). **Correct reversible causes first** — refractory hypocalcemia is often driven by low magnesium.³⁰

Drug	Dose	Route & notes
Phenobarbital	20 mg/kg IV load (further 10–20 mg/kg to ~40 mg/kg total)	First-line regardless of etiology (superior to levetiracetam, NEOLEV2). Use the shortest course; discontinue early once acute provoked seizures are controlled. ⁸⁸
Levetiracetam	40–60 mg/kg (max 4.5 g)	IV; second-line — preferred when there is cardiac disease. ³⁰

Drug	Dose	Route & notes
Phenytoin / fosphenytoin	20 mg/kg (PE)	IV; second-line; favored for suspected sodium-channelopathy (e.g. KCNQ2, SCN2A). ³⁰
Midazolam infusion	0.05–0.4 mg/kg/h (after 0.2 mg/kg load)	IV; refractory seizures. ³⁰
Lidocaine (lignocaine)	specialist infusion (load, then taper)	IV; refractory. Avoid with phenytoin and in cardiac disease (proarrhythmic). ³⁰
Pyridoxine	100 mg IV	Trial under EEG + cardiorespiratory monitoring (apnea / cerebral depression); for vitamin B6-dependent epilepsy or unexplained refractory seizures.
Pyridoxal-5-phosphate (PLP)	30 mg/kg/day ÷ doses	If pyridoxine fails and PNPO deficiency is suspected.
Folinic acid	3–5 mg/kg/day	Add-on for pyridoxine-dependent epilepsy with incomplete response.

Drug	Dose	Route & notes
Dextrose 10% / calcium gluconate 10% / magnesium sulfate	2 mL/kg / 0.5–1 mL/kg slow / 0.1–0.2 mmol/kg	IV; correct hypoglycemia, hypocalcemia, hypomagnesemia.

✦ PEARLS

- **Phenobarbital remains first-line** (superior to levetiracetam for electrographic control, NEOLEV2 RCT), but use the shortest course and **stop early** once acute provoked seizures are controlled — often before discharge.⁸⁸
- Refractory with no cause found: trial **pyridoxine**, then **pyridoxal-5-phosphate** if PNPO is suspected, then add **folinic acid** — these treat vitamin-responsive epilepsies that anticonvulsants miss.³⁰

⚠ JITTERY IS NOT SEIZING

- Jitteriness stops with gentle holding and has no eye or autonomic features; true seizures do not stop with restraint.³⁰
- Don't miss glucose, calcium, magnesium, or infection.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → NICU with monitoring/aEEG.
- **Discharge** — not acutely; under neurology with a plan.
- **Refer** — neurology and metabolic/genetics for the cause.

Vitamin K Deficiency Bleeding (VKDB)

MINUTES

AAP 2022

Preventable with one injection at birth — the late form presents as catastrophic intracranial hemorrhage.

▲ RECOGNIZE (THREE WINDOWS)

- **Early** (0–24 h) — usually maternal enzyme-inducing drugs (anticonvulsants, rifampicin, warfarin).²⁵⁶
- **Classic** (day 2–7) — bruising, GI or umbilical bleeding in a baby who missed prophylaxis.²⁵⁶
- **Late** (2–12 weeks) — exclusively breastfed, no/oral-only prophylaxis, or malabsorption/cholestasis; up to

half present with **intracranial hemorrhage** (bulging fontanelle, seizures, pallor, poor feeding).²⁵⁶

⚠️ ANY BLEEDING NEONATE

- Check a coagulation screen (prolonged PT/aPTT with normal platelets and fibrinogen suggests VKDB), give vitamin K without waiting for confirmation, and image the brain if any neurological sign.²⁵⁶
- Investigate for **cholestasis/malabsorption** in late VKDB.²⁵⁶

📌 WORKUP

- **Coagulation screen — prolonged PT** (most sensitive) and aPTT with **normal platelets and fibrinogen** (distinguishes from DIC); rapid correction after vitamin K confirms.²⁵⁶
- CBC / film.
- **Neuroimaging** (cranial ultrasound / CT) if any neurological sign — late VKDB commonly presents as intracranial hemorrhage.²⁵⁶

Setting	Dose	Route & notes
Prophylaxis at birth	Phytonadione (vitamin K ₁) 1 mg (0.5 mg if < 1500 g)	IM — single dose; the reliable route ²⁵⁶
Active bleeding	Vitamin K ₁ 1–2 mg slow IV	Plus FFP or prothrombin complex concentrate for serious/intracranial bleeding ²⁵⁶

◆ COUNSELING

- IM prophylaxis at birth prevents both classic and late VKDB; purely oral regimens are less protective, especially in breastfed infants and cholestasis.²⁵⁶
- Reinforce with families who decline the injection.

▲ ADMIT · REFER

- **Admit** any bleeding neonate; PICU/neurosurgery for intracranial hemorrhage.
- **Refer** — gastroenterology if underlying cholestasis/malabsorption.

Neonatal Abstinence Syndrome

ASSESS

ESC-NOW • NEJM 2023

The jittery, inconsolable, poorly feeding newborn after in-utero drug exposure.

RECOGNIZE

- Tremors/jitteriness, irritability and a high-pitched cry, hypertonia, poor feeding/uncoordinated suck, vomiting/loose stools, sweating, sneezing, and disturbed sleep — typically within the first days after opioid exposure (later for long-acting agents).²⁵⁷

EXCLUDE MIMICS; WATCH FOR SEIZURES

Hypoglycemia, sepsis, hypocalcemia, hyperthyroidism and hypoxic-ischemic encephalopathy can look similar — check them. **Seizures** occur in severe withdrawal and warrant pharmacological treatment and escalation.²⁵⁷

NON-PHARMACOLOGICAL CARE FIRST

- The **Eat-Sleep-Console** approach — rooming-in, breastfeeding where appropriate, swaddling, low stimulation, skin-to-skin —

reduces the need for medication and shortens stay.²⁵⁷

- Reserve opioids (morphine or methadone) for infants who cannot feed, sleep or be consoled despite optimal supportive care.²⁵⁷

▲ WORKUP

- Primarily **clinical** — the **Eat-Sleep-Console** assessment (feeds ≥ 1 oz / breastfeeds · sleeps ≥ 1 h · consoled ≤ 10 min).
- Confirm exposure with **maternal history + maternal/infant toxicology** (urine, meconium, or umbilical cord).²⁵⁷
- Exclude mimics: glucose, calcium, sepsis screen and thyroid studies as indicated.

▲ ADMIT · REFER

- **Admit**/observe for scoring; escalate if seizures or failure of supportive care.
- **Refer** — neonatology, social work and safeguarding as indicated.

Neonatal Dosing by Corrected Age (PMA)

DOSING

NeoFax / Lexicomp / BNFc • confirm against unit for

The dose per kg barely changes — corrected age changes the **interval**.

THE ONE IDEA

- **Postmenstrual age (PMA) = gestational age at birth + weeks since birth.** Immature kidneys clear renally-eliminated drugs slowly, so the more premature the baby and the younger the postnatal age, the **longer the interval** between doses.
- For aminoglycosides and vancomycin, the dose stays roughly the same per kg while the gap between doses stretches.²⁵⁹

△ LEVELS GOVERN, NOT A FIXED SCHEDULE

- Any published regimen is a **starting point** only.
- After the first doses, the **drug level** sets the interval for gentamicin, amikacin and vancomycin — draw a trough just before the next dose and do not give it until the level is acceptable. **Extend the interval**

(do not cut the dose) for renal impairment / oliguria, perinatal asphyxia or therapeutic cooling, and concurrent indomethacin / ibuprofen or vancomycin.²⁵⁹

- Confirm every value against your unit's formulary and a live NeoFax / BNF for Children.

▲ **DOSES – USE A NEONATAL REFERENCE**

- Neonatal and premature dosing is **PMA- and postnatal-age-dependent**, and aminoglycoside/vancomycin regimens are titrated to **drug levels** — so specific doses and intervals are **not reproduced here**.
- Use a maintained neonatal dosing reference — **NeoFax, Lexicomp**, or the **BNF for Children** — for gentamicin, amikacin, vancomycin and β -lactam dosing by PMA and postnatal age, and confirm every value against your **unit protocol / NICU formulary**.

◆ **EMPIRIC EARLY-ONSET SEPSIS — AMPICILLIN + GENTAMICIN** (ampicillin covers group B *Streptococcus* and *Listeria*; gentamicin adds Gram-negative cover and synergy).

Avoid ceftriaxone in neonates (bilirubin displacement, calcium precipitation).²⁵⁵

NeoFax • AAP Red Book

✦ CAFFEINE CITRATE – THE QUINTESSENTIAL NEONATAL DRUG

- Apnea of prematurity: **load 20 mg/kg**, then **5 mg/kg once daily** maintenance (up to 10 mg/kg/day for refractory apnea).²⁵⁸
- Always prescribe as **citrate** — caffeine base is half the number (20 mg citrate = 10 mg base).²⁵⁸

Caffeine • CAP trial / FDA label

◆ SOURCES

- NeoFax / Micromedex (gentamicin, amikacin, vancomycin, β -lactams); vancomycin AUC target — ASHP/IDSA/PIDS consensus 2020; ANMF Neonatal Medicines consensus 2024; AAP Red Book (empiric regimens); CAP trial (Schmidt, NEJM 2006) and FDA Cafcit label (caffeine).
- Neonatal dosing varies by unit and drug-monitoring protocol — this is a

starting point, not a substitute for
your local formulary.

Convulsive Status Epilepticus (> 1 month of age)

MINUTES

Seizure $\geq$ 5 minutes or repeated without recovery — treat by the clock.

0–5 MIN · STABILIZE (CLOCK STARTS AT SEIZURE ONSET)

- **Time is counted from the start of the seizure, not from arrival.** A child who reaches you already seizing $\geq$ 5 min gets the benzodiazepine **immediately** — airway, oxygen and IV/IO access proceed **in parallel**, never before the drug.^{11,297}
- Airway + oxygen 10–15 L/min via non-rebreather mask · IV/IO · monitor.
- **Check glucose** (treat if low); send sodium, calcium, magnesium, glucose.¹¹

5 min from onset · 1st line



**BENZODIAZEPINE – MAY REPEAT ONCE
AFTER 5 MIN**

Lorazepam IV/IO, or diazepam IV/IO;

no IV access → do not wait for one:

intramuscular midazolam (preferred),
buccal/intranasal midazolam, or **rectal
diazepam.**^{11,297}

~15–20 min • 2nd line

**ESETT: THE THREE ARE EQUALLY
EFFECTIVE**

Levetiracetam (preferred — fewer side
effects and easy administration),
valproate, or fosphenytoin/phenytoin
(phenobarbital is an alternative).¹⁰

~40 min • refractory

3RD LINE • PICU Refractory status

epilepticus: urgent

**PICU/anesthesia/neurology (anesthetic
infusion, continuous EEG, find the
cause). Intubation is a team decision
whenever needed — not reserved for
this step.**²⁰⁸

FIRST LINE – BENZODIAZEPINES

Drug	Dose	Route & notes
Lorazepam	0.1 mg/kg (max 4 mg)	IV/IO
Diazepam	0.2–0.3 mg/kg (max 10 mg) IV/IO rectal 0.5 mg/kg (max 20 mg)	IV/IO; rectal when there is no IV access — an accepted first-line route ¹¹
Midazolam	0.1 mg/kg IV (max 4 mg) 0.2 mg/kg IM/buccal/intranasal (max 10 mg)	IM/buccal/intranasal preferred when no IV access ¹¹

SECOND LINE – CHOOSE ONE

Drug	Dose	Route & notes
Levetiracetam	40–60 mg/kg (max 4.5 g)	IV; often preferred for safety ¹⁰
Valproate	40 mg/kg (max 3 g)	IV; avoid if metabolic/mitochondrial disease or liver failure suspected. Avoid in suspected mitochondrial/POLG disease, significant liver disease, or thrombocytopenia/coagulopathy
Fosphenytoin	20 mg PE/kg (max 1.5 g)	IV (PE = phenytoin equivalents)
Phenytoin	20 mg/kg (max 1.5 g)	IV slow, cardiac monitoring — use if fosphenytoin unstocked

Drug	Dose	Route & notes
Phenobarbital	15–20 mg/kg IV	Alternative second-line; monitor for respiratory depression and hypotension

◆ DRAVET SYNDROME

- Known/suspected Dravet syndrome (prolonged febrile hemiclonic seizures in infancy): prefer **LEVETIRACETAM** or **VALPROATE**; sodium-channel blockers (phenytoin/fosphenytoin) are avoided as **MAINTENANCE** but are **NOT** contraindicated for acute status epilepticus.²⁰⁹
- Phenobarbital is an alternative.

THIRD LINE – REFRACTORY (PICU)

Drug	Dose	Route & notes
Midazolam infusion	0.2 mg/kg load → 0.05–0.4 mg/kg/h	IV; titrate to EEG ²⁰⁹
Ketamine / thiopental / propofol	specialist dosing	IV infusion titrated to EEG; propofol short-term only (propofol infusion syndrome — monitor labs) ²⁰⁹

Drug	Dose	Route & notes
Pyridoxine	100 mg IV	Trial in refractory seizures in infants ²⁰⁹ < 2 yr
Dextrose 10% (if low glucose)	2 mL/kg	IV/IO

ADMIT · DISCHARGE · REFER

- **Admit** — all → monitored bed; refractory or needing infusions → PICU.
- **Discharge** — not from the acute episode; a first seizure that stops may go home after workup, neurology follow-up.
- **Refer** — neurology (EEG, cause); **PICU for refractory status, or airway/hemodynamic compromise.**

Convulsive status epilepticus – treat by the clock

Stabilize → benzodiazepine at 5 min → second-line at ~20 min → refractory (PICU) at ~40 min.



SEIZURE ≥ 5 MIN = STATUS EPILEPTICUS

Two benzodiazepine doses maximum, then move to second-line — don't keep repeating.

0–5 min

1

STABILIZE

Airway + O₂ 10–15 L/min · IV/IO · monitor. Check glucose (treat if low); send Na, Ca, Mg.

5 min

2

BENZODIAZEPINE (1st line)

Lorazepam 0.1 mg/kg IV (max 4) — may repeat once. No IV → IM/buccal/intranasal midazolam 0.2 mg/kg.

~15–20 min

3

SECOND-LINE AGENT

Levetiracetam 40–60 mg/kg, valproate 40 mg/kg, or fosphenytoin 20 mg PE/kg — ESETT: equally effective.

~40 min

4

REFRACTORY — PICU

Anesthetic infusion, continuous EEG, find the cause. Involve PICU, anesthesia and neurology.

Febrile Seizure

HOURS

6 MO–5 YR

Seizure with fever, no CNS infection or prior afebrile seizure, and no pre-existing neurologic abnormality.

SIMPLE VS COMPLEX

- **Simple:** generalized, under 15 minutes, once in 24 hours — benign; reassure, treat the fever source, antipyretics for comfort; no routine EEG, imaging, or LP.¹⁷³
- **Complex:** focal, over 15 minutes, or recurs within 24 hours — evaluate more carefully.¹⁷³

✦ PEARL – NOT A SIMPLE FEBRILE SEIZURE

- Not a simple febrile seizure if any are present: **focal** features, duration **> 15 min** or recurrence within 24 h (complex), age **< 6 months or > 6 years**, incomplete recovery, or a child unwell between events.¹⁷³
- A febrile seizure is, by definition, a seizure with fever in a child **without** CNS infection, metabolic disturbance, or prior afebrile seizures.²⁹⁶ When any of the features above is present, **exclude those first** — meningitis/encephalitis (see **Bacterial Meningitis & Encephalitis**), hypoglycemia, electrolyte disturbance, and an underlying (non-febrile) epilepsy — before labelling it a *complex febrile seizure*.

⚠ LOWER THE THRESHOLD FOR LUMBAR PUNCTURE

- Consider meningitis with any meningism or history/exam suggesting CNS infection; in an infant 6–12 months **under-immunized against Hib or**

pneumococcus (or of unknown immunization status); or in a child **pretreated with antibiotics** (may mask meningitis).¹⁷³

- Age < 12 months alone is no longer an indication — AAP 2011 replaced the 1996 advice to “strongly consider” LP in every infant under 12 months; LP is now an *option* only for the under-immunized 6–12-month-old or the antibiotic-pretreated child.¹⁷³
- A seizure lasting ≥ 5 minutes is status epilepticus (see [Convulsive Status Epilepticus](#)) — treat as above.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — complex features, prolonged/focal, incomplete recovery, or a source needing admission.
- **Discharge** — simple febrile seizure, well child, normal exam, reliable carers with safety-net advice.
- **Refer** — neurology only for complex, recurrent, or atypical seizures — not routine for simple febrile seizures.

Bacterial Meningitis & Encephalitis

MINUTES

> 2 MO

Fever plus altered consciousness, neck stiffness, or a bulging fontanelle — do the LP first, but never delay antibiotics for it.

RECOGNIZE & ACT

- **Infants (non-specific):** fever, poor feeding, lethargy or decreased interaction, vomiting, irritability, $\pm$ rash; inconsolable or worsening irritability with progressive lethargy are red flags. **Nuchal rigidity is uncommon in infants.**¹⁰⁶
- **Older children:** headache, neck pain/stiffness, photophobia, vomiting, impaired consciousness, seizures, **non-blanching rash.**¹²²
- Take **blood cultures** (weight-based volumes), then give **empiric IV antibiotics immediately** — do not delay if LP is deferred.²¹⁴
- Add **acyclovir** if encephalitis features (behavior change, focal neurology, seizures). **Encephalitis criteria:** altered mental status $\geq$ 24 h plus $\geq$ 2 of — fever, new seizures,

focal neurology, CSF pleocytosis, abnormal neuroimaging, or abnormal EEG.²⁶⁶

- Watch for **SIADH** and **raised ICP** (see [Raised ICP](#)).

EMPIRIC & TARGETED THERAPY – MENINGITIC DOSES

Drug	Dose	Route & notes
Ceftriaxone	50 mg/kg/dose q12h (or 100 mg/kg/dose once daily; max 4 g/day (2 g/dose if q12h))	IV (IM if IV not immediate); first-line empiric. Covers <i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>H. influenzae</i> . ²¹⁴
Cefotaxime	50–75 mg/kg/dose q6h (max 3 g/dose)	IV; alternative to ceftriaxone — preferred in neonates . ¹⁰⁶
Vancomycin	15–20 mg/kg/dose q6h; target trough 15–20 mg/L (AUC/MIC-guided) for CNS infection (max 1 g/dose)	IV; add empirically for possible cephalosporin-resistant pneumococcus. ²¹⁴
Ampicillin	75 mg/kg/dose q6h (or 50 mg/kg/dose q4h; max 12 g/day)	IV; add for <i>Listeria</i> in neonates / immunocompromised / at risk. ¹⁰⁶

Drug	Dose	Route & notes
Meropenem	40 mg/kg/dose q8h (max 2 g/dose)	IV; alternative if a 3rd-gen cephalosporin is contraindicated — get ID advice.
Dexamethasone	0.15 mg/kg/dose q6h (max 10 mg/dose)	IV; within 4 h of the first antibiotic — see steroid note. Not in neonates. ²¹⁴
Acyclovir	≤ 4 mo: use neonatal HSV dosing (20 mg/kg/dose every 8 h). > 4 mo–< 12 y: 15–20 mg/kg/dose every 8 h (Red Book target 20; balance against nephrotoxicity). ≥ 12 y: 10 mg/kg/dose every 8 h. Duration 21 days (CNS)	IV, infuse over 1 h with adequate hydration + renal monitoring; add if HSV encephalitis suspected. Beyond the neonatal period the higher 20 mg/kg dose is not routine — increased neurotoxicity / nephrotoxicity (Lexicomp).¹⁶¹

◆ DURATION BY PATHOGEN

- *N. meningitidis* and *H. influenzae* 7 days; *S. pneumoniae* 10–14 days; group B *Streptococcus* 14–21 days; *Listeria* ≥ 21 days; aerobic Gram-negative bacilli 21 days (or ≥ 2 weeks after the first sterile CSF in neonates) —

confirm the final duration with infectious diseases.²¹⁴

- Confirm the pathogen and tailor to response.

◆ STEROIDS – DEXAMETHASONE

- Give 0.15 mg/kg/dose q6h just before, with, or **within 4 h** of the first antibiotic (earlier is better).
Continue 4 days if Hib or *S. pneumoniae* is confirmed.
- Reduces hearing loss, especially in Hib.²¹⁴

DURATION, TARGETED THERAPY & FOLLOW-UP

- When susceptibilities return, narrow therapy (penicillin-susceptible pneumococcus or meningococcus → penicillin G or ampicillin; GBS → penicillin/ampicillin + gentamicin). **IV duration: *S. pneumoniae* 10–14 d**
- Hib 7–10 d
- *N. meningitidis* 5–7 d
- GBS 14–21 d (longer if cerebritis/ventriculitis). **Formal audiology before discharge** for every child.¹⁰⁶

- Repeat LP is not routine — reserve it for failure to improve, resistant pneumococcus, or selected neonates/complex cases after ID advice; also repeat the LP at the end of therapy for confirmed HSV encephalitis.²¹⁴
- Arrange **chemoprophylaxis** for close contacts of meningococcal (and Hib per criteria) via public health.²¹⁴

⚠ DEFER THE LP (BUT NEVER THE ANTIBIOTICS) IF

- reduced or fluctuating consciousness, focal signs, signs of raised ICP, coagulopathy, or skin infection over the site (stabilize a cardiorespiratorily unstable child before any imaging) — image (contrast CT and/or MRI) first.²¹⁴
- A purpuric / non-blanching rash with fever is meningococcal sepsis until proven otherwise¹²² (see [The Rash](#), [Septic Shock](#)).

▀ ADMIT · DISCHARGE · REFER

- **Admit** — all → hospital; reduced consciousness, shock, seizures, or raised ICP → PICU.

- **Discharge** — not from the acute illness; complete the course with audiology and neurodevelopmental follow-up.
- **Refer** — infectious diseases/neurology; PICU for complications; public health for notification and contact prophylaxis.

Brain Tumor – New Presentation

HOURS

HeadSmart / RCPCH

Persistent morning headache and vomiting with a visual, coordination, or behavioral sign — image the whole neuraxis, don't LP, and call neurosurgery.

▶ RECOGNIZE – HEADSMART RED FLAGS

- **Raised ICP:** recurrent **headache worse on waking or lying down**, **persistent vomiting (classically early-morning)**, and **papilledema** or a falling conscious level. ²⁶³
- **Visual / eye signs:** new squint or double vision, **nystagmus**, reduced acuity, an **abnormal head tilt**, or abnormal eye movements. ²⁶³

- **Motor / coordination: ataxia** or unsteady gait, new focal weakness or facial droop, and in a young child **regression / loss of milestones.** ²⁶³
- **Age-specific:** infants — **enlarging head, bulging fontanelle, “sunset” eyes, developmental regression;** older children / teens — **behavior or school change, fatigue, delayed or arrested puberty, polyuria/polydipsia** (diabetes insipidus), and **precocious puberty** — associated with intracranial (pineal / suprasellar) germ-cell tumors. ²⁶³
- A new **focal seizure**, or any of the above that is **persistent or progressive.** ²⁶³

⚠ **DO NOT LP**

With a mass or **obstructive hydrocephalus**, an LP can precipitate **herniation** — **image first.** Don't dismiss persistent morning vomiting as gastroenteritis. ²⁶³

◆ **DIAGNOSE**

- **MRI brain and whole spine with contrast** is the test of choice (spine imaging stages

leptomeningeal / "drop" metastases); **contrast CT** if MRI is not immediately available or to exclude acute hydrocephalus / hemorrhage.

- Look actively for **obstructive hydrocephalus** — it is a neurosurgical emergency.²⁶³

▲ ACUTE MANAGEMENT

- **Position first:** head of bed up **30°**, neck midline (no venous obstruction), avoid hypotonic fluids and hyperthermia; treat seizures — the same first-line measures as for raised ICP (see **Raised ICP**).
- **Dexamethasone** for symptomatic peritumoral (vasogenic) edema / raised ICP — specialist-directed; commonly **0.25–0.5 mg/kg/dose**, and for severe edema or impending herniation a **1–2 mg/kg IV load then 1–1.5 mg/kg/day divided q6h** (max **16 mg/day**). Add a **PPI (e.g., omeprazole)** for gastric protection.
- **Call neurosurgery urgently**; treat hydrocephalus (EVD / CSF diversion) as directed.²⁶³

- **Transfer to PICU** if drowsy or dropping conscious level, and manage as **Raised ICP** in the interim.

▀ ADMIT · DISCHARGE · REFER

- **Admit** — any suspected tumor with raised ICP, hydrocephalus, a focal deficit, or a reduced conscious level.
- **Discharge** — not during the acute presentation; expedite imaging and specialist review.
- **Refer** — pediatric **neurosurgery / neuro-oncology** at a specialist center for resection and hydrocephalus management; involve oncology.

Brain Abscess

HOURS

NEUROSURGICAL

Fever, headache, focal deficit, or new seizures — image with contrast MRI, treat by source, don't LP.

▀ RECOGNIZE

- **Subacute:** headache (most common), fever, **focal neurological deficit**, new-onset seizures, and features of **raised ICP** (vomiting,

altered consciousness, papilledema). The classic triad (fever + headache + focal deficit) occurs in a **minority**.¹⁰⁷

- **The source predicts the organisms:** contiguous — **otitis/mastoiditis, sinusitis, dental**, meningitis; hematogenous — **cyanotic congenital heart disease** (right-to-left shunt), endocarditis, pulmonary; direct — penetrating trauma or neurosurgery; **immunocompromise** (add fungi, *Nocardia*, *Toxoplasma*).¹⁰⁷

◆ **DIAGNOSE**

- **Contrast MRI with diffusion-weighting** is the test of choice — abscess shows **restricted diffusion** (a tumor usually does not); contrast CT if MRI unavailable.¹⁰⁸
- Take **blood cultures**. **LP only after neurosurgical evaluation** — low yield, herniation risk with a focal mass; abscess drainage and culture is best.¹⁰⁸
- Contrast MRI + neurosurgery + IV antibiotics by source.

- Urgent neurosurgery + ID; empiric therapy follows source and local resistance.

DECISION – WHAT IS THE LIKELY SOURCE?

OTOGENIC / SINUS / DENTAL / HEMATOGENOUS

Ceftriaxone (or cefotaxime) + metronidazole + vancomycin. Covers the *Streptococcus anginosus* (milleri) group and anaerobes.¹⁰⁸

POST-TRAUMA / POST-NEUROSURGICAL

Vancomycin + an antipseudomonal beta-lactam (cefepime, ceftazidime, or meropenem) — covers *S. aureus* (incl. MRSA), Gram-negatives and *Pseudomonas*.¹⁰⁸

EMPIRIC AGENTS – PEDIATRIC DOSES

Drug	Dose	Route & notes
Ceftriaxone	50 mg/kg/dose q12h (or 100 mg/kg/dose once daily; max 4 g/day (2 g/dose if q12h))	IV; streptococci + Gram-negatives. Cefotaxime is an alternative.
+ Metronidazole	7.5 mg/kg/dose q6h (max 500 mg)	IV; anaerobic cover for otogenic / sinus / dental sources.

Drug	Dose	Route & notes
+ Vancomycin	15 mg/kg/dose q6h; trough 15–20 mg/L (max 1 g/dose)	IV; add for <i>S. aureus</i> / MRSA or severe disease.
Meropenem (OR — alternative)	40 mg/kg/dose q8h (max 2 g)	IV; cephalosporin allergy, resistant Gram-negatives, or suspected <i>Nocardia</i> .
± Vancomycin	15 mg/kg/dose q6h (max 1 g/dose)	Add to meropenem where <i>S. aureus</i> / MRSA is likely.

SURGERY, DURATION & ADJUNCTS

- **Neurosurgical aspiration or excision** for diagnosis and for potential abscess drainage, or if not responding — send pus for aerobic, anaerobic, fungal and mycobacterial studies. **IV antibiotics ≈ 4–8 weeks**, guided by clinical response and serial imaging; involve **infectious diseases**.¹⁰⁸ **Treat the source** (mastoid, sinus, dental). **Urgent PICU consult for ICP management**; give **anticonvulsants** for seizures.
- Reserve **steroids** for significant edema / mass effect — they can reduce antibiotic penetration.¹⁰⁸

⚠ DON'T LP RIGHT AWAY – WATCH FOR HERNIATION

- With a focal mass, an LP can precipitate herniation and rarely changes management¹⁰⁷ — image first.
- Rapid deterioration, a large abscess with midline shift, or impending herniation → emergency neurosurgery & PICU consult (see [Raised ICP / The Comatose Child](#)).

▀ ADMIT · DISCHARGE · REFER

- **Admit** — all → for treatment and monitoring; potential abscess drainage.
- **Discharge** — after prolonged IV therapy and imaging response, with follow-up.
- **Refer** — neurosurgery (drainage) and infectious diseases; find the source (ENT/dental/cardiac).

Raised ICP / The Comatose Child

MINUTES

Protect the brain: airway, glucose, and reverse herniation fast.

▶ RECOGNIZE HERNIATION

- Falling conscious level (GCS / AVPU), unequal or fixed pupils, abnormal posturing, and the **Cushing triad** (hypertension + bradycardia + irregular breathing) = impending herniation.⁸²
- Coma work-up — **DEFG: "Don't Ever Forget Glucose."** Sort causes with **AEIOU-TIPS**: alcohol/toxins, epilepsy, insulin/glucose, opioids, uremia/metabolic, trauma, infection, psychiatric, stroke/shock.

STABILIZE

- Airway (add C-spine control if trauma; use an appropriately sized collar), oxygen, ventilation.
- Check glucose; treat seizures.
- Position the head midline and elevate the head of the bed to 30^{82°}.
- Keep normothermic and normoglycemic; avoid hyponatremia — keep sodium in the normal-to-high range.⁸²
- **Call PICU.**



LOWER ICP

Maintain oxygenation and **normocapnia** (avoid routine hyperventilation); give hyperosmolar therapy; urgent CT + neurosurgery.⁸²

Drug	Dose	Route & notes
Hypertonic saline 3%	3–5 mL/kg over 10–15 min	IV; for raised ICP / herniation ⁸²
Mannitol	0.5–1 g/kg over 10–15 min	IV; alternative (caution if hypovolemic/hypotensive) ⁸²

Drug	Dose	Route & notes
Dextrose 10% — hypoglycemia only	2–5 mL/kg (0.2–0.5 g/kg); recheck glucose	IV/IO only to correct documented hypoglycemia (“don't ever forget glucose”); use the smallest effective dose.

⚠️ FLUIDS – AVOID FREE WATER

In suspected raised ICP, **avoid dextrose-containing and hypotonic fluids** as a bolus or maintenance — once the glucose is metabolized the leftover free water worsens cerebral edema.

Maintain with **isotonic 0.9% saline**.⁸²

The one exception is **documented hypoglycemia**, which injures the brain and must be corrected.

⚠️ DO NOT DELAY TRANSFER

Do not delay transfer to PICU/neurosurgery; repeat hyperosmolar therapy is specialist-guided.

⚠️ HYPERVENTILATION

Brief hyperventilation **only** as a bridge for impending herniation; sustained hypocapnia causes cerebral ischemia.⁸²

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → PICU.
- **Discharge** — not from the acute episode.
- **Refer** — neurosurgery/neurology and the primary-cause specialty (tumor, infection, metabolic, trauma).

Guillain-Barré Syndrome / Acute Flaccid Paralysis

MINUTES

EAN/PNS 2023

Ascending weakness that can reach the diaphragm — the emergency is respiratory failure and autonomic instability.

▲ RECOGNIZE

- Progressive, fairly **symmetric ascending weakness** with **reduced/absent reflexes**, often after a GI or respiratory infection; a refusal to walk in a young child.²⁶⁴
- CSF (if done) shows **high protein with normal cell count** (albuminocytologic dissociation) —

may be normal in the first week; MRI may show nerve-root enhancement.²⁶⁴

⚠ THE TWO THINGS THAT KILL

- **Respiratory failure** — track vital capacity/effort; rising CO₂ and a falling forced vital capacity (roughly < 15–20 mL/kg) mean elective intubation, not a trial of NIV. **Autonomic instability** — labile blood pressure, arrhythmia, ileus; put the child on a cardiac monitor.²⁶⁴
- Rapid progression or bulbar weakness (choking, weak cough) = ICU.²⁶⁴

✦ DON'T BE REASSURED BY A NORMAL EARLY TAP

- Weakness can outrun the CSF and the reflexes early on.
- If the picture fits, admit and monitor breathing serially rather than discharging on a single normal test.
- Always exclude a compressive cord lesion (a sensory level, back

pain or sphincter loss points to the cord, not GBS).²⁶⁴

▲ WORKUP

- **Lumbar puncture** — albuminocytologic dissociation (high protein, normal cells); **may be normal in the first week.**²⁶⁴
- **Nerve conduction studies / EMG** (subtype; may be normal early — repeat).
- **Serial bedside spirometry** — FVC (and NIF/MIP) **every 4–6 h** in the progressive phase, plus bulbar assessment;²⁶⁴ act on the **"20/30/40" thresholds** (FVC < 20 mL/kg, MIP < 30, MEP < 40 cmH₂O) or a **> 30% fall** from baseline — these predict the need for ventilation.²⁹¹
- **Spirometry is unreliable under ~5–6 years** or with facial/bulbar weakness (poor mouth seal) — there, rely on **serial clinical assessment:** single-breath count, cough strength, paradoxical breathing, ability to lie flat, and the blood-gas/CO₂ trend.²⁶⁴
- **Continuous ECG/BP** for dysautonomia.²⁶⁴

- **Spinal MRI with contrast** (nerve-root enhancement; excludes a compressive cord lesion).
- Anti-ganglioside antibodies (GQ1b for Miller-Fisher); **stool for poliovirus/enterovirus** in acute flaccid paralysis.²⁶⁴

Treatment	Dose	Notes
IV immunoglobulin	2 g/kg total, over 2–5 days	First-line; equivalent to plasma exchange — a second course is not recommended in non-responders (SID-GBS) ²⁶⁴
Plasma exchange	Per center protocol	Alternative — not combined with IVIG ²⁶⁴
Corticosteroids	Not recommended	No benefit in GBS ²⁶⁴

▲ ADMIT · REFER

- **Admit** all suspected cases for respiratory and cardiac monitoring.
- **Call** neurology; critical care early if bulbar/respiratory involvement or rapid progression. Acute flaccid paralysis is **notifiable** (polio surveillance) — send stool.

The Child with Neuromuscular Disease

MINUTES

DMD Care 2018 • CHEST 2023

A known neuromuscular disease rewrites every emergency — the weak cough, the anesthetic that can kill, the heart, and the steroids.

▶ RESPIRATORY – THE USUAL KILLER

- It is the **weak cough, not the lungs**: a “simple” chest infection tips into **hypercapnic respiratory failure**. Support ventilation with **NIV (bilevel)** and clear secretions with **mechanical insufflation–exsufflation (cough assist)** — not oxygen alone.²⁶⁵
- **The oxygen saturation lies** — it can be normal while CO₂ climbs. Track **capnography / blood-gas pCO₂**, not SpO₂.

⚠ ANESTHETIC / RSI HAZARD – FLAG BEFORE ANY SEDATION

- **Succinylcholine is contraindicated** in dystrophinopathies and many myopathies (hyperkalemic cardiac

arrest), and **volatile agents** (sevoflurane, isoflurane, desflurane) risk **anesthesia-induced rhabdomyolysis / hyperkalemia** (see [Rhabdomyolysis](#)). **Safe approach:** total IV anesthesia (propofol, opioid, benzodiazepine) + a **non-depolarizing** blocker (rocuronium) reversible with sugammadex.²⁶⁵

- Check ionized K⁺, ECG and CK after any procedure.²⁶⁵

⚠ DO NOT GIVE UNCONTROLLED OXYGEN

- High-flow O₂ masks a rising CO₂ and can abolish respiratory drive²⁶⁵ → **CO₂ narcosis and arrest.**
- Titrate O₂ only **with** ventilatory support and CO₂ monitoring — correct the hypoventilation, not the number.²⁶⁵

▀ CARDIAC

- **Duchenne / Becker** → dilated cardiomyopathy and arrhythmia (may be silent — reduced mobility hides symptoms). **Emery-Dreifuss /**

LGMD1B → progressive **AV / conduction block** and sudden death (may need pacing).²⁶⁵

- Any acute deterioration: **12-lead ECG + echocardiogram**; troponin and BNP/NT-proBNP if acute myocardial injury or heart failure is suspected.²⁶⁵

▀ **ADRENAL – STRESS-DOSE STEROIDS**

- A child on **chronic daily steroids** (prednisone/deflazacort for Duchenne) **cannot mount a stress response**. Acute illness, vomiting, trauma, surgery or shock needs **stress-dose steroids** to prevent adrenal crisis — **never stop steroids abruptly**²⁶⁵ (see [**Adrenal Crisis**](#)).

▀ **MYASTHENIC CRISIS**

- Worsening bulbar/respiratory weakness → treat with **IVIG or plasma exchange** and support the airway.²⁶⁵ Follow **FVC and NIF**, not SpO₂.
- **Avoid drugs that worsen myasthenia**: aminoglycosides, macrolides, fluoroquinolones, **IV magnesium**, beta-blockers, and procainamide.²⁶⁵

✦ TWO NUMBERS THAT MISLEAD, ONE THAT DOESN'T

- In neuromuscular weakness a normal **SpO₂** and a normal **respiratory rate** are falsely reassuring — the child fails from **CO₂ retention and a failing cough**.
- The honest bedside numbers are **pCO₂ (capnography/gas)** and **FVC / NIF**.
- A metabolic/mitochondrial myopathy adds a fourth: start **IV dextrose (D10) early** — fasting drives catabolism, hypoglycemia and rhabdomyolysis.²⁶⁵

▶ WORKUP

- **VBG/ABG with pCO₂ + continuous capnography** — the key respiratory measure; **FVC and NIF** where the child can cooperate.
- Electrolytes incl. **potassium, magnesium, calcium; creatine kinase**; renal function; **glucose**; lactate.
- **12-lead ECG**; troponin + BNP/NT-proBNP; echocardiogram if cardiac concern.

- Chest X-ray ± sputum culture if infective; swallow/bulbar assessment.²⁶⁵

Situation	Dose	Route & notes
Stress steroids — tolerating oral	≈ triple the maintenance steroid dose during the illness	PO; for the child on chronic steroids (Duchenne)
Stress steroids — unwell / vomiting / shock	Hydrocortisone ≈ 2 mg/kg (or 50 mg/m ²) IV, then continued	IV/IM; treat as adrenal crisis ²⁶⁵ — do not wait
Myasthenic crisis	IVIg 2 g/kg over 2–5 days or plasma exchange	Plus airway support; treat the trigger/infection
Metabolic myopathy — catabolic stress	IV dextrose (D10) early	Prevents fasting-driven decompensation / rhabdomyolysis

◆ LOCAL & ANTICIPATORY

- Ask for the child's **EMERGENCY / ANESTHESIA ALERT CARD** and neuromuscular care plan on arrival, and involve the **NEUROMUSCULAR TEAM** early.

- Keep succinylcholine and volatile agents off the plan for dystrophinopathies, and flag the diagnosis to anesthesia/critical care before any transfer, sedation or intubation.²⁶⁵

▲ ADMIT · REFER

- **Admit** — any respiratory decompensation, myasthenic/adrenal crisis, new cardiac signs, or a child who needs assisted airway clearance.
- **Call** neuromuscular/neurology, cardiology and critical care/anesthesia; new acute flaccid weakness without a known diagnosis → see **Guillain-Barré / Acute Flaccid Paralysis**.

Orthopedic &

12

Musculoskeletal

The limping child · fractures & NAI · SCFE · compartment syndrome · bone & joint infection

Orthopedic Emergencies – Call Now

MINUTES

Immediate orthopedic ($\pm$ vascular/spinal) consultation — do not wait for imaging.

⚠ IMMEDIATE ORTHOPEDIC CONSULTATION

- Septic arthritis
- Compartment syndrome
- Open fracture
- SCFE
- Joint dislocation
- Fracture with pulseless limb
- Unstable pelvic injury
- Spine injury with neurological deficit¹³⁶

Osteoarticular Infections

HOURS

RECOGNIZE

Acute hematogenous osteomyelitis (AHO) and septic arthritis (SA) — consider in any limp, refusal to use a limb, or pseudoparalysis.

CLINICAL PRESENTATION

- **Pseudoparalysis / limp / refusal to weight-bear**, or focal limb pain — often the only symptom. Fever supports diagnosis but may be absent.⁶³
- **Osteomyelitis**: point tenderness over the **metaphysis** of a long bone ± localized swelling/erythema; fluctuance suggests a periosteal abscess.⁶³
- **Septic arthritis**: joint effusion, swelling, and pain on any movement of the isolated joint; the limb is held in a position of comfort.⁶³
- *S. aureus* → acute, rapidly progressive, higher CRP (MRSA more severe). *Kingella kingae* (young children) → milder, indolent, lower markers.⁶³
- **Chronic recurrent multifocal osteomyelitis (CRMO/CNO)** — consider with recurrent bone pain,

multifocal lesions, sterile cultures, and poor response to antibiotics.⁶³

▶ MOST COMMON SITES & PATHOGENS

- **Bone:** metaphysis of long bones — **femur, tibia, humerus**. Pelvic bones are easily missed (vague pain, mimics abdomen).⁶³
- **Joint: hip and knee** most common, then ankle, elbow, shoulder. SA + adjacent osteomyelitis is common ($\approx$ 32% of SA), especially with *S. aureus* and longer symptom duration.⁶³
- **Pathogens:** *S. aureus* overall most common; *K. kingae* dominates under $\sim$ 4 yr (inoculate fluid into blood-culture bottles). Add *Salmonella* in sickle cell disease; *H. influenzae* type b if under-immunized.⁶³

◆ KOCHER CRITERIA – HIP SA VS TRANSIENT SYNOVITIS

- Four predictors, one point each:
**NON-WEIGHT-BEARING; FEVER $> 38.5^{\circ}\text{C}$;
ESR $> 40\text{ MM/H}$; WHITE-CELL COUNT $> 12,000/\text{ML}$.**

- Many add **CRP > 20 MG/L** as a strong independent predictor.
- A risk-stratification aid for the hip, not a rule-out.⁶⁶
- More positive criteria substantially raise septic-arthritis probability, but always interpret alongside CRP, ultrasound, and clinical judgment.⁶⁶

KOCHER – PROBABILITY OF SEPTIC ARTHRITIS

Predictors present	Approx. probability of SA
0 of 4	< 0.2 %
1 of 4	~ 3 %
2 of 4	~ 40 %
3 of 4	~ 93 %
4 of 4	~ 99 %

⚠ WORK-UP

- Blood cultures **before** antibiotics.
CRP at baseline (sensitive, falls fast — best for monitoring; ESR less useful).⁶³
- Plain films to exclude mimics;
ultrasound for hip/shoulder effusion; **MRI is the imaging modality of choice** (add contrast if abscess or soft-tissue extension is suspected); **MRI with gadolinium** is the most sensitive/specific test,

defining abscess, adjacent osteomyelitis, and soft-tissue extent.⁶³

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected septic arthritis/osteomyelitis → IV antibiotics after cultures/aspiration. **Delay antibiotics until a joint aspirate is obtained whenever clinically feasible.**
- **Discharge** — not acutely; step down to oral with monitoring and orthopedic follow-up.
- **Refer** — orthopedics (aspiration/washout) and infectious diseases.

Osteoarticular Infections — Management

HOURS

CULTURES FIRST

Cultures first, then empiric IV cefazolin; involve orthopedics early for any septic joint or bony abscess.

DECISION

Osteomyelitis or septic arthritis?

OSTEOMYELITIS (AHO)

Most respond to **antibiotics alone**. Surgery (drainage/debridement) for a **subperiosteal/intraosseous abscess**, or if not improving in the first few days — repeat imaging and reconsider debridement.⁶³

SEPTIC ARTHRITIS

Orthopedic emergency — urgent **aspiration ± arthroscopic/open drainage & irrigation**. Hip and shoulder are especially time-critical (avascular-necrosis risk). Send synovial fluid for cell count, Gram stain, culture (blood-culture bottle) + PCR.⁶³

Empiric therapy after cultures: IV cefazolin — covers MSSA and *K. kingae*.⁶³

ANTIBIOTIC THERAPY

Scenario	Drug & dose	Notes
Empiric — first-line IV	Cefazolin 37.5–50 mg/kg/dose IV q6–8h (max 2 g/dose)	Covers MSSA and <i>K. kingae</i> — the two commonest pathogens. Start after blood cultures. ⁶³
Under 4 yr / unimmunized	Cefuroxime 50 mg/kg/dose IV q8h (max 1.5 g/dose)	Broadens to cover <i>H. influenzae</i> in unimmunized young children. ⁶³
MSSA — confirmed	Continue cefazolin, or cloxacillin 37.5–50 mg/kg/dose IV q6h (cloxacillin max 2 g/dose)	Cloxacillin narrows further but irritates veins. ⁶³
MRSA — suspected/confirmed	Add/use vancomycin 15 mg/kg/dose IV q6h (max 1 g/dose; level-guided)	Where MRSA prevalent or known carrier. Note: <i>K. kingae</i> is resistant to vancomycin, clindamycin & cloxacillin. ⁶³

Scenario	Drug & dose	Notes
Oral step-down — MSSA / culture-negative	Cephalexin 40–50 mg/kg/dose PO TID (max 6 g/day)	First-line oral. Cloxacillin 25 mg/kg/dose QID is an alternative (poor taste). ⁷⁸
Oral step-down — MRSA	Clindamycin, TMP-SMX, or linezolid	Per susceptibilities, with infectious-diseases input; MRSA usually needs longer before switching. ⁷⁸

◆ IV → ORAL SWITCH & DURATION

- Switch to oral when **CLINICALLY IMPROVED** (afebrile, using the limb) and **CRP FALLING** ($\approx$ 50% drop, often < 20–30 mg/L) with assured adherence — even after prior bacteremia.⁷⁸ **DURATION (CRP-GUIDED): AHO 3–4 WEEKS**
- isolated SA **2–3 WEEKS**
- SA with adjacent osteomyelitis **4 WEEKS**
- hip/shoulder SA⁶³ **4–6 WEEKS.**
- Randomized data (Peltola) support short courses once response is good and CRP normalizes early.⁶⁵

⚠ FOLLOW-UP

- Confirm clinical resolution and a **normal/normalizing CRP** before stopping.
- End-of-therapy radiographs mainly if the **growth plate** was involved or a large lytic lesion present — these and any physeal involvement need **orthopedic follow-up**.
- Routine follow-up MRI/CT is not needed — reserve for complications or non-improvement.⁶³

◆ **DIFFICULT IV ACCESS?** Reassess for an early IV→oral switch at every line failure rather than re-sticking — see [IV → Oral Switch & the DIVA Patient](#) (Quick Reference).

The Limping Child

ASSESS

APPROACH

A limp is a symptom — use age and red flags to separate the benign from septic joint, SCFE, malignancy, and abuse.

DEFINE

- A **limp** is any deviation from normal age-appropriate gait; pain present in ~80%. Broad differential — separate benign causes from the **can't-miss** ones: septic joint, osteomyelitis, SCFE, malignancy, and inflicted injury.¹³⁶

APPROACH – AGE NARROWS THE DIFFERENTIAL

- **Toddler (1–3 y):** toddler's fracture, transient synovitis, developmental dysplasia of the hip, inflicted injury.
- **Child (4–10 y):** transient synovitis, Legg-Calvé-Perthes, septic arthritis, reactive/juvenile idiopathic arthritis.
- **Adolescent (> 10 y):** SCFE, apophysitis/overuse, sports injury.¹³⁶
- **Spinal:** consider discitis, vertebral osteomyelitis, or a spinal tumor in a child refusing to walk with a **normal**

hip exam — especially with back pain or spinal tenderness.¹³⁶

- **Malignancy:** persistent night pain, pain out of proportion to the exam, constitutional symptoms, or failure to improve¹³⁶ → **urgent investigation.**
- History: pain, trauma, fever, systemic or night symptoms. Examine gait (antalgic vs Trendelenburg), hip, knee, spine, neurology — **knee pain often means hip disease.**¹³⁶
- **Any age — look beyond the hip:** consider a **first presentation of juvenile idiopathic arthritis (JIA)**, which occurs across **all age groups**, and **examine every joint** — not only the painful one (a commonly missed step that reveals oligo- or polyarticular disease).¹³⁶

⚠ **WORK-UP**

Plain radiographs of the involved site — **both hips (add a frog-leg view for SCFE)** if the exam does not localize. **FBC, CRP/ESR, blood culture** if febrile or systemically unwell; ultrasound for a hip effusion; MRI if osteomyelitis or malignancy is suspected.¹³⁶

▀ RED FLAGS

- **Fever + refusal to weight-bear + raised inflammatory markers** → septic joint/osteomyelitis. Normal early CRP/ESR/white count do **not** exclude osteoarticular infection — reassess if symptoms persist or worsen. ¹³⁶
- **Night pain, weight loss, pallor, bruising, or hepatosplenomegaly** ¹³⁶ → malignancy/leukemia.
- **Multiple or unexplained fractures, or a fracture in a non-ambulatory infant** ¹³⁹ → inflicted injury.
- **Adolescent hip or knee pain with a limp** ¹³⁸ → SCFE until excluded.

✦ PEARL – POINT TO THE HIP

Isolated knee or thigh pain in a child is hip pathology until proven otherwise — SCFE and Perthes both refer pain to the knee. ¹³⁶

▀ ADMIT · DISCHARGE · REFER

- **Admit** — septic arthritis, osteomyelitis, fracture, or systemic illness; unable to weight-bear with red flags.

- **Discharge** — transient synovitis in a well child with safety-net advice and review.
- **Refer** — orthopedics for red flags; oncology/rheumatology as indicated.

Transient Synovitis vs Septic Arthritis

HOURS

KOCHER

The commonest atraumatic limp — but rule out the septic joint first; fever + non-weight-bearing + high CRP means aspirate.

▀ DEFINE

- **Transient synovitis** — self-limiting, usually post-viral hip inflammation and the commonest cause of an atraumatic limp (3–8 years); symptoms usually resolve within **7–10 days**. Must be separated from **septic arthritis**, a surgical emergency.⁶⁶

▲ APPROACH – READ THE CHILD

- **Transient synovitis:** a well child, low-grade or no fever, **weight-bears with discomfort**, mildly raised markers, often a recent upper-respiratory infection.
- **Septic arthritis:** an unwell/febrile child who **refuses to move the joint**, severe pain on any movement, the limb held in flexion/abduction/external rotation.⁶⁶

◆ KOCHER CRITERIA – HIP: SEPTIC ARTHRITIS VS TRANSIENT SYNOVITIS

- One point each: **NON-WEIGHT-BEARING; FEVER > 38.5 °C; ESR > 40 MM/H; WBC > 12,000/ML.**
- Many add **CRP > 20 MG/L** as a strong independent predictor.
- Probability of septic arthritis: 0/4 < 0.2%
- 1/4 ≈ 3%
- 2/4 ≈ 40%
- 3/4 ≈ 93%
- 4/4 ≈ 99%.
- A risk aid, not a rule-out.⁶⁶

⚠ WORK-UP

- FBC, **CRP**, ESR, blood culture; hip **ultrasound** (an effusion is seen in both and does not distinguish).⁶⁶
- If septic arthritis is suspected → **urgent ultrasound-guided aspiration** (cell count, Gram stain, culture into a blood-culture bottle).⁶⁶

▀ MANAGEMENT

- **Septic arthritis** — orthopedic emergency: urgent joint drainage plus IV antibiotics after cultures (see [Osteoarticular Infections](#)).
- **Transient synovitis** — rest, NSAIDs, review in 24–48 h; settles over days. Re-evaluate promptly if it does not improve or markers rise.⁶⁶

✦ PEARL – WHEN IN DOUBT, ASPIRATE

Fever + refusal to weight-bear + a raised CRP is a septic joint until an aspirate proves otherwise — do not label it transient synovitis by default.⁶⁶

▲ ADMIT · DISCHARGE · REFER

- **Admit** — any suspicion of septic arthritis (Kocher criteria, raised markers) → aspiration.
- **Discharge** — transient synovitis, afebrile, mobilizing, with review.
- **Refer** — orthopedics for aspiration/washout if septic.

Nursemaid's Elbow

ASSESS

1–4 YR

A pulled, pronated arm the child will not use — reduce by hyperpronation; no film needed if the story is classic.

▲ DEFINE

- **Nursemaid's elbow (radial head subluxation, 'pulled elbow')** — the annular ligament slips over the radial head after axial traction on an extended, pronated arm.¹³⁷
Commonest elbow injury in children aged 1–4 years.

▲ APPROACH & WORK-UP

- Typical story: sudden **pull or lift** on the arm. Child holds the arm slightly flexed and **pronated**, refuses to use it, with **no swelling, bruising, or deformity** and minimal tenderness.¹³⁷
- **No imaging needed** with a classic history and a normal-looking arm. Image if the story is atypical, or there is swelling, focal bony tenderness, or deformity.¹³⁷

⚠ RED FLAGS

- Swelling, bruising, point bony tenderness, or deformity → suspect a fracture and radiograph.
- An inconsistent or implausible history should prompt consideration of inflicted injury.

▲ MANAGEMENT – REDUCE IT

- **Hyperpronation** (support the elbow, firmly pronate the forearm) — a meta-analysis shows it is **more effective and less painful** than supination-flexion — do it first.¹³⁷

- Alternative: **supination-flexion** (supinate the forearm, then fully flex the elbow). A palpable click often signals success.¹³⁷
- Child usually resumes normal use within 5–30 minutes. If reduction fails, try the other maneuver; if still unsuccessful, immobilize, image, reassess.¹³⁷

✦ PEARL – NO FILM NEEDED IF IT WORKS

- With a classic pull mechanism and no swelling, reduce without imaging — if the child uses the arm normally afterwards, diagnosis is confirmed and no radiograph is required.¹³⁷
- Do not attempt repeated reductions if there is swelling, bruising, focal tenderness, or an atypical history — image and consider safeguarding.¹³⁷

▲ ADMIT · DISCHARGE · REFER

- **Admit** — not required.
- **Discharge** — after successful reduction and return of normal arm use.

- **Refer** — orthopedics only if reduction fails, or for swelling/deformity suggesting fracture (image first).

Slipped Capital Femoral Epiphysis (SCFE)

HOURS

CAN'T MISS

Peri-pubertal limp with hip or knee pain — stop weight-bearing and get frog-leg views of both hips.

DEFINE

- **Slipped capital femoral epiphysis (SCFE)** — the proximal femoral epiphysis slips through the physis. Peri-pubertal; **obesity is the strongest risk factor**; up to $\frac{1}{3}$ bilateral. Consider endocrinopathy if the child is young (< 10 years), underweight, or atypical.¹³⁸

APPROACH & WORK-UP

- Limp with **hip, groin, thigh, or knee pain**; the leg lies **externally rotated** with limited internal rotation and

obligate external rotation on hip flexion (Drehmann sign).¹³⁸

- **Stable** (able to weight-bear, with or without walking aids) vs **unstable** (cannot bear weight — high avascular-necrosis risk).¹³⁸
- **AP + frog-leg lateral radiographs of BOTH hips** for a stable slip; for an **unstable** slip avoid forced frog-leg positioning and use AP + **cross-table lateral** to prevent further displacement. Klein's line fails to intersect the epiphysis on the affected side. The frog-leg view is most sensitive.¹³⁸

⚠ **RED FLAGS**

- An **unstable slip** (non-weight-bearing) carries high avascular-necrosis risk.
- Missed or delayed diagnosis → progression and permanent deformity.¹³⁸
- Assess both hips.

▀ **MANAGEMENT**

- **Make the child non-weight-bearing immediately** and refer urgently to orthopedics for **in-situ screw**

fixation. ¹³⁸

- **Do not attempt reduction** — forceful manipulation increases avascular-necrosis risk. ¹³⁸
- Consider prophylactic contralateral fixation in high-risk patients (young age, obesity, endocrinopathy). ¹³⁸

✦ **PEARL – FROG-LEG BOTH HIP**

Any obese adolescent with a limp or knee pain gets frog-leg views of both hips and is kept off the leg until SCFE is excluded — knee pain is the classic decoy. ¹³⁸

▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — all → non-weight-bearing, urgent surgery.
- **Discharge** — post-operative.
- **Refer** — orthopedics urgently (surgical fixation); endocrinology if bilateral or atypical.

Fractures – Patterns & Non-Accidental Red Flags

ASSESS

SAFEGUARDING

Know the common breaks — and the fractures that mandate a child-protection assessment.

DEFINE

- Common pediatric fracture patterns and — critically — the fractures that trigger a **child-protection** assessment. Children's growing bone breaks differently from adults'. ¹³⁹

APPROACH – COMMON ACCIDENTAL PATTERNS

- **Buckle/torus** (distal radius, stable), **greenstick**, and **toddler's fracture** (spiral distal tibia in an ambulant toddler). ¹³⁹
- **Physeal (Salter-Harris)** fractures cross the growth plate (SH-II commonest) and risk growth arrest — always classify and follow up. ¹³⁹
- Radiograph the **joint above and below**; obtain comparison views when needed. ¹³⁹

- **Supracondylar humeral fractures** need careful neurovascular assessment — document radial pulse, capillary refill, and **median (including anterior interosseous), radial, and ulnar** nerve function before and after splinting/reduction. A pulseless or poorly perfused hand needs **immediate orthopedic consultation**.¹³⁹

⚠ RED FLAGS – INFLICTED INJURY

- Escalate to safeguarding when the **injury is inconsistent** with the stated mechanism or the child's development — especially a **long-bone (femur/tibia) fracture in a non-ambulatory infant**, delayed presentation, multiple fractures of different ages, or **rib, metaphyseal 'corner', scapular, or spinous-process** fractures.¹³⁹
- In children < 2 years with a suspicious fracture, obtain a **skeletal survey**.¹³⁹
- Associated **bruising, patterned burns, retinal hemorrhage**, or other unexplained injuries further raise suspicion of abuse.¹³⁹

▀ MANAGEMENT

- Immobilize, provide analgesia, and refer to orthopedics for displaced, physeal, or unstable fractures.
- Where abuse is suspected, follow the mandatory **child-protection pathway** — evaluation and reporting are not optional.¹³⁹

◆ KUWAIT – REPORTING IS MANDATORY

- Under the **CHILD RIGHTS LAW NO. 21 OF 2015**, children are protected from all forms of violence, abuse, exploitation, and neglect, and **PHYSICIANS ARE LEGALLY OBLIGED TO REPORT** suspected child maltreatment.
- Refer through the **KUWAIT NATIONAL CHILD PROTECTION PROGRAM (KNCPP), MINISTRY OF HEALTH**, via the national **CHILD PROTECTION HOTLINE 147**.
- Ensure the child's immediate safety, document findings objectively, and involve the hospital child-protection team.

✦ **PEARL – IF THEY CAN'T WALK, BE SUSPICIOUS**

A fracture in a child who is not yet walking is highly suspicious for abuse until proven otherwise — do the skeletal survey and involve safeguarding.¹³⁹

✦ **PEARL – DOCUMENT THE NEUROVASCULAR EXAM**

Every **supracondylar fracture** needs documented radial pulse, capillary refill, and median (including anterior interosseous), radial, and ulnar nerve function **before and after** splinting or reduction. **A pulseless hand is an orthopedic emergency, even if it remains pink.**¹³⁹

Compartment Syndrome

MINUTES

LIMB-THREATENING

Rising compartment pressure starves muscle and nerve — in children the 3 A's beat the 5 P's; fasciotomy is the fix.

▴ DEFINE

- **Acute compartment syndrome** — rising pressure in a fascial compartment cuts perfusion →

muscle/nerve ischemia.
Limb-threatening; **clinical** diagnosis.
Most follow fractures (tibia, forearm),
tight casts, crush, or reperfusion.¹⁴⁰

⚠ **RED FLAGS – THE 3 A'S IN CHILDREN**

- The adult '5 Ps' (pain out of proportion, pain on passive stretch, paresthesia, pallor, pulselessness) are **unreliable in children**.¹⁴⁰
- Suspect from the **3 A's — rising Analgesia requirement, Anxiety, and Agitation** — appearing on average ~7 hours before any neurovascular change.¹⁴⁰
- Tensely swollen compartment = most consistent sign.
Pulselessness and paralysis are late.¹⁴⁰

⚠ **WORK-UP**

- Clinical diagnosis.
- Reserve compartment-pressure measurement for the obtunded or equivocal patient ($\Delta P < 30$ mmHg from diastolic is concerning). **Do**

not delay fasciotomy to obtain pressures.¹⁴⁰

▀ MANAGEMENT

- **Remove or split all constricting casts and dressings down to skin;** keep the limb at heart level (do not elevate).¹⁴⁰
- Analgesia, oxygen, correct hypotension. Dense regional/nerve blocks can mask ischemic pain — in high-risk limb injuries prefer low-concentration blocks and stay vigilant (breakthrough or escalating pain, rising analgesic needs).¹⁴⁰
- **Emergency surgical fasciotomy without delay** — the definitive treatment.¹⁴⁰

✦ PEARL – ESCALATING PAIN = CALL SURGERY

A child with a limb injury whose analgesia needs keep climbing has compartment syndrome until proven otherwise — split the cast and call surgery.¹⁴⁰

▀ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected → emergency.

- **Discharge** — post-operative.
- **Refer** — orthopedic (lower limb) / plastic surgery (upper limb) IMMEDIATELY (fasciotomy); do not delay for pressure measurement in a clear case.

Open Fracture HOURS

ANTIBIOTICS EARLY

Skin breach over bone — early IV antibiotics, tetanus cover, saline dressing, and urgent orthopedics.

DEFINE

- **Open (compound) fracture** — fracture with an overlying skin breach communicating with bone; high infection risk. Graded Gustilo-Anderson I–III by wound size, contamination, and soft-tissue injury. ¹⁴¹

APPROACH & WORK-UP

- Visible wound/bleeding over the fracture, exposed bone, or gross deformity. **Assess neurovascular status distal to the injury.** ¹⁴¹

- Remove gross contamination, photograph the wound once, then **cover with a saline-soaked sterile dressing**; radiograph; check **tetanus status**.¹⁴¹

⚠ RED FLAGS

- A **pulseless, pale, cold limb** needs urgent reduction/vascular assessment.
- Gross contamination, type-III wounds, and evolving **compartment syndrome** raise the stakes.¹⁴¹

MANAGEMENT

Drug	Dose	Route & notes
Antibiotics — timing	First IV dose as soon as possible (ideally < 1 h, at latest < 3 h)	The clock starts at injury, not arrival. ¹⁴¹
Type I-II	Cefazolin 30 mg/kg/dose IV (max 2 g) every 8 h	First-generation cephalosporin — gram-positive cover. ¹⁴¹

Drug	Dose	Route & notes
Type III	Add gentamicin 5– 7 mg/kg/dose IV once daily; add benzylpenicillin for farm/soil contamination (anti-clostridial cover)	Add gram-negative cover for high-grade/contaminated wounds. ¹⁴¹
Farm / soil contamin ation	Add penicillin	Anti-clostridial cover.

THEN

- **Tetanus prophylaxis** per immunization status; analgesia; realign and splint the limb. ¹⁴¹
- **Urgent orthopedic referral** for formal debridement, irrigation, and stabilization in the operating room. ¹⁴¹

PEARL – ANTIBIOTICS ON THE CLOCK

Give the first cephalosporin within an hour of injury, update tetanus, cover with a saline dressing, call orthopedics — early antibiotics drive the infection rate down. ¹⁴¹

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all open fractures and significant limb injuries.
- **Discharge** — not acutely; post-operative/reduction.
- **Refer** — orthopedics (early antibiotics, tetanus, debridement); vascular surgery if perfusion is threatened.

Fracture with Pulseless Limb

MINUTES

LIMB-THREATENING

An absent pulse needs urgent assessment; a pale, cool, poorly perfused limb needs emergent restoration of circulation.

▲ DEFINE

- A fracture with an absent distal pulse may indicate arterial compression, entrapment, or injury.^{298,299}

▲ APPROACH

- Obtain urgent radiographs.

- Record injury time, pulses, capillary refill, skin temperature and color, and individual peripheral-nerve function; compare with the opposite limb.
- Reassess and document after realignment, splinting, reduction, and surgery.
- Persistent suspected arterial injury requires consultant-directed vascular imaging, usually **CT angiography**, coordinated so that emergency revascularization is not delayed.²⁹⁹

RED FLAGS

- Pallor, coolness, delayed refill, worsening pain, progressive sensory or motor deficit, or deteriorating perfusion → **immediate orthopedic and vascular escalation.**
- Increasing pain or swelling also warrants reassessment for **compartment syndrome.**

MANAGEMENT

- Splint in the position of deformity in the emergency room.
- Emergent reduction and pinning in the operating room.

- Reassess circulation immediately post-reduction; vascular exploration if the limb remains cold and pulseless.²⁹⁸
- **Supracondylar fracture with a pink, warm hand** — arrange urgent reduction and fixation. If the hand remains well perfused afterwards, an absent radial pulse alone does not mandate brachial-artery exploration; the orthopedic consultant documents the decision and arranges close inpatient neurovascular observation.²⁹⁸

◆ **PEARL – THE PINK HAND IS STILL AN EMERGENCY**

A pink pulseless hand remains an orthopedic emergency; its perfusion after reduction determines the need for immediate vascular exploration.²⁹⁸

▲ **ADMIT · DISCHARGE · REFER**

- **Admit** — all fractures presenting with a pulseless limb.
- **Discharge** — only after definitive management and consultant confirmation that perfusion is restored.

- **Refer** — orthopedics immediately; vascular surgery immediately for ischemia or suspected persistent arterial injury.

Joint Dislocation MINUTES

LIMB-THREATENING

Reduce traumatic dislocations promptly; compromised perfusion or threatened skin needs immediate intervention.

▀ DEFINE

- Traumatic loss of joint alignment, with or without an associated fracture.

▀ APPROACH

- Assess the mechanism, associated injuries, deformity, skin condition, and distal neurovascular function; document findings before and after reduction.
- Obtain joint radiographs to identify associated fractures — unless imaging would delay treatment of vascular compromise or threatened skin.

- Obtain post-reduction imaging and repeat the neurovascular examination.

⚠ RED FLAGS

- Absent or diminished pulses, impaired perfusion, neurological deficit, an open injury, or threatened skin → **immediate orthopedic assessment.**

▀ MANAGEMENT

- Analgesia, then prompt reduction by an orthopedic surgeon under appropriate sedation or anesthesia with monitoring — emergency room or operating theatre depending on hospital capability.
- Immobilize after reduction and obtain post-reduction radiographs.
- Definitive management depends on the joint, injury pattern, and any associated fracture.

✦ PEARL – THE HIP CLOCK

A delay in reduction > 6 hours greatly increases the risk of avascular necrosis in traumatic hip dislocation.³⁰⁰

▲ ADMIT · DISCHARGE · REFER

- **Admit** — hip or true knee (not patellar) dislocations; open, irreducible, unstable, or neurovascularly compromised injuries; associated fractures needing inpatient treatment.
- **Discharge** — selected uncomplicated injuries after stable reduction, satisfactory imaging, normal neurovascular findings, and arranged follow-up; advise return for increasing pain, swelling, or numbness.
- **Refer** — orthopedics immediately; vascular surgery immediately for ischemia or suspected persistent arterial injury.

Unstable Pelvic Injury

MINUTES

LIFE-THREATENING

Treat a suspected unstable pelvic injury as major trauma — stabilization and hemorrhage control begin before radiological confirmation.

DEFINE

- A pelvic-ring injury with mechanical instability, usually anterior and posterior ring disruption. Mechanical and hemodynamic instability are distinct: a child may initially maintain circulation despite an unstable fracture.³⁰¹

APPROACH

- Pediatric trauma primary survey (see [Major Trauma — Primary Survey](#)).
- Look for pelvic pain, deformity, limb-length asymmetry, and shock.
- Avoid pelvic springing and repeated stability testing; minimize unnecessary movement.

- Apply a pelvic binder as soon as possible; arrange targeted CT once sufficiently stable, with the trauma team — imaging must not delay stabilization or hemorrhage control.

⚠ RED FLAGS

- Persistent tachycardia, poor perfusion, hypotension, altered consciousness, or deterioration despite resuscitation → major hemorrhage.
- Open injury, blood at the urethral meatus, or lower-limb neurological abnormality → immediate specialist assessment.

▀ MANAGEMENT

- Activate the pediatric trauma and major-hemorrhage pathways; resuscitate while arranging urgent orthopedic, pediatric-surgical, anesthetic, and intensive-care support.
- A sheet applied over the greater trochanters serves as a pelvic binder when an unstable pelvic fracture is suspected — do not delay application for imaging.

- Ongoing hemodynamic instability despite resuscitation and binding needs immediate hemorrhage control — operative intervention or angioembolization; orthopedics determines temporary and definitive pelvic stabilization.
- Seek urological or pediatric-surgical advice before urethral catheterization if urethral injury is suspected.
- Open injuries require antibiotics and tetanus assessment.

✦ PEARLS

- An apparently reduced pelvis on imaging obtained with a binder on does not establish fracture stability.
- A negative FAST (Focused Assessment with Sonography for Trauma) examination does not exclude pelvic bleeding.

▀ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected or confirmed unstable pelvic injuries; may need intensive care.

- **Discharge** — only after definitive stabilization, mobility, and a follow-up plan.
- **Refer** — triage to a pediatric-capable major trauma service; coordinate transfer and hemorrhage control with orthopedics and pediatric surgery.

Spine Fracture with Neurological Deficit

MINUTES

Protect spinal alignment, support oxygenation and circulation, and obtain immediate spinal-surgical advice.

DEFINE

- Traumatic cervical, thoracic, or lumbar fracture or dislocation with new motor, sensory, or sphincter dysfunction suggesting spinal cord, conus medullaris, or cauda equina injury. The same emergency pathway applies when neurological findings occur without a visible

fracture (SCIWORA — spinal cord injury without radiographic abnormality).

▲ APPROACH

- Pediatric trauma primary survey with strict cervical-spine immobilization and log-roll precautions (see [Major Trauma — Primary Survey](#)).
- Document the time and evolution of weakness, sensory loss, and bladder or bowel symptoms; repeat the neurological assessment after transfers and any deterioration.
- Urgent radiographs and/or CT for injury characterization; urgent MRI for suspected cord or ligamentous injury.
- Do not delay referral to pediatric orthopedic spine or neurosurgery in order to complete imaging.

▲ RED FLAGS

- Progressive weakness, a sensory level, saddle sensory loss, new urinary retention or incontinence, or respiratory compromise → immediate escalation.

- Hypotension with bradycardia may be neurogenic shock — but hemorrhage must also be excluded.

▀ MANAGEMENT

- Maintain age-appropriate spine immobilization and minimize unnecessary movement.
- Analgesia and monitored resuscitation; correct hypoxemia and hypotension promptly.
- Involve intensive care early for mean arterial pressure support — in older children a lower limit of **75–80 mmHg** and upper limit of **90–95 mmHg**, maintained for **3–7 days** according to injury severity and neurological recovery; there is no established MAP target for young children.³⁰²
- The spine surgeon urgently determines the role of surgical decompression and stabilization, aiming for intervention **within 24 hours** of injury when clinically feasible to optimize neurological recovery.³⁰³

- Do not give high-dose methylprednisolone routinely for traumatic spinal cord injury.³⁰⁴

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all; PICU for cord injury, neurogenic shock, or respiratory compromise.
- **Discharge** — not acutely.
- **Refer** — pediatric orthopedic spine or neurosurgery immediately; pediatric-capable major trauma service.

Pediatric Trauma & TBI

Primary survey & resuscitation · secondary survey & decisions · traumatic brain injury

Trauma Team & Roles MINUTES

Know who leads before the patient arrives — role confusion costs the golden hour.

WHO LEADS – SET LOCALLY

- **Trauma team leader (TTL)** — identified by each hospital's own trauma pathway; the designation differs between hospitals and this guide does not prescribe it. Whoever leads runs the primary survey and resuscitation and keeps the overview; the airway is managed by the most skilled airway operator present. ³¹⁰
- **Pediatric surgery** (where available; otherwise general surgery) — the surgical survey, hemorrhage source control, and performance of any operation. **The decision to operate, to image first, or to transfer is a team decision**, based on shared

situational awareness of the child's physiology and trajectory — not a single-specialty call.^{310,100}

- **No trauma team on site (private hospital, peripheral facility):** stabilize with the primary survey (cABCDE), control hemorrhage, secure oxygenation and access, keep the child warm, and **transfer to the nearest trauma-capable center while resuscitating** — call the receiving team early. Do not delay transfer for imaging or specialist review that cannot change management locally.¹⁰⁰

⚠️ AGREE THE PATHWAY IN ADVANCE

Confirm the trauma call, role allocation, and transfer route locally before the patient arrives — front-door role confusion costs time in the golden hour.

Major Trauma – Primary Survey & Resuscitation

MINUTES

Stop the bleeding, protect the brain, keep them warm — and read the physiology, because children compensate then crash.

▲ HOW CHILDREN DIFFER – ANTICIPATE IT

- **Large head, thin scalp:** head injury is common; scalp-wound blood loss can be significant in a small child.¹⁰⁰
- **Compliant chest wall:** serious lung injury (pulmonary contusion) occurs **without** rib fractures — a normal-looking chest wall does not exclude it.¹⁰⁰
- **Less-protected solid organs:** liver and spleen injure easily in blunt abdominal trauma.¹⁰⁰
- **Great reserve, then sudden collapse:** children hold blood pressure by tachycardia and vasoconstriction, so **hypotension is a late, pre-terminal sign** — act on tachycardia, cool peripheries, and prolonged capillary refill long before the BP falls.¹⁰⁰

- **High surface-area-to-mass:** they cool fast — hypothermia drives coagulopathy. Hypoxia is the commonest cause of traumatic arrest.¹⁰⁰

<C> CATASTROPHIC HEMORRHAGE

Control visible bleeding first: direct pressure, wound packing, a **limb tourniquet (record the exact time applied)** for uncontrolled extremity bleeding, and a **pelvic binder** for a suspected pelvic fracture.²⁶⁸

A – AIRWAY + C-SPINE

- Manual in-line stabilization; open (using a **jaw-thrust maneuver**) and clear the airway.
- Anticipate a difficult airway (large occiput flexes the neck; large tongue). In younger children — especially **< 2 yr** — a **shoulder roll** helps: maintain C-spine precautions and avoid over-extension or flexion.
- Secure the airway if GCS ≤ 8 or it is not protected.¹⁰⁰

B — BREATHING

- Oxygen 10–15 L/min via non-rebreather mask. **Tension pneumothorax is a clinical diagnosis — treat before imaging** (needle then finger thoracostomy). Both decompression sites are correct: **4th–5th intercostal space anterior to the mid-axillary line (ATLS)**, or the **2nd intercostal space, mid-clavicular line (PALS)** — prefer the 2nd space, mid-clavicular line in **younger children and infants**¹⁰⁰
- Seal an open pneumothorax; suspect massive hemothorax and pulmonary contusion.¹⁰⁰

C — CIRCULATION

- Two large-bore IV lines; go early to **intraosseous** if access fails.
- Find the bleeding — **chest, abdomen, pelvis, long bones, and the floor**. Resuscitate with **blood, not repeated crystalloid**; consider tranexamic acid within 3 h.²⁷⁰
Activate the massive transfusion protocol when required.¹⁰⁰

D – DISABILITY

GCS or AVPU (alert / voice / pain / unresponsive), pupils, and a **glucose** (do not miss hypoglycemia). Manage raised ICP with **urgent neurosurgery consult** — see the Traumatic Brain Injury card.⁸²

E – EXPOSURE / ENVIRONMENT

Fully undress to examine and log-roll, then **cover and actively warm** — every degree of cooling worsens clotting.²⁶⁸

RESUSCITATION – BLOOD, NOT CRYSTALLOID

Drug / product	Dose	Notes
Packed red cells	10 mL/kg boluses, reassess after each	First-line for hemorrhagic shock; balanced 1:1:1 (RBC:plasma:platelets) in massive hemorrhage; activate the hospital massive transfusion protocol (pre-agreed with the blood bank) ²⁶⁹
Crystalloid (warmed)	≤ one 20 mL/kg bolus, only if blood not yet available	Each further crystalloid bolus → more coagulopathy, ventilator and ICU days ¹⁰⁰

Drug / product	Dose	Notes
Tranexa mic acid (TXA)	15 mg/kg (max 1 g) IV over 10 min, then 2 mg/kg/h; > 40 kg → 1 g then 1 g over 8 h	Within 3 h of injury ²⁷⁰

◆ RESUSCITATION REASONING

- Estimated blood volume ~70 mL/kg in a child (80–90 in an infant), so small losses matter.¹⁰⁰
- Move to blood early and limit crystalloid; transfuse to a **HEMOGLOBIN GOAL OF 7 G/DL. TARGET A SYSTOLIC BP NO LESS THAN $70 + 2 \times \text{AGE (YR)}$ MMHG** (1–10 yr band; use the age-specific systolic thresholds in the Shock card for other ages). **ADDRESS ONGOING LOSSES. AVOID HYPOTENSION.**^{6,100}

△ THE LETHAL TRIAD – AND REASSESS

- Hypothermia, acidosis, and coagulopathy feed each other: keep the child warm, give blood early, correct coagulopathy.²⁶⁸

- If the patient deteriorates at any point, **go back to A** and work through again.¹⁰⁰

Major Trauma – Secondary Survey & Decisions

MINUTES

Once the primary survey is controlled — find the injuries and make the calls.

▲ SECONDARY SURVEY – HEAD-TO-TOE + AMPLE HISTORY (ALLERGIES, MEDICATIONS, PAST HISTORY, LAST MEAL, EVENTS)

- Start **only after <C>ABCDE is controlled** and resuscitation underway. Examine front and back (log-roll): scalp and face, neck, chest, abdomen, pelvis and perineum, back, limbs, and a focused neurological exam.¹⁰⁰
- **AMPLE** history: **A**llergies, **M**edications, **P**ast history, **L**ast meal, **E**vents/mechanism. Mechanism predicts the injuries — ask for it explicitly.¹⁰⁰

DECISION

Abdomen — does this child need a CT?

PECARN (Pediatric Emergency Care Applied Research Network)

intra-abdominal injury rule — 7

features:

abdominal wall trauma / seatbelt sign

GCS $\leq$ 14

abdominal tenderness

chest-wall trauma

abdominal pain

decreased breath sounds

vomiting.⁹⁹

NONE present

Very low risk for injury needing intervention — CT can usually be safely avoided; observe with serial examination.⁹⁹

ANY present / unstable

Not low-risk — image (CT if stable) and involve surgery. A **negative FAST does not exclude** intra-abdominal injury in children (FAST = Focused Assessment with Sonography for Trauma — bedside ultrasound for free fluid in the abdomen and pericardium).⁹⁹

CHEST

CXR for hemo/pneumothorax and pulmonary contusion; place a **chest drain** for a hemothorax or significant pneumothorax. (*Tension pneumothorax is already decompressed in the primary survey.*)¹⁰⁰

PELVIS

Confirm the binder is correctly positioned and **image** (pelvic X-ray ± CT); do **not** repeatedly spring the pelvis.¹⁰⁰ (*The binder is applied in the primary survey.*)

C - SPINE

- Clear with the **PECARN** 3-tier rule (clinical clearance / radiograph / CT).
- **SCIWORA** (spinal cord injury without radiographic abnormality) is commoner in children — a normal X-ray does not exclude cord injury; keep the spine immobilized and call neurosurgery/spine urgently if there is any neurological deficit (weakness, sensory change, or bladder/bowel dysfunction), even with normal imaging.⁸¹

LIMBS & ANALGESIA

Splint long-bone fractures and check distal pulses. Give **timely analgesia** (intranasal or IV opioid) — pain is routinely undertreated in injured children.¹⁰⁰

⚠ ALWAYS CONSIDER INFLICTED INJURY

- Be alert when the history is inconsistent with the injury or the child's development, presentation is delayed, or the pattern is suspicious (bruising or fracture in a non-mobile infant, torn frenulum, multiple injuries of different ages).
- Document carefully and escalate through local safeguarding.¹³⁹

◦ ESCALATE & TRANSFER EARLY

- With no in-house pediatric surgery, involve the trauma team, anesthesia, intensive care and the receiving **trauma center early** — do not wait for the patient to deteriorate.¹⁰⁰
- Package for transfer: airway secured, hemorrhage controlled, the child warm, and blood en route.

- Do **not** delay transfer for a CT that will not change what you do before the child leaves.¹⁰⁰

Traumatic Brain Injury MINUTES

Decide who needs a CT; in severe injury, stop the second insult.

▲ IMAGING DECISION – PECARN

- Use the **PECARN** rule to select children for CT after blunt head trauma and safely avoid imaging in the very-low-risk group. It is **age-stratified** — use the correct **age column** below. PECARN was not derived or validated for suspected **abusive (inflicted)** head trauma — keep a low threshold to image regardless of the rule when abuse is suspected.⁸¹

PECARN HEAD-CT – BY AGE GROUP

Risk tier	< 2 years	≥ 2 years
CT recommended (high risk)	Any of: GCS ≤ 14 , altered mental status / agitation / slow response, or a palpable skull fracture . ⁸¹	Any of: GCS ≤ 14 , altered mental status, or signs of a basilar skull fracture . ⁸¹
CT vs observe (intermediate)	Any of: occipital / parietal / temporal scalp hematoma , LOC (loss of consciousness) ≥ 5 s , severe mechanism , or not acting normally per the parent ⁸¹ — shared decision-making + observation .	Any of: LOC , vomiting , severe mechanism , or severe headache — ⁸¹ shared decision-making + observation .
No CT (low risk)	None of the above.	None of the above.

◆ DECISION TOOLS

- PECARN pediatric head-CT rule —

mdcalc.com



- Severe-TBI management — Brain Trauma Foundation pediatric

guideline, 3rd ed — doi.org



SEVERE TBI (GCS $\leq$ 8)

- Secure the airway.
- Keep **SpO₂ $\geq$ 94%** and normocapnia (PaCO₂ 35–40 mmHg).
- Avoid hypotension — **target SBP no less than $70 + 2 \times \text{age}$ mmHg (1–10 yr band; age-specific thresholds otherwise).**⁶ Higher blood-pressure targets may be beneficial in isolated severe TBI, but the threshold, and whether to target SBP or MAP, are not established — set targets with neurosurgery and the receiving intensive care unit (PICU or ICU, per hospital policy).³¹¹
- Head midline, **elevate the head of the bed 30°**; treat pain, agitation, glucose, and temperature.⁸²



IMPENDING HERNIATION – BRIDGE TO DEFINITIVE CARE

- **Recognize:** new pupil asymmetry or a fixed dilated pupil, Cushing response (bradycardia + hypertension + irregular breathing), extensor posturing, or a falling GCS.
- **Bridge:** head up 30° and midline; secure oxygenation and normocapnia; avoid hypotension; give **one hyperosmolar bolus** (3% saline 2–5 mL/kg, or mannitol 0.5–1 g/kg if not hypovolemic); brief hyperventilation **only** for signs of herniation, as a temporizing measure.⁸²
- **Hand over** for urgent CT and neurosurgical review; ongoing intracranial-pressure and perfusion management is intensive-care therapy (PICU or ICU, per hospital policy) and is beyond the scope of this guide.

Agent	Dose	Notes
Hypertonic saline 3%	2–5 mL/kg bolus (impending herniation)	First-line hyperosmolar bridge; infusion regimens are intensive-care therapy ⁸²
Mannitol	0.5–1 g/kg over 15–20 min	Alternative; caution if hypovolemic or hypotensive ⁸²
Levetiracetam	20–55 mg/kg IV (max 4.5 g), then 10 mg/kg BID	Prophylaxis to reduce early post-traumatic seizures in severe TBI — a short course (~7 days); phenytoin / fosphenytoin an alternative ⁸²

⚠ PREVENT THE SECOND INSULT

- Hypoxia and hypotension each independently worsen outcome — avoid both.
- Do **not** routinely hyperventilate (vasoconstriction lowers cerebral blood flow).
- Avoid hyperthermia and treat seizures promptly.⁸²

▀ ADMIT · DISCHARGE · REFER

- **Admit** — all moderate/severe TBI and any abnormal CT to intensive care (PICU or ICU, according to each

hospital's internal policy) with neurosurgical review. Until the child leaves the department, care is led by the consultant responsible at the patient's location — in the emergency department, the trauma team leader.

- **Discharge** — minor head injury meeting a validated low-risk rule, observed, with head-injury advice.
- **Refer** — neurosurgery for surgical lesions or deterioration; safeguarding if abuse is suspected.

Child Protection & Safeguarding

ASSESS

AAP • NICE NG76

The injury that doesn't fit the history — recognizing it once may prevent a fatal second event.

▶ RECOGNIZE (RED-FLAG PATTERNS)

- **Any bruise in a not-yet-mobile infant** is a sentinel injury until proven otherwise. Use **TEN-4-FACEs**: bruising of **T**orso, **E**ars, **N**eck in a child ≤ 4 years, **any** bruising < 4

months, or of the **Frenulum**, **Angle of jaw**, **Cheek**, **Eyelid**, **Sclera**, or **patterned marks**.²⁶⁷

- History that is **absent, inconsistent, or doesn't match the developmental stage or the injury**; delayed presentation; injuries of different ages.¹³⁹

⚠ **ABUSIVE HEAD TRAUMA**

- An infant with unexplained irritability, vomiting, apnea, seizures or reduced consciousness may have **abusive head trauma** — the triad of subdural hemorrhage, **retinal hemorrhages** and encephalopathy.¹³⁹
- Image the brain and arrange a dilated fundoscopy; it is easily mistaken for sepsis or a feeding problem.¹³⁹

✦ **EXAMINE, DOCUMENT, ESCALATE**

- Fully undress and examine; document injuries factually with sizes and a body map (photograph per policy).

- In children < 2 years with suspected physical abuse, arrange a **skeletal survey** and consider coagulation studies to exclude mimics.¹³⁹
- Care for the child first; you do not need to be certain to refer.¹³⁹

◆ **LOCAL & LEGAL** Follow Kuwait MOH child-protection procedures — involve the hospital child-protection team/social work and make the mandatory report; ensure the child is in a safe place and that siblings are considered.¹⁴⁸

▴ **WORKUP**

- **Skeletal survey** — all children < **24 months**; **2–5 years selectively** (radiographs of areas of concern; full survey if multiple or occult injuries are suspected); not useful > 5 years. **Repeat limited survey at ~2 weeks** when the first is negative or equivocal — it reveals healing rib and metaphyseal fractures.^{316,139}
- **Neuroimaging** — **immediate non-contrast head CT** for any neurological symptom or sign, scalp swelling, or a low hemoglobin (the strongest predictors of an abnormal

CT: head swelling OR 46.7, neurological impairment OR 20.6, low hemoglobin OR 11.8).³¹⁸

Neurologically normal: CT head is still usually appropriate < 24

months — occult intracranial injury is found in up to 37% of high-risk infants (rib or multiple fractures, facial injury, age < 6 months).^{316,317,319}

Where rapid non-sedated MRI is available and expertly reported, **MRI may replace CT as first-line** in a stable infant whose injuries are unrelated to the head.³¹⁸ **MRI brain ± whole spine at day 2–5** whenever CT shows hemorrhage, parenchymal injury or skull fracture, or neurology persists.^{317,316}

- **Dilated indirect ophthalmoscopy by ophthalmology — within 24 h, no later than 72 h** (hemorrhages fade): for intracranial hemorrhage or injury on imaging, unexplained coma, seizures or altered consciousness in a child < 5 years, visible eye injury, or coagulopathy. **Not a screening test** in the neurologically normal infant — image the brain first.³²⁰
- **Bleeding-disorder evaluation** when bruising or intracranial hemorrhage is what raises the concern: **PT,**

aPTT, CBC with platelets, von Willebrand antigen and activity, factor VIII and IX activity.

Unnecessary when injuries are patterned (hand marks; ear, neck or genital bruising) or clearly non-hemorrhagic (fractures, burns). After transfusion, delay testing until transfused products have cleared; TEG/ROTEM are not acceptable for this purpose.³²¹

- **Occult abdominal injury — screen every child < 5 years: AST, ALT, lipase, urinalysis.**

Contrast-enhanced abdominal CT if AST or ALT > 80 IU/L or abdominal bruising, tenderness or distension — a quarter of injuries with raised transaminases have no abdominal signs.^{322,316}

▴ **ADMIT · REFER**

- **Admit** for safety and investigation whenever abuse is a realistic concern.
- **Call/Refer** — child-protection team, social work, and safeguarding lead; police per local law.

Massive Transfusion & Blood Products

MINUTES

Eur. trauma guideline 2023

Balanced-ratio resuscitation with the per-kg doses — and the two silent killers, ionized hypocalcemia and the cold, acidotic child.

▲ ACTIVATE EARLY – ANY OF

- **> 40 mL/kg** of blood products in 24 h · **~1 blood volume** ($\approx$ 70–80 mL/kg) in 24 h · or **> 20 mL/kg red cells in 2 h** with ongoing bleeding.¹⁰⁰
- Uncontrolled hemorrhage with instability, a high **pediatric age-adjusted shock index (SIPA)**, or clinical judgment — **activate the protocol early, not late.**¹⁰⁰

Component (aim 1:1:1)	Dose	Notes
Packed red cells	10–20 mL/kg	Emergency: uncrossmatched O-negative until type-specific is ready ²⁶⁸
Fresh frozen plasma	10–20 mL/kg	Give in a 1:1:1 ratio with red cells & platelets ²⁶⁹

Component (aim 1:1:1)	Dose	Notes
Platelets	10–20 mL/kg (often 10)	
Cryo-precipitate	5–10 mL/kg	For fibrinogen < 1.5 g/L (or give fibrinogen concentrate) ²⁶⁸
Low-titer O whole blood	Per protocol	Emerging in pediatric trauma ¹⁰⁰ — observational evidence

⚠ THE TWO SILENT KILLERS

Ionized hypocalcemia — citrate in stored products chelates calcium, worsening coagulopathy and cardiac contractility; keep **ionized Ca** $\geq$ ~1.1 mmol/L and recheck after every ~30–40 mL/kg. **The lethal triad** — hypothermia + acidosis + coagulopathy: **warm every product and the child**, correct acidosis, and treat coagulopathy early.²⁶⁸

Adjunct	Dose	Notes
Tranexamic acid (TXA)	15 mg/kg IV (max 1 g) over 10 min, then 2 mg/kg/h for 8 h (max 1 g)	Give within 3 h of injury; consider in TBI with GCS < 12 before intracranial bleed is excluded ²⁷⁰
Calcium gluconate	50–100 mg/kg IV (max 2–3 g)	Peripheral-safe

Adjunct	Dose	Notes
Calcium chloride	20 mg/kg IV (max 1 g)	Central line preferred ²⁶⁸ — 3× the elemental calcium of gluconate

▴ TARGETS

- **Fibrinogen $\geq 1.5\text{--}2.0$ g/L · platelets $> 50 \times 10^9/\text{L}$ (> 100 with TBI or ongoing bleeding) · **INR < 1.5 .****

✦ DON'T DROP THE PRESSURE

- In children, and especially with TBI or threatened perfusion, maintain age-appropriate blood pressure throughout resuscitation — **target SBP no less than $70 + 2 \times \text{age (yr)}$ mmHg**, address ongoing losses, avoid hypotension.^{6,268}
- Use **viscoelastic testing (TEG — thromboelastography / ROTEM — rotational thromboelastometry; bedside whole-blood clotting tests)** to guide which component to give, where available, and watch for **hyperkalemia** with large-volume or irradiated red cells.²⁶⁸

▲ **WORKUP – SEND AT ACTIVATION, REPEAT EVERY 30-60 MIN**

- CBC (hemoglobin, platelets), **fibrinogen**, PT/INR ± aPTT, **ionized calcium**, blood gas with lactate / base deficit, type & crossmatch; **TEG/ROTEM** if available; monitor temperature and potassium.²⁶⁸

◆ **LOCAL & EVIDENCE**

- Activate your hospital massive-hemorrhage protocol and follow its ratios and dosing.
- Pediatric ratios and thresholds are largely **EXTRAPOLATED FROM ADULT RCTS** (PROPPR, CRASH-2/CRASH-3); the pediatric TXA signal (PED-TRAX) is observational.^{269,270}
- Verify every dose against local protocol.
- Sources: European guideline on major bleeding & coagulopathy after trauma, 6th ed. (Rossaint 2023); ATOMAC pediatric MTP; PED-TRAX; Transfusion Ontario pediatric appendix.²⁶⁸

Renal & Urology

AKI · HUS · hypertensive emergency · acute scrotum · foreskin emergencies · priapism · hematuria

Acute Kidney Injury HOURS

Abrupt fall in kidney function — rising creatinine and/or falling urine output.

RECOGNIZE & DEFINE

- Suspect in any child with oliguria/anuria, rising creatinine, edema or fluid overload, or unexplained electrolyte disturbance.¹⁵⁹
- **KDIGO (Kidney Disease: Improving Global Outcomes) stages AKI 1–3** — Stage 1 is the threshold that diagnoses AKI (see the table below).¹⁵⁹

KDIGO STAGING – GRADE BY THE WORSE OF CREATININE OR URINE OUTPUT¹⁵⁹

Stage	Serum creatinine	Urine output
1	1.5–1.9× baseline, or a rise ≥ 26.5 $\mu\text{mol/L}$ (≥ 0.3 mg/dL)	< 0.5 mL/kg/h for 6–12 h
2	2.0–2.9× baseline	< 0.5 mL/kg/h for ≥ 12 h
3	$\geq 3.0\times$ baseline, or ≥ 353.6 $\mu\text{mol/L}$ (≥ 4.0 mg/dL), or dialysis started, or – if age < 18 y – eGFR < 35 mL/min/1.73 m ²	< 0.3 mL/kg/h for ≥ 24 h, or anuria ≥ 12 h

FIND THE CATEGORY – IT DRIVES TREATMENT

PRE-RENAL • impaired perfusion (commonest)

Hypovolemia, shock, sepsis — poor perfusion. → **Restore perfusion:** if volume loss, give fluid; if another cause, treat that cause. Reversible if treated early.¹⁵⁹

HUS (see [Hemolytic Uremic Syndrome](#)), glomerulonephritis, acute tubular necrosis, nephrotoxins. → **Supportive care; treat the cause.** ¹⁵⁹

Posterior urethral valves, stones, neurogenic bladder. → **Insert a urinary catheter first**, then image. After relief, watch for **post-obstructive diuresis** (urine > 3.5–5 mL/kg/h) — replace fluid/electrolytes with nephrology. ¹⁵⁹

► **MANAGE – VOLUME STATUS DRIVES FLUID THERAPY**

Assess volume status first, then treat by category:

- **Hypovolemic / intravascularly deplete** (cool peripheries, tachycardia, prolonged capillary refill, low BP, weight loss) — resuscitate with **isotonic crystalloid**.
- **Euvolemic** (well-perfused, no edema, no fluid-overload signs) — match intake to **insensible losses plus measured output**; do **not** apply routine 4-2-1 (Holliday–Segar) maintenance during established AKI. ¹⁵⁹

- **Hypervolemic / overloaded** (edema, hypertension, raised JVP, pulmonary crackles or a gallop, weight gain) — **restrict fluids**; diuretics do not aid recovery but may be used for established hypervolemia.¹⁵⁹

Stop nephrotoxins (NSAIDs, aminoglycosides, contrast) and adjust drug doses. Watch potassium, sodium, acid–base; **avoid potassium in maintenance fluids unless treating hypokalemia.**¹⁵⁹ Involve nephrology for KDIGO stage 2–3 or worsening AKI; arrange structured follow-up (stage 2–3, AKI $\geq$ 7 days, incomplete recovery, pre-existing CKD, or recurrent AKI).¹⁵⁹

🚩 WHEN TO CALL NEPHROLOGY / PICU FOR DIALYSIS – THE "AEIOU" TRIGGERS

Refer urgently for dialysis / CRRT when any of these will **not** correct with medical treatment:

- **A — Acidosis:** severe metabolic acidosis (pH < 7.1) despite bicarbonate / ventilation.¹⁵⁹
- **E — Electrolytes:** hyperkalemia that keeps rising or recurs despite full medical therapy.
- **I — Intoxication:** a dialyzable poison / drug (salicylate, methanol, ethylene glycol, lithium, theophylline).¹⁵⁹

- **O — Overload:** fluid overload / pulmonary edema — **or severe hypertension driven by volume overload** — not responding to fluid restriction + diuretics (especially if it limits feeding or ventilation).
- **U — Uremia:** symptomatic — encephalopathy, pericarditis, or bleeding.¹⁵⁹

The clinical picture — not the creatinine or urea number — decides.

Recognize early and call; the specialist chooses the modality (hemodialysis, CRRT, or peritoneal dialysis).¹⁵⁹

⚠ CROSS-REFERENCE

Treat life-threatening hyperkalemia immediately — see [Acute Electrolyte Emergencies](#) (Endocrine & Metabolic).

▀ ADMIT · DISCHARGE · REFER

- **Admit** — all but mild pre-renal AKI; oliguria, hyperkalemia, fluid overload, or acidosis → high-dependency.
- **Discharge** — resolving pre-renal AKI with normalizing function and follow-up.

- **Refer** — nephrology for KDIGO stage 2–3, no clear cause, or need for dialysis.

Hemolytic Uremic Syndrome HOURS

Triad: microangiopathic hemolytic anemia + thrombocytopenia + acute kidney injury.

RECOGNIZE

- **Typical (STEC):** preceded by bloody diarrhea; pallor, reduced urine, edema, petechiae, irritability.²⁷¹
- Blood film: fragmented red cells (schistocytes), low platelets, negative direct antiglobulin test, high lactate dehydrogenase.²⁷¹
- **Atypical (non-diarrheal) HUS:**
STEC-negative; any age. **Primary** — inherited or antibody-mediated complement dysregulation (low C3, family history, relapsing course).
Secondary — invasive *S. pneumoniae* (pneumonia/empyema, meningitis, bacteremia; **direct antiglobulin test positive; avoid plasma-containing products** — use washed red cells and platelets; treat the infection), HIV, drugs

(chemotherapy, calcineurin inhibitors, ionizing radiation), systemic disease (lupus, antiphospholipid syndrome, scleroderma), and cobalamin C defect (methylmalonic aciduria with homocystinuria — check homocysteine and methylmalonic acid in infants). The underlying condition is usually already known; this guide details the complement-mediated form.²⁷¹

▲ WORKUP

- CBC + film, U&E/creatinine, LDH, haptoglobin, direct Coombs (negative).
- **Send C3, C4, CH50 and ADAMTS13** (ADAMTS13 = the von Willebrand factor–cleaving enzyme; **severely low confirms TTP (thrombotic thrombocytopenic purpura), normal points to HUS**)²⁷¹ **before** any plasma therapy or eculizumab. **If available, also send anti-factor-H antibodies and complement-gene screening** (*CFH*, *CD46/MCP*, *CFI*, *C3*, *CFB*, *THBD* and *DGKE*) to confirm complement dysregulation, guide prognosis and estimate recurrence risk, including after transplantation

— results take weeks and must not delay treatment. Not currently available in Kuwait: store samples and send out.^{271,323}

- Stool culture for *E. coli* O157 and Shiga-toxin PCR/antigen when diarrhea is present.²⁷¹

SUPPORTIVE CARE IS THE MAINSTAY

- Careful fluid and electrolyte management — in suspected STEC-HUS, early isotonic volume expansion before oligo-anuria may reduce kidney injury; once oliguric or fluid-overloaded, restrict fluids and involve nephrology.²⁷¹
- Manage the acute kidney injury (see [Acute Kidney Injury](#)); transfuse packed red cells for symptomatic anemia or Hb < 6–7 g/dL (platelets only for active bleeding or a procedure); dialysis as needed — dialysis does not change STEC-HUS prognosis.²⁷¹
- Monitor closely through the **5–14-day peak period**: daily CBC, creatinine, electrolytes and LDH, plus urine output, blood pressure and weight.²⁷¹



DECISION

**Typical (STEC) or atypical
(complement-mediated)?**

TYPICAL • STEC

Supportive care only. Benefit of eculizumab in STEC-HUS is unproven — not routine. The only placebo-controlled RCT (100 children) showed no improvement in the acute course, dialysis or extrarenal disease; use, if any, is a nephrology decision in severe neurological involvement. ^{271,325}

Complement inhibition (eculizumab/ravulizumab) — nephrology directed. **Meningococcal risk is > 1,000-fold and vaccination alone does not prevent it:** give **MenACWY + MenB $\geq$ 2 weeks before the first dose**; if treatment cannot wait, vaccinate and start **antibiotic prophylaxis (e.g., penicillin V) for the duration of therapy**. Boosters: MenACWY every 5 years (every 3 years if < 7 years at the last dose); MenB 1 year after the series, then every 2–3 years while on treatment. **Any fever = emergency blood cultures and IV ceftriaxone** — 14 of 16 US cases occurred in vaccinated patients, most from non-groupable strains.^{326,327}

⚠️ AVOID

- Antibiotics in STEC (may increase toxin release — generally avoided), antimotility agents (loperamide), and NSAIDs.²⁷¹
- Do not transfuse platelets unless active bleeding or a procedure is planned.²⁷¹

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → monitoring and supportive care; anuria/severe → high-dependency. **PICU** for neurologic symptoms or hemodynamic instability; involve **nephrology** early for atypical HUS.
- **Discharge** — not acutely; after hematologic and renal recovery.
- **Refer** — nephrology (dialysis); atypical HUS → a **nephrology specialist** for complement inhibition / eculizumab.

Hypertensive Emergency MINUTES

Severe hypertension **with** new or progressing end-organ damage (encephalopathy, seizures, facial palsy, severe headache; heart failure or chest pain; acute kidney injury; retinopathy or blurred vision).

▲ ASSESS FOR END-ORGAN DAMAGE

- **Brain:** headache, vomiting, altered mental status, seizures, visual symptoms, focal deficits.

- **Heart:** tachycardia, gallop, pulmonary edema, heart failure, chest pain (older children).
- **Kidney:** AKI, oliguria, hematuria, proteinuria.
- **Eyes:** blurred vision, retinal hemorrhages, papilledema.⁴⁰

▀ WORKUP

- CBC, electrolytes, urea, creatinine.
- Urinalysis ± urine protein.
- ECG.
- Chest radiograph if heart failure suspected.
- Echocardiography if LV dysfunction suspected.
- Fundoscopy if available.
- Renal ultrasound ± Doppler when a renal cause is suspected.
- CT/MRI brain only if neurological symptoms or concern for intracranial pathology.⁴⁰

✦ PEARL

In children, severe hypertension is **secondary until proven otherwise** — renal disease is the commonest cause.⁴⁰

LOWER GRADUALLY – CONTROLLED DESCENT

- A true hypertensive emergency needs PICU/high-dependency monitoring with a titratable IV infusion under specialist guidance.
- Reduce blood pressure by no more than ~25% in the first 8 hours, then normalize over 24–48 hours.
- Abrupt drops risk cerebral, renal, and retinal ischemia.⁴⁰



TREAT COMPLICATIONS

Seizures → benzodiazepine. Fluid overload (e.g., acute glomerulonephritis) → furosemide. Monitor in PICU.⁴⁰

MEDICATIONS – TITRATABLE IV AGENTS PREFERRED

Drug	Dose	Route & notes
Labetalol	bolus 0.2–1 mg/kg (max 40 mg); infusion 0.25–3 mg/kg/h	IV; preferred first-line (best option) in Kuwait ; avoid in asthma, heart failure ⁴⁰
Hydralazine	0.1–0.5 mg/kg/dose (max 20 mg)	IV/IM every 4 h; second-line ; may cause reflex tachycardia

Drug	Dose	Route & notes
Sodium nitropruside	0.3–8 mcg/kg/min	IV; reserve — cyanide risk with prolonged use/renal impairment ⁴⁰
Esmolol	50–300 mcg/kg/min (after load)	IV; short-acting; may cause bradycardia/bronchospasm
Furosemi de	1–2 mg/kg (max 40–80 mg)	IV; if fluid overloaded (e.g., post-streptococcal glomerulonephritis)

◆ **NOTE** Oral or sublingual nifedipine can cause a precipitous, uncontrolled fall and is **NOT** recommended for a true hypertensive emergency.⁴⁰

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all true emergencies → PICU with a titratable infusion.
- **Discharge** — not from the emergency; controlled blood pressure with a plan.
- **Refer** — nephrology/cardiology for cause and long-term control.

Nephrotic Syndrome HOURS

Heavy proteinuria, low albumin, and edema — usually steroid-responsive, but watch the complications. Doses follow the International Pediatric Nephrology Association (IPNA) 2023 guideline.

► RECOGNIZE

- **The triad:** nephrotic-range proteinuria (urine protein/creatinine > 200 mg/mmol, or $\geq 3+$ on dipstick), **hypoalbuminemia < 30 g/L (< 3.0 g/dL)**, and edema (periorbital → generalized, ascites, scrotal/labial); usually with hyperlipidemia.⁵⁰
- In a child **1–10 years** without the atypical features below, it is almost always steroid-responsive minimal-change disease — treat without a biopsy.⁵⁰

CONFIRM & ASSESS

- Confirm proteinuria (first-morning-void protein/creatinine ratio preferred over a 24-hour collection); send albumin, renal function and electrolytes, lipids; weigh, record blood pressure, assess volume status.
- Assess clinically for active infection before commencing high-dose corticosteroids; investigate according to symptoms and risk factors rather than performing routine infection screening in an otherwise well child.⁵⁰



FIRST PRESENTATION

- Oral **prednisolone 60 mg/m²/dose (≈ 2 mg/kg/dose, max 60 mg)** once daily for 4–6 weeks, then 40 mg/m² (max 40 mg) on alternate days for a further 4–6 weeks (total 8–12 weeks)
- Add a no-added-salt diet (no fluid or protein restriction) and ensure pneumococcal vaccination; routine antibiotic prophylaxis is not recommended — instead treat suspected infection promptly, including spontaneous bacterial peritonitis.⁵⁰

⚠️ COMPLICATIONS TO PRE-EMPT

- **Hypovolemia** can hide behind gross edema — abdominal pain, cool peripheries, tachycardia, oliguria; needs urgent senior/nephrology input; treat with **fluid resuscitation if needed and 20% albumin plus a diuretic** (avoid diuretic monotherapy). **Hypervolemia** presents differently — edema with **strong peripheral pulses and warm peripheries**; treat with diuretics and close reassessment.⁵⁰

- Watch for **spontaneous bacterial peritonitis** (pneumococcal) and other sepsis, **venous or arterial thromboembolism** (these children are hypercoagulable), and **acute kidney injury (AKI)**.⁵⁰

▲ ADMIT · DISCHARGE · REFER

- **Admit** — first presentation, hypovolemia, peritonitis or sepsis, thromboembolism, severe edema with respiratory compromise, AKI, or a relapse not responding to oral steroid.
- **Discharge** — steroid responsiveness is **not** required before discharge, provided close OPD follow-up is arranged: a well child with no complications, able to tolerate oral steroids, with the family able to monitor first-morning urine and act on a written plan.
- **Refer** — Nephrology for any atypical feature (age < 1 or > 12 years, persistent hypertension, gross hematuria, low complement (C3), renal impairment), steroid resistance,

or frequent relapse — these may need a biopsy or second-line therapy.⁵⁰

Circumcision – Kuwait Practice

ASSESS

≥ 1 WK • Ped SURGERY

Near-universal in Kuwait — screen first, always give analgesia, and know the complications.

▀ BEFORE YOU CIRCUMCISE – CHECKLIST

- **Examine the genitalia first. Do NOT circumcise** if hypospadias, epispadias, chordee, ambiguous genitalia / suspected DSD, micropenis, or buried / webbed penis — the foreskin may be needed for repair → refer to pediatric surgery / urology.¹¹²
- Contraindicated in an **unstable or sick infant** — confirm normal feeding, elimination and thermoregulation; ensure vitamin K given.¹¹²
- Family history of a bleeding disorder → coagulation studies / defer.¹¹²

- Informed parental consent. In Kuwait usually performed by **pediatric (or general) surgery from ~1 week of age** (and later in infancy / childhood).

◆ ANALGESIA – NEVER OMIT

- **Dorsal penile nerve block** or subcutaneous **ring block** (lidocaine 1% without epinephrine, ≤ 1 mL, 3–4 mg/kg) — success ~90–100%, complications <0.1%.¹¹³
- Add topical EMLA, oral sucrose, non-nutritive sucking and acetaminophen.
- Sucrose or EMLA **alone are not enough**; combination analgesia is best.¹¹³

METHODS

Gomco clamp, Mogen clamp, or Plastibell (ring) device — each with its own technique and complication profile.¹¹²

WOUND CARE

- Clean gently with warm water at each nappy change; avoid soap and wipes (they sting).
- **Gomco / Mogen:** apply petroleum jelly (or antibiotic ointment) to the glans and on the nappy for ~5–7 days to stop the wound sticking. A moist **yellow coating** (granulation) on the glans is **normal healing**, not infection.¹¹²
- **Plastibell:** the ring falls off by itself in ~5–10 days — **do not pull it**; sponge-bathe until it separates¹¹²; seek review if it slips onto the shaft or has not fallen by 10–14 days.
- Heals in ~7–10 days. After healing, gently retract the skin at the groove to prevent adhesions / skin bridges.¹³²

⚠ **COMPLICATIONS – WHEN TO SEEK URGENT REVIEW**

- **Bleeding** more than a few drops (commonest — direct pressure and a compressive dressing around the penis, then pediatric surgery review + coagulation workup)

- spreading redness up the shaft, pus or fever = **wound infection**
- no urine for ~8 h = **retention**¹³²
- Plastibell ring migrated with a **dusky glans** = constriction (emergency)
- trapped / concealed penis, adhesions, skin bridges, or late **meatal stenosis** (deflected stream)¹³² → pediatric surgery / urology.

Acute Scrotum

MINUTES

6-HOUR WINDOW

Sudden scrotal pain is testicular torsion until proven otherwise — a clinical diagnosis; do not wait for imaging.

▀ DEFINE

- **Acute scrotum** = sudden scrotal pain/swelling. **Testicular torsion until proven otherwise** — a time-critical surgical emergency. Bimodal peaks: the neonate and the peri-pubertal boy.¹³¹
- Differential: **torsion of the appendix testis** (blue-dot sign, gradual, milder), **epididymo-orchitis** (dysuria,

gradual, older/sexually active), trauma, idiopathic scrotal edema, incarcerated inguinal hernia.¹³¹

▀ APPROACH – FAVORS TORSION

- **Sudden severe pain** ± nausea/vomiting, **high-riding testis**, **horizontal lie**, **absent cremasteric reflex**, diffuse tenderness.¹³¹
- The **TWIST score** (hard testis, swelling, nausea/vomiting, absent cremasteric reflex, high-riding testis) risk-stratifies but does not replace surgical judgment.¹³¹

⚠ WORK-UP

- **Torsion is a clinical diagnosis — do not delay surgical exploration for imaging.** Color Doppler ultrasound (absent intratesticular flow) is useful only when the diagnosis is equivocal **and** will not delay surgery.¹³¹
- Urinalysis: pyuria or bacteriuria supports bacterial epididymitis — but a **normal urinalysis does not exclude epididymitis, and an**

abnormal one does not exclude torsion; send a culture.³²⁴

⚠ **RED FLAGS – THE CLOCK**

- Testicular salvage is **time-dependent:** near-universal if detorsion occurs within ~6 hours of onset, falling sharply thereafter.¹³¹
- Neonatal torsion and any delayed presentation need urgent senior/urology input.

▀ **MANAGEMENT**

- **Immediate urology/surgical referral** for scrotal exploration, detorsion, and **bilateral orchidopexy.**¹³¹
- If surgery is delayed, a trained clinician may attempt **manual detorsion** (usually 'opening a book' — medial-to-lateral) **only if experienced** as a temporizing measure while immediate exploration is arranged, guided by pain relief; success does not remove the need for surgery and orchidopexy.¹³¹
- **Consult surgery/urology at once and confirm the diagnosis.** Torsion → surgery. Once torsion is excluded:

appendix-testis torsion → NSAID

analgesia and rest;

epididymo-orchitis → NSAIDs, rest and scrotal support, with **antibiotics only if urinalysis or urine culture shows bacterial infection** (send the culture before treating — prepubertal epididymitis is usually non-bacterial).³²⁴

✦ **PEARL – ABDOMINAL PAIN CAN BE A TORSION**

Examine the testes in every boy with acute lower-abdominal or groin pain — torsion frequently refers pain to the abdomen and is missed when the scrotum is not examined.¹³¹

▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — torsion (confirmed or not excluded) → emergency exploration; incarcerated hernia; epididymo-orchitis only if systemically unwell, an infant, or unable to take oral analgesia.³²⁴
- **Discharge** — once the surgical/urology team has excluded torsion: appendix-testis torsion and epididymo-orchitis go home on NSAIDs ± antibiotics per urine results, review in 48–72 h;

prepubertal epididymitis warrants outpatient renal-tract ultrasound (underlying urogenital anomaly in ~25–30%).³²⁴

- **Refer** — surgery/urology IMMEDIATELY to confirm the diagnosis (torsion → exploration within 6 h).¹³¹

◆ **SEE ALSO** [Testicular Torsion](#) — the time-critical diagnosis to exclude first.

Testicular Torsion MINUTES

ADOLESCENT & NEONATE

Salvage window ~6 hours. A boy's abdominal pain → examine the scrotum.

▲ RECOGNIZE

- Sudden severe scrotal/lower-abdominal pain, nausea/vomiting, swelling. High-riding/horizontal testis, **absent cremasteric reflex**, negative Prehn sign.¹³¹

**Clinical suspicion = urgent exploration.
Don't delay for imaging.**



ACT

- **Immediate urology / pediatric surgery** → exploration, detorsion + bilateral orchidopexy.
- Color Doppler ultrasound **only if it will not delay** surgery.
- Manual detorsion may temporize **only in experienced hands** — do not attempt if untrained (can worsen the torsion), and never let it delay surgical exploration.¹³¹

▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — all → emergency exploration.
- **Discharge** — post-operative.
- **Refer** — pediatric surgery/urology IMMEDIATELY (explore within 6 h); do not delay for imaging.

◆ **SEE ALSO** [Acute Scrotum](#) for the full differential.

Paraphimosis

HOURS

REDUCE PROMPTLY

A retracted foreskin trapped behind the glans — edema then ischemia; reduce promptly.

DEFINE

- **Paraphimosis** — a retracted foreskin trapped behind the glans, cannot be reduced. The constricting band causes progressive glans and distal-foreskin edema, then venous → arterial compromise. A urological emergency.¹³²
- Commonly **iatrogenic** — the foreskin is left retracted after catheterization, cleaning, or examination.¹³²

APPROACH & WORK-UP

- Painful, swollen glans with an edematous 'collar' of foreskin proximal to it. Diagnosis is clinical.
- Distinguish from balanoposthitis (foreskin/glans inflammation without a trapped retracted ring).

RED FLAGS

Increasing edema, dusky glans, or pain out of proportion signal arterial

compromise and impending necrosis —
**reduce promptly; escalate to urology if
reduction fails.** ¹³²

MANAGEMENT

▴ REDUCE IT

- **Analgesia first** — penile block or procedural analgesia.
- **Reduce edema** — sustained manual compression of the glans ($\pm$ a wrap such as saline-soaked gauze, ice, or an osmotic agent such as granulated sugar) for several minutes. ¹³²
- **Reduce the foreskin** — push the glans back with the thumbs while drawing the foreskin forward over it with the fingers.
- **If manual reduction fails** — urology for a dorsal-slit or emergency reduction. ¹³²

✦ PEARL – PREVENTION IS THE CURE

Always return the foreskin to its natural forward position after any catheterization, cleaning, or examination, and teach carers never to force-retract a child's foreskin. ¹³²

▲ ADMIT · DISCHARGE · REFER

- **Admit** — if reduction fails or there is tissue compromise.
- **Discharge** — after successful reduction with follow-up.
- **Refer** — urology for failed reduction or recurrent episodes.

Balanitis / Balanoposthitis & Phimosis

ASSESS

COMMON

Glans/foreskin inflammation and non-retractile foreskin; physiologic phimosis is normal, needs no treatment.

▲ DEFINE

- **Balanitis** = inflammation of the glans; **balanoposthitis** = glans + foreskin. **Phimosis** = non-retractile foreskin — **physiologic** (normal in young boys) or **pathologic** (scarred).
- **Physiologic phimosis is normal:** non-retractile in most infants, becomes retractile with age ($\approx 99\%$ by 17 years). Needs no treatment.¹³²

▲ APPROACH & WORK-UP

- Balanitis: red, swollen, tender glans/foreskin ± discharge, dysuria. Usually hygiene-related or candidal; consider sexually transmitted causes in adolescents.
- Distinguish physiologic phimosis (soft, healthy, pink opening) from **pathologic** (scarred white fibrotic ring — balanitis xerotica obliterans/lichen sclerosus).¹³²
- Clinical diagnosis; swab if recurrent/purulent; urinalysis if voiding symptoms.

▲ RED FLAGS

Pathologic phimosis (white scarred ring / BXO), recurrent balanoposthitis, **ballooning or difficulty voiding**¹³², or **urinary retention** — refer to urology.

MANAGEMENT

Drug	Dose	Route & notes
Balanitis — general	Hygiene, warm saline soaks	Retract gently only as far as comfortable; avoid irritants.
Balanitis — bacterial/ candidal	Topical/oral antibiotic or topical antifungal	Guided by likely organism; treat/screen for STI in adolescents.
Physiologic phimosis	No treatment; do not force-retract	Reassure; resolves with age.
Symptomatic / early pathologic c phimosis	Topical betamethasone 0.05% applied twice daily for 4–6 weeks	≈ 80% success; avoids or defers circumcision. ¹³²
Pathologic c phimosis / BXO / recurrent	Refer for circumcision	Definitive for scarred foreskin or lichen sclerosis. ¹³²

◆ PEARL – HANDS OFF THE YOUNG FORESKIN

Never forcibly retract a young boy's foreskin — it tears, bleeds, and scars, converting harmless physiologic phimosis into pathologic phimosis.¹³²

▀ ADMIT · DISCHARGE · REFER

- **Admit** — rarely — severe infection or urinary retention.

- **Discharge** — most, with hygiene and topical treatment.
- **Refer** — urology for pathological phimosis, recurrent balanitis, or balanitis xerotica obliterans.

Priapism

MINUTES

> 4 H = EMERGENCY

Painful rigid erection > 4 hours; ischemic priapism is a penile compartment syndrome — relieve fast.

DEFINE

- **Priapism** — persistent erection > 4 hours unrelated to stimulation.
Ischemic (low-flow, painful, rigid corpora — penile compartment syndrome, a true emergency) versus **non-ischemic** (high-flow, usually post-traumatic, not urgent).¹³³
- Most pediatric priapism is **ischemic and associated with sickle cell disease**; short self-limiting 'stuttering' episodes often precede a major event.¹³³

▲ APPROACH & WORK-UP

- Rigid, painful corpora cavernosa with a **soft glans and corpus spongiosum** → ischemic. Ask about sickle cell disease and drugs.
- **Cavernosal blood gas** distinguishes: ischemic = dark, hypoxic, acidotic blood. **Color Doppler ultrasound** also separates the two (ischemic = absent/low cavernosal flow; non-ischemic = normal/high flow). **Do not delay treatment for imaging.**¹³³

▲ RED FLAGS – THE CLOCK

- Smooth-muscle edema and atrophy begin as early as ~6 hours; risk of permanent erectile dysfunction rises with duration.¹³³
- Ischemic priapism > 4 hours is a time-sensitive emergency needing immediate urology (and hematology in sickle cell disease) — do not delay urologic intervention for sickle-cell measures.¹³³

MANAGEMENT

▶ RELIEVE THE ERECTION FIRST

- **In sickle cell disease, do not delay penile intervention for exchange transfusion** — urologic relief comes first; sickle measures (hydration, oxygen, analgesia) run concurrently.²⁹⁴
- **Corporal aspiration ± saline irrigation plus intracavernosal phenylephrine** (urology-directed), repeated as needed, with blood-pressure and heart-rate monitoring.¹³³
- Surgical shunting only after aspiration + sympathomimetic have failed.¹³³

Drug	Dose	Notes
Phenylephrine (intracavernosal)	100–200 mcg/dose every 3–5 min (dilute to 100–500 mcg/mL); max ~1 mg	Monitor BP and HR; use with caution — reduced dose in children. ¹³³

✦ PEARL – CALL UROLOGY NOW

Acute ischemic priapism is an emergency; in sickle cell disease relieve

the erection first and give sickle-specific care alongside — not instead.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — ischemic priapism → urgent intervention.
- **Discharge** — after detumescence with follow-up.
- **Refer** — urology (aspiration/irrigation) and hematology in sickle cell disease; do not delay urology.

Gross Hematuria

ASSESS

APPROACH

Confirm true hematuria, then sort glomerular from non-glomerular — that split directs the entire work-up.

▲ DEFINE

- **Gross hematuria** — visibly bloody or tea/cola-colored urine. Confirm **true hematuria** (dipstick positive **and** ≥ 5 RBC/high-power field on microscopy) — exclude mimics:

foods (beetroot), drugs (rifampin), myoglobin, urate, menstrual contamination.¹³⁴

▲ APPROACH – GLOMERULAR VS NON-GLOMERULAR

- **Glomerular** — brown/cola-colored, dysmorphic RBCs or RBC casts, proteinuria, hypertension, edema → glomerulonephritis (post-infectious, IgA nephropathy, Alport). **Recurrent painless gross hematuria** with a fever, viral illness or exercise → most likely **IgA nephropathy**; hematuria with **hearing impairment** (or a family history of kidney failure) → **Alport syndrome** — send genetics first.¹³⁴
- **Non-glomerular** — bright red ± clots → urologic: UTI, hypercalciuria/stones, trauma, benign urethrorrhagia, rarely tumor. **Timing helps:** blood at the **start** of the stream → urethra; at the **end** → bladder. **Benign urethrorrhagia** — prepubertal boys, spots of blood at the end of or between voids, resolves spontaneously.¹³⁵

- Commonest causes overall: UTI and hypercalciuria; gross hematuria without glomerular features is **usually benign**.¹³⁵

⚠ WORK-UP

- Urinalysis with **microscopy** (casts, dysmorphic RBCs), urine culture, **blood pressure**; urine protein:creatinine and calcium:creatinine ratios; serum creatinine/electrolytes; **C3/C4 and anti-DNase B** if glomerulonephritis is suspected; **renal-bladder ultrasound** — stones, polycystic kidney disease, trauma.¹³⁴
- Image for trauma or obstructing clot as indicated.

▀ RED FLAGS

- Hypertension, edema, oliguria, or rising creatinine → glomerulonephritis with acute kidney injury.¹³⁴
- Flank pain with clots (stones), significant trauma, a palpable mass, or systemic symptoms.

▀ MANAGEMENT

- Treat the underlying cause.
Nephrology referral for glomerular features (hypertension, proteinuria, casts, reduced GFR); **urology** for structural/urologic causes.¹³⁴
- Isolated gross hematuria with normal blood pressure, no proteinuria, and normal ultrasound can usually be observed.¹³⁵

✦ PEARL – DIPSTICK + MICROSCOPY + BLOOD PRESSURE

Those three bedside steps triage most children — color and RBC morphology sort glomerular from non-glomerular and direct the whole work-up.¹³⁴

▀ ADMIT · DISCHARGE · REFER

- **Admit** — clot retention, significant bleeding, hypertension, or acute nephritis with impairment.
- **Discharge** — well child with a benign cause and outpatient workup.
- **Refer** — nephrology (glomerular) or urology (structural, stones, or mass).

Acute Post-infectious Glomerulonephritis (PIGN)

ASSESS

2–12 YR

The commonest acute nephritis in children — cola urine, puffy eyes and a high blood pressure 1–6 weeks after an infection, classically streptococcal.

RECOGNIZE

- **Acute nephritic syndrome** — cola/tea-colored urine, periorbital/leg edema, **hypertension**, oliguria; onset **1–3 weeks after pharyngitis** or **3–6 weeks after impetigo** (skin infection is the commoner trigger in hot climates). Group A streptococcus is the classic cause, but other bacteria, viruses, fungi and parasites cause the same immune-mediated nephritis — hence **post-infectious** (infection-related) GN. ^{328,329}
- Ask about recent sore throat, skin sores or scabies; examine the skin, blood pressure and volume status.

▲ WORK-UP

- Urinalysis and microscopy — dysmorphic RBCs, **RBC casts**, proteinuria (usually sub-nephrotic); urine protein:creatinine ratio.
- Creatinine, urea, electrolytes (potassium), albumin, CBC.
- **Low C3 with normal C4** supports the diagnosis; **anti-DNase B** confirms recent streptococcal infection (positive in nearly all; ASO titer adds little); throat or skin swab if there is a current infection.^{328,329}
- Chest X-ray if breathless or overloaded; renal ultrasound only if the picture is atypical.

MANAGE — SUPPORTIVE CARE

- **Manage fluid balance** (strict input/output, daily weight, fluid restriction if oliguric or overloaded) and a **no-added-salt diet**.
- **Treat hypertension** — furosemide 1–2 mg/kg IV/oral for edema, overload or hypertension is usually enough; add a calcium-channel blocker if still hypertensive. Severe hypertension with headache, seizures or visual change → [Hypertensive Emergency](#).³²⁸
- **Antibiotics only for a current infection** (e.g., active pharyngitis or impetigo) — they do not alter the nephritis and are not given for the nephritis itself.³²⁸
- **No corticosteroids or immunosuppression** for uncomplicated disease.³²⁸

⚠ RED FLAGS

- **Hypertensive encephalopathy** (headache, vomiting, seizures, altered consciousness) — about 1 in 10 admissions.³²⁹
- **Pulmonary edema**, hyperkalemia, or a **rapidly rising creatinine** → possible crescentic

glomerulonephritis — nephrology the same day (see [Acute Kidney Injury](#)).^{328,329}

◆ EXPECTED COURSE

Diuresis and normal blood pressure within 1–2 weeks; C3 normalizes by 8 weeks; microscopic hematuria may persist up to a year. Recurrence is rare.³²⁸

▀ ADMIT · DISCHARGE · REFER

- **Admit** — hypertension, oliguria, edema/overload, raised creatinine or hyperkalemia.
- **Discharge** — normal blood pressure and urine output; check blood pressure and urinalysis at 1–2 weeks, C3 at 6–8 weeks.
- **Refer** — nephrology if creatinine keeps rising, nephrotic-range proteinuria, C3 still low at **12 weeks** (consider C3 glomerulopathy), or hematuria/proteinuria beyond 1 year — these may need a biopsy.³²⁸

Muscle breakdown → tea-colored urine, a very high CK, and the twin threats of hyperkalemia and AKI.

▀ RECOGNIZE

- Myalgia, weakness and **dark (tea-/cola-colored) urine** that is **dipstick-positive for blood but shows few red cells** (myoglobin), with **creatinine kinase > 5× the upper limit**. Triggers: viral myositis (influenza), trauma/crush, extreme exertion or heat, seizures, toxins/drugs, and metabolic myopathies.²⁷²

⚠ TWO IMMEDIATE DANGERS

- **Hyperkalemia** — get an ECG and treat at once (see [Acute Electrolyte Emergencies](#)). **Acute kidney injury** — start **early, generous IV fluids** to maintain a good urine output.²⁷²
- Also watch calcium/phosphate and for **compartment syndrome** in a swollen, painful limb. **Routine urinary alkalization and**

mannitol are not recommended.²⁷²

✦ CHASE THE CAUSE

Recurrent or exertion-out-of-proportion rhabdomyolysis, or a family history, suggests an **underlying metabolic myopathy** — refer for metabolic work-up once the acute episode is controlled.²⁷²

▀ WORKUP

- **Creatine kinase** (diagnostic > 5× the upper limit of normal (ULN); **serial** to track the trend; AKI risk climbs > 5000 U/L)²⁷².
- **Potassium + ECG**, calcium, phosphate.
- Renal function, venous blood gas/bicarbonate; urinalysis (**heme-positive with few RBCs** = myoglobinuria) + urine myoglobin.²⁷²
- Uric acid, LDH, AST; metabolic/genetic screen if recurrent or exertion-disproportionate.²⁷²

▲ ADMIT · REFER

- **Admit** — significant CK elevation, any renal impairment or hyperkalemia.
- **Refer** — nephrology if AKI; surgery for compartment syndrome.

Respiratory, Allergy & Immunology

15

Airway, breathing, anaphylaxis, allergy & immunology

Anaphylaxis MINUTES

Acute multisystem allergic reaction — epinephrine first, always.

▲ RECOGNIZE & DIAGNOSE

- **Diagnose anaphylaxis (GA²LEN 2024 criteria) when any one applies:**
 - 1 · **No known allergen** — acute skin/mucosal signs **plus** respiratory **or** cardiovascular involvement.
 - 2 · **Likely or known allergen exposure** — any **two** of:

skin/mucosa · respiratory ·
cardiovascular · severe GI symptoms
(severe crampy pain, repeated
vomiting).

3 · Known allergen exposure —
respiratory involvement **alone**
(non-inhaled allergen) **or**
cardiovascular involvement **alone**
(hypotension, poor perfusion,
collapse) — skin signs may be
absent. ³³¹

- **If in doubt, give epinephrine.**
Epinephrine is given immediately for
suspected anaphylaxis and may be
given on clinical judgment before the
criteria are fully met — the risk of a
missed dose exceeds the risk of an
unnecessary one. ^{331,9}
- Skin: urticaria, flushing, angioedema,
pruritus (absent in up to 20%). ⁹
- Breathing: stridor, wheeze,
hoarseness, falling SpO₂, cyanosis,
increased work of breathing, signs of
respiratory distress, persistent
cough.
- Circulation: poor perfusion,
hypotension, collapse — **reduced BP
= low systolic for age or > 30% drop
from baseline.** Infants: floppiness,
drooling, pallor, sudden behavior
change. ⁹

- GI: severe crampy abdominal pain or repeated vomiting — counts toward the diagnosis, especially after a likely allergen.⁹
- **Not anaphylaxis:** repeated vomiting 1–4 h after a food **without** skin, respiratory or cardiovascular signs → think food protein-induced enterocolitis syndrome (FPIES) — fluids and ondansetron, not epinephrine (see [Cow's Milk Protein Allergy](#)).

1ST LINE

Intramuscular epinephrine
immediately — there is no
contraindication⁹



THEN

- Lay supine, legs up (sit if breathing is hard; never stand suddenly).
- Oxygen 10–15 L/min via non-rebreather mask.
- Stridor, voice change or tongue/lip swelling = impending airway obstruction → call anesthesia/ENT now for early, controlled intubation (see [Difficult Airway](#)).
- Fluid bolus for hypotension.
- Repeat epinephrine every 5–15 min if needed.⁹

refractory after ≥ 2 IM doses

2ND–3RD LINE Refractory after repeated IM epinephrine → urgent PICU/anesthesia support; IV epinephrine infusion only by an experienced team with monitoring · IV fluids · glucagon if on a beta-blocker.⁹

MEDICATIONS – EACH GIVEN SEPARATELY

Drug	Dose	Route & notes
Epinephrine (adrenaline) — 1st line	0.01 mg/kg (max 0.5 mg)	IM, 1 mg/mL (1:1000), anterolateral thigh; repeat every 5–15 min ⁹
Epinephrine autoinjector (limited availability)	0.1 / 0.15 / 0.3 mg	Some emergency departments dispense on senior-staff authorization; otherwise via the allergy clinic. 0.1 (7.5–14 kg) · 0.15 (15–25 kg) · 0.3 (>25 kg). If the 0.1 mg device is unavailable, the 0.15 mg device may be used down to 7.5 kg.
Epinephrine infusion (refractory)	0.1–1 mcg/kg/min (max 10 mcg/min)	IV infusion; titrate to effect under PICU supervision with continuous cardiac monitoring ³³³
Balanced crystalloid	20 mL/kg	IV bolus for hypotension, repeat
Salbutamol (for wheeze)	2.5–5 mg nebulized	Adjunct only — does not replace epinephrine

Drug	Dose	Route & notes
Diphenhydramine (for skin symptoms)	1 mg/kg/dose (max 50 mg/dose)	IV/IM; adjunct only, not first-line. May use cetirizine (age-appropriate dose) as a non-sedating alternative ⁹
Glucagon	20–30 mcg/kg (max 1 mg)	IV over 5 min, only if on a beta-blocker and refractory

⚠ OBSERVE EVERY CHILD BEFORE ANY DISCHARGE DECISION – NO EARLY DISCHARGE FROM THE EMERGENCY DEPARTMENT

- **Minimum 2 h** after full resolution — **only when every criterion is met:** good response to a single IM dose within 5–10 min, symptoms completely resolved, the child leaves with **two in-date autoinjectors** dispensed and demonstrated, and a trained adult supervising. **Observe longer (4–6 h) whenever possible**, and always when any criterion is unmet or the department cannot dispense autoinjectors.^{332,9}
- **Minimum 6 h** if two IM doses were needed or there is a history of a biphasic reaction.³³²

- **Minimum 12 h (usually admit)** if more than two doses were needed, **history of severe asthma**, anaphylaxis with **severe respiratory compromise**, possible continuing allergen absorption, presentation at night/out of hours, or the family cannot recognize and respond to deterioration.³³²
- Routine corticosteroids are no longer recommended (consider glucocorticoids for persistent bronchospasm, or severe symptoms requiring more than 2 doses of IM epinephrine or IV epinephrine).⁹

▲ **ADMIT · OBSERVE · DISCHARGE · REFER**

- **Admit** — more than two epinephrine doses or an epinephrine infusion; ongoing or biphasic symptoms; **moderate–severe respiratory distress** or hypotension at any point; airway involvement; history of severe asthma; any child in the 12-h observation tier.^{332,9}
- **Observe (emergency department / short stay) then discharge** — resolution after one or two IM doses,

the observation tier above completed, no respiratory distress at review.

- **Discharge checklist (Kuwait pathway) — Home epinephrine:** where the emergency department stocks autoinjectors, a senior physician dispenses **two devices with hands-on training** before discharge; where it does not, refer to the allergy clinic with the supply pathway (imported device) documented and teach the family the warning signs and to call emergency services at the first sign. Written action plan; allergen-avoidance advice; non-sedating antihistamine for residual urticaria only; no routine steroid.⁹
- **Refer** — Allergy/Immunology for every case; **PICU** for refractory anaphylactic shock despite repeated IM epinephrine requiring an epinephrine infusion, or airway compromise requiring definitive airway management.

Anaphylaxis – epinephrine first

Recognize (GA²LEN criteria) → IM epinephrine immediately → fluids & repeat → refractory (PICU).



IM EPINEPHRINE FIRST, ALWAYS

There is no contraindication — give it before antihistamine or steroid.

recognize

1

RECOGNIZE — GA²LEN CRITERIA

- 1 No known allergen: skin/mucosa + respiratory or cardiovascular
 - 2 Likely allergen: any 2 of skin/mucosa - respiratory - cardiovascular - severe GI
 - 3 Known allergen: respiratory alone or cardiovascular alone
- If in doubt, give epinephrine.**

now

2

IM EPINEPHRINE

0.01 mg/kg (max 0.5 mg) of 1:1000 in the anterolateral thigh. Lay supine, legs up; O₂ 10–15 L/min.

5–15 min

3

FLUIDS · REPEAT

Fluid bolus for hypotension. Repeat IM epinephrine every 5–15 min as needed.

≥ 2 doses

4

REFRACTORY — PICU

Urgent PICU/anesthesia. IV epinephrine infusion by experts only; glucagon if on a beta-blocker.

Acute Asthma

MINUTES

GINA 2026

Grade severity continuously: speech, work of breathing, SpO₂, silent chest. Doses and steps follow the Global Initiative for Asthma (GINA) 2026 strategy.

GRADE SEVERITY FIRST

- Use a validated score — the **PRAM (Pediatric Respiratory Assessment Measure)** (validated 2–17 yr) or **PASS (Pediatric Asthma Severity Score)** — with SpO₂ on room air,



work of breathing, and ability to speak or feed; reassess and re-score after each step.^{79,80}

✦ PEARL – WHEN IT ISN'T ASTHMA

- Doubt the diagnosis and look for a mimicker when wheeze is present from the **first weeks of life**, is **monophonic or focal**, fails to respond to adequate bronchodilator and steroid, or comes with faltering growth, clubbing, or chronic wet cough.
- Think: inhaled **foreign body**, tracheo-/bronchomalacia or a **vascular ring, cystic fibrosis** or primary ciliary dyskinesia, bronchiolitis in the infant, a **cardiac** cause (heart failure), aspiration/gastroesophageal reflux disease (GERD), and — in adolescents — inducible laryngeal obstruction (vocal cord dysfunction).⁸

1ST LINE

- **Oxygen** only if $\text{SpO}_2 < 92\%$, then target 94–98% (93–95% in adolescents) — avoid routine over-oxygenation.
- **Inhaled salbutamol** every 20 min × 3 in the first hour, reassessing before each repeat.
- **Systemic corticosteroid** for a **moderate–severe** exacerbation or inadequate response to salbutamol (oral preferred, as effective as IV). **Not routine for mild**; avoid routine oral steroids for mild viral wheeze in preschoolers.^{8,91}



2ND LINE

MODERATE–SEVERE

- Add **ipratropium** × 3.
- Severe or poor response → **IV magnesium sulfate**.
- **Call PICU**.



life-threatening / not responding



Not responding /
life-threatening → call **PICU** urgently.
IV salbutamol or aminophylline
infusion, heliox-driven or high-output
continuous nebulization, non-invasive
ventilation, and intubation are
PICU-directed.⁸

BRONCHODILATORS

Drug	Dose	Route & notes
Salbutamol (inhaler + spacer)	≤5 yr: up to 6 puffs >5 yr: up to 12 puffs	100 mcg/puff; every 20 min × 3 in the first hour (one puff at a time, 4–5 breaths each)
Salbutamol (nebulized)	2.5–5 mg	Every 20 min × 3
Ipratropium bromide (nebulized)	250 mcg	Every 20 min × 3, moderate–severe
Magnesium sulfate	40–50 mg/kg (max 2 g)	IV over 20 min, age ≥ 2 yr
Aminophylline	5 mg/kg load, then infusion	IV infusion — PICU-directed ; 3rd line, monitor levels

Drug	Dose	Route & notes
Salbutamol IV	bolus then infusion	IV — PICU-directed ; 3rd line, cardiac monitoring; watch for hypokalemia and lactic acidosis
Salbutamol — continuous (high-output put nebulizer)	≈ 0.5 mg/kg/hr (commonly 10–20 mg/hr)	Continuous nebulization for severe/life-threatening; large-volume high-output nebulizer , O ₂ - or heliox-driven; PICU-directed , cardiac + K ⁺ monitoring
Heliox (helium–oxygen $\geq 70:30$)	as driving / breathing gas	Severe/refractory or impending respiratory failure — lowers airway resistance and improves distal aerosol delivery; a temporizing bridge. Needs low fraction of inspired oxygen (FiO ₂ tolerance (effect requires $\geq 70\%$ helium). PICU-directed

◆ HELIOX & HIGH-OUTPUT NEBULIZATION – WHERE THEY FIT

- In **SEVERE OR REFRACTORY** asthma (or impending respiratory failure), a **HELIOX** (helium–oxygen $\geq 70:30$)

driving or breathing gas reduces turbulent airway resistance and improves distal drug delivery, and a **HIGH-OUTPUT / LARGE-VOLUME CONTINUOUS NEBULIZER** sustains bronchodilator delivery over hours — both are **TEMPORIZING BRIDGES** and a critical-care decision, not first-line.^{179,180,181}

- Benefit is seen mainly in the **MOST SEVERE** subgroup, not routine exacerbations; heliox needs $\geq 70\%$ helium, so it is unsuitable when a high FiO₂ is required.

CORTICOSTEROIDS – CHOOSE ONE; ORAL FIRST-LINE

Drug	Dose	Route & notes
Prednisolone	1–2 mg/kg once daily (max 40 mg)	PO, first-line ⁸ , 3–5 days
Dexamethasone	0.3–0.6 mg/kg (max 12–16 mg)	PO/IV, 1–2 doses; oral alternative
Methylprednisolone	1–2 mg/kg load, then 0.5–1 mg/kg every 6 h	IV agent when oral is not tolerated
Hydrocortisone	4 mg/kg every 6 h (max 100 mg/dose)	IV alternative when oral is not tolerated (equivalent to methylprednisolone)

◆ STEROID CHOICE & AVAILABILITY

- Oral prednisolone (or dexamethasone) is first-line and as effective as IV; reserve IV steroid for the child who cannot take oral.
- When IV is needed, methylprednisolone and hydrocortisone are both acceptable — evidence does not clearly favor one.
- Note that hydrocortisone's mineralocorticoid effect can add to the hypokalemia caused by high-dose salbutamol, and all systemic steroids can raise blood pressure and glucose.
- All four are widely stocked.⁸

⚠ CLINICAL PEARLS & RED FLAGS

A **silent chest**, a falling respiratory rate with rising drowsiness, or a normal/rising PaCO₂ in a tiring child is **pre-arrest** — not improvement. **Call PICU urgently:** decisions on sedation (e.g., ketamine), non-invasive ventilation, imaging, and intubation are made by the critical-care team, aiming to maximize urgent treatment before any intubation decision.⁸

▲ ADMIT · DISCHARGE · REFER

- **Admit** — ongoing oxygen requirement, incomplete response after first-hour therapy, salbutamol needed more often than every 3–4 h, or prior PICU admission/intubation.
- **Discharge** — $\text{SpO}_2 \geq 92\%$ on room air, salbutamol stretched to $\geq$ every 4 h, feeding and talking, inhaler and spacer technique confirmed; discharge prescription = salbutamol via spacer (2–6 puffs every 4 h, weaning over days), the oral steroid course, and **an inhaled corticosteroid controller started or stepped up** (GINA: an exacerbation is a failure of controller treatment — no child ≥ 6 years leaves on salbutamol alone; preschool wheeze by phenotype); written action plan; review within 1–2 weeks.⁸
- **Refer** — Pulmonology for any PICU-level escalation, ≥ 2 admissions per year, or diagnostic doubt; PICU for impending respiratory failure; Allergy/Immunology for difficult-to-control atopic disease.

Croup

HOURS

6 MO–6 YR

Barking cough, hoarse voice, inspiratory stridor; worse at night. Grade severity with the **Westley score** (0–17: ≤ 2 mild · 3–5 moderate · 6–11 severe · ≥ 12 impending respiratory failure), summing level of consciousness, cyanosis, stridor, air entry, and retractions.¹⁸

KEEP CALM

Agitation worsens obstruction — examine on a parent's lap, minimal handling.



DECISION

Stridor at rest / respiratory distress?

MILD

Single steroid dose; discharge with clear return-precaution advice (safety-netting).¹⁸

MODERATE–SEVERE

Steroid + nebulized epinephrine; observe ≥ 2 –4 hours for rebound; oxygen if hypoxic.¹⁶⁶

Drug	Dose	Route & notes
Dexamethasone	0.6 mg/kg (max 16 mg)	PO/IM/IV, single dose, all severities ¹⁸
Budesonide (nebulized)	2 mg	If oral steroid not tolerated
Epinephrine (nebulized)	L-epinephrine (adrenaline) 1:1000 (1 mg/mL) 0.5 mL/kg (max 5 mL) [racemic 2.25% 0.05 mL/kg, max 0.5 mL]	Moderate–severe; observe for rebound ¹⁶⁶

◆ L-EPINEPHRINE VS RACEMIC

- Both are equally effective.
- Racemic epinephrine 2.25% is a 50:50 mix of D- and L-isomers and only the L-isomer is active, so **0.5 mL of 2.25% racemic $\approx$ 5 mL of 1:1000 L-epinephrine (adrenaline).**
- L-epinephrine (the single active isomer) is cheaper and more widely available — racemic is often unavailable outside the USA.
- Do not confuse racemic 2.25% with 1:1000 or 1:10 000

epinephrine — different formulations.¹⁶⁶

◆ **HELIOX IN SEVERE CROUP**

- In **SEVERE** croup with significant distress **NOT SETTLING** after dexamethasone and nebulized epinephrine, **HELIOX** (helium–oxygen) may temporize while the steroid takes effect or transfer/airway support is arranged — in small trials its effect was comparable to oxygen plus epinephrine.
- It is a **BRIDGE, NOT A SUBSTITUTE**, and the evidence is very limited (low certainty).
- Needs $\geq 70\%$ helium, so it is unsuitable when a high oxygen requirement is present. **INVOLVE ICU/ANESTHESIA EARLY.**¹⁷⁸

⚠ **DIFFERENTIALS**

- Drooling + toxic + no cough → epiglottitis (do not examine the throat).
- Sudden choking → inhaled foreign body.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — stridor at rest persisting after a steroid $\pm$ nebulized epinephrine, any oxygen requirement, or a repeat epinephrine dose needed — observe ≥ 2 –4 h after epinephrine for rebound before any discharge.
- **Discharge** — mild croup (no stridor at rest, no distress) after a single steroid dose, feeding well, with reliable carers and safety-netting.
- **Refer** — **early ICU / anesthesia consult** for impending obstruction or failure to respond; ENT if an alternative airway diagnosis is suspected. If intubation is considered, do it in a controlled setting (PICU or operating room) with anesthesia and ENT present and difficult-airway equipment ready — see [Difficult Airway](#).

◆ **SEE ALSO** [Pediatric ENT Emergencies](#) (epiglottitis, bacterial tracheitis) in Section 07.

Retropharyngeal Abscess HOURS

1–5 YR

Deep neck-space infection after recent upper-respiratory illness — fever, neck pain, refusal to move the neck, drooling; guard the airway.

RECOGNIZE

- Peak age 1–5 yr (median ~3 yr), usually after upper-respiratory infection.
- Symptoms: fever, sore throat, odynophagia/dysphagia, neck pain and swelling, muffled (“hot-potato”) voice, trismus, and drooling.
- Signs: neck tenderness, limited neck extension or torticollis, and cervical lymphadenopathy. **Neck pain and refusal to move or extend the neck** are the most useful clues.¹⁶⁷

GUARD THE AIRWAY

- Stridor, respiratory distress, tripod posture, or pooling of secretions signal an airway threat (it mimics epiglottitis).
- Keep the child calm and sitting upright, give oxygen, do **not**

examine the throat or lie the child flat, and involve ENT + anesthesia for a controlled airway before any manipulation.¹⁶⁷

◆ IMAGING

- Lateral neck radiograph is a screening test when the diagnosis is equivocal and there is no airway obstruction (widened prevertebral soft tissue — > 7 mm at C2 or > 14 mm at C6). **Contrast-enhanced CT of the neck is the preferred, confirmatory test** — it separates a drainable abscess (rim-enhancing hypodensity) from retropharyngeal cellulitis/phlegmon and maps the extent.¹⁶⁷
- No specific laboratory test is usually required.

DECISION

Airway compromise, or a large / loculated abscess?

YES

Secure the airway with ENT / anesthesia, then **surgical incision & drainage**; send pus for culture.¹⁶⁸

A 48-hour trial of IV antibiotics first — many children resolve without surgery. Re-image and drain if not improving.¹⁶⁸

Drug	Dose	Route & notes
Ceftriaxone + clindamycin — first line	Ceftriaxone 50 mg/kg once daily (max 2 g); clindamycin 10 mg/kg/dose every 8 h (max 900 mg/dose)	IV; together cover group A streptococcus, oral anaerobes, Gram-negatives and MRSA (methicillin-resistant <i>Staphylococcus aureus</i>) — Kuwait Pediatric Council antibiotic policy. ^{330,167}
Amoxicillin-clavulanate — second line	30 mg/kg/dose (amoxicillin component) every 8 h IV (max 1.2 g/dose)	IV alternative when the first line is unsuitable. No oral amoxicillin-clavulanate step-down at standard dose — it does not cover <i>S. pneumoniae</i> adequately; choose the oral step-down agent with infectious diseases/ENT per the Council policy. ³³⁰

ADMIT · DISCHARGE · REFER

- **Admit** — all suspected retropharyngeal abscess for IV antibiotics and observation; a

monitored bed if there is any airway concern.

- **Discharge** — not from the acute phase; step to an oral agent chosen with infectious diseases/ENT (per the Council policy) once afebrile, improving and swallowing, with ENT follow-up.
- **Refer** — **ENT** for drainage and **anesthesia** for airway control; a screening lateral neck film only if it will not delay a needed CT or airway management.

Difficult Airway MINUTES

Oxygenate first, then work the three zones — optimize (yellow) → intubate (green) → front-of-neck (red); change one thing with every attempt.

▲ ANTICIPATE & PREPARE

- **Predictors:** craniofacial anomaly / syndrome (Pierre Robin, Treacher Collins, Goldenhar, Down), limited mouth opening or neck movement, deep-neck infection or airway swelling (retropharyngeal abscess, epiglottitis, angioedema, burns),

stridor, a large tongue, or a previously documented difficult airway.¹⁷⁰

- **Anticipated** — manage in a **controlled setting (PICU / OR)** with **ENT + anesthesia back-up available. Unanticipated** (ineffective bag-mask ventilation, or a failed first direct / video laryngoscopy (DL / VL) attempt) — **call for help now.**¹⁷⁰
- **Prepare:** assign the team (1st + 2nd airway managers, assistant, respiratory therapist, drug nurse, circulating nurse); a **time-out** to agree **Plan A + B** (1st / 2nd intubator and technique); check equipment and drugs; full monitoring; assess the airway (history + exam).

⚠️ OXYGENATION IS THE GOAL – NOT THE TUBE

- Pre-oxygenate fully and keep **apneic oxygenation** (nasal cannula / high-flow nasal cannula (HFNC)) running through every attempt.
- Change **one variable** with each attempt — operator, device, blade, position, or adjunct — rather than repeating the same one. **If in**

doubt, take it out and
oxygenate.¹⁷⁰

THE THREE-ZONES APPROACH (KUWAIT ALGORITHM)

✦ FOR THE GENERAL PEDIATRICIAN

Recognize the predictors, keep the child oxygenated (yellow zone) and **call anesthesia/ENT early**. The green and red zones below are the airway team's work — shown so you can anticipate and assist, not to lead.

● **YELLOW ZONE · OPTIMIZE** — make bag-mask ventilation effective:
position (head tilt / chin lift, jaw thrust; **shoulder roll if < 2 yr**), suction, **100% FiO₂**, change mask size / adjust the seal, **two-person bag-mask ventilation (BMV)**, insert a nasogastric tube (NGT). Add an **oropharyngeal / nasopharyngeal airway (OPA / NPA)**. Add a **supraglottic airway (laryngeal mask airway, LMA)** if ventilation is still ineffective.¹⁷⁰

saturated & stable



● **GREEN ZONE · INTUBATE** — only when **saturation is at its best and the patient is stable**. Videolaryngoscopy (first choice in an anticipated difficult airway); **max 4 attempts total (DL + VL)**. Between attempts change **operator** or **blade** (straight / curved / hyperangulated), add a **bougie** or a **flexible intubation scope (FIS)** — via the nose, an oral bite-block, or through a supraglottic airway — or try a **smaller endotracheal tube (ETT)**. In infants **< 5 kg** a standard-geometry (Macintosh-style) blade often beats a hyperangulated one.¹⁷¹ **If oxygenation falls or the patient deteriorates → go back to the yellow zone** and re-optimize.¹⁷⁰

can't intubate, can't ventilate



● **RED ZONE · CAN'T INTUBATE, CAN'T VENTILATE (CICV)** — declare it early; **reverse the muscle relaxant** (sugammadex if rocuronium was used) and reassess; keep **apneic O₂** (NC / HFNC) and **treat laryngospasm. ENT available:** rigid bronchoscopy ventilation or tracheostomy. **ENT not available:** needle cricothyrotomy (younger child), or percutaneous / "slash" surgical cricothyrotomy (older child). **Call for help if it has not arrived.** ¹⁷⁰

✦ **CONFIRM PLACEMENT**

On successful intubation, confirm the tube **primarily with continuous waveform capnography (EtCO₂)**; support with auscultation, chest rise, and a chest X-ray.¹

▀ **HAVE READY (DIFFICULT-AIRWAY TROLLEY)**

Videolaryngoscope (standard + hyperangulated blades), a range of tubes, bougie and stylets, supraglottic airways (LMA) in graded sizes, a **flexible intubation scope (FIS)**, and a **front-of-neck-access / cricothyrotomy**

kit — checked and immediately available before induction.¹⁷⁰

Bronchiolitis

HOURS

< 2 YR

Viral (RSV): coryza → cough, fast breathing, wheeze/crackles, feeding difficulty.

▲ SUPPORTIVE CARE IS THE TREATMENT

- Nasal suction; small frequent feeds; NG or IV fluids if intake under 50–75% or dehydrated.
- Give oxygen for persistent $\text{SpO}_2 < 90\%$ (consider $< 92\%$ in young infants, prematurity, cardiac/lung disease, or severe work of breathing). Escalate early to a local HFNC/CPAP-capable team or PICU if work of breathing rises, apnea recurs, or hypoxemia persists.¹⁷

✦ PEARL – FIRST WHEEZE VS THE MIMICKERS

A **first** episode of wheeze with coryza in a child under 12 months in the bronchiolitis season is bronchiolitis.

Recurrent or episodic wheeze, a strong atopic or family history, or wheeze that responds well to a bronchodilator points

instead to viral-induced wheeze or pre-school asthma; **persistent focal signs** raise a retained foreign body or a structural airway lesion; and an infant who is failing to feed with hepatomegaly or a murmur may have **heart failure**, not bronchiolitis.¹⁷

✗ NOT ROUTINELY RECOMMENDED

Bronchodilators, corticosteroids, antibiotics, chest physiotherapy, routine chest X-ray or viral testing.¹⁷ **Red flags:** apnea, grunting, severe recession, SpO₂ < 90%, dehydration, age < 6–12 weeks, comorbidity.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — SpO₂ persistently < 90% (< 92% in high-risk infants), apnea, intake under 50–75% or dehydration, marked work of breathing, — especially if high-risk (age < 8–12 wk, prematurity, or cardiopulmonary disease).
- **Discharge** — maintaining saturations in air, feeding adequately, work of breathing settling, carers able to cope with clear safety-netting.
- **Refer** — PICU (for escalation to high-flow nasal cannula or continuous positive airway pressure

(CPAP)) for rising work of breathing, recurrent apnea, or hypoxia despite oxygen.

◆ **SEE ALSO** [Community-Acquired Pneumonia](#), [Complicated Pneumonia / Empyema](#), [Pertussis](#) and [BRUE](#) in Section 04.

Choking (Foreign-Body Obstruction)

MINUTES

Sudden cough/stridor/one-sided wheeze with a choking history.

DECISION

Effective cough?

EFFECTIVE

Encourage coughing, stay, monitor closely.

INEFFECTIVE + CONSCIOUS

Infant: 5 back blows + 5 chest thrusts (no abdominal thrusts). **Child:** 5 back blows + 5 abdominal thrusts. Repeat.¹

becomes unconscious



Start CPR. Check the mouth before breaths; remove the object only if visible. No blind finger sweeps.¹

⚠ REMOVE NOW

- Foreign body in the larynx or trachea (central airway), stridor at rest, severe distress or cyanosis, or complete obstruction.
- If complete obstruction persists after basic life support → call anesthesia/ENT/PICU immediately; advanced airway maneuvers (direct laryngoscopy with Magill forceps, pushing the object distally to intubate, needle cricothyroidotomy as a last resort) should be attempted only by experienced clinicians.¹

▀ STABLE CHILD – ROUTE BY SITE

- No respiratory distress and normal oxygen saturation: refer by location — **above the clavicle to ENT, below the clavicle to chest surgery or pulmonology** for bronchoscopy.
- A likely bronchial foreign body (one-sided wheeze or reduced air entry) needs rigid bronchoscopy by

an experienced anesthesia and surgical team under optimal conditions, ideally **within 24 hours**; delay beyond 24 h increases complications.²⁴

▲ ADMIT · DISCHARGE · REFER

- **Admit** — after removal, observe for airway injury/aspiration; any hypoxia, stridor, or incomplete relief → ED/PICU.
- **Discharge** — fully recovered, effective cough and normal exam/SpO₂.
- **Refer** — ENT/anesthesia for suspected retained foreign body → rigid bronchoscopy.

Obstructive Sleep Apnea

ASSESS

AAP • ERS

Flagged here for its acute peri-operative airway risk (post-adenotonsillectomy compromise, opioid/sedation sensitivity). Habitual snoring and witnessed apnea; in children it looks like hyperactivity, not sleepiness.

RECOGNIZE

- **Obstructive sleep apnea (OSA)** — habitual snoring (> 3 nights/week), witnessed apnea or gasping, restless sleep, mouth breathing, enuresis; by day **hyperactivity, inattention and poor concentration rather than sleepiness**. Peak 2–8 years (adenotonsillar hypertrophy).¹²⁶
- **High-risk groups:** obesity, Down syndrome, craniofacial anomalies, neuromuscular disease, Prader-Willi, achondroplasia, mucopolysaccharidoses, sickle cell disease.¹²⁶
- **Work-up is elective:** polysomnography (PSG; the apnea-hypopnea index, AHI, grades severity) or sleep-clinic referral; ECG/echocardiography if cor

pulmonale or pulmonary hypertension is suspected. Chronic treatment (adenotonsillectomy, CPAP, intranasal steroid, weight management) is decided in clinic, not in the emergency setting.¹²⁶

▀ RED FLAGS

- **Severe OSA** (AHI ≥ 10 or nadir SpO₂ < 80%), nocturnal desaturations, or hypoventilation.
- **Cor pulmonale / pulmonary hypertension**, failure to thrive — complex, high-risk: refer to a specialist.
- **Cardiorespiratory failure cannot await elective evaluation** — treat urgently.¹²⁶

⚠ PERI-OPERATIVE & ACUTE CARE – THE BEDSIDE PRIORITY

- **Post-adenotonsillectomy respiratory compromise:** anesthesia to assess and consult PICU for any need for respiratory support — transient compromise does not necessarily require admission or overnight monitoring.

- Higher-risk features (age < 3 y, severe OSA on PSG, obesity, craniofacial/neuromuscular disease or Down syndrome, cor pulmonale, recent upper-respiratory infection, or failure to thrive) lower the threshold for a monitored bed.
Opioids: avoid codeine and tramadol after tonsillectomy/adenoidectomy (FDA — CYP2D6 ultra-rapid metabolizer deaths).¹²⁶
- In children with moderate-to-severe OSA, minimize opioids, use opioid-sparing analgesia where possible, and monitor closely — they are at higher risk of opioid-induced respiratory depression; **reduce other opioid doses by ~50% in severe OSA** (opioid hypersensitivity from chronic intermittent hypoxia) and titrate with monitoring.¹²⁶
- Position ramped, keep airway adjuncts and CPAP ready.

✦ PEARL – IT LOOKS LIKE ADHD

A snoring child who is hyperactive and inattentive may have OSA, not just a

behavior problem — ask about snoring and witnessed apnea at every visit.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — severe OSA with desaturation or cor pulmonale, and high-risk post-adenotonsillectomy (age < 3, severe OSA, comorbidity).
- **Discharge** — mild/uncomplicated for outpatient sleep study and ENT follow-up.
- **Refer** — ENT (adenotonsillectomy) and sleep medicine; cardiology if pulmonary hypertension.

Cow's Milk Protein Allergy

ASSESS

INFANT · WAO / CPS 2024

The commonest infant food allergy — IgE-mediated (fast) or non-IgE (delayed, cell-mediated).

▲ RECOGNIZE THE TWO PATTERNS

- **IgE-mediated (minutes–2 h):** urticaria, angioedema, vomiting, wheeze, anaphylaxis.²⁷

- **Non-IgE (hours–days):** eczema flare, reflux/regurgitation, colic, blood or mucus in stool, loose stools or constipation, faltering growth. Distinct from lactose intolerance (an enzymatic, non-immune condition).¹⁹⁷
- Diagnosis is clinical: exclude cow's milk protein from the diet; a planned reintroduction confirms **non-IgE**-mediated disease. For **IgE-mediated** reactions, strictly avoid cow's milk protein and use an extensively hydrolyzed (EHF) or amino acid formula (AAF).^{27,293}

▀ **NON-IGE SUBTYPES – RECOGNIZE THE PATTERN**

- **FPIAP (proctocolitis)** — a well-appearing infant with blood/mucus streaks in the stool, onset **2–8 wks. Most common** subtype; up to **50%** occur in exclusively breastfed infants.¹⁹⁷
- **FPE (enteropathy)** — diarrhea, emesis, abdominal distension, failure to thrive; steatorrhea in up to **80%**. Uncommon — the **only subtype needing gastroscopy + biopsy**

(villous atrophy, crypt hyperplasia) to exclude celiac disease, infection, or cystic fibrosis.¹⁹⁷

- **FPIES (enterocolitis)** — profuse **repetitive vomiting** ± bloody diarrhea, pallor, lethargy; ~**15%** develop hypovolemic shock. Rare (0.015–0.7%); the severe end of the spectrum.¹⁹⁷
- Diagnosis is clinical in FPIAP and FPIES — symptom resolution on elimination confirms it.¹⁹⁷

MANAGEMENT – BY FEEDING & SUBTYPE

Situation	First choice	Notes
IgE-mediated — anaphylaxis	Amino acid formula (AAF)	Use AAF for anaphylactic reactions to cow's milk protein. ²⁹³
IgE-mediated — milder immediate reaction	Extensively hydrolyzed formula (EHF)	Step up to AAF if EHF fails (not tolerated or symptoms persist).
IgE-mediated, breastfed	Continue breastfeeding	If the infant is symptom-free on breast milk, no maternal elimination is needed ; eliminate maternal dairy only if the infant reacts to breast milk. ³³⁴

Situation	First choice	Notes
FPIAP, breastfed	Maternal dairy + soy elimination, 2 wks	Continue breastfeeding; dietitian. ¹⁹⁷
FPIAP, formula-fed	Extensively hydrolyzed formula (EHF)	Switch to AAF only if symptoms persist 2–4 wks. Soy formula an option > 6 mo if soy is tolerated.
FPE, breastfed	Maternal dairy + soy elimination, 2–4 wks	If no improvement (or formula-fed) → AAF directly. Monitor growth and nutrition; resolution takes weeks.
FPE, formula-fed	Amino acid formula (AAF)	First-line (not EHF), given malabsorption severity.
FPIES, breastfed	Continue breastfeeding / oral rehydration	Rare in this group. If reacting: maternal dairy + soy elimination; if unresolved, discontinue breastfeeding → EHF.
FPIES, formula-fed	EHF first; AAF if no improvement in 2 wks	10–20% need AAF from the start; AAF preferred if failing to thrive. Strictly avoid all forms (incl. heated/baked) of the trigger.

Situation	First choice	Notes
Acute FPIES with shock	Medical emergency	Aggressive IV fluids, steroids, ondansetron (monitor QT). Mimics sepsis — CBC, blood gas, chemistry to gauge severity. ¹⁹⁷

⚠️ AVOID

- **Goat's / sheep's milk** (high cross-reactivity) and partially hydrolyzed / "comfort" or "HA" formulas.
- **Refer to a specialist** (allergy / pediatric GI) rather than escalating formula empirically; involve a dietitian to protect calcium and growth.¹⁹⁷

✦ RE-CHALLENGE

Timing and setting (home vs supervised) are decided by the specialist.

▀ ADMIT · DISCHARGE · REFER

- **Admit** — acute FPIES with shock/dehydration.
- **Discharge** — most, with dietary elimination and advice.

- **Refer** — allergy/gastroenterology and a dietitian; anaphylaxis → allergy urgently; unclear diagnosis or FPE → pediatric GI.

SOURCE

Canadian Pediatric Society position statement — Sathya & Fenton, Nutrition & Gastroenterology Committee (non-IgE-mediated CMPA), *Pediatr Child Health* 2024;29(6):382; WAO DRACMA 2024 (IgE-mediated); ESPGHAN 2024.

Antibiotic Allergy Assessment & Pretesting

ASSESS

Most allergy labels are false. Take a history, don't routinely skin-test, and de-label when you can.

▲ THE LABEL IS USUALLY WRONG

- About **10% of people** carry a penicillin/beta-lactam allergy label, but **~90% are not truly allergic**: only **5–10%** of children with a “penicillin allergy label” have a true penicillin allergy.¹¹⁰

- Most labels accumulate from **non-allergic events** — GI upset, headache, family history, or a viral rash during treatment.
- In children, a **maculopapular rash is more often viral than allergic.**¹¹⁰
- Routine pre-dose intradermal **"test dosing" is not standard of care, not validated, and should be abandoned** — it delays therapy and misleads (positive predictive value of skin testing only ~50%; it misses dose-independent and enteric reactions).¹⁰⁹

◆ WHY MISLABELING HARMS

A beta-lactam label drives **second-line agents** (vancomycin, clindamycin, fluoroquinolones) that are less effective, broader, and costlier — linked to more *C. difficile* (~+23%), **MRSA** (~+13%), **vancomycin-resistant enterococci (VRE)** (~+30%), longer stays, more surgical-site infections, ICU admission, aminoglycoside-related acute kidney injury, and a ~50-min delay to the first ED antibiotic dose. **No reaction history — or a beta-lactam tolerated since — give the drug. No skin test required.**¹⁰⁹

▲ RISK-STRATIFY THE HISTORY – SEVERITY, TIMING, DRUG CLASS, AND ANY BETA-LACTAM TOLERATED SINCE

◆ READING THE TIERS

- Assign the **highest** tier any feature reaches — one high-risk feature makes the patient high-risk.
- Risk rises with reaction **severity** (intolerance → hives → anaphylaxis / severe delayed), **immediacy** (delayed benign → immediate IgE), and **recency** (remote → recent). **None / Low** → give the drug or a direct oral challenge; **Medium** → skin test ± challenge, pick a non-cross-reacting beta-lactam meanwhile; **High** → avoid the culprit, refer, do not challenge.
- The pediatric category method is preferred; adult scores (e.g., the PEN-FAST penicillin-allergy score) are not robustly validated in children. **The table below applies to a child with a reported penicillin allergy (cephalosporin allergy excluded).**¹¹⁰

Risk	History (any of)	Action
History not consistent with allergy	No previous allergic reaction; side effects / intolerance only (GI upset, headache); a reaction history but tolerated the antibiotic since the index event; family history only.	De-label the record. Give penicillin if indicated.
Low	Non-IgE-mediated reaction; delayed maculopapular rash starting $\geq$ 3 days into the course.	De-label the record. Give penicillin if indicated.
Medium	Urticaria (hives) or other immediate, IgE-type reaction without anaphylaxis ; a "rash" where hives cannot be excluded.	Refer to an allergist. Avoid penicillin. See the side-chain chart below for prescribing cephalosporins. ¹¹⁰

Risk	History (any of)	Action
High	Anaphylaxis; recurrent reactions or reactions to multiple beta-lactams; prior positive skin test; any severe delayed reaction — Stevens-Johnson syndrome / toxic epidermal necrolysis (SJS/TEN), drug reaction with eosinophilia and systemic symptoms (DRESS), serum sickness, hemolytic anemia, nephritis, hepatitis.	Do not test or challenge. Refer to an allergist. Avoid penicillins and cephalosporins. ¹¹⁰

△ WHEN TO INVOLVE AN ALLERGIST

Any **type-1 (immediate / IgE-mediated)** reaction — urticaria, angioedema, wheeze, or anaphylaxis — prompts **allergist referral**; more nuanced than the

table alone, which is a triage aid, not a substitute for specialist assessment.¹⁰⁹

BETA-LACTAM CROSS-ALLERGY – BY SIDE-CHAIN STRUCTURE

◆ HOW TO USE THIS CHART

- **Prescribing a cephalosporin for a penicillin-allergic child:** choose one that does **not** share a similar side-chain structure with penicillins (a "✓" against the penicillin columns below).
- Cross-reactivity follows **side-chain similarity** — not the shared beta-lactam ring.
Cefazolin has a unique side chain and is safe even after penicillin anaphylaxis.¹⁰⁹ Use only for **immediate / IgE** histories — **never** for SJS/TEN/DRESS, where all beta-lactams are avoided.¹⁰⁹

▴ CHILDREN WITH A CEPHALOSPORIN ALLERGY

- **Immediate hypersensitivity** (urticaria, angioedema, anaphylaxis) — **avoid all cephalosporins and penicillins**; use an alternative and refer to allergy.

- **Inpatient / ICU** — avoid all cephalosporins and penicillins and use alternatives; consult **Allergy / ID** if a penicillin or cephalosporin is the recommended first-line treatment.
- **Severe cutaneous reaction (SJS/TEN/DRESS)** — **avoid all penicillins, cephalosporins, and carbapenems**; use an alternative and consult allergy or ID.¹¹⁰

SOURCE

- Kuwait Pediatric Council practice point on skin testing before antibiotic administration (Table 1 modified from Wong et al., *Pediatr Child Health* 2020, and Devchand & Trubiano, *Aust Prescr* 2019); CHOP Penicillin Drug Allergy Clinical Pathway (rev. Nov 2025); BC PACE Beta-lactam Allergy Delabeling Guideline & Toolkit (2021); De-Labeling Penicillin Allergies in the Pediatric ED, *Children* 2025 ([PMC12730042](https://pubmed.ncbi.nlm.nih.gov/38127300/)); Khan et al., Drug Allergy: 2022 Practice Parameter Update, *JACI*.



- Apply to local antibiogram; verify doses before use.

⚠ **NEVER TEST THESE**

Anaphylaxis, SJS/TEN, DRESS, acute generalized exanthematous pustulosis (AGEP), serum sickness, cytopenias, interstitial nephritis, or hepatitis — no skin test, no challenge; refer to an allergist.¹⁰⁹

Hereditary Angioedema – Acute Attack

MINUTES

Bradykinin-mediated swelling that does **not** respond to epinephrine, antihistamine, or steroid.

▀ **RECOGNIZE**

- Recurrent angioedema **without urticaria or itch**; laryngeal swelling, abdominal attacks; family history; known C1-esterase inhibitor deficiency.²⁷³
- If anaphylaxis cannot be excluded, give intramuscular epinephrine first (see [Anaphylaxis](#)) — failure to

respond supports the diagnosis.²⁷³

Airway is the danger — early anesthetics/ENT for any laryngeal involvement²⁷³



SPECIFIC THERAPY

Give a hereditary angioedema (HAE)-specific agent early; earlier treatment means a milder, shorter attack.²⁷³

Drug	Dose	Route & notes
C1-esterase inhibitor concentrate	20 units/kg	IV; best safety evidence in children — first-line ²⁷³
Icatibant	Weight-based (max 30 mg): 10 mg (12–25 kg) 15 mg (26–40 kg) 20 mg (41–50 kg) 25 mg (51–65 kg) 30 mg (>65 kg)	SC (subcutaneous); bradykinin B2-receptor antagonist; ready-to-use prefilled syringe. Not recommended < 2 years or < 12 kg — no dose established (EMA label, licensed from 2 years; the US FDA label covers adults ≥ 18 years only) ²⁷³

Drug	Dose	Route & notes
Fresh frozen plasma	10 mL/kg	IV; only if specific agents unavailable ²⁷³

▲ ADMIT · DISCHARGE · REFER

- **Admit** — laryngeal or severe attacks
→ a monitored bed.
- **Discharge** — after resolution with
on-demand therapy and a written
plan.
- **Refer** — immunology/allergy
(diagnosis, prophylaxis).

When to Suspect Primary Immunodeficiency (IEI)

ASSESS

Inborn errors of immunity — think of it before
the next live vaccine or transfusion.

▲ WHEN TO SUSPECT – THE SPUR PATTERN

- **Severe, Persistent, Unusual**
(opportunistic organism or unusually
severe course), or **Recurrent**
infection.¹¹⁷

- **Family history** of immunodeficiency or unexplained early infant death, **parental consanguinity, failure to thrive**, infections needing **intravenous antibiotics** or admission.¹¹⁷
- Classic thresholds: ≥ 2 pneumonias or ≥ 8 ear infections in a year, ≥ 2 deep-seated infections (abscess, osteomyelitis), persistent oral/skin thrush after age 1, or little response to ≥ 2 months of antibiotics.¹¹⁷
- **Non-infectious red flags** (increasingly recognized): autoimmune cytopenias — especially chronic, refractory, or multi-lineage — early or treatment-resistant inflammatory bowel disease (< 2 years), granulomas, lymphoproliferation, severe eczema.¹¹⁶

OVERVIEW – INFECTION PATTERN → LIKELY DEFECT

Infection pattern	Think	Typical clues
Recurrent sinopulmonary (encapsulated bacteria) , onset mostly after 6 months	Antibody deficiency	Agammaglobulinemia, common variable immunodeficiency (CVID), IgA deficiency; low immunoglobulins
Early severe viral/ fungal/ opportunistic infection, chronic diarrhea, failure to thrive	Combined / T-cell	Severe combined immunodeficiency (SCID): lymphopenia, may have absent thymic shadow — a pediatric emergency (below)

Infection pattern	Think	Typical clues
Deep abscesses, unusual organisms (Serratia, Burkholderia, Aspergillus), poor wound healing	Phagocyte defect	Chronic granulomatous disease; leukocyte adhesion deficiency (delayed cord separation)
Recurrent Neisseria (meningococcal) or encapsulated-organism infection, or lupus-like disease	Complement deficiency	Early-onset, recurrent or familial meningococcal disease; lupus-like autoimmunity (early classical-pathway defects) ¹¹⁶

⚠ SUSPECTED SCID IS A PEDIATRIC EMERGENCY

- Infant in the first months of life with severe or persistent infection, chronic diarrhea, failure to thrive, oral thrush, or an **absolute lymphocyte count $< 2.5\text{--}3 \times 10^9/\text{L}$**

(or a positive newborn T-cell receptor excision circle (TREC) screen) may have severe combined immunodeficiency.¹¹⁸

- Diagnose **before** live vaccines, non-irradiated blood, or overwhelming infection — allows a life-saving transplant.¹¹⁸
- Phone pediatric immunology the same day.¹¹⁸

✦ FIRST-LINE SCREEN

- Full blood count **with differential** (absolute lymphocyte and neutrophil counts), immunoglobulins (IgG, IgA, IgM, IgE), lymphocyte subsets (T/B/NK by flow cytometry), complement (CH50/AH50 — total classical / alternative-pathway complement activity), and specific antibody responses (vaccine titers, isohemagglutinins).
- Add a neutrophil oxidative-burst test (dihydrorhodamine) if a phagocyte defect is suspected.
Save a pre-treatment serum sample before giving intravenous immunoglobulin (IVIG).¹¹⁶

- **To diagnose infections do not rely on serology** — antibody production is impaired and IVIG transfers donor antibody; use culture, PCR or antigen detection.

⚠ ACUTE DO / DON'T WHEN IEI IS SUSPECTED

- **Do not** give live vaccines (BCG, oral polio, rotavirus, MMR, varicella, yellow fever) — they can cause fatal vaccine-strain disease.¹¹⁸
- **If blood products are needed, use only irradiated, CMV-negative, leukoreduced products** (prevents transfusion-associated graft-versus-host disease).³³⁵
- Do not delay empiric antibiotics for fever — treat any infection promptly and aggressively; avoid non-essential steroids/immunosuppression before a diagnosis.¹¹⁶

⚠ SUSPECTED SCID – EMERGENCY MANAGEMENT, UNDER PEDIATRIC IMMUNOLOGY DIRECTION

- **A medical emergency:**
immunology the same day;

confirmatory testing (lymphocyte subsets $\pm$ TREC) at a center able to interpret it. ^{336,335}

- **Protective isolation** of infant and family from the moment of suspicion (single room, strict hand hygiene; laminar-flow unit where available). ^{336,335}
- **Treat any infection promptly and aggressively.** *Pneumocystis jirovecii* pneumonia (PJP) is frequent — chest X-ray $\pm$ bronchoalveolar lavage; if suspected or proven, co-trimoxazole (trimethoprim-sulfamethoxazole) 20 mg/kg/day of trimethoprim IV in divided doses. ³³⁶
- **Prophylaxis:** co-trimoxazole 5 mg/kg/day of trimethoprim (from 4–6 weeks of age — in a younger neonate immunology chooses the agent, because sulfonamides displace bilirubin); fluconazole 5 mg/kg/day; palivizumab in the RSV season. ^{336,335}
- **IVIG replacement** 400 mg/kg every 3 weeks. ^{336,335}
- Only **irradiated, CMV-negative, leukoreduced** blood products; **no live vaccines.** ^{336,335}

- Growth failure → parenteral nutrition.³³⁶
- Plan **hematopoietic cell transplantation** (or gene therapy) immediately once the diagnosis is established.^{336,335}

REFER

- Pediatric immunology for all suspected inborn errors of immunity — the same day for suspected SCID, any opportunistic infection, or profound lymphopenia.
- IVIG replacement and HLA-typing proceed under specialist direction.¹¹⁶

▲ ADMIT · DISCHARGE · REFER

- **Admit** — as dictated by the presenting infection.
- **Discharge** — per the acute problem, with the workup arranged.
- **Refer** — immunology; avoid live vaccines and use irradiated blood products until cleared.

Tension Pneumothorax & Needle Decompression

MINUTES

APLS/ATLS • NAEMSP 2024

Sudden distress, absent breath sounds and shock — treat before the chest X-ray.

RECOGNIZE (CLINICAL)

- Rapidly worsening respiratory distress and **hypoxia** with **hypotension, unilateral absent/reduced breath sounds**, hyper-resonance, distended neck veins, and (late) tracheal deviation — often after trauma, asthma, positive-pressure ventilation, or in a ventilated neonate.¹

⚠️ DECOMPRESS NOW

- Tension pneumothorax is a **clinical diagnosis** — if the child is peri-arrest, **do not wait for a chest X-ray**.
- Perform immediate **needle (or finger) decompression**, then place a chest drain.¹⁰⁰

- In a ventilated patient who suddenly desaturates and drops their pressure, think of it first.¹

▲ WORKUP

- **Clinical diagnosis — do not wait for imaging.** Decompress first.
- Confirm **after** decompression with chest X-ray or bedside **lung ultrasound (eFAST — extended Focused Assessment with Sonography for Trauma, adding lung views)**; definitive chest drain follows.¹

Step	Landmark	Notes
Needle decompression	2nd intercostal space, mid-clavicular line – <i>or</i> 4th/5th intercostal space anterior axillary line	Over the top of the rib; use a cannula long enough for chest-wall thickness ¹⁰⁰
Definitive	Chest tube (tube thoracostomy)	Follows decompression; size for age

✦ NEONATAL CLUE

In a ventilated or term neonate, sudden deterioration with a shift of the apex and unilateral hyper-resonance suggests tension pneumothorax; **transillumination** can help at the cot-side while you prepare to decompress.¹

▀ ADMIT · REFER

- **Admit** — all pneumothoraces needing drainage; PICU/NICU.
- **Call** critical care ± surgery.

16

Rheumatologic

Cytokine storm (HLH/MAS) & Kawasaki vasculitis

Cytokine Storm — HLH / MAS HOURS

Hyperinflammation: unrelenting fever that looks like sepsis but does not respond.

▀ **RECOGNIZE (TREAT ON SUSPICION – DON'T WAIT FOR ALL CRITERIA; IMMUNOSUPPRESSIVE**

DOSING IS

HEMATOLOGY/RHEUMATOLOGY/PICU-DIRECTED)

- Persistent high fever, cytopenias (≥ 2 lines), hepatosplenomegaly, **very high ferritin**, high triglycerides, low fibrinogen, high lactate dehydrogenase. In MAS complicating sJIA, a **falling or normal ESR alongside a rising CRP/ferritin** (consumptive hypofibrinogenemia) is an early clue^{215,216}.
- Macrophage activation syndrome is the same process complicating systemic juvenile idiopathic arthritis, lupus, or infection. Overlaps with multisystem inflammatory syndrome (post-viral hyperinflammation^{216,218}).

ACT

Send **ferritin** (the key clue) + **CBC, ESR, CRP, fibrinogen, triglycerides, renal & liver function** (full panel).
Phone hematology/rheumatology urgently. Supportive care for organ failure; PICU²¹⁵.



IMMUNOSUPPRESSION – SPECIALIST-DIRECTED

- Corticosteroid (pulse methylprednisolone) is the accessible first step¹¹⁵.
- Anakinra (interleukin-1 antagonist) is increasingly used early, especially in MAS¹¹⁵.
- Etoposide + dexamethasone (HLH protocol) for primary or severe HLH²¹⁵.
- Emapalumab (anti-interferon- γ) — FDA-approved for MAS in Still's disease refractory to glucocorticoids, and for primary HLH¹⁵⁴.

Drug	Dose	Route & notes
Methylprednisolone	Pulse 30 mg/kg/dose IV once daily (max 1 g) × 3–5 days, then 2 mg/kg/day ÷ q6h-BID if needed	IV; accessible first-line ¹¹⁵
Anakinra	2–10 mg/kg/day (higher in refractory)	IV/subcut; interleukin-1 receptor antagonist ¹¹⁵

Drug	Dose	Route & notes
Etoposide + dexamethasone	HLH-2004 protocol	Specialist (hematology)-directed for primary/severe HLH ²¹⁵
Second-line (MAS)	Tocilizumab, cyclosporine, or PLEX (plasma exchange)	Specialist (rheumatology)-directed for MAS or HLH secondary to a rheumatic cause ¹¹⁵ .
Emapalumab	Primary HLH: 1 mg/kg/dose IV twice weekly (with dexamethasone). HLH/MAS in Still's / sJIA: 6 mg/kg once, then 3 mg/kg/dose IV every 3 days through day 16, then 3 mg/kg/dose IV twice weekly. Titrate to max 10 mg/kg/dose.	Anti-interferon- γ antibody; specialist-directed. FDA-approved for primary HLH and for HLH/MAS in Still's disease (inadequate glucocorticoid response or recurrent MAS). Screen and monitor for infection (TB, CMV, other ¹⁵⁴).

ADMIT · DISCHARGE · REFER

- **Admit** — all → high-dependency/PICU.
- **Discharge** — not acutely; under hematology/rheumatology.

- **Refer** — hematology/rheumatology urgently (immunosuppression) and PICU.

Kawasaki Disease HOURS

Medium-vessel vasculitis; leading cause of acquired heart disease in children.

RECOGNIZE

- Fever ≥ 5 days + ≥ 4 of: bilateral non-exudative conjunctivitis, oral mucosal changes (cracked/red lips, strawberry tongue, diffuse oropharyngeal erythema), polymorphous rash, extremity changes (swelling/peeling), cervical lymphadenopathy > 1.5 cm³⁵.
- **Incomplete Kawasaki** is common in infants < 6 months and older children — fewer features but high coronary risk; use inflammatory markers and echocardiography³⁵.

✦ PEARL – THINK INCOMPLETE KAWASAKI

- Suspect **incomplete Kawasaki** in any infant < 6 months with ≥ 5 **days of unexplained fever** even

without the full clinical criteria — this is the group most often missed and at the highest risk of coronary aneurysms³⁵.

- Check inflammatory markers and arrange echocardiography; treat on a compatible picture rather than waiting for every feature.

AHA 2017 pathway: fever ≥ 5 days + 2–3 clinical criteria (or an infant with ≥ 7 days of unexplained fever) → check CRP/ESR; if CRP ≥ 3 mg/dL and/or ESR ≥ 40 mm/h, look for ≥ 3 supplementary lab findings (anemia, platelets $\geq 450 \times 10^9/L$ after day 7, albumin ≤ 3 g/dL, raised ALT, WBC $\geq 15 \times 10^9/L$, sterile pyuria) or a positive echocardiogram³⁵ → treat.

Treat within 10 days of fever onset to protect the coronaries³⁵



FIRST-LINE

- Single high-dose IVIG + aspirin; baseline echocardiogram³⁵.
- Discuss adjunct corticosteroids with cardiology/rheumatology for high-risk patients (infants < 1 year, coronary changes, high inflammatory markers, or Kawasaki disease shock syndrome)³⁵ — AHA 2024 defines high risk as **age < 6 months or baseline LAD/RCA z-score ≥ 2.5** (CRP ≥ 13 mg/dL and Asian ancestry also predict aneurysms) and supports intensified initial therapy for this group.²⁹⁵.

Drug	Dose	Route & notes
IVIG	2 g/kg single infusion over 10–12 h	IV. IVIG-resistant fever > 36 h → second IVIG 2 g/kg, or corticosteroids, or a biologic (infliximab, now dosed 10 mg/kg IV over 2 h) ^{35,295}
Aspirin	Low-dose 3–5 mg/kg/day once daily; or moderate-to-high dose 30–50 up to 80–100 mg/kg/day until afebrile, then low-dose	PO. AHA guidance continues to recommend moderate-to-high-dose aspirin in the acute phase (until afebrile), then low-dose antiplatelet dosing. RCTs and meta-analyses show low-dose 3–5 mg/kg/day from diagnosis gives equivalent coronary outcomes and IVIG resistance with fewer adverse effects, and many centers use it from the outset — either regimen is acceptable ^{35,114} . Continue low-dose to 6–8 weeks / until echo normal

Drug	Dose	Route & notes
Methyl- prednisol one → prednisol one (high-risk)	2 mg/kg/day, IV then oral, tapered over 2–3 weeks	Adjunct to IVIG in high-risk patients or Kawasaki disease shock syndrome ³⁵

⚠ NOTE

- Single-dose methylprednisolone with IVIG is **not** recommended as routine primary therapy³⁵.
- Avoid ibuprofen with aspirin (antagonizes antiplatelet effect³⁵).

⚠ WATCH FOR

Macrophage activation syndrome

(cytokine storm — see [HLH / MAS](#)) and

Kawasaki disease shock syndrome

(hypotension / poor perfusion) — both signal severe, high-risk disease; escalate, add adjunctive steroids, and involve rheumatology/cardiology + PICU^{35,216}.

▴ ADMIT · DISCHARGE · REFER

- **Admit** — all → for IVIG and monitoring.

- **Discharge** — after defervescence and echo, with a plan for repeat echo (at ~1 week and 4–6 weeks)²⁹⁵ and aspirin.
- **Refer** — cardiology (echo) and rheumatology; IVIG-resistant cases urgently.

IgA Vasculitis (Henoch-Schönlein Purpura)

ASSESS

SHARE 2019

Palpable purpura on the legs and buttocks — the commonest childhood vasculitis; the kidney is the long game.

RECOGNIZE (PURPURA + ≥ 1)

- **Palpable purpura** (mandatory), lower-limb/buttock predominant, with normal platelets and coagulation, **plus** at least one of: diffuse abdominal pain · arthritis/arthralgia · biopsy showing IgA deposition · renal involvement (hematuria/proteinuria)¹²⁴.

△ TWO THINGS THAT TURN IT INTO AN

EMERGENCY

- **Intussusception** — usually **ileo-ileal** (easily missed on routine imaging — use ultrasound²¹⁷) in a child with severe/colicky abdominal pain or GI bleeding.
- And **testicular pain/swelling** can mimic torsion²¹⁷.
- Severe abdominal pain, GI bleeding, scrotal pain or a rapidly rising creatinine/BP all need admission²¹⁷.

✦ THE KIDNEY CAN DECLARE ITSELF LATE

- Nephritis may appear **weeks to months** after the rash.²¹⁷
- Check **blood pressure and urinalysis at every visit** and continue surveillance (weekly early, then spacing out to ~6–12 months) even after the rash resolves²¹⁷.
- Persistent proteinuria²¹⁷ → nephrology.

WORKUP

- **Urinalysis + urine protein:creatinine ratio** and **blood pressure** at every visit — nephritis screen (monitor for 6–12 months even if initially normal²¹⁷).
- Serum **creatinine / eGFR**²¹⁷.
- CBC + coagulation — platelets are **normal/high** in IgA vasculitis (excludes ITP/HUS¹²⁴).
- **Renal biopsy** only for significant or persistent proteinuria or impaired function²¹⁷.

Problem	Approach	Notes
Joint / mild pain	Acetaminophen; NSAIDs if no renal involvement or GI bleeding	Most cases are supportive only ²¹⁷
Severe abdominal pain / orchitis	Prednisolone 1–2 mg/kg/day (max 60 mg), short course with taper	Eases pain; does not prevent nephritis ²¹⁷
Nephritis (significant)	Nephrology-directed (steroids ± immunosuppression)	Do not manage severe nephritis alone ²¹⁷

▲ ADMIT · REFER

- **Admit** — severe abdominal pain, GI bleeding, suspected intussusception, scrotal involvement, hypertension or acute kidney injury.
- **Refer** — nephrology for persistent proteinuria/hypertension; surgery if intussusception.

MIS-C (Multisystem Inflammatory Syndrome)

MINUTES

ACR 2022 · CDC

Post-COVID hyperinflammation that overlaps Kawasaki and toxic shock — and can crash the heart.

▲ RECOGNIZE (CRITERIA)

- **Fever $\geq$ 3 days** in a child/adolescent, **plus $\geq$ 2 systems** involved (rash/conjunctivitis, hypotension/shock, cardiac dysfunction, coagulopathy, prominent GI symptoms), **raised inflammatory markers** (CRP, ferritin,

D-dimer), **evidence of SARS-CoV-2** or recent exposure, and **no alternative diagnosis**^{219,218}.

⚠ IT LOOKS LIKE SEVERAL KILLERS AT ONCE

- Overlaps **Kawasaki disease, toxic shock syndrome and sepsis** — treat the shock, take cultures and cover for sepsis until excluded²¹⁸.
- Prominent abdominal pain can mimic appendicitis. **Cardiac:** myocardial dysfunction, arrhythmia and **coronary artery aneurysms** — get an early echocardiogram and troponin/BNP²¹⁸.

▴ WORKUP

- **Inflammatory:** CRP, ESR, **ferritin**, procalcitonin, CBC (lymphopenia), fibrinogen²¹⁸, **D-dimer**.
- **Cardiac:** troponin, BNP/NT-proBNP, ECG and echocardiogram (coronary arteries + function)²¹⁸.
- **SARS-CoV-2** PCR and serology; **blood cultures** to exclude sepsis²¹⁸.
- CMP, LDH, triglycerides, coagulation.

Treatment	Dose	Notes
IV immunoglobulin	2 g/kg	Cornerstone; assess cardiac function before large-volume infusion ²¹⁸
Corticosteroids	Methylprednisolone per protocol (moderate–high dose for severe)	With IVIG in moderate/severe disease ²¹⁸
Aspirin / thromboprophylaxis	Per Kawasaki/coronary protocol ²¹⁸	Cardiology-guided

ADMIT · REFER

- **Admit** all; PICU for shock or cardiac involvement.
- **Call** pediatric rheumatology/infectious diseases and cardiology; follow up coronary imaging.

Malrotation / Midgut Volvulus

MINUTES**OFTEN < 1 MO**

Bilious (green) vomiting in a well infant — the "drop-everything" abdomen.

RECOGNIZE

Bilious vomiting ± distension/pain;
bloody stool and shock are late
(ischemia)²²¹.



RESUSCITATE & CONFIRM

- Nil by mouth, NG decompression, IV fluids, bloods.
- **Urgent upper-GI contrast**
(abnormal duodeno-jejunal junction, "corkscrew") ± Doppler ultrasound²²¹.



CONSULT FIRST

- Consult **pediatric surgery immediately** — for definitive care and **safe transport**.
- Do not image at a center that will not operate; if surgery is unavailable on-site, **transfer without delaying for contrast studies**.

Immediate surgery — Ladd procedure. Peritonitis or shock → straight to the operating room, do not delay for imaging²²¹.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected → emergency, nil by mouth, fluids.
- **Discharge** — not acutely; post-operative.
- **Refer** — pediatric surgery IMMEDIATELY (bilious vomiting = volvulus until excluded²²¹); urgent upper-GI contrast.

Intussusception

HOURS

6 MO–3 YR

Telescoping bowel; the classic triad is usually incomplete.

RECOGNIZE

- Intermittent colicky pain (drawing up the legs), vomiting, lethargy between episodes.
- Sausage-shaped mass; "redcurrant jelly" stool is **late**. Unexplained lethargy may be the only clue²²².

CONFIRM

Ultrasound — "target/doughnut" sign (highly sensitive and specific in experienced hands)²²². Resuscitate with IV fluids first.



DECISION

Stable, no peritonitis/perforation?

YES

Non-operative air or contrast enema reduction — perform only at a center with **pediatric surgical backup** where the child will be definitively treated (perforation needs immediate surgery)²²².

NO / failed

Operative reduction; resect non-viable bowel²²².

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → for reduction and observation.
- **Discharge** — after successful reduction and an observation period without recurrence.
- **Refer** — pediatric surgery and interventional radiology (air/contrast enema); surgery if reduction fails or peritonitis.

Hypertrophic Pyloric Stenosis

HOURS

3-6 WK

Progressive non-bilious projectile vomiting in a hungry infant.

RECOGNIZE & CONFIRM

- "Olive" mass, visible gastric peristalsis, dehydration. Labs: **hypochloremic hypokalemic metabolic alkalosis**.²²³ Ultrasound: muscle $\geq 3-4$ mm, channel $\geq 14-16$ mm²²³.

① CORRECT FIRST

Rehydrate and normalize chloride, potassium and bicarbonate **before** surgery. NG if vomiting²²³.



② Pyloromyotomy once biochemically stable. The electrolytes are the emergency, not the surgery²²³.

ADMIT · DISCHARGE · REFER

- **Admit** — all → correct fluids/electrolytes before surgery.
- **Discharge** — post-operative, feeding established.

- **Refer** — pediatric surgery (pyloromyotomy) after biochemical correction — it is not a surgical emergency until corrected.

Acute Appendicitis HOURS

Most common surgical abdomen; young children atypical, high perforation risk.

RECOGNIZE

- Periumbilical pain migrating to the right lower quadrant, anorexia, fever, vomiting; guarding/rebound, pain on movement¹⁵¹.
- Clinical scores (Alvarado / Pediatric Appendicitis Score) + ultrasound first-line; CT or MRI selectively¹⁵¹.

MANAGE

Nil by mouth · IV fluids · analgesia for pain · **urgent surgical evaluation** → appendectomy.

MIMICS

Examine the testes (torsion), consider DKA (see **Diabetic Ketoacidosis**),

lower-lobe pneumonia, UTI, ovarian torsion¹⁵¹.

⚠ DO NOT BYPASS SURGICAL EVALUATION

- **Antibiotics** — not for *suspected* appendicitis before surgical assessment; started before diagnosis is confirmed, they partially treat and mask it (blunted WBC/neutrophils, softer signs, more perforation¹⁵¹).
- Start once diagnosis is confirmed and surgical plan agreed — or immediately if the child is septic or perforated. **Analgesia** — never leave a child in pain, but stay opioid-sparing while awaiting surgical review: give non-opioid analgesia first (IV or oral acetaminophen); reserve opioids for severe pain as a single titrated dose, with no repeat opioid until the surgeon has assessed¹⁵¹.
- Never give parenteral opioids then discharge an undifferentiated abdomen.
- Analgesia does not reduce diagnostic accuracy — never a reason to delay evaluation¹⁵¹.

ANTIBIOTICS – ONCE DIAGNOSIS CONFIRMED AND SURGICAL PLAN AGREED (OR NOW, IF SEPTIC/PERFORATED)

Drug	Dose	Route & notes
Piperacillin-tazobactam	100 mg/kg/dose (piperacillin component) IV every 6–8 h (max 4 g/dose)	Single-agent cover including anaerobes; common first choice ¹⁵¹ .
Ceftriaxone + metronidazole	Ceftriaxone 50–75 mg/kg/dose IV once daily (max 2 g) + metronidazole 10 mg/kg/dose IV every 8 h (max 500 mg/dose)	Alternative combination; give the first dose promptly once diagnosed ¹⁵¹ .

◆ DURATION

- Uncomplicated appendicitis: perioperative prophylaxis only — a single pre-operative dose, up to 24 h¹⁵¹.
- Complicated or perforated: continue ~4–5 days after adequate source control, or until afebrile with a normalizing white count and returning bowel function (STOP-IT trial)¹⁶³.

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → for surgery and IV antibiotics.
- **Discharge** — post-operative per pathway.
- **Refer** — pediatric surgery; interventional radiology for an abscess (drainage) when non-operative.

Incarcerated Inguinal Hernia

HOURS

Irreducible tender groin or scrotal lump — reduce it or operate.

▲ RECOGNIZE

- Firm, tender, **irreducible swelling** in the groin extending to the scrotum or labia; irritability and vomiting (may mimic bowel obstruction²²⁴).
- Common in infancy, especially preterm. **Overlying skin erythema or a toxic child suggests strangulation²²⁴.**

REDUCE

- If no signs of strangulation: analgesia or sedation, head-down (Trendelenburg) position, gentle sustained manual reduction (taxis) by experienced hands²²⁴.
- If it does not reduce on the first attempt, stop and consult pediatric surgery — repeated attempts worsen edema and make later reduction harder²²⁴.
- Once reduced, admit for repair on the same admission — re-incarceration risk is high²²⁴.

⚠ DO NOT REDUCE – STRAIGHT TO SURGERY

Skin erythema, systemic toxicity, peritonitis, bloody stools, or prolonged incarceration — signal ischemic bowel or gonad → emergency exploration²²⁴.

▶ ADMIT · DISCHARGE · REFER

- **Admit** — all → for reduction/observation or surgery.
- **Discharge** — after successful reduction, for early elective repair.

- **Refer** — pediatric surgery urgently; emergency if irreducible or strangulated.

18

Toxicology & Environmental

Poisoning · burns · heat & drowning

Poisoning & Toxic Ingestion HOURS

Resuscitate, identify the agent, call for advice, give the antidote.

☎ **Kuwait Poison Control Center (KPCC): 1804774 — 24/7.** Call early for any exposure: agent-specific advice, decontamination, antidotes, and disposition.³⁴

▴ GENERAL APPROACH (ABCDE FIRST)

- Stabilize airway, breathing, circulation; give oxygen; check glucose and a 12-lead ECG (QRS width, QT interval).
- Identify the agent, amount, time, and intent; in any deliberate ingestion send **acetaminophen and salicylate levels**.³¹

- Decontamination: activated charcoal 1 g/kg (max 50 g) **only** if airway protected and within ~1 h — not for hydrocarbons, caustics, alcohols, iron, or lithium.³⁴
- Call KPCC **1804774**; give the specific antidote below; observe for delayed toxicity.³⁴

Poison	Antidote / treatment	Notes
Acetaminophen (acetaminophen)	N-acetylcysteine – per Rumack-Matthew nomogram ³¹	IV: 150 mg/kg over 1 h → 50 mg/kg over 4 h → 100 mg/kg over 16 h
Opioids	Naloxone 0.1 mg/kg (max 2 mg)	IV/IM/IN, repeat; infusion if needed
Tricyclic antidepressant	Sodium bicarbonate 1–2 mEq/kg	IV for QRS > 100 ms or arrhythmia
Beta-blocker	Glucagon + high-dose insulin/euglycemia	+ vasopressors
Calcium-channel blocker	Calcium + high-dose insulin/euglycemia	+ vasopressors
Iron	Deferoxamine	IV for severe toxicity
Organophosphate / carbamate	Atropine + pralidoxime	Titrate atropine to drying of secretions

Poison	Antidote / treatment	Notes
Sulfonylurea	Octreotide + dextrose	For recurrent hypoglycemia
Methanol / ethylene glycol	Fomepizole	+ urgent dialysis
Carbon monoxide	100% oxygen ± hyperbaric	High suspicion with heaters/fires
Cyanide (smoke inhalation, nitroprusside)	Hydroxocobalamin 70 mg/kg IV/IO (max 5 g), may repeat once; ± sodium thiosulfate 0.25 g/kg IV (max 12.5 g); do not delay oxygenation, airway control, or burn/PICU transfer while arranging the antidote	First-line; give early in fire victims (lactate ≥ 8 mmol/L). Do not co-infuse hydroxocobalamin and thiosulfate in the same line
Methemoglobinemia	Methylene blue 1–2 mg/kg	IV

Poison	Antidote / treatment	Notes
Local anesthetic toxicity (LAST)	Lipid emulsion 20% – <70 kg: 1.5 mL/kg bolus over 2–3 min, then 0.25 mL/kg/min; ≥70 kg: 100 mL bolus, then 200–250 mL over 15–20 min	Call resuscitation/PICU early and keep 20% lipid emulsion immediately available wherever regional anesthesia is used. Repeat bolus and/or double infusion if unstable; max ~12 mL/kg. Avoid vasopressin and calcium-channel/beta-blockers. epinephrine boluses ≤1 mcg/kg only

⚠ AVOID

- Routine gastric lavage and ipecac; flumazenil (seizure risk) is generally avoided.³⁴
- Several agents have **delayed or biphasic** toxicity (acetaminophen, iron, sustained-release products, sulfonylureas) — observe and recheck.³¹

▀ ADMIT · DISCHARGE · REFER

- **Admit** — symptomatic, a significant/unknown ingestion, or a delayed-toxicity agent (observe).

- **Discharge** — asymptomatic after the appropriate observation window with a non-toxic exposure.
- **Refer** — the poison center/toxicology; psychiatry for intentional ingestion; safeguarding as needed.

Caustic (Corrosive) Ingestion HOURS

Acid or alkali burns — protect the airway; do no harm.

▲ RECOGNIZE

- Drooling, dysphagia, painful swallowing, oral or lip burns, chest or abdominal pain. Stridor or respiratory distress means airway injury.
- The **absence of oral burns does not exclude esophageal injury**. Alkali (drain cleaner) causes deep liquefactive necrosis; acid causes coagulative necrosis.²⁷⁴

⚠ DO NOT

No induced emesis, no gastric lavage, **no activated charcoal**, no neutralizing or

forced dilution — these re-expose the esophagus and worsen injury.²⁷⁴

DO

- Airway first — anticipate swelling and intubate early if any compromise (by anesthesia/ENT/PICU)
- Avoid blind nasogastric-tube insertion unless directed after endoscopic/surgical assessment.
- Keep nothing by mouth; give IV fluids, analgesia, antiemetics.
- Chest and abdominal X-ray for free air. **If esophageal perforation is suspected (free air, severe chest or abdominal pain, or sepsis), start IV fluids and consult pediatric surgery immediately.** Contact poison control.²⁷⁴

◆ ENDOSCOPY

- Upper endoscopy within **12–24 h**, once airway secure, grades the injury — earlier may miss it, later than 48 h raises perforation risk.²⁷⁴
- Watch for delayed strictures over the following weeks.

- A swallowed button battery is a separate emergency²⁵ — see **Button (Lithium) Battery Ingestion.**

▲ ADMIT · DISCHARGE · REFER

- **Admit** — all symptomatic or significant exposures → nil by mouth, plan imaging/endoscopy.
- **Discharge** — asymptomatic minor exposure after observation.
- **Refer** — ENT/GI/surgery (endoscopy) and PICU; do not blind-pass a nasogastric tube.

Burns HOURS

Cool it, size it, replace fluid to urine output — and protect the airway.

▲ ASSESS

- Estimate % total body surface area (TBSA) with a Lund-Browder chart (the child's palm including fingers ≈ 1%); assess depth and mechanism.²⁷⁵
- Suspect **inhalation injury** (facial burns, soot, hoarseness/stridor, enclosed-space fire) → early

airway/anesthetic involvement¹⁴⁶
(see Smoke Inhalation & Carbon Monoxide Poisoning).

FIRST AID

Stop the burning; cool with running water for 20 min (**not** ice); remove clothing/jewelry; cover; analgesia.²⁷⁵



FLUIDS (TBSA > 10%)

- Modified Parkland — Lactated Ringer's **3–4 mL/kg × %TBSA** over 24 h, ½ in the first 8 h from the time of burn, **plus** maintenance (4-2-1, with dextrose in young children)
- 20 mL/kg bolus if shocked.
- Titrate to urine output 0.5–1 mL/kg/h.²⁷⁵



REFER TO A BURNS CENTER

> 10% TBSA, full-thickness, face/hands/feet/perineum/joints, electrical or chemical, inhalation injury, or suspected inflicted injury.²⁷⁵

⚠️ AVOID OVER-RESUSCITATION ("FLUID CREEP")

- The formula is only a starting estimate — titrate to urine output, not to the number.²⁷⁵
- Don't apply ice or home remedies.

⚠️ RAISE SUSPICION FOR INFLICTED INJURY

- A significant share of pediatric burns are inflicted (up to ~1 in 5 in burns-unit series).¹⁷²
- Red flags: a **well-demarcated** burn with **uniform depth** and **no splash marks**; a symmetric immersion ("stocking/glove") pattern with a clear upper margin and skin-fold sparing; burns in normally **protected sites** (groin, perineum, buttocks, back, thighs); delayed presentation or a history that does not fit the burn pattern or the child's developmental stage.
- Look for other injuries and old fractures; safeguard.¹⁷²

▲ ADMIT · DISCHARGE · REFER

- **Admit** — large or deep burns, airway/inhalation injury, electrical/chemical burns, or a safeguarding concern → burns center.
- **Discharge** — small superficial burns with dressing and review.
- **Refer** — a burns unit per criteria; ophthalmology/surgery as needed.

Heat Stroke MINUTES

A clinical diagnosis — core > 40°C + CNS dysfunction after heat exposure. Cool fast; antipyretics do not work.

▲ DIAGNOSE CLINICALLY

- **Core temperature > 40°C** with **CNS dysfunction** after environmental heat exposure.²⁷⁶
- **CNS:** confusion, delirium, irritability, hallucinations, ataxia, slurred speech, seizures, coma. **CVS:** tachycardia, hypotension,

dehydration/shock. **GI:** nausea, vomiting ($\pm$ diarrhea). **Other:** hot skin, tachypnea.

INVESTIGATIONS – DO NOT DELAY COOLING

RBS (hypoglycemia) · CBC · electrolytes (Na/K/Ca) · urea & creatinine · LFTs · coagulation (DIC) · blood gas (metabolic acidosis $\pm$ early respiratory alkalosis) · creatine kinase (rhabdomyolysis) · urinalysis (myoglobinuria).²⁷⁶

MANAGEMENT

- **1 · ABCDE** — secure airway, support breathing and circulation.
- **2 · Immediate rapid cooling (do not delay). Cold-water immersion** preferred if feasible (best for exertional heat stroke). Otherwise **evaporative cooling** — remove clothing, spray/mist tepid water, high-flow fans, mesh stretcher/hammock — plus **ice packs to neck, axillae, and groin**. Room-temperature 0.9% saline for dehydration/shock (avoid overload). **Continuous rectal core-temperature monitoring.**²⁷⁶

- **3 · Stop active cooling at core 38–39°C** to avoid overshoot hypothermia.²⁷⁶
- **4 · No antipyretics** (acetaminophen/ibuprofen) — ineffective in heat stroke and may worsen liver, renal, or GI injury.²⁷⁶ Treat shivering cautiously with IV midazolam or lorazepam; treat seizures per the pediatric status epilepticus protocol (see **Convulsive Status Epilepticus**).

⚠ **DISPOSITION**

- Children with true heat stroke (CNS dysfunction) go to a PICU or high-dependency/PICU-capable setting depending on severity and organ injury — thermal instability and organ injury (rhabdomyolysis, AKI, DIC) can evolve over days.²⁷⁶
- High-yield in Kuwait's climate.

▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — all with CNS dysfunction or organ injury → high-dependency/PICU.

- **Discharge** — mild heat exhaustion, resolved and hydrated with normal mentation.
- **Refer** — PICU; nephrology for rhabdomyolysis/AKI.

Drowning MINUTES

Respiratory impairment from submersion/immersion — ventilate first; rescue breaths are the key intervention.

▀ INITIAL MANAGEMENT (ABCDE)

- **Airway & breathing:** start **rescue breathing immediately** (the single most important intervention); give 100% oxygen; support ventilation as needed. **A Heimlich maneuver and postural drainage have no role in drowning** — do not perform them; they delay ventilation and risk aspiration.¹⁷⁴
- **Circulation:** check pulse; if no pulse, start CPR per the **PALS cardiac-arrest algorithm**. In the ED, **check the rhythm and give electrical therapy with a manual**

defibrillator if the rhythm is shockable (an AED is a prehospital substitute).¹⁷⁴

- **Temperature:** remove wet clothing; if core < 35°C, prevent further heat loss and begin active rewarming.¹⁷⁴
- **Post-ROSC:** manage per PALS post-cardiac-arrest care.

OBSERVE ALL – 6–8 H

- Every submersion patient, even if asymptomatic: continuous pulse oximetry with serial respiratory and neurological exams¹⁷⁴ (delayed pulmonary edema can appear within 4–8 h)
- No routine chest X-ray or laboratory tests — reserve for symptomatic or hypoxic children.¹⁷⁴

STAYS WELL

Asymptomatic with a normal exam, normal vitals, and normal SpO₂ on room air at 6–8 h → **discharge** with return precautions (cough, breathing difficulty, vomiting, fever, or lethargy).¹⁷⁴

Any respiratory symptoms, an oxygen requirement, or altered mental status → **admit**; give oxygen and support and obtain a chest X-ray / blood gas as indicated.

⚠ **ADMIT**

- Respiratory distress or increased work of breathing
- persistent or worsening symptoms
- altered mental status
- need for supplemental oxygen or ventilatory support.

◆ **NOT RECOMMENDED**

- Heimlich maneuver
- postural drainage
- routine chest X-ray for discharge decisions
- glucocorticoids
- routine antibiotics
- routine cervical-spine immobilization unless trauma (e.g., a diving injury) is suspected.¹⁷⁴

▲ ADMIT · DISCHARGE · REFER

- **Admit** — any symptomatic patient, or asymptomatic with risk features → observe.
- **Discharge** — asymptomatic with a normal exam and SpO₂ on room air after 6–8 h observation, with return precautions.
- **Refer** — PICU for respiratory compromise; safeguarding as needed.

Smoke Inhalation & Carbon Monoxide Poisoning

MINUTES

FIRE / SMOKE

Fire injures three ways — airway, carbon monoxide, cyanide. Look for and treat all three.

▲ DEFINE

- Fire/smoke injures by three mechanisms: **(1) thermal / chemical airway injury, (2) carbon monoxide (CO), (3) hydrogen cyanide (HCN).**

Suspect all three after any **enclosed-space fire**, especially with altered mental status.¹⁴⁶

▀ APPROACH – THE AIRWAY FIRST

- Signs of inhalation injury: **facial/neck burns, singed nasal hairs, soot in the mouth or sputum, carbonaceous sputum, hoarseness, stridor, oropharyngeal edema or blistering.**
- Airway injury is **progressive** — edema worsens over hours. Reassess continuously.¹⁴⁶

⚠ SECURE THE AIRWAY EARLY

- Deep facial/neck burns, oropharyngeal blistering or edema, hoarseness/stridor, or large cutaneous burns → **intubate early** (call PICU/anesthesia urgently — expert airway control) before edema makes the airway impassable.¹⁴⁶
- Avoid over-aggressive IV fluids — they worsen airway edema.¹⁴⁶

CARBON MONOXIDE

RECOGNIZE & MEASURE

- CO binds hemoglobin ~**200–250× more avidly than oxygen** → tissue hypoxia plus direct cellular toxicity; **symptoms and COHb level correlate poorly.**¹⁴⁴
- **Pulse oximetry reads falsely normal** (it cannot distinguish carboxyhemoglobin) — measure **COHb by co-oximetry** (blood gas). COHb > 2% (non-smoker) or > 9% (smoker) supports the diagnosis.¹⁴⁴
- Symptoms: headache, nausea, dizziness, confusion, syncope, seizure, cardiac ischemia, coma.

◆ HYPERBARIC OXYGEN (HBO) – AVAILABLE LOCALLY

- High-flow **100% OXYGEN VIA A NON-REBREATHING** (or ventilator) is first-line and cuts COHb half-life from ~4–5 h to ~60–90 min.
- With a chamber on site, **CONSIDER HBO** for severe poisoning: **COHB > 25%** (> 20% in pregnancy), **LOSS OF CONSCIOUSNESS/SYNCOPE**, neurologic signs, **CARDIAC ISCHEMIA**, severe metabolic acidosis, or **PREGNANCY** — ideally within 6 h.^{142,145}

- HBO speeds elimination and may reduce delayed neurocognitive sequelae; trial evidence is mixed, so weigh case by case.¹⁴⁵
- In children benefit is extrapolated from adult trials (under-16s excluded from the pivotal RCT); let CO exposure duration, COHb level, neurologic signs, and lactate guide the decision.¹⁴⁴

⚠ **CYANIDE – DON'T MISS IT**

- Suspect HCN co-poisoning in an enclosed-space fire with soot, **altered mental status, hemodynamic instability, or severe lactic acidosis (lactate $\geq$ 8–10 mmol/L).**
- Treat **empirically — do not wait for levels.**¹²¹
- Use **hydroxocobalamin**; avoid nitrite-based kits in smoke victims (induced methemoglobinemia worsens hypoxia).¹²¹

MANAGEMENT

Intervention	Dose	Notes
Oxygen	100% via non-rebreather (or intubate)	Continue until asymptomatic and COHb normal. ¹⁴²
Hydroxocobalamin (suspected cyanide)	70 mg/kg/dose IV (max 5 g)	Empiric in smoke inhalation with shock or lactate ≥ 10 mmol/L; repeat once if needed. ¹²¹
HBO (severe CO)	2.5–3.0 ATA, per protocol	Refer to the hyperbaric unit for the criteria above. ¹⁴²

▲ ALSO

- Cardiac monitoring + ECG, serial lactate and COHb, cautious fluid resuscitation; ICU/burns referral.
- See also **Burns** and **Poisoning & Toxic Ingestion** for concurrent burn and antidote management.

◆ PEARL – THE OXIMETER LIES

- A normal SpO₂ does not exclude CO poisoning — measure carboxyhemoglobin.
- In a fire victim with shock and a high lactate, give

hydroxocobalamin without waiting for confirmation.¹²¹

▲ ADMIT · DISCHARGE · REFER

- **Admit** — respiratory symptoms, altered mentation, or a significant carboxyhemoglobin → oxygen and monitoring.
- **Discharge** — asymptomatic with normal levels after observation.
- **Refer** — hyperbaric/PICU for severe CO; burns/ENT for airway injury.

Envenomation — Scorpion Sting & Snakebite

MINUTES

WHO 2016 · regional

Two regional bites: scorpion = an autonomic storm; snake = local tissue injury and coagulopathy.

▲ SCORPION STING

- Severe local pain, then an **autonomic storm**: sweating, salivation, vomiting, hypertension and tachycardia, agitation, and in

severe cases **pulmonary edema, myocarditis and shock** (young children at highest risk).²⁷⁷

- Supportive care and analgesia; **prazosin** for autonomic/cardiovascular toxicity; benzodiazepines for agitation. Give **antivenom** for systemic envenomation per local protocol.²⁷⁷

▀ SNAKEBITE

- Look for **local** swelling/necrosis and **systemic** effects — coagulopathy/bleeding, neurotoxicity (ptosis, bulbar/respiratory weakness) or rhabdomyolysis, depending on species.²⁷⁸
- First aid: reassure, **immobilize the limb** at heart level, remove constrictions, and move to hospital. Give species-appropriate **antivenom** for systemic envenomation and monitor the airway and clotting.²⁷⁸

⚠ DON'T MAKE IT WORSE

- **No** tourniquets, incision, suction or ice on snakebites — they increase harm.²⁷⁸

- Watch for delayed coagulopathy and compartment syndrome; recheck clotting serially.
- Anticipate **anaphylaxis to antivenom** and have epinephrine ready.²⁷⁸

◆ **LOCAL** Use the antivenom stocked for the species in your region and follow the national/Poison-Control protocol; contact the Poison Control Center early for identification and dosing.

▀ **WORKUP**

- **Snakebite: 20-minute whole-blood clotting test**, CBC/platelets, PT/INR + fibrinogen, D-dimer, renal function + potassium, CK, urinalysis (myoglobin), ECG, **serial limb marking**, group & crossmatch.²⁷⁸
- **Scorpion:** continuous cardiac/BP monitoring, ECG, **troponin + echocardiogram** if cardiac signs, glucose, electrolytes, blood gas (pulmonary edema).²⁷⁷

ADMIT · REFER

- **Admit/observe** all systemic envenomation and any child with progressing local signs.
- **Call** critical care and Poison Control; surgery for compartment syndrome/necrosis.



Quick Reference

Weight · airway · vital signs · master doses

Time-critical emergencies – quick reference

Solid bar = knowable clock / deadline · dashed bar = unknown onset / prognosis. See each condition's card for detail.

The Golden Window

What must be done — and by when — for the guide's time-critical emergencies. The intervention above each bar is what the window governs.

• **Red "now"** = a step that starts immediately, before any window applies (e.g. 100% O₂ in CO poisoning).

■ **Solid** = clock starts at a knowable moment (recognition / arrival) — a real deadline to beat.

■ ■ ■ **Dashed** = onset usually unknown — the window is prognosis; act on suspicion, do not treat it as a countdown.



Solid bar windows are consensus targets for starting the named intervention; dashed bar figures are prognostic. Always act as early as possible. Immediate ABC resuscitation is assumed for every row. Built from the guide's own time-critical cards.

When to Admit

General triggers first, then condition-specific thresholds.

▲ ANY ACUTELY UNWELL CHILD

Toxic or septic appearance; shock or unstable vital signs; hypoxia ($\text{SpO}_2 < 92\%$); unable to tolerate oral fluids or medicines; moderate to severe dehydration; fever under 3 months; diagnostic uncertainty in an unwell child; or unreliable follow-up.

Condition	Admit if
Febrile young infant	Any under 28 days; 29–90 days unless low-risk on screening; ill-appearing at any age.
UTI	Under 3 months; toxic; vomiting or unable to take oral; no response by 48 h or a resistant organism; known uropathy, obstruction, chronic kidney disease, or transplant; suspected renal abscess; acute kidney injury.
Gastroenteritis / dehydration	Shock or severe dehydration; failed oral or nasogastric rehydration; persistent vomiting; significant

	electrolyte disturbance; under 6 months with heavy losses; possible surgical abdomen.
Bronchiolitis	Apnea; SpO ₂ persistently under 90–92%; feeding under 50–75% or dehydration; severe respiratory distress; under 3 months or comorbidity (prematurity, congenital heart disease, chronic lung disease).
Croup	Stridor at rest persisting after a corticosteroid; repeated nebulized epinephrine; hypoxia, exhaustion, or toxic appearance.
Asthma	Incomplete response to bronchodilators; ongoing oxygen need; needing bronchodilator more often than 1–2 hourly; silent chest, exhaustion, or altered consciousness (→ critical care).
Pneumonia	SpO ₂ under 92%; moderate to severe distress; under 3–6 months; not tolerating oral; dehydration; effusion or empyema; or failed oral antibiotics.
Nephrotic syndrome	First presentation; hypovolemia (abdominal pain, cool peripheries, hypotension, oliguria); peritonitis or sepsis; thromboembolism; severe

edema with respiratory compromise; acute kidney injury; relapse not responding to oral steroids.

BRUE

Under 2 months; prematurity; more than one event; event needed CPR or vigorous stimulation; or a concerning history or examination (higher-risk).

Rabies Post-Exposure Prophylaxis

HOURS

WHO 2018

Dog, cat, or bat bite/scratch — wash, risk-stratify, give vaccine ± immunoglobulin same day.

▲ FIRST – WOUND CARE (MOST IMPORTANT)

- Wash/flush the wound **~15 minutes** with soap and running water, then apply a virucidal (povidone-iodine or 70% alcohol).²⁷⁹
- Avoid early suturing; if unavoidable, suture loosely **after** immunoglobulin infiltration.²⁷⁹

- Give **tetanus prophylaxis**; add antibiotics for a deep or contaminated bite (e.g., amoxicillin-clavulanate).

WHO category	Exposure	Action
I	Touching/feedingNo PEP — wash only licks on intact skin	
II	Nibbling of uncovered skin; minor scratch/abrasion without bleeding	Wound care + vaccine
III	Transdermal bite/scratch, lick on broken skin, mucous-membrane contamination, or any bat contact ²⁷⁹	Wound care + vaccine + immunoglobulin

▀ VACCINE & IMMUNOGLOBULIN

- **Never previously vaccinated** — cell-culture vaccine IM on days **0, 3, 7, 14** (add day 28 if immunocompromised); the WHO 1-week 2-site intradermal schedule (days 0, 3, 7) is an accepted dose-sparing option.²⁷⁹

- **Previously fully vaccinated** — 2 doses only (days 0 and 3); **no immunoglobulin.**²⁷⁹
- **Rabies immunoglobulin** (category III, or category II if immunocompromised) — human RIG **20 IU/kg** (equine 40 IU/kg) as a single day-0 dose; infiltrate as much as possible into and around the wound, give any remainder IM at a site distant from the vaccine.²⁷⁹
- Repeat exposure within 3 months of a complete course → wound care only (no vaccine or RIG).²⁷⁹

⚠ DO NOT DELAY

- Start PEP the same day — rabies is almost always fatal once symptomatic.²⁷⁹
- Immunoglobulin can blunt live vaccines (separate MMR/varicella by ~4 months).²⁷⁹
- Follow the Kuwait MOH / local rabies protocol and notify public health.

IV → Oral Switch & the DIVA Patient

ASSESS

In difficult access, every lost cannula is a switch decision — not an automatic re-stick.

WHY IT MATTERS IN DIVA

- In **difficult intravenous access (DIVA)**, each re-cannulation means pain, delay, distress, and line risk. Reassess switch eligibility at **every line failure** rather than reflexively re-sticking — for many infections early switch is as effective as completing the IV course.⁹⁷

DIVA SCORE – PREDICT DIFFICULT ACCESS

Variable (add the points)	Points
History of prematurity	3
Age: < 1 yr / 1–2 yr / ≥ 3 yr	3 / 1 / 0
Vein not palpable (after tourniquet)	2
Vein not visible (after tourniquet)	2

◆ INTERPRETATION

- Score **≥ 4** (max 10) → > 50% chance of first-attempt cannulation failure.⁹⁷

- Use it early: escalate to an ultrasound-guided / senior inserter or vein-finder — and in a child improving and able to take oral, favor an **IV → ORAL SWITCH** over repeated sticks.
- The simpler **DIVA-3** (age + vein visibility + palpability, dropping prematurity) performs as well.⁹⁷

▲ SWITCH CRITERIA – ALL SHOULD BE MET (COMS)

- **Clinically improving** and responding to treatment.
- **Oral route works** — functioning gut, tolerating intake, not vomiting, no malabsorption.
- **Markers settling** — defervescing, hemodynamically stable, no ongoing sepsis (± falling CRP/white count).
- **Suitable oral agent** — adequate spectrum and bioavailability.

DECISION

Deep-seated / high-risk infection?

SWITCH EARLY

Uncomplicated CAP, UTI/pyelonephritis, uncomplicated skin/soft-tissue infection, and acute osteomyelitis/septic arthritis (early oral non-inferior, with fewer line complications).

KEEP IV

Bacterial meningitis/brain abscess, endocarditis, *S. aureus* bacteremia, febrile neutropenia (until low-risk/recovery criteria), undrained deep abscess, septic shock until stabilized.

◆ LEAN ON HIGH ORAL-BIOAVAILABILITY AGENTS

- Near-IV exposure, ideal in DIVA: ciprofloxacin, levofloxacin, linezolid, metronidazole, fluconazole, trimethoprim-sulfamethoxazole, clindamycin, doxycycline, rifampicin.
- Beta-lactams (e.g., high-dose amoxicillin) are usable but dose adequately for their lower oral absorption.

⚠️ DIVA RULE OF THUMB

- Do not place a PICC or midline solely to finish an IV course that an oral agent could complete.
- Weigh the procedural burden of central access against the switch criteria above.

Weight & Airway Sizing

Quick estimates — a length-based tape is more reliable.

Estimate	Formula
Weight (1–5 yr)	$(\text{age} + 4) \times 2 \text{ kg}$
Weight (6–12 yr)	$(\text{age} \times 3) + 7 \text{ kg}$
Endotracheal tube (uncuffed)	$(\text{age} \div 4) + 4 \text{ mm}$
Endotracheal tube (cuffed)	$(\text{age} \div 4) + 3.5 \text{ mm}$
Tube depth at lip	$\text{tube size} \times 3 \text{ cm}$
Hypotension (< 1 yr)	$\text{systolic BP} < 70 \text{ mmHg}$
Hypotension (1–10 yr)	$\text{systolic BP} < 70 + (2 \times \text{age}) \text{ mmHg}$

Hypotension (> 10
yr)

systolic BP < 90
mmHg

◆ GROWTH CHARTS

- Plot weight, length/height, and head circumference — [WHO standards](#)

[\(birth–5 yr\)](#)



- [CDC charts \(2–19 yr\)](#)



Oxygen Delivery Systems & Cylinder Duration

Device → flow → approximate FiO_2 → when to escalate; and how long a cylinder will last.

OXYGEN DELIVERY SYSTEMS

Device	Flow · FiO_2	Notes & escalation
LOW Nasal cannula (low-flow)	Infant 1-2 L/min Child up to 4 L/min Older child up to 6 L/min FiO_2 ~24-45%	Variable — FiO_2 rises with flow, falls with high inspiratory demand / mouth-breathing. Above ~2 L/min the dry gas irritates and dries the nasal mucosa — add humidification (and warming) at these flows.
LOW Simple face mask	5-10 L/min FiO_2 ~40-60%	Minimum 5 L/min to flush CO_2 (avoid rebreathing). Variable performance.
HIGH O_2 Non-rebreather (reservoir) mask	10-15 L/min keep the bag inflated FiO_2 ~60-90%	Highest FiO_2 without support — resuscitation / emergency.

Device	Flow · FiO ₂	Notes & escalation
FIXED Venturi (air-entrainment) mask	2–15 L/min per valve fixed FiO₂ 24 / 28 / 31 / 35 / 40 / 60%	Delivers a precise FiO₂ regardless of breathing pattern — when a controlled FiO ₂ matters.
SUPPORT High-flow nasal cannula (HFNC)	≈ 2 L/kg/min (to device max, up to ~60 L/min) FiO₂ 21–100% (titratable)	Warmed & humidified; adds 1–3 cmH₂O PEEP + deadspace washout. Can be run on the ward where a clear written escalation protocol exists (defined limits, monitoring, and criteria to step up); otherwise HDU / PICU.
SUPPORT CPAP / BiPAP (NIV)	Device-set flow set FiO₂	Non-invasive support for rising work of breathing, recurrent apnea, or hypoxaemia despite O ₂ . PICU-directed.

🕒 READING THE TABLE

- **LOW** low-flow, **VARIABLE** FiO₂ · **HIGH O₂** / **FIXED** high or **CONTROLLED** FiO₂ · **SUPPORT** also provides **PRESSURE / WASHOUT** → escalation.
- **TARGET SPO₂**: give O₂ for SpO₂ < 90% (consider < 92% in young infants, prematurity, or cardiac / lung disease); titrate to target and wean.

- **INFANTS:** if prongs/mask are not tolerated use **BLOW-BY**; a **HEAD BOX / HOOD** (5–15 L/min) gives up to ~80–90% FiO₂. Neonatal resuscitation: T-piece or self-inflating bag **WITH RESERVOIR**.
- Low-flow FiO₂ figures are estimates — the faster the child breathes, the lower the true delivered FiO₂.

READING THE PULSE OXIMETER – PRACTICAL NOTES

▶ GET A RELIABLE NUMBER

- **Trust it only with a good trace.**
Confirm a clean plethysmograph **waveform** and that the oximeter's **pulse rate matches** the palpated / ECG rate — a damped or absent trace means the number is unreliable.
- **Weak signal = poor perfusion.** Cold, shock, vasoconstriction or hypotension drop the signal (under-reads or fails) — warm the hand, try another site (toe, earlobe), check the perfusion index if shown.
- **Motion and bright / ambient light** cause erratic or false readings — settle the limb, shield the probe, use motion-tolerant sensors.

- **Nail polish, henna, dirt** — remove them or turn the probe sideways. In **darker skin** oximeters can **overestimate** true saturation (occult hypoxemia) — correlate with the clinical picture and a blood gas if in doubt.

⚠ WHEN THE NUMBER LIES – FALSELY REASSURING

- **Carbon monoxide poisoning:** a standard oximeter reads **carboxyhemoglobin as oxyhemoglobin**, so SpO₂ looks **normal / high despite real hypoxia**. Suspect it by history; confirm with **co-oximetry / ABG** and give **100% O₂ regardless of the SpO₂**.
- **Methemoglobinemia** pins SpO₂ around **~85%** whatever the true saturation (a “saturation gap” vs the ABG) — suspect it with cyanosis **unresponsive to O₂** (topical anesthetics, dapsone).
- **Severe anemia / blood loss:** SpO₂ measures **saturation, not oxygen content** — it can read normal while oxygen delivery is inadequate.

✦ **SPO₂ IS NOT PAO₂ – AND IT LAGS**

- On the dissociation curve, **SpO₂ 90% $\approx$ PaO₂ 60 mmHg**; **below 90%** a small SpO₂ fall means a **large PaO₂ drop** — act quickly. The curve is **flat near the top**, so “100%” hides the reserve and **cannot detect hyperoxia** (important in preterm / neonates).
- SpO₂ **lags** real change by seconds (longer if poorly perfused) — a falling number can already be a **late** sign.
- **Neonates — pre- vs post-ductal:** right hand = pre-ductal, foot = post-ductal; a difference **> 3%** (or lower post-ductal) suggests a right-to-left shunt / **critical CHD** (CCHD screening).

OXYGEN CYLINDER – CONTENT & HOW LONG IT LASTS

Step	Equation
Usable duration (quick · psi)	(gauge psi – 200) × cylinder factor ÷ flow (L/min)
Usable content (metric)	capacity (L) × (gauge bar – 14) ÷ full pressure (bar)

Duration (metric)	$\frac{\text{usable content (L)}}{\text{flow (L/min)}}$
----------------------	---

⊙ WHAT THE TERMS MEAN

- **PSI** = pounds per square inch, a unit of gas pressure (what the cylinder gauge reads); **1 BAR ≈ 14.5 PSI**.
- **CYLINDER FACTOR** = litres of oxygen released per 1 psi of pressure (= the cylinder's full capacity ÷ its full pressure). Multiply the **USABLE** pressure (gauge – 200 psi safe residual) by this factor to get the litres available.

⊙ SAFE RESIDUAL & CYLINDER DATA

- **SAFE RESIDUAL ≈ 200 PSI (~14 BAR)** — a cylinder is treated as **EMPTY** at this pressure; **SUBTRACT IT FROM THE GAUGE** before calculating, and do not run lower.
- **CONTENT WHEN FULL (BOC):** CD **460 L** (230 bar) · D **340 L** (137 bar) · E **680 L** (137 bar) · F **1360 L** (137 bar). US E ≈ 625–680 L at ~2000 psi. 1 bar ≈ 14.5 psi. **CHECK THE LABEL – CAPACITY/PRESSURE VARY BY SUPPLIER.**
- **CYLINDER FACTORS (L/PSI):** D 0.16 · E 0.28 · M 1.56 · G 2.41 · H/K 3.14.

✦ WORKED EXAMPLES

- **Imperial:** E cylinder, gauge **1800 psi**, flow **6 L/min** → $(1800 - 200) \times 0.28 \div 6 \approx$ **75 min**.
- **Metric:** E cylinder, gauge **100 bar**, flow **4 L/min** → $680 \times ((100 - 14) \div 137) \approx$ **427 L** → $\div 4 \approx$ **107 min**.
- **High flow drains fast:** at 30–60 L/min a full E cylinder lasts only **~10–20 min** — use wall O₂ or a large cylinder (and an air blender) for transport.
- An **interactive calculator** is in the **Bedside Calculators** section (app / web).

⚠ SAFETY MARGIN

Do not use a cylinder below the **safe residual (~200 psi / ~14 bar)** — treat it as empty. Change or refill at about **~500 psi / ~35 bar** to keep a working reserve, and for transport carry $\geq 2\times$ the calculated requirement (account for device / blender flush).

🕒 BEST CLINICAL RESOURCES (OPEN-ACCESS)

- **OPEN CRITICAL CARE – OXYGEN DELIVERY DEVICES** (evidence-based,

WHO-aligned): opencriticalcare.org



- **WHO – OXYGEN THERAPY FOR CHILDREN**

(2016 manual): who.int



Normal Vital Signs by Age

Reference ranges (awake, at rest).

Age	Heart rate /min	Respiratory rate /min	Systolic BP mmHg
Newborn	100–180	40–60	60–90
Infant < 1 yr	100–160	30–53	72–104
1–2 yr	90–150	22–37	86–106
3–5 yr	80–140	20–28	89–112
6–11 yr	70–120	18–25	97–120
12+ yr	60–100	12–20	110–131

◆ **BLOOD-PRESSURE PERCENTILES** Age-, sex-, and height-based BP centiles (elevated BP $\geq$ 90th; hypertension $\geq$ 95th) — [AAP 2017](#)

[BP tables](#)



Scales, Equations & Synonyms

The scores, formulas, and drug names you reach for at the bedside.

Glasgow Coma Scale — child values in brackets (best 15, worst 3)

Pts	Eye	Verbal (<2 y)	Motor (infant)
6	—	—	Obeys (spontaneous movement)
5	—	Oriented (coos, babbles)	Localizes (withdraws to touch)
4	Spontaneous	Confused (irritable cry)	Withdraws to pain
3	To voice	Wrong words (cries to pain)	Abnormal flexion

2	To pain	Sounds only (moans)	Extension
1	None	None	None

APGAR — score at 1 and 5 minutes (≤ 3 low, 4–6 moderate, 7–10 reassuring)

Sign	0	1	2
Appearance	blue / pale	body pink, limbs blue	all pink
Pulse	absent	< 100	≥ 100
Grimace	none	grimace	cry / cough
Activity (tone)	limp	some flexion	active
Respiration	absent	slow, irregular	strong cry

Dehydration — severe means shock; give 20 mL/kg

Sign	Mild ~3–5%	Moderate ~6–9%	Severe $\geq 10\%$
General	alert, thirsty	restless, irritable	lethargic, floppy
Eyes / mucosa	normal, moist	sunken, dry	deeply sunken, parched

Skin pinch	instant	< 2 s	> 2 s
Cap refill / pulse	normal	prolonged	> 3 s, weak, cool
Urine	normal	reduced	minimal or none

Quick equations

For	Formula
Maintenance fluid	4·2·1: 4 mL/kg/h first 10 kg + 2 next 10 + 1 each kg over 20 (Holliday–Segar)
Fluid deficit (mL)	% dehydration × weight (kg) × 10
Corrected Na (high glucose)	add 2.4 mmol/L per 5.5 mmol/L (100 mg/dL) glucose above 5.5 mmol/L (Hillier; the older 1.6 factor underestimates at high glucose)
Corrected Ca	measured + 0.02 × (40 – albumin g/L) mmol/L
Anion gap	Na – (Cl + HCO ₃); normal 8–12 mmol/L
Serum osmolality	2 × Na + glucose + urea (mmol/L). Osmolar gap = measured – calculated osmolality; > 10 suggests an unmeasured osmole (methanol, ethylene glycol, ethanol, mannitol).

Expected pCO ₂ (met. acidosis)	Winter: $1.5 \times \text{HCO}_3 + 8 \pm 2$ (mmHg) — the expected respiratory compensation for a metabolic acidosis; measured PaCO ₂ above the range = added respiratory acidosis, below = added respiratory alkalosis.
QTc (Bazett)	$\text{QT} \div \sqrt{\text{RR}}$ (s); prolonged > 0.46 s
eGFR (Schwartz)	$36.2 \times \text{height (cm)} \div \text{creatinine } (\mu\text{mol/L})$
Glucose infusion rate	$\text{mg/kg/min} = \text{dextrose\%} \times \text{rate (mL/h)} \div (6 \times \text{weight kg})$

Names & synonyms

Generic (US)	Also known as
Epinephrine	epinephrine
Norepinephrine	norepinephrine
Albuterol	salbutamol
Acetaminophen	acetaminophen
Furosemide	frusemide
Lidocaine	lignocaine
Acyclovir	acyclovir
Amoxicillin-clavulanate	co-amoxiclav
Valproic acid	sodium valproate

Regular insulin	soluble insulin
Normal saline (0.9% NaCl)	isotonic saline
Lactated Ringer's	Hartmann's, compound sodium lactate
Succinylcholine	suxamethonium
Chlorpheniramine	chlorphenamine

Procedural Sedation, RSI & Analgesia

Quick weight-based doses — prepare airway, suction, and monitoring first.

◆ NON-EMERGENT NEONATAL INTUBATION – PREMEDICATE

- Analgesic/vagolytic premedication is standard of care (not optional) for **all non-emergent** neonatal intubations; the only exception is delivery-room resuscitation or a life-threatening airway without IV access.¹⁵⁸
- Use a vagolytic + fast-acting opioid + short-acting relaxant:
atropine 0.01–0.02 mg/kg + fentanyl 2 mcg/kg slow over 2–5

min + **succinylcholine 2 mg/kg**
(avoid succinylcholine in
hyperkalemia,
malignant-hyperthermia risk, or
neuromuscular disease) — CPS. ¹⁵⁸

ANALGESIA

Drug	Dose	Route & notes
Acetaminophen	15 mg/kg/dose (max 1 g/dose)	PO/PR/IV; mild–moderate pain
Ibuprofen	10 mg/kg/dose (max 400 mg/dose)	PO; mild–moderate pain (avoid in dehydration, renal impairment, GI bleeding)
Morphine	0.05–0.1 mg/kg	IV, titrate; for moderate–severe pain
Fentanyl	1–2 mcg/kg IV; 1.5 mcg/kg intranasal	Rapid onset; useful when no IV access

PROCEDURAL SEDATION

Procedural sedation and RSI by credentialed clinicians only — airway-rescue capability, full monitoring, and an institutional sedation policy.

Drug	Dose	Route & notes
Ketamine	1–2 mg/kg IV; 4–5 mg/kg IM	Dissociative; usually preserves airway reflexes/respiratory drive — but airway obstruction, laryngospasm, and apnea can still occur
Midazolam	0.05–0.1 mg/kg IV; 0.3–0.5 mg/kg buccal/intranasal	Anxiolysis; respiratory depression with opioids

RAPID SEQUENCE INTUBATION

Step	Dose	Route & notes
Induction	Ketamine 1–2 mg/kg, propofol 2–3 mg/kg, etomidate 0.3 mg/kg, or midazolam 0.1–0.3 mg/kg (± fentanyl 1–2 mcg/kg as pre-treatment)	Choose by hemodynamics
Paralytic	Rocuronium 1–1.2 mg/kg, or succinylcholine 1–2 mg/kg	IV

Step	Dose	Route & notes
Atropine (pre-treatment)	0.02 mg/kg	IV in young infants if bradycardia risk

⚠ BEFORE ANY SEDATION/RSI

- Have oxygen, suction, bag-mask, the right tube size $[(\text{age} \div 4) + 4]$ uncuffed], monitoring, and a reversal plan ready.
- See Weight & Airway Sizing.

Emergency Drug Doses at a Glance

MINUTES

Weight-based resuscitation doses — use measured weight or a length-based tape when possible (formulas are estimates); cross-check weight, concentration, and local resuscitation chart before every dose.

Drug · indication	Dose	Notes
-------------------	------	-------

Epinephrine — cardiac arrest	0.01 mg/kg (0.1 mL/kg of 1:10 000) IV/IO	Repeat every 3–5 min; max 1 mg/dose. Endotracheal: 0.1 mg/kg (0.1 mL/kg of 1:1000)
Epinephrine — anaphylaxis	0.01 mg/kg (0.01 mL/kg of 1:1000) IM	Max 0.5 mg; anterolateral thigh; repeat every 5–15 min
Amiodarone — VF / pulseless VT	5 mg/kg IV/IO bolus	May repeat up to twice; max 300 mg/dose. Perfusing tachycardia: give over 20–60 min
Adenosine — SVT	0.1 mg/kg rapid push (max 6 mg), then 0.2 mg/kg (max 12 mg)	Immediate saline flush; use line closest to the heart
Atropine — bradycardia	0.02 mg/kg IV/IO	Min 0.1 mg; max 0.5 mg/dose; may repeat once
Calcium chloride 10% —	20 mg/kg (0.2	central line preferred (calcium chloride is

hyperkalemia, Ca-blocker toxicity, hypocalcemia	mL/kg) IV/IO	sclerosing — high extravasation risk); calcium gluconate 10% 60 mg/kg (0.6 mL/kg) is preferred for peripheral IV
Dextrose — hypoglycemia	0.25–0.5 g/kg	D10W 2.5–5 mL/kg (neonate/infant); recheck glucose
Magnesium sulfate — torsades, severe asthma	25–50 mg/kg IV/IO (max 2 g)	Rapid in torsades; over 15–20 min in asthma
Naloxone — opioid toxicity	0.1 mg/kg IV/IM/IN (max 2 mg)	Titrate to breathing; re-dose (short half-life)
Sodium bicarbonate	1 mEq/kg IV/IO	TCA toxicity (QRS > 100 ms), severe acidosis, hyperkalemia
Lidocaine — VF/pVT (alternative to amiodarone)	1 mg/kg IV/IO	Then infusion 20– 50 mcg/kg/min
Isotonic fluid bolus — shock	20 mL/kg	10 mL/kg in neonates, cardiac disease, DKA, or severe

malnutrition;
reassess after
each

Defibrillation
— VF/pVT

2 J/kg →
4 J/kg →
≥ 4 J/kg

Max 10 J/kg or
adult dose (≈ 200
J biphasic / 360 J
monophasic)

Synchronized
cardioversion
— unstable
SVT/VT

0.5–1
J/kg → 2
J/kg

Sedate if
conscious and
time permits

Concentrations & pitfalls

- Epinephrine 1:10 000 = 0.1 mg/mL (IV/IO) versus 1:1000 = 1 mg/mL (IM/endotracheal) — confirm the ampoule before every dose.
- Amiodarone is 150 mg/3 mL.
- Use a length-based (Broselow) tape when weight is unknown.
- Source: American Heart Association / AAP Pediatric Advanced Life Support (PALS), 2025.

Resuscitation Dose Calculator

MINUTES

PALS 2025

Enter the weight — doses compute live and cap at the maximum per dose (works in app and web version).

Weight kg

Length/height cm

Drug · indication	Rule (per kg · max)	Dose for this child
Epinephrine — arrest (IV/IO)	0.01 mg/kg · max 1 mg	—
Epinephrine — anaphylaxis (IM)	0.01 mg/kg · max 0.5 mg	—
Amiodarone — VF/pVT	5 mg/kg · max 300 mg	—
Adenosine — 1st dose	0.1 mg/kg · max 6 mg	—
Adenosine — 2nd dose	0.2 mg/kg · max 12 mg	—

Drug · indication	Rule (per kg · max)	Dose for this child
Atropine — bradycardia	0.02 mg/kg · min 0.1, max 0.5 mg	—
Calcium gluconate 10%	60 mg/kg (0.6 mL/kg) · max 2 g	—
Dextrose 10% — hypoglycemia	2.5–5 mL/kg (0.25–0.5 g/kg)	—
Magnesium sulfate	25–50 mg/kg · max 2 g	—
Naloxone — opioid	0.1 mg/kg · max 2 mg	—
Sodium bicarbonate 8.4%	1 mEq/kg (1 mL/kg)	—
Lidocaine — VF/pVT alt	1 mg/kg · max 100 mg	—
Isotonic fluid bolus — shock	10–20 mL/kg	—
Defibrillation — VF/pVT	2 → 4 J/kg · max ≈ 200 J	—

Drug · indication	Rule (per kg · max)	Dose for this child
Synchromized cardioversion	0.5–1 → 2 J/kg	–

◆ RSI – RAPID SEQUENCE INTUBATION

Drug · role	Rule (per kg)	Dose for this child
Fentanyl — premed /analgesia	1–2 mcg/kg (slow IV)	–
Ketamine — induction	1–2 mg/kg IV (4–5 IM)	–
Etomidate — induction	0.3 mg/kg IV	–
Propofol — induction	2–3 mg/kg IV	–
Midazolam — induction alt	0.1–0.3 mg/kg IV	–

Drug · role	Rule (per kg)	Dose for this child
Rocuro nium — paralyti c	1–1.2 mg/kg IV	–
Succiny lcholine — paralyti c	1–2 mg/kg IV	–

Choose one induction agent (by hemodynamics) + one paralytic. Atropine premed (0.02 mg/kg, max 0.5 mg) is in the table above — no minimum dose when used as intubation premedication (the paradoxical-bradycardia rationale for a 0.1 mg floor has been debunked). Credentialed clinicians only — airway rescue ready.

VASOACTIVE INFUSIONS – WEIGHT-BASED

PREP (VERIFY LOCALLY)

Infusion	Make up to 50 mL	Rate → dose
Epinephrine— / Norepinephrine		1 mL/h = 0.1 mcg/kg/min · usual 0.05–1 mcg/kg/min (0.5–10 mL/h)
Milrinone —		1 mL/h = 0.1 mcg/kg/min · usual 0.25–0.75 mcg/kg/min (2.5–7.5 mL/h); load 50 mcg/kg over 30–60 min
Dopamine — / Dobutamine		1 mL/h = 1 mcg/kg/min · usual 5–20 mcg/kg/min (5–20 mL/h)

Central line for these concentrations;
for larger children use your unit's
standard fixed-concentration chart.

⚠ ALWAYS CROSS-CHECK

- Confirm weight, ampoule concentration, and the local resuscitation chart before every dose.
- If weight is unknown, entering length gives a **Broselow length-based estimate** — confirm with a scale.

- Static PDF / Word / EPUB copies show the rules and maxima but do not calculate; use the app or web (HTML) file for live values.

Resuscitation Doses by Weight

MINUTES

PALS 2025

Pre-computed arrest doses — pick the column nearest the weight. Works in every format, including print.

Drug · rule (per dose) · kg ▶	5	10	20	30	50
Epinephrine (adrenaline) 0.01 mg/kg IV/IO — mg (mL 1:10 000)	0.05 (0.5)	0.1 (1)	0.2 (2)	0.3 (3)	0.5 (5)
Amiodarone 5 mg/kg (arrest) — mg	25	50	100	150	250

Drug · rule (per dose) · kg▶	5	10	20	30	50
Adeno sine 0.1 mg/kg (1st, max 6) — mg	0.5	1	2	3	5
Atropin e 0.02 mg/kg (0.1– 0.5) — mg	0.1	0.2	0.4	0.5	0.5
Calciu m glucon ate 10% 0.6 mL/kg (max 20) — mL	3	6	12	18	20
Dextro se 10% 2 mL/kg — mL	10	20	40	60	100

Drug · rule (per dose) · kg▶	5	10	20	30	50
Fluid bolus 20 mL/kg (10 if neonat e/cardi ac/DK A) — mL	100	200	400	600	1000
Defibril lation 4 J/kg (max 200) — J	20	40	80	120	200

Numbers are per single dose, already capped. Pick the nearest column and adjust to the child's actual weight; for an exact figure use the interactive Resuscitation Dose Calculator above (app / web).

Anticoagulation – VTE Prophylaxis & Treatment

ASSESS

Enoxaparin dosing and anti-Xa targets in children.

Regimen	Dose	Monitoring / notes
---------	------	--------------------

Enoxaparin — treatment, < 2 months	1.5 mg/kg/dose SC every 12 h	Anti-Xa target 0.5–1.0 IU/mL, drawn 4–6 h post-dose (after the 3rd–4th dose). Treatment anticoagulation is hematology-guided with anti-Xa monitoring; adjust for renal function, age, weight, and bleeding risk ¹²⁰
Enoxaparin — treatment, ≥ 2 months	1.0 mg/kg/dose SC every 12 h	Infants often need more: term up to 1.7, preterm up to 2 mg/kg/dose
Enoxaparin — prophylaxis, < 2 months	0.75 mg/kg/dose SC every 12 h	Anti-Xa 0.1–0.3 IU/mL if monitored
Enoxaparin — prophylaxis, ≥ 2 months	0.5 mg/kg/dose SC every 12 h	Routine anti-Xa not required

Before starting

- Baseline complete blood count, renal function, and coagulation screen.
- Avoid with active bleeding, platelets $< 50 \times 10^9/L$, or within 24 h of major surgery or a neuraxial procedure.

- Reduce dose if creatinine clearance < 30 mL/min.¹²⁰

Reversal & alternatives

- Protamine partially reverses enoxaparin (about 1 mg per 1 mg given in the prior 8 h; max 50 mg) — anti-Xa activity is only partly neutralized.
- Consider unfractionated heparin when rapid reversibility or severe renal impairment is needed.¹²⁰
- Source: ASH 2018 and CHEST pediatric antithrombotic guidelines.

Maximum Local Anesthetic Doses

ASSESS

Single-dose ceilings — count total mg/kg across all sites. Use lower limits in neonates/infants, liver disease, or low cardiac output; count the total across all injection sites and when agents are combined.

Agent	Max single dose	Notes
-------	-----------------	-------

Lidocaine (plain)	4.5 mg/kg (max 300 mg)	Onset fast; ~30–60 min duration
Lidocaine + epinephrine	7 mg/kg (max 500 mg)	Longer duration; caution on end-arteries (digits, pinna, penis) ¹¹⁹
Bupivacaine ± epinephrine	2.5 mg/kg (single-shot; epinephrine does not raise the pediatric ceiling)	Most cardiotoxic; never IV or Bier block; avoid in young infants ¹¹⁹
Levobupivacaine	2.5 mg/kg	Less cardiotoxic than bupivacaine
Ropivacaine	2 mg/kg	Lower cardiotoxicity; useful for infusions
Mepivacaine	4.4 mg/kg (max ~300 mg)	Duration between lidocaine and bupivacaine

CONTINUOUS INFUSION CAPS

Bupivacaine, levobupivacaine, ropivacaine: 0.2 mg/kg/h (< 3 mo), 0.3 mg/kg/h (3 mo–1 yr), up to 0.4 mg/kg/h (> 1 yr).¹¹⁹

Toxicity (LAST)

- Perioral numbness, tinnitus, and agitation progress to seizures, then arrhythmia or arrest.
- Stop injection, secure airway and oxygen, treat seizures with a benzodiazepine, and give 20% lipid emulsion (see [Poisoning & Toxic Ingestion](#)).
- Dilute to the least concentrated effective solution, aspirate before injecting, and inject slowly in divided doses; reduce maximums in neonates.¹¹⁹
- Source: ESRA/ASRA 2018 pediatric recommendations; ASRA 2020 LAST advisory.

Vital signs – normal vs red zone by age

The escalation threshold, not just the normal range. Ranges: PALS/APLS.

Vital Signs — Normal vs Red Zone by Age					
The number that should trigger escalation. Red-zone values are concern thresholds, not just "abnormal."					
Compensated pediatric shock is often normotensive — tachycardia + poor perfusion precede hypotension, which is a late, pre-arrest sign.					
Age	Normal HR	HR red zone	Normal RR	RR red zone	Hypotension SBP
< 1 month	100–205	<100 or >205	30–60	>60	<60
1–12 months	100–190	<100 or >190	30–53	>60	<70
1–2 years	98–140	<90 or >140	22–37	>40	<70+(2×age)
2–5 years	80–120	<80 or >120	20–28	>34	<70+(2×age)
5–12 years	75–118	<60 or >120	18–30	>30	<70+(2×age)
> 12 years	60–100	<60 or >100	12–20	>20	<90
Hypotension (5th-centile SBP): 70 + (2 × age in years) for 1–10 y; < 90 mmHg for ≥ 10 y. Example — 4-year-old: < 78 mmHg.					
Septic/shock red flags (any): capillary refill > 2–3 s or flush, mottled/cold or bounding pulses, altered mental status, lactate > 2 mmol/L, reduced urine output.					
Ranges: PALS/ANNA and APLS. Bradycardia and bradypnea in a sick child are ominous late signs — treat as pre-arrest.					

Kuwait Childhood Immunization Schedule

REFERENCE

MOH Kuwait 2025

The routine schedule at a glance — and the ages a missed dose should prompt catch-up.

Age	Vaccine	Route
Pregnant mother	Tdap – one dose in this pregnancy, and every subsequent pregnancy	IM
Within 24 h of birth	Hepatitis B (HBV) – 1st dose	IM

Age	Vaccine	Route
End of 2 months	Hexa (1st) • Pneumococcal (1st) • Rotavirus (1st)	IM • IM • Oral
End of 3 months	BCG	ID
End of 4 months	Hexa (2nd) • Pneumococcal (2nd) • Rotavirus (2nd)	IM • IM • Oral
End of 6 months	Hexa (3rd) • Pneumococcal (3rd) • Rotavirus (3rd)*	IM • IM • Oral
End of 12 months	OPV (1st) • MMR (1st) • Varicella (1st) • Meningococcal conjugate (ACWY)	Oral • SC • SC • IM
At 18 months	OPV (2nd) • Hexa booster • Pneumococcal booster	Oral • IM • IM
End of 2 years	MMRV (1st)	SC
End of 3.5 years	DTaP booster	IM

Age	Vaccine	Route
4–6 years (preschool)	Check immunization status before school admission	—
10–12 years (school)	Tdap booster	IM
16–18 years (school)	Tdap booster	IM

◆ VACCINE KEY

- **HEXA** = diphtheria, tetanus, acellular pertussis, hepatitis B, inactivated poliovirus (IPV) & Hib
- **MMRV** = measles, mumps, rubella, varicella
- **DTAP/TDAP** = diphtheria, tetanus, acellular pertussis
- **OPV** = oral polio
- * rotavirus is 2 or 3 doses according to the vaccine type used.

✦ WHAT CHANGED IN THE 2025 SCHEDULE

- **Hexa** replaces the pentavalent (adds IPV)
- **MMRV** at 2 years replaces the old separate 2nd MMR + varicella

- **DTaP** replaces DPT at 3.5 years
- **Tdap** replaces Td at 10–12 and 16–18 years and is now given in **every pregnancy**
- the old OPV booster at 3.5 years and the girls-only MMR booster at 10–12 years were dropped.

✦ **USE EVERY EMERGENCY VISIT AS A CATCH-UP TRIGGER**

- An unwell or injured child is a chance to check the card.
- A missed **MMR/MMRV** in an outbreak, an incomplete primary series in a young infant with a fever, or no **tetanus-containing** cover before a dirty wound all change management on the spot.

◆ **SOURCE**

- Ministry of Health, State of Kuwait — **KUWAIT CHILDHOOD IMMUNIZATION SCHEDULE 2025** (Public Health Sector).
- Transcribed from the official MOH schedule.

Think Here – Consanguinity & Gulf-Prevalent Conditions

REFERENCE

Gulf pre-test probability

The same presentation carries a different differential in Kuwait — recessive disease is not rare here.

▲ WHY THE BASELINE DIFFERS

- Consanguineous marriage is common in Kuwait (population studies report roughly **half** of unions, first-cousin predominant). Homozygosity sharply raises the pre-test probability of **autosomal-recessive** disease.

✦ RAISE THESE ON THE DIFFERENTIAL EARLIER THAN THE TEXTBOOK WOULD

- **Inborn errors of metabolism** in the encephalopathic / acidotic / hypoglycemic neonate or the child who decompensates with fasting or intercurrent illness — check gas, glucose, ammonia, lactate and start **IV dextrose** early. **Hemoglobinopathies** (sickle cell, β -thalassemia) and **G6PD deficiency** in the pale,

jaundiced, or acutely hemolyzing child.

- A **recessive syndrome** when a sick infant has an affected sibling or known parental consanguinity.

◆ LOCAL PROGRAMMES & SOURCES

- Kuwait runs premarital screening and expanded newborn screening; a family history of consanguinity or an affected sibling is itself a red flag.
- Sources: autosomal-recessive disorders in Kuwait (Teebi/Al-Awadi series); IEM & consanguinity reviews. **[VERIFY LOCAL SCREENING PANEL].**

High-Alert Drugs & Fluid Safety

EVERY SHIFT

ISMP 2024 • AAP 2018

The commonest pediatric harm is not the missed diagnosis — it's the decimal point.

▲ HIGH-ALERT DRUGS – WHERE A DECIMAL KILLS

- **Insulin** (units, U-100 vs U-500) · **opioids** (mcg vs mg) · **concentrated KCl / K-phosphate** (never IV push) ·

heparin / LMWH (1,000 vs 10,000 units/mL) · **chemotherapy** (intrathecal vincristine = fatal) · **digoxin / milrinone** · **magnesium sulfate** (mg vs mmol vs mEq) · **hypertonic saline / dextrose** · **neuromuscular blockers** · **oral methotrexate** (weekly, not daily).

- **Epinephrine concentration: 1 mg/mL** (old 1:1,000, IM) vs **0.1 mg/mL** (old 1:10,000, IV) — the classic 10-fold pediatric error. State the concentration, never the ratio.

⚠ THE THREE ERROR MODES TO DOUBLE-CHECK

- **10-fold / decimal** errors
- **mg vs mL vs microgram vs unit** confusion
- **concentration mix-ups.**
- For any high-alert drug: independent double-check of dose, concentration and pump rate.

▀ SODIUM – CORRECT WITHIN THE SAFE RAILS

- **Hyponatremia:** raise Na ≤ 8 mmol/L per 24 h (≈ 0.5 mmol/L/h) to avoid **osmotic demyelination. Acute symptomatic (seizing):** 3% saline 2–

5 mL/kg (max ~100–150 mL) to raise Na ~4–6 mmol/L and stop the seizure.

- **Hypernatremia:** lower Na $\leq$ **10–12 mmol/L per 24 h** ($\approx$ 0.5 mmol/L/h) to avoid **cerebral edema** (pediatric correction-rate evidence is evolving).

▀ FLUIDS & POTASSIUM

- **Maintenance fluids (AAP 2018):** for children **28 days–18 years**, use **isotonic** fluid (0.9% saline or a balanced solution) with KCl and dextrose — hypotonic maintenance caused iatrogenic hyponatremia. Volume still by Holliday-Segar.
- **IV potassium:** peripheral $\leq$ **40–60 mmol/L**; ward rate 0.25–0.5 mmol/kg/h (max 10 mmol/h); PICU up to **1 mmol/kg/h** with **continuous ECG**. Never IV push.

✦ EXTRAVASATION – STOP, ASPIRATE, AND KNOW THE TWO ANTIDOTES

- Vesicants: calcium, concentrated potassium, **vasopressors**, hypertonic dextrose/saline, TPN, chemotherapy.

- Immediately **stop the infusion, aspirate, elevate.**
- Antidotes: **hyaluronidase** (most hyperosmolar/calcium/TPN infiltrations) and **phentolamine** (vasopressor extravasation, e.g. norepinephrine/dopamine).

◆ SOURCE

- ISMP High-Alert Medications (Acute Care 2024); AAP maintenance IV fluids (Feld et al., Pediatrics 2018); CJASN hyponatremia 2024; pediatric potassium administration guidance; J Infusion Nursing 2020.
- Confirm concentration/rate ceilings against local protocol.

Decision-Rules Engine – Cutoff → Action

REFERENCE

Validated pediatric rules

The rules you already trust, each with its exact operating point and what to do next — in one place.

Rule	Operating point → action
PECARN head — < 2 y	GCS ≤14 / AMS / palpable skull fracture → CT. Occipital-parietal-temporal hematoma, LOC ≥5 s, severe mechanism, or not acting normally → observe vs CT. None → no CT.
PECARN head — ≥ 2 y	GCS ≤14 / AMS / basilar skull-fracture signs → CT. LOC, vomiting, severe mechanism, or severe headache → observe vs CT. None → no CT.
PECARN c-spine (2024)	GCS 3–8 / abnormal ABC / focal deficit → CT. AMS, substantial head/torso injury, midline neck pain or tenderness → radiographs. None → clinical clearance.
Kocher (septic hip)	Non-weight-bearing · fever >38.5 · ESR >40 · WBC >12k. Probability by count: 1≈3% · 2≈40% · 3≈93% · 4≈99%. ≥3 → aspirate / ortho; 2 → strongly consider.

Rule	Operating point → action
Pediatric Appendicitis Score (/10)	≤ 3 (or ≤ 5) low → observe/discharge; 4–6 equivocal → imaging/observe; ≥ 7 high → surgical consult.
Westley croup (/17)	≤ 2 mild → dexamethasone, home; 3–7 moderate → dexamethasone ± nebulized epinephrine, observe; 8–11 severe → epinephrine + steroid, admit; ≥ 12 → airway / ICU.
PRAM asthma (/12)	0–4 mild · 5–8 moderate · 9–12 severe → escalates bronchodilator frequency, systemic steroid, and admission/ICU.
PASS asthma	Pediatric Asthma Severity Score – grades work of breathing, wheezing / air entry, and SpO ₂ / ability to speak or feed; higher = more severe. An alternative to PRAM to guide bronchodilator frequency, systemic steroid, and admission; reassess after each step.
Bacterial Meningitis Score	1 pt each: CSF Gram+ · CSF ANC ≥ 1000 · CSF protein ≥ 80 · blood ANC $\geq 10k$ · seizure. Score 0 = very low risk (NPV ~100%). Excludes <2 mo, prior antibiotics, purpura, immunosuppression, shunt.

Rule	Operating point → action
Febrile infant ≤ 60–90 d	Step-by-Step low risk = well · >21 d · normal UA · PCT <0.5 · CRP ≤20 · ANC ≤10k → discharge with follow-up. PECARN ≤60 d: negative UA + ANC ≤4090 + PCT ≤1.71 → may forgo LP/antibiotics.
Gorelick dehydration (/10)	≥3 signs → ≥5% dehydration; ≥7 → ≥10%. (4-sign subset: cap refill, tears, mucous membranes, ill appearance.)
Glasgow-Blatchford d (UGI bleed)	Score 0 → low risk, outpatient. Caveat: adult-derived, not validated in children – adjunct only.

◆ SOURCES

- PECARN head (Kuppermann, Lancet 2009) & c-spine (2024); Kocher 1999 (Caird CRP modification); Samuel PAS 2002; Westley 1978; PRAM; Nigrovic BMS 2007; Step-by-Step (Gomez/Mintegi 2016) & PECARN febrile (Kuppermann 2019); Gorelick 1997; Glasgow-Blatchford 2000.
- Cross-link each row to its card on insertion.

How This Guide Is Maintained

GOVERNANCE

Issued June 2026 • review June 2028

A reference is only trusted if it shows its currency, its reviewers, and a way to correct it.

▲ CURRENCY & REVIEW

- **Issued** June 2026 • **scheduled review** June 2028, or sooner if a major guideline changes. Each clinical section carries a **"reviewed / next review"** stamp so the reader can see the section's currency at a glance.
- Content is written to the **most recent high-impact guideline** for each topic; the guideline body and year appear on every card.

▲ REVIEWERS & GOVERNANCE

- Prepared for the **Kuwait Pediatric Council** and reviewed by its named faculty panel (see the Reviewers page). Section leads sign off their section before release.

⚠ **REPORT AN ERROR**

- Found a wrong dose, a broken link, or an out-of-date threshold?
Report it — corrections are logged and issued between scheduled reviews.
- Use the feedback channel on the Reviewers page (a QR / email link is placed here in the published edition).
- Every erratum is dated and credited in the change log below.

▀ **CHANGE LOG – THIS EDITION**

- **New clinical cards** — immune thrombocytopenia, IgA vasculitis, pertussis, Guillain–Barré / acute flaccid paralysis, new-diagnosis oncologic emergencies, hyperglycemic hyperosmolar state, vitamin K deficiency bleeding, envenomation, MIS-C, tension pneumothorax, child protection, neonatal abstinence, thyroid storm, rhabdomyolysis, toxic megacolon, neuromuscular disease, massive transfusion.

- **New reference tools** — Kuwait 2025 immunization schedule, high-alert drug & fluid safety, decision-rules engine, symptom front-door navigator; bilirubin nomogram + calculator and age-banded vital-sign red zones.
- **Corrections** — US spelling & drug nomenclature throughout; bilirubin escalation defined as within 2 mg/dL of the exchange line (AAP 2022); immunization schedule updated to the MOH 2025 decree.
- **Editorial** — clickable Contents restored in every PDF; workup block standardized; version label removed from published copies.

◆ HOW TO CITE

- Kuwait Pediatric Council. *Pediatric & Neonatal Common & Emergency Care Guide*. Issued June 2026.
- Decision support only — confirm against Ministry of Health protocols and specialist advice.

Shortest Antibiotic Durations

An improving, immunocompetent child — how few days is actually enough.

⚠ THESE DURATIONS ASSUME

- An **immunocompetent** child who is **improving** (afebrile / clearly better by 48–72 h), with **source control** and **no bacteremia, abscess, or foreign material**.
- Confirm organism and susceptibility; extend if not improving.

Infection	Shortest course	Condition
Community-acquired pneumonia (outpatient, non-severe) (<u>card</u>)	5 days	Improving by day 3–4 (SCOUT-CAP). 3 days is emerging for mild disease, not yet standard.
Acute otitis media (<u>card</u>)	< 2 y: 10 days • ≥ 2 y: 5–7 days	A 5-day course was inferior in under-2s; full 10 days for < 2 y and for severe/perforated.

Infection	Shortest course	Condition
GAS pharyngitis (<u>card</u>)	10 days (penicillin/amoxicillin)	Unchanged — needed for eradication and rheumatic-fever prevention.
Cystitis (afebrile, lower UTI) (<u>card</u>)	3–5 days	Afebrile lower-tract infection.
Febrile UTI / pyelonephritis	7–10 days (5 only if clearly improving)	Evidence conflicts — STOP trial: 5 d non-inferior; SCOUT-UTI: more failures at 5 d. Shared decision-making.
Cellulitis / soft-tissue (uncomplicated) (<u>card</u>)	5 days	Extend only if not improving at day 5.
Acute osteomyelitis (<u>card</u>)	~3 weeks total; IV→oral at 2–4 days	Non-MRSA, good response, falling CRP before the oral switch (Peltola).
Septic arthritis	10 days total; early oral switch	Uncomplicated, non-MRSA/non-Salmonella, CRP normalizing (Peltola).

Infection	Shortest course	Condition
Bacterial meningitis (<u>card</u>)	Meningococcus 5–7 d • Hib 7–10 d • pneumococcus 10–14 d	By organism (GBS 14–21 d; gram-negative/Listeria 21 d).
Perforated appendicitis (post-op) (<u>card</u>)	Often 3–5 days	Adequate source control; stop when afebrile ~24 h and tolerating diet.
Neonatal culture-negative "rule-out" (<u>card</u>)	Stop at 36–48 h	Blood culture negative and infant well — most pathogens grow by 36 h.

✦ THE STEWARDSHIP HABIT

- Set a **stop or review date** when you start.
- For most improving outpatient infections, **shorter is now standard** — the exceptions are the under-2 ear, the strep throat, and febrile UTI, where the longer course still holds.

◆ SOURCES

- SCOUT-CAP (JAMA Peds 2022) & CAP-IT (2021); AAP AOM 2013; IDSA GAS 2012 / CDC; STOP (Pediatrics

2024) & SCOUT-UTI (JAMA Peds 2023); IDSA SSTI 2014; Peltola osteomyelitis (PIDJ 2010) & septic arthritis (CID 2009); AAP Red Book (meningitis durations); pediatric appendicitis pathways.

- Confirm against your current Red Book / local antibiogram.

Weight ruler – Broselow-Luten color zones

Find the weight → color zone → read across. Doses are per-kg (AHA PALS); ETT sizes are the standardized zone values. Confirm with a length-based tape and your formulary.

Weight Ruler — Broselow-Luten color zones

Use this weight ruler to find the color zone for a child's weight. Use the child's actual weight, not the parent's weight. Use the child's actual weight, not the parent's weight. Use the child's actual weight, not the parent's weight.

Zone 1: 0-10 kg, Zone 2: 10-20 kg, Zone 3: 20-30 kg, Zone 4: 30-40 kg, Zone 5: 40-50 kg, Zone 6: 50-60 kg, Zone 7: 60-70 kg, Zone 8: 70-80 kg, Zone 9: 80-90 kg, Zone 10: 90-100 kg

Color Zone	ETT Size (mm)	ETT Volume (mL)	ETT Length (cm)	ETT Weight (kg)	ETT Volume (mL)	ETT Length (cm)	ETT Weight (kg)	ETT Volume (mL)	ETT Length (cm)	ETT Weight (kg)
Zone 1 (0-10 kg)	4.0-5.0	10-15	10-15	10-15	10-15	10-15	10-15	10-15	10-15	10-15
Zone 2 (10-20 kg)	5.0-6.0	15-20	15-20	15-20	15-20	15-20	15-20	15-20	15-20	15-20
Zone 3 (20-30 kg)	6.0-7.0	20-25	20-25	20-25	20-25	20-25	20-25	20-25	20-25	20-25
Zone 4 (30-40 kg)	7.0-8.0	25-30	25-30	25-30	25-30	25-30	25-30	25-30	25-30	25-30
Zone 5 (40-50 kg)	8.0-9.0	30-35	30-35	30-35	30-35	30-35	30-35	30-35	30-35	30-35
Zone 6 (50-60 kg)	9.0-10.0	35-40	35-40	35-40	35-40	35-40	35-40	35-40	35-40	35-40
Zone 7 (60-70 kg)	10.0-11.0	40-45	40-45	40-45	40-45	40-45	40-45	40-45	40-45	40-45
Zone 8 (70-80 kg)	11.0-12.0	45-50	45-50	45-50	45-50	45-50	45-50	45-50	45-50	45-50
Zone 9 (80-90 kg)	12.0-13.0	50-55	50-55	50-55	50-55	50-55	50-55	50-55	50-55	50-55
Zone 10 (90-100 kg)	13.0-14.0	55-60	55-60	55-60	55-60	55-60	55-60	55-60	55-60	55-60

Use this weight ruler to find the color zone for a child's weight. Use the child's actual weight, not the parent's weight. Use the child's actual weight, not the parent's weight.

Use this weight ruler to find the color zone for a child's weight. Use the child's actual weight, not the parent's weight. Use the child's actual weight, not the parent's weight.

Bedside Calculators



Offline calculators — every result shows its formula and source. Decision support only; confirm against the drug chart and local protocol before administration.

Infusion calculator – dose ↔ rate

Weight

kg

Drug

Epinephrine (adrenaline)



Typical: 0.05–1 mcg/kg/min

**Your preparation – enter the actual concentration
(nothing is assumed)**

4 mg / 50 mL

1 mg / 50 mL

Drug amount

mg

in volume

mL

= concentration

mcg/mL

Target dose

dose

unit

mcg/kg/min



or

Known rate

mL/hr

Enter weight, your preparation, and a dose *or* a rate.

Rate (mL/hr) = dose $\times$ weight (kg) $\div$ concentration (mcg/mL), with dose in mcg/kg/hr ($\times 60$ if per-minute; $\times 1000$ if mg/kg/hr). Concentrations are **never assumed** — presets are common preparations you can edit, or type your own. Dose ranges are titrated to response. **Source:** AHA/AAP PALS 2025; pediatric Surviving Sepsis Campaign (Weiss, 2020).

Maintenance fluids & bolus

Weight

kg

Enter weight.

Maintenance — Holliday–Segar: 4-2-1 mL/kg/hr (100-50-20 mL/kg/day) for the first 10 / next 10 / each further kg. Use an **isotonic** fluid with glucose (AAP 2018). Bolus 10–20 mL/kg balanced crystalloid; reassess after each. **Source:** Holliday & Segar 1957; AAP Clinical Practice Guideline 2018; PALS 2025.

PUCAI – Pediatric Ulcerative Colitis Activity Index

Abdominal pain

None (0) 

Rectal bleeding

None (0)

Stool consistency (most stools)

Formed (0) 

Stools per 24 h

0–2 (0) 

Nocturnal stools (waking the child)

No (0) 

Activity level

No limitation (0) 

PUCAI **0** — remission (< 10).

Six clinical items, no laboratory values; total 0–85. **< 10** remission · **10–34** mild · **35–64** moderate · **≥ 65** severe = acute severe colitis (admit, IV corticosteroids). On IV steroids: **day 3 > 45** → plan second-line therapy; **day 5 ≥ 65** → start second-line; **35–65** on day 5 → continue steroids 2–5 more days with daily scoring; **< 35** → responding. **Source:** Turner 2007 (Gastroenterology); ESPGHAN/ECCO acute severe colitis guideline 2025 (Assa et al., JPGN).

Oxygen cylinder duration

Cylinder

— size —



Gauge pressure

now

Units

psi



Flow

L/min

Choose a cylinder, then enter the gauge pressure and flow.

Usable duration = (gauge pressure – safe residual) × cylinder factor ÷ flow. **Safe residual 200 psi (~14 bar)** — treat as empty; change / refill at ~500 psi (~35 bar).

Factors (L/psi): D 0.16 · E 0.28 · M 1.56 · G 2.41 · H/K 3.14 (US/CGA; metric cylinders vary — check the label). **Source:** Egan's Fundamentals of Respiratory Care 12e; AARC; BOC cylinder data.

Electrolyte & lab corrections

Corrected sodium (for hyperglycemia)

Measured Na

mmol/L

Glucose

mmol/L

—

Anion gap

Na

mmol/L

Cl

mmol/L

HCO₃

mmol/L

Albumin

g/L (opt)

—

Corrected calcium

Total Ca

mmol/L

Albumin

g/L

—

Serum osmolality (calculated)

Na

mmol/L

Glucose

mmol/L

Urea

mmol/L

—

Corrected Na = measured + $2.4 \times (\text{glucose} - 5.5) \div 5.5$ (Hillier 1999). Anion gap = $\text{Na} - (\text{Cl} + \text{HCO}_3^-)$, ref 8–16; albumin-corrected + $2.5 \times (40 - \text{albumin g/L}) \div 10$. Corrected Ca (mmol/L) = measured + $0.02 \times (40 - \text{albumin g/L})$. Osmolality = $2 \times \text{Na} + \text{glucose} + \text{urea}$ (mmol/L). **Source:** Hillier, Am J Med 1999; Payne 1973; standard SI formulae.

QTc (corrected QT)

QT interval

ms

Heart rate

bpm

Enter QT (ms) and heart rate.

Bazett: $QTc = QT \div \sqrt{RR}$. Fridericia: $QTc = QT \div RR^{1/3}$, where $RR (s) = 60 \div HR$. Prolonged > 450 ms (> 460 ms in females/infants); > 500 ms = high risk of torsades. Bazett over-corrects at extremes of rate.

Source: Bazett 1920; Rautaharju/AHA 2009.

Glasgow Coma Scale

Eye (1–4)

4 Spontaneous



Verbal (1–5)

5 Oriented / coos-babbles



Motor (1–6)

6 Obeys / spontaneous



GCS **15** / 15 (E4 V5 M6) — minor

Pediatric verbal responses shown for pre-verbal children. GCS $\leq$ 8 signals a threatened airway → involve PICU/anesthesia. **Source:** Teasdale & Jennett 1974; pediatric modification (James 1986).

Westley croup score

Level of consciousness

Normal



Cyanosis

None



Stridor

None



Air entry

Normal



Retractions

None



Westley score **0** — **mild**

≤ 2 mild · 3–5 moderate · 6–11 severe · ≥ 12 impending respiratory failure. Give dexamethasone 0.15–0.6 mg/kg for all; nebulized epinephrine for moderate–severe. **Source:** Westley 1978.

APGAR

Appearance

2 All pink



Pulse

2 ≥ 100



Grimace

2 Cry/cough



Activity

2 Active



Respiration

2 Strong cry



APGAR **10** / 10

Scored at 1 and 5 min (continue every 5 min to 20 if < 7). APGAR does not guide resuscitation — do not delay NRP steps to assign it. **Source:** Apgar 1953; NRP 8th ed / AHA-AAP 2025.

Endotracheal tube – size & depth

Age (years)

yr

Enter age (≥ 1 year). For neonates use weight-based sizing.

Cuffed ID (mm) = $\text{age} \div 4 + 3.5$; uncuffed = $\text{age} \div 4 + 4$ (≥ 1 yr). Depth at lip (cm) = $\text{age} \div 2 + 12$, or $3 \times \text{tube ID}$. Confirm with capnography and CXR. Neonate: ID $\approx 2.5\text{--}3.5$ by weight; depth $\approx \text{weight (kg)} + 6$ cm.

Source: APLS; AHA/AAP PALS 2025.



References

Primary guidelines & landmark trials
(Vancouver)

Sources

Tap to view the full reference list.

1. AHA & AAP. Part 8: Pediatric Advanced Life Support: 2025 Guidelines for CPR and ECC. Circulation. 2025.
[doi:10.1161/CIR.0000000000001368](https://doi.org/10.1161/CIR.0000000000001368)
2. Appelboam A, Reuben A, Mann C, et al. Postural modification to the standard Valsalva maneuver for emergency treatment of supraventricular tachycardias (REVERT): a randomized controlled trial. Lancet. 2015;386(10005):1747–1753.
[doi:10.1016/S0140-6736\(15\)61485-4](https://doi.org/10.1016/S0140-6736(15)61485-4)
3. Magor NRE, Elhalafawy SEH, Ads SEM, Abdelfattah ME, Melika SWM. Efficacy of modified versus standard Valsalva maneuvers on clinical outcomes and satisfaction of children with paroxysmal supraventricular tachycardia: a randomized controlled trial. BMC Pediatr. 2025;25(1):1003. [doi:10.1186/s12887-025-06396-9](https://doi.org/10.1186/s12887-025-06396-9)

4. AHA & AAP. Part 5: Neonatal Resuscitation: 2025 Guidelines for CPR and ECC. Circulation. 2025.
[doi:10.1161/CIR.0000000000001367](https://doi.org/10.1161/CIR.0000000000001367)
5. Oster ME, Pinto NM, Pramanik AK, et al; AAP Section on Cardiology and Cardiac Surgery, Section on Hospital Medicine, Committee on Fetus and Newborn. Newborn Screening for Critical Congenital Heart Disease: A New Algorithm and Other Updated Recommendations: Clinical Report. Pediatrics. 2025;155(1):e2024069667.
[doi:10.1542/peds.2024-069667](https://doi.org/10.1542/peds.2024-069667)
6. Weiss SL, Peters MJ, Alhazzani W, et al. Surviving Sepsis Campaign International Guidelines for the Management of Septic Shock and Sepsis-Associated Organ Dysfunction in Children. *Pediatr Crit Care Med*. 2020;21(2):e52–e106.
[doi:10.1097/PCC.0000000000002198](https://doi.org/10.1097/PCC.0000000000002198)
7. Glaser N, Fritsch M, Priyambada L, et al. ISPAD Clinical Practice Consensus Guidelines 2022: Diabetic ketoacidosis and hyperglycemic hyperosmolar state. *Pediatr Diabetes*. 2022;23(7):835–856.
[doi:10.1111/pedi.13406](https://doi.org/10.1111/pedi.13406)
8. Global Initiative for Asthma (GINA). Global Strategy for Asthma Management and Prevention, 2026 update. ginasthma.org
9. Golden DBK, Wang J, Waserman S, et al. Anaphylaxis: a 2023 practice parameter update. *Ann Allergy Asthma Immunol*. 2024;132(2):124–176.
[doi:10.1016/j.anai.2023.09.015](https://doi.org/10.1016/j.anai.2023.09.015)

10. Kapur J, Elm J, Chamberlain JM, et al. Randomized trial of three anticonvulsant medications for status epilepticus (ESETT). *N Engl J Med*. 2019;381(22):2103–2113. [doi:10.1056/NEJMoa1905795](https://doi.org/10.1056/NEJMoa1905795)
11. Glauser T, Shinnar S, Gloss D, et al. Evidence-based guideline: treatment of convulsive status epilepticus in children and adults (AES). *Epilepsy Curr*. 2016;16(1):48–61. [PMID:26900382](https://pubmed.ncbi.nlm.nih.gov/26900382/)
12. Kemper AR, Newman TB, Slaughter JL, et al. Clinical Practice Guideline Revision: Management of Hyperbilirubinemia in the Newborn Infant 35 or More Weeks of Gestation (AAP). *Pediatrics*. 2022;150(3):e2022058859. [doi:10.1542/peds.2022-058859](https://doi.org/10.1542/peds.2022-058859)
13. Forny P, Hörster F, Ballhausen D, et al. Guidelines for the diagnosis and management of methylmalonic acidemia and propionic acidemia: first revision. *J Inherit Metab Dis*. 2021;44(3):566–592. [doi:10.1002/jimd.12370](https://doi.org/10.1002/jimd.12370)
14. Lazzerini M, Wanzira H. Oral zinc for treating diarrhea in children. *Cochrane Database Syst Rev*. 2016;(12):CD005436. [doi:10.1002/14651858.CD005436.pub5](https://doi.org/10.1002/14651858.CD005436.pub5)
15. Speiser PW, Arlt W, Auchus RJ, et al. Congenital Adrenal Hyperplasia Due to Steroid 21-Hydroxylase Deficiency: An Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab*. 2018;103(11):4043–4088. [doi:10.1210/jc.2018-01865](https://doi.org/10.1210/jc.2018-01865)

16. Pantell RH, Roberts KB, Adams WG, et al. Evaluation and Management of Well-Appearing Febrile Infants 8 to 60 Days Old (AAP). *Pediatrics*. 2021;148(2):e2021052228. [doi:10.1542/peds.2021-052228](https://doi.org/10.1542/peds.2021-052228)
17. Ralston SL, Lieberthal AS, Meissner HC, et al. Diagnosis, Management, and Prevention of Bronchiolitis (AAP). *Pediatrics*. 2014;134(5):e1474–e1502. [doi:10.1542/peds.2014-2742](https://doi.org/10.1542/peds.2014-2742)
18. Westley CR, Cotton EK, Brooks JG. Nebulized racemic epinephrine by IPPB for the treatment of croup: a double-blind study. *Am J Dis Child*. 1978;132(5):484–487. [doi:10.1001/archpedi.1978.02120300044008](https://doi.org/10.1001/archpedi.1978.02120300044008)
19. Ake JA, Jelacic S, Ciol MA, et al. Relative nephroprotection during *Escherichia coli* O157:H7 infections: association with intravenous volume expansion. *Pediatrics*. 2005;115(6):e673–e680. [PMID 15930195](https://pubmed.ncbi.nlm.nih.gov/15930195/)
20. Squires RH, Dhawan A, Alonso E, et al. Intravenous N-acetylcysteine in pediatric patients with nonacetaminophen acute liver failure: a placebo-controlled clinical trial. *Hepatology*. 2013;57(4):1542–1549. [PMC3509266](https://pubmed.ncbi.nlm.nih.gov/23509266/)
21. Eroglu Y, Emerick KM, Whittington PF, Alonso EM. Octreotide therapy for control of acute gastrointestinal bleeding in children. *J Pediatr Gastroenterol Nutr*. 2004;38(1):41–47. [doi:10.1097/00005176-200401000-00011](https://doi.org/10.1097/00005176-200401000-00011)

22. Villanueva C, Colomo A, Bosch A, et al. Transfusion strategies for acute upper gastrointestinal bleeding. *N Engl J Med*. 2013;368(1):11–21.
[doi:10.1056/NEJMoa1211801](https://doi.org/10.1056/NEJMoa1211801)
23. Alfaraidi H, Alissa A, Alsaiari A, et al. Efficacy and safety of empiric omeprazole continuous infusion in critically ill children with gastrointestinal bleeding. *Front Pediatr*. 2024;12:1270911.
[doi:10.3389/fped.2024.1270911](https://doi.org/10.3389/fped.2024.1270911)
24. Kramer RE, Lerner DG, Lin T, et al; NASPGHAN Endoscopy Committee. Management of ingested foreign bodies in children: a clinical report of the NASPGHAN Endoscopy Committee. *J Pediatr Gastroenterol Nutr*. 2015;60(4):562–574.
[doi:10.1097/MPG.0000000000000729](https://doi.org/10.1097/MPG.0000000000000729)
25. Mubarak A, Benninga MA, Broekaert I, et al. Diagnosis, management, and prevention of button battery ingestion in childhood: a European Society for Paediatric Gastroenterology Hepatology and Nutrition position paper. *J Pediatr Gastroenterol Nutr*. 2021;73(1):129–136.
[doi:10.1097/MPG.00000000000003048](https://doi.org/10.1097/MPG.00000000000003048)
26. Kuppermann N, Ghetti S, Schunk JE, et al. (PECARN) Clinical trial of fluid infusion rates for pediatric DKA. *N Engl J Med*. 2018;378(24):2275–2287.
[doi:10.1056/NEJMoa1716816](https://doi.org/10.1056/NEJMoa1716816)

27. Meyer R, Venter C, Bognanni A, et al. World Allergy Organization (WAO) Diagnosis and Rationale for Action against Cow's Milk Allergy (DRACMA) guideline update – VII – Milk elimination and reintroduction in the diagnostic process of cow's milk allergy. *World Allergy Organ J.* 2023;16(7):100785. [doi:10.1016/j.waojou.2023.100785](https://doi.org/10.1016/j.waojou.2023.100785)
28. Rosen R, Vandenplas Y, Singendonk M, et al. Pediatric Gastresophageal Reflux Clinical Practice Guidelines: Joint NASPGHAN–ESPGHAN Recommendations. *J Pediatr Gastroenterol Nutr.* 2018;66(3):516–554. [doi:10.1097/MPG.0000000000001889](https://doi.org/10.1097/MPG.0000000000001889)
29. Waterfield T, Maney JA, Lyttle MD, et al. Validating clinical practice guidelines for management of children with non-blanching rashes (PiC): a prospective, multicentre cohort study. *Lancet Infect Dis.* 2021;21(4):569–577. [doi:10.1016/S1473-3099\(20\)30474-6](https://doi.org/10.1016/S1473-3099(20)30474-6)
30. Pressler RM, Abend NS, Auvin S, et al. Treatment of seizures in the neonate: ILAE Task Force guidelines and consensus recommendations. *Epilepsia.* 2023;64(10):2550–2570. [doi:10.1111/epi.17745](https://doi.org/10.1111/epi.17745)
31. Dart RC, Mullins ME, Matoushek T, et al. Management of acetaminophen poisoning in the US and Canada: a consensus statement. *JAMA Netw Open.* 2023;6(8):e2327739. [doi:10.1001/jamanetworkopen.2023.27739](https://doi.org/10.1001/jamanetworkopen.2023.27739)

32. Tieder JS, Bonkowsky JL, Etzel RA, et al. Brief Resolved Unexplained Events (formerly ALTE) and evaluation of lower-risk infants: AAP clinical practice guideline. *Pediatrics*. 2016;137(5):e20160590.
[doi:10.1542/peds.2016-0590](https://doi.org/10.1542/peds.2016-0590)
33. Tieder JS, Sullivan E, Stephans A, et al; Brief Resolved Unexplained Event Research and Quality Improvement Network. Risk Factors and Outcomes After a Brief Resolved Unexplained Event: A Multicenter Study. *Pediatrics*. 2021;148(1):e2020036095.
[doi:10.1542/peds.2020-036095](https://doi.org/10.1542/peds.2020-036095)
34. Kuwait Poison Control Center (KPCC), Ministry of Health — 24/7 toxicology hotline: 1804774.
35. McCrindle BW, Rowley AH, Newburger JW, et al. Diagnosis, Treatment, and Long-Term Management of Kawasaki Disease (AHA). *Circulation*. 2017;135(17):e927–e999.
[doi:10.1161/CIR.0000000000000484](https://doi.org/10.1161/CIR.0000000000000484)
36. Lehrnbecher T, Robinson PD, Ammann RA, et al. Guideline for the Management of Fever and Neutropenia in Pediatric Patients With Cancer and HCT Recipients: 2023 Update. *J Clin Oncol*. 2023;41(9):1774–1785.
[doi:10.1200/JCO.22.02224](https://doi.org/10.1200/JCO.22.02224)
37. Coiffier B, Altman A, Pui CH, et al. Guidelines for the management of pediatric and adult tumor lysis syndrome. *J Clin Oncol*. 2008;26(16):2767–2778.
[doi:10.1200/JCO.2007.15.0177](https://doi.org/10.1200/JCO.2007.15.0177)

38. Brandow AM, Carroll CP, Creary S, et al. American Society of Hematology 2020 guidelines for sickle cell disease: management of acute and chronic pain. *Blood Adv.* 2020;4(12):2656–2701.
[doi:10.1182/bloodadvances.2020001851](https://doi.org/10.1182/bloodadvances.2020001851)
39. Srivastava A, Santagostino E, Dougall A, et al. WFH Guidelines for the Management of Hemophilia, 3rd edition. Hemophilia. 2020;26(Suppl 6):1–158.
[doi:10.1111/hae.14046](https://doi.org/10.1111/hae.14046)
40. Flynn JT, Kaelber DC, Baker-Smith CM, et al. Clinical Practice Guideline for Screening and Management of High Blood Pressure in Children and Adolescents (AAP). *Pediatrics.* 2017;140(3):e20171904.
[doi:10.1542/peds.2017-1904](https://doi.org/10.1542/peds.2017-1904)
41. Gerson R, Malas N, Feuer V, Silver GH, Prasad R, Mroczkowski MM. Best practices for evaluation and treatment of agitated children and adolescents (BETA) in the emergency department: consensus statement of the American Association for Emergency Psychiatry. *West J Emerg Med.* 2019;20(2):409–418.
[doi:10.5811/westjem.2019.1.41344](https://doi.org/10.5811/westjem.2019.1.41344)
42. Le Saux N, Robinson JL; Canadian Pediatric Society, Infectious Diseases and Immunization Committee. Management of acute otitis media in children six months of age and older. *Pediatr Child Health.* 2016;21(1):39–50 (reaffirmed Nov 2024).
[doi:10.1093/pch/21.1.39](https://doi.org/10.1093/pch/21.1.39)

43. Castelli Gattinara G, Bergamini M, Simeone G, et al. Antibiotic treatment of acute and recurrent otitis media in children: an Italian intersociety consensus. Ital J Pediatr. 2025;51(1):50. [doi:10.1186/s13052-025-01894-z](https://doi.org/10.1186/s13052-025-01894-z)
44. Lieberthal AS, Carroll AE, Chonmaitree T, et al. The Diagnosis and Management of Acute Otitis Media (AAP). Pediatrics. 2013;131(3):e964–e999. [doi:10.1542/peds.2012-3488](https://doi.org/10.1542/peds.2012-3488)
45. Linder JA, Watson ME Jr, Wessels MR, et al. 2025 Clinical Practice Guideline Update by the Infectious Diseases Society of America on Group A Streptococcal (GAS) Pharyngitis: Risk Assessment Using Clinical Scoring Systems in Children and Adults. Clin Infect Dis. 2025. [doi:10.1093/cid/ciaf668](https://doi.org/10.1093/cid/ciaf668)
46. Shulman ST, Bisno AL, Clegg HW, et al. Clinical Practice Guideline for the Diagnosis and Management of Group A Streptococcal Pharyngitis: 2012 Update (IDSA). Clin Infect Dis. 2012;55(10):e86–e102. [doi:10.1093/cid/cis629](https://doi.org/10.1093/cid/cis629)
47. Mattoo TK, Shaikh N, Nelson CP. Contemporary Management of Urinary Tract Infection in Children. Pediatrics. 2021;147(2):e2020012138. [doi:10.1542/peds.2020-012138](https://doi.org/10.1542/peds.2020-012138)

48. Subcommittee on Urinary Tract Infection. Reaffirmation of AAP Clinical Practice Guideline: The Diagnosis and Management of the Initial UTI in Febrile Infants and Young Children 2–24 Months of Age. Pediatrics. 2016;138(6):e20163026.
[doi:10.1542/peds.2016-3026](https://doi.org/10.1542/peds.2016-3026)
49. National Institute for Health and Care Excellence. Urinary tract infection in under 16s: diagnosis and management (NG224). 2022. [nice.org.uk/guidance/ng224](https://www.nice.org.uk/guidance/ng224)
50. Trautmann A, Boyer O, Hodson E, et al; International Pediatric Nephrology Association. IPNA clinical practice recommendations for the diagnosis and management of children with steroid-sensitive nephrotic syndrome. Pediatr Nephrol. 2023;38(3):877–919.
[doi:10.1007/s00467-022-05739-3](https://doi.org/10.1007/s00467-022-05739-3)
51. Abman SH, Hansmann G, Archer SL, et al. Pediatric Pulmonary Hypertension: Guidelines From the American Heart Association and American Thoracic Society. Circulation. 2015;132(21):2037–2099.
[doi:10.1161/CIR.0000000000000329](https://doi.org/10.1161/CIR.0000000000000329)
52. State of Kuwait. Law No. 70 of 2020 on the Practice of the Medical Profession and its Auxiliary Professions, the Rights of Patients and Health Facilities. Official Gazette (Kuwait Al-Youm), 2020.
53. State of Kuwait. Law No. 14 of 2019 on Mental Health. Official Gazette (Kuwait Al-Youm), 2019.

54. Abu-El-Haija M, Kumar S, Quiros JA, et al. Management of Acute Pancreatitis in the Pediatric Population: A Clinical Report From the NASPGHAN Pancreas Committee. J Pediatr Gastroenterol Nutr. 2018;66(1):159–176. [doi:10.1097/MPG.0000000000001715](https://doi.org/10.1097/MPG.0000000000001715)
55. Fawaz R, Baumann U, Ekong U, et al. Guideline for the Evaluation of Cholestatic Jaundice in Infants: Joint Recommendations of NASPGHAN and ESPGHAN. J Pediatr Gastroenterol Nutr. 2017;64(1):154–168. [doi:10.1097/MPG.0000000000001334](https://doi.org/10.1097/MPG.0000000000001334)
56. Thomson M, Amil-Dias J, Assa A, et al. A management strategy for nonvariceal gastrointestinal bleeding in children: an ESPGHAN position paper. J Pediatr Gastroenterol Nutr. 2026;82(3):895–915. [doi:10.1002/jpn3.70278](https://doi.org/10.1002/jpn3.70278)
57. Squires RH Jr, Shneider BL, Bucuvalas J, et al. Acute liver failure in children: the first 348 patients in the pediatric acute liver failure study group. J Pediatr. 2006;148(5):652–658. [doi:10.1016/j.jpeds.2005.12.051](https://doi.org/10.1016/j.jpeds.2005.12.051)
58. Le Saux N, Robinson JL; Canadian Pediatric Society, Infectious Diseases and Immunization Committee. Uncomplicated pneumonia in healthy Canadian children and youth: Practice points for management. Updated 2024. cps.ca
59. Chibuk T, Cohen E, Robinson JL, Mahant S, Hartfield D; Canadian Pediatric Society. Pediatric complicated pneumonia: Diagnosis and management of empyema. Updated 2024. cps.ca

60. Bradley JS, Byington CL, Shah SS, et al. The management of community-acquired pneumonia in infants and children older than 3 months of age (PIDS/IDSA). Clin Infect Dis. 2011;53(7):e25–e76. [doi:10.1093/cid/cir531](https://doi.org/10.1093/cid/cir531)
61. Penner J, Fanella S; Canadian Pediatric Society, Infectious Diseases and Immunization Committee. Management of community-associated bacterial skin and soft tissue infections in children and prevention of recurrence. 2026. cps.ca
62. Hao R, Saleh M, Liang T, et al. The prevalence of serious bacterial infections in neutropenic immunocompetent febrile children. Am J Emerg Med. 2021;45:1–6. [doi:10.1016/j.ajem.2021.02.017](https://doi.org/10.1016/j.ajem.2021.02.017)
63. Woods CR, Bradley JS, Chatterjee A, et al. Clinical practice guideline by PIDS and IDSA: 2021 guideline on diagnosis and management of acute hematogenous osteomyelitis in pediatrics. J Pediatric Infect Dis Soc. 2021;10(8):801–844. [doi:10.1093/jpids/piab027](https://doi.org/10.1093/jpids/piab027)
64. Woods CR, Bradley JS, Chatterjee A, et al. Clinical practice guideline by PIDS and IDSA: 2023 guideline on diagnosis and management of acute bacterial arthritis in pediatrics. J Pediatric Infect Dis Soc. 2024;13(1):1–59. [doi:10.1093/jpids/piad089](https://doi.org/10.1093/jpids/piad089)

65. Peltola H, Pääkkönen M, Kallio P, Kallio MJT; OM-SA Study Group. Prospective, randomized trial of 10 days versus 30 days of antimicrobial treatment, including a short-term course of parenteral therapy, for childhood septic arthritis. Clin Infect Dis. 2009;48(9):1201–1210. [doi:10.1086/597582](https://doi.org/10.1086/597582)
66. Kocher MS, Zurakowski D, Kasser JR. Differentiating between septic arthritis and transient synovitis of the hip in children: an evidence-based clinical prediction algorithm. J Bone Joint Surg Am. 1999;81(12):1662–1670. [doi:10.2106/00004623-199912000-00002](https://doi.org/10.2106/00004623-199912000-00002)
67. Averbuch D, Lehrnbecher T, Cesaro S, et al. Empirical and targeted antimicrobial therapy in patients with febrile neutropenia and hematological malignancy or after hematopoietic cell transplantation: recommendations from the 10th European Conference on Infections in Leukemia (ECIL-10). Lancet Infect Dis. 2026;26(6):e232–e247. [doi:10.1016/S1473-3099\(25\)00619-X](https://doi.org/10.1016/S1473-3099(25)00619-X)
68. Jones GL, Will A, Jackson GH, et al. Guidelines for the management of tumor lysis syndrome in adults and children with hematological malignancies on behalf of the British Committee for Standards in Hematology. Br J Hematol. 2015;169(5):661–671. [doi:10.1111/bjh.13403](https://doi.org/10.1111/bjh.13403)

69. Labonté S, Bergeron É, Laverdière I, et al. Rasburicase real-world usage for the prevention and treatment of tumor lysis syndrome in adults and children. *Ann Pharmacother.* 2025.
[doi:10.1177/10600280251391871](https://doi.org/10.1177/10600280251391871)
70. Zailani MAH, et al. Population screening for glucose-6-phosphate dehydrogenase deficiency using quantitative point-of-care tests: a systematic review. *Front Genet.* 2023;14:1098828.
[doi:10.3389/fgene.2023.1098828](https://doi.org/10.3389/fgene.2023.1098828)
71. Gwarzo I, Coleman KD, McKinley K, et al. Opioid timeliness in the emergency department and hospitalizations for acute sickle cell pain. *JAMA Pediatr.* 2025;179(11):1194–1202.
[doi:10.1001/jamapediatrics.2025.2967](https://doi.org/10.1001/jamapediatrics.2025.2967)
72. Yates AM, Aygun B, Liem R, et al. Health supervision for children and adolescents with sickle cell disease: clinical report. *Pediatrics.* 2024;154(2):e2024066842.
[doi:10.1542/peds.2024-066842](https://doi.org/10.1542/peds.2024-066842)
73. Kracht MJ, et al. A proposed algorithm for evaluation and management of pediatric hemophilia patients who present to the emergency department with head trauma. *Pediatr Blood Cancer.* 2022;69(9):e29811.
[doi:10.1002/pbc.29811](https://doi.org/10.1002/pbc.29811)
74. Kendel NE, et al. Increasing optimal coagulation factor dosing in the pediatric emergency department: update to a quality improvement study. *Hemophilia.* 2024;30(2):367–374. [doi:10.1111/hae.14927](https://doi.org/10.1111/hae.14927)

75. Wirrell EC, Hood V, Knupp KG, et al. International consensus on diagnosis and management of Dravet syndrome. *Epilepsia*. 2022;63(7):1761–1777. [doi:10.1111/epi.17274](https://doi.org/10.1111/epi.17274)
76. McMullan BJ, Andresen D, Blyth CC, et al; ANZPID-ASAP group. Antibiotic duration and timing of the switch from intravenous to oral route for bacterial infections in children: systematic review and guidelines. *Lancet Infect Dis*. 2016;16(8):e139–e152. [doi:10.1016/S1473-3099\(16\)30024-X](https://doi.org/10.1016/S1473-3099(16)30024-X)
77. Keren R, Shah SS, Srivastava R, et al; Pediatric Research in Inpatient Settings Network. Comparative effectiveness of intravenous vs oral antibiotics for postdischarge treatment of acute osteomyelitis in children. *JAMA Pediatr*. 2015;169(2):120–128. [doi:10.1001/jamapediatrics.2014.2822](https://doi.org/10.1001/jamapediatrics.2014.2822)
78. Zaoutis T, Localio AR, Leckerman K, et al. Prolonged intravenous therapy versus early transition to oral antimicrobial therapy for acute osteomyelitis in children. *Pediatrics*. 2009;123(2):636–642. [doi:10.1542/peds.2008-0596](https://doi.org/10.1542/peds.2008-0596)
79. Chalut DS, Ducharme FM, Davis GM. The Preschool Respiratory Assessment Measure (PRAM): a responsive index of acute asthma severity. *J Pediatr*. 2000;137(6):762–768. [doi:10.1067/mpd.2000.110121](https://doi.org/10.1067/mpd.2000.110121)

80. Ducharme FM, Chalut D, Plotnick L, et al. The Pediatric Respiratory Assessment Measure: a valid clinical score for assessing acute asthma severity from toddlers to teenagers. *J Pediatr*. 2008;152(4):476–480. [doi:10.1016/j.jpeds.2007.08.034](https://doi.org/10.1016/j.jpeds.2007.08.034)
81. Kuppermann N, Holmes JF, Dayan PS, et al; Pediatric Emergency Care Applied Research Network (PECARN). Identification of children at very low risk of clinically-important brain injuries after head trauma: a prospective cohort study. *Lancet*. 2009;374(9696):1160–1170. [doi:10.1016/S0140-6736\(09\)61558-0](https://doi.org/10.1016/S0140-6736(09)61558-0)
82. Kochanek PM, Tasker RC, Carney N, et al. Guidelines for the Management of Pediatric Severe Traumatic Brain Injury, Third Edition: Update of the Brain Trauma Foundation Guidelines, Executive Summary. *Pediatr Crit Care Med*. 2019;20(3):280–289. [doi:10.1097/PCC.0000000000001736](https://doi.org/10.1097/PCC.0000000000001736)
83. Thornton PS, Stanley CA, De León DD, Harris D, Haymond MW, Hussain K, et al. Recommendations from the Pediatric Endocrine Society for Evaluation and Management of Persistent Hypoglycemia in Neonates, Infants, and Children. *J Pediatr*. 2015;167(2):238–245. [doi:10.1016/j.jpeds.2015.03.057](https://doi.org/10.1016/j.jpeds.2015.03.057)
84. Adamkin DH; Committee on Fetus and Newborn. Postnatal Glucose Homeostasis in Late-Preterm and Term Infants. *Pediatrics*. 2011;127(3):575–579. [doi:10.1542/peds.2010-3851](https://doi.org/10.1542/peds.2010-3851)

85. Harris DL, Weston PJ, Signal M, Chase JG, Harding JE. Dextrose gel for neonatal hypoglycemia (the Sugar Babies Study): a randomized, double-blind, placebo-controlled trial. *Lancet*. 2013;382(9910):2077–2083.
[doi:10.1016/S0140-6736\(13\)61645-1](https://doi.org/10.1016/S0140-6736(13)61645-1)
86. Neu J, Walker WA. Necrotizing enterocolitis. *N Engl J Med*. 2011;364(3):255–264.
[doi:10.1056/NEJMra1005408](https://doi.org/10.1056/NEJMra1005408)
87. Walsh MC, Kliegman RM. Necrotizing enterocolitis: treatment based on staging criteria. *Pediatr Clin North Am*. 1986;33(1):179–201. [doi:10.1016/S0031-3955\(16\)34975-6](https://doi.org/10.1016/S0031-3955(16)34975-6)
88. Sharpe C, Reiner GE, Davis SL, et al; NEOLEV2 Investigators. Levetiracetam Versus Phenobarbital for Neonatal Seizures: A Randomized Controlled Trial. *Pediatrics*. 2020;145(6):e20193182.
[doi:10.1542/peds.2019-3182](https://doi.org/10.1542/peds.2019-3182)
89. Burstein B, Lirette MP, Beck C, Chauvin-Kimoff L, Chan K; Canadian Pediatric Society, Acute Care Committee. Management of well-appearing febrile young infants aged $\leq$ 90 days. *Pediatr Child Health*. 2024;29(1):50–66.
[doi:10.1093/pch/pxad085](https://doi.org/10.1093/pch/pxad085)

90. Guarino A, Ashkenazi S, Gendrel D, Lo Vecchio A, Shamir R, Szajewska H; ESPGHAN/ESPID. Evidence-Based Guidelines for the Management of Acute Gastroenteritis in Children in Europe: Update 2014. *J Pediatr Gastroenterol Nutr.* 2014;59(1):132–152.
[doi:10.1097/MPG.0000000000000375](https://doi.org/10.1097/MPG.0000000000000375)
91. Gill PJ, Buba M, Krueger C, Wolter NE, Reginald YA; Canadian Pediatric Society, Hospital Pediatrics Section. Preseptal and orbital cellulitis in otherwise healthy children and youth. Ottawa: CPS; 2026. cps.ca
92. Krueger C, Nguyen ELV, Mahant S, et al. Association of empiric antibiotic selection and clinical outcomes in hospitalized children with severe orbital infections: a retrospective cohort study. *Arch Dis Child.* 2024;109(11):932–940.
[doi:10.1136/archdischild-2023-326175](https://doi.org/10.1136/archdischild-2023-326175)
93. Kornelsen E, Mahant S, Parkin P, et al. Corticosteroids for periorbital and orbital cellulitis. *Cochrane Database Syst Rev.* 2021;4(4):CD013535.
[doi:10.1002/14651858.CD013535.pub2](https://doi.org/10.1002/14651858.CD013535.pub2)
94. Leonard JC, et al; Pediatric Emergency Care Applied Research Network (PECARN). PECARN prediction rule for cervical spine imaging of children presenting to the emergency department with blunt trauma: a multicentre prospective observational study. *Lancet Child Adolesc Health.* 2024;8(7):482–490. [doi:10.1016/S2352-4642\(24\)00104-4](https://doi.org/10.1016/S2352-4642(24)00104-4)

95. Lee J, Kim Y, Kim YJ, et al. Impact of empirical and definitive antibiotics on pediatric febrile urinary tract infection caused by ESBL-producing Enterobacterales. *Pathogens*. 2025;14(11):1103.
[doi:10.3390/pathogens14111103](https://doi.org/10.3390/pathogens14111103)
96. McGrath BA, Bates L, Atkinson D, Moore JA; National Tracheostomy Safety Project. Multidisciplinary guidelines for the management of tracheostomy and laryngectomy airway emergencies. *Anesthesia*. 2012;67(9):1025–1041.
[doi:10.1111/j.1365-2044.2012.07217.x](https://doi.org/10.1111/j.1365-2044.2012.07217.x)
97. Yen K, Riegert A, Gorelick MH. Derivation of the DIVA score: a clinical prediction rule for the identification of children with difficult intravenous access. *Pediatr Emerg Care*. 2008;24(3):143–147.
[doi:10.1097/PEC.0b013e3181666f32](https://doi.org/10.1097/PEC.0b013e3181666f32)
98. Riker MW, Kennedy C, Winfrey BS, Yen K, Dowd MD. Validation and refinement of the difficult intravenous access score. *Acad Emerg Med*. 2011;18(11):1129–1134.
[doi:10.1111/j.1553-2712.2011.01205.x](https://doi.org/10.1111/j.1553-2712.2011.01205.x)
99. Holmes JF, Lillis K, Monroe D, et al; PECARN. Identifying children at very low risk of clinically important blunt abdominal injuries. *Ann Emerg Med*. 2013;62(2):107–116.e2.
[doi:10.1016/j.annemergmed.2012.11.009](https://doi.org/10.1016/j.annemergmed.2012.11.009)

100. Russell RT, Esparaz JR, Beckwith MA, et al. Pediatric traumatic hemorrhagic shock consensus conference recommendations. J Trauma Acute Care Surg. 2023;94(1S):S2–S10. [doi:10.1097/TA.0000000000003805](https://doi.org/10.1097/TA.0000000000003805)
101. Spasovski G, Vanholder R, Allolio B, et al; Hyponatremia Guideline Development Group. Clinical practice guideline on diagnosis and treatment of hyponatremia. Eur J Endocrinol. 2014;170(3):G1–G47. [doi:10.1530/EJE-13-1020](https://doi.org/10.1530/EJE-13-1020)
102. Shah SA, Ayus JC, Moritz ML. A survey of hospital pharmacy guidelines for the administration of 3% sodium chloride in children. Children (Basel). 2022;9(1):57. [doi:10.3390/children9010057](https://doi.org/10.3390/children9010057)
103. Cohen E, Kuo DZ, Agrawal R, et al. Children with medical complexity: an emerging population for clinical and research initiatives. Pediatrics. 2011;127(3):529–538. [doi:10.1542/peds.2010-0910](https://doi.org/10.1542/peds.2010-0910)
104. Kuo DZ, Houtrow AJ; Council on Children With Disabilities. Recognition and management of medical complexity. Pediatrics. 2016;138(6):e20163021. [doi:10.1542/peds.2016-3021](https://doi.org/10.1542/peds.2016-3021)
105. Homan M, Hauser B, Romano C, et al. Percutaneous endoscopic gastrostomy in children: an update to the ESPGHAN position paper. J Pediatr Gastroenterol Nutr. 2021;73(3):415–426. [doi:10.1097/MPG.0000000000003207](https://doi.org/10.1097/MPG.0000000000003207)

106. Le Saux N; Canadian Paediatric Society, Infectious Diseases and Immunization Committee. Guidelines for the management of suspected and confirmed bacterial meningitis in Canadian children older than one month of age. *Paediatr Child Health*. 2014;19(3):141–146. [doi:10.1093/pch/19.3.141](https://doi.org/10.1093/pch/19.3.141)
107. Brouwer MC, Tunkel AR, McKhann GM 2nd, van de Beek D. Brain abscess. *N Engl J Med*. 2014;371(5):447–456. [doi:10.1056/NEJMra1301635](https://doi.org/10.1056/NEJMra1301635)
108. Bodilsen J, D'Alessandris QG, Humphreys H, et al; ESCMID Study Group for Infections of the Brain (ESGIB). European Society of Clinical Microbiology and Infectious Diseases guidelines on diagnosis and treatment of brain abscess in children and adults. *Clin Microbiol Infect*. 2024;30(1):66–89. [doi:10.1016/j.cmi.2023.08.016](https://doi.org/10.1016/j.cmi.2023.08.016)
109. Khan DA, Banerji A, Blumenthal KG, et al. Drug allergy: A 2022 practice parameter update. *J Allergy Clin Immunol*. 2022;150(6):1333–1393. [doi:10.1016/j.jaci.2022.08.028](https://doi.org/10.1016/j.jaci.2022.08.028)
110. Wong T, Atkinson A, t'Jong G, et al; Canadian Paediatric Society. Beta-lactam allergy in the paediatric population. *Paediatr Child Health*. 2020;25(1):62–71. [doi:10.1093/pch/pxz179](https://doi.org/10.1093/pch/pxz179)

111. Fraser N, Davies BW, Cusack J. Neonatal omphalitis: a review of its serious complications. *Acta Paediatr.* 2006;95(5):519-522. [doi:10.1111/j.1651-2227.2006.tb02277.x](https://doi.org/10.1111/j.1651-2227.2006.tb02277.x)
112. American Academy of Pediatrics Task Force on Circumcision. Male circumcision. *Pediatrics.* 2012;130(3):e756-e785. [doi:10.1542/peds.2012-1990](https://doi.org/10.1542/peds.2012-1990)
113. Sharara-Chami R, Lakissian Z, Charafeddine L, Milad N, El-Hout Y. Combination Analgesia for Neonatal Circumcision: A Randomized Controlled Trial. *Pediatrics.* 2017;140(6):e20171935. [doi:10.1542/peds.2017-1935](https://doi.org/10.1542/peds.2017-1935)
114. Safar H, Alsaber A, Al-Shammari A, et al. The Effectiveness of No or Low-Dose versus High-Dose Aspirin in Treating Acute Kawasaki Disease: A Systematic Review and Meta-Analysis. *Clin Pract.* 2024;14(4):1421-1435. [doi:10.3390/clinpract14040105](https://doi.org/10.3390/clinpract14040105)
115. Bindoli S, De Matteis A, Mitrovic S, Fautrel B, Carmona L, De Benedetti F. Efficacy and safety of therapies for Still's disease and macrophage activation syndrome: a systematic review informing the EULAR/PreS guidelines. *Ann Rheum Dis.* 2024;83(12):1731-1747. [doi:10.1136/ard-2024-225854](https://doi.org/10.1136/ard-2024-225854)

116. Inborn errors of immunity (primary immunodeficiencies): definition, classification and diagnostic approach. *Allergy Asthma Clin Immunol.* 2025;21. [doi:10.1186/s13223-024-00938-z](https://doi.org/10.1186/s13223-024-00938-z)
117. Aden A, Jemide I, Islam T, et al. Clinical warning signs in detecting inborn errors of immunity in children: a diagnostic accuracy systematic review. *J Allergy Clin Immunol Pract.* 2026;14(4):920–936.e1. [doi:10.1016/j.jaip.2026.01.026](https://doi.org/10.1016/j.jaip.2026.01.026)
118. Buckley RH. The long quest for neonatal screening for severe combined immunodeficiency. *J Allergy Clin Immunol.* 2012;129(3):597–604. [doi:10.1016/j.jaci.2011.12.964](https://doi.org/10.1016/j.jaci.2011.12.964)
119. Neal JM, Neal EJ, Weinberg GL. American Society of Regional Anesthesia and Pain Medicine Local Anesthetic Systemic Toxicity checklist: 2020 version. *Reg Anesth Pain Med.* 2021;46(1):81–82. [doi:10.1136/rapm-2020-101986](https://doi.org/10.1136/rapm-2020-101986)
120. American Society of Hematology 2018 guidelines for management of venous thromboembolism: treatment of pediatric venous thromboembolism. *Blood Adv.* 2018;2(22):3292–3316. [doi:10.1182/bloodadvances.2018024786](https://doi.org/10.1182/bloodadvances.2018024786)
121. Hydroxocobalamin for injection: US prescribing information. US Food and Drug Administration; 2018. accessdata.fda.gov

122. Thompson MJ, Ninis N, Perera R, et al. Clinical recognition of meningococcal disease in children and adolescents. *Lancet*. 2006;367(9508):397-403.
[doi:10.1016/S0140-6736\(06\)67932-4](https://doi.org/10.1016/S0140-6736(06)67932-4)
123. De Rose DU, Bosco A, Braguglia A, et al. Neonatal herpes simplex virus infection: from the maternal infection to the child outcome. *J Med Virol*. 2023;95(8):e29024.
[doi:10.1002/jmv.29024](https://doi.org/10.1002/jmv.29024)
124. Ozen S, Pistorio A, Iusan SM, et al. EULAR/PRINTO/PRES criteria for Henoch-Schönlein purpura, childhood polyarteritis nodosa, childhood Wegener granulomatosis and childhood Takayasu arteritis: Ankara 2008. Part II: final classification criteria. *Ann Rheum Dis*. 2010;69(5):798–806.
[doi:10.1136/ard.2009.116657](https://doi.org/10.1136/ard.2009.116657)
125. Pediatric herpes zoster: should I be concerned for immunodeficiency? A review. *Front Pediatr*. 2025;13:1561339.
[doi:10.3389/fped.2025.1561339](https://doi.org/10.3389/fped.2025.1561339)
126. Marcus CL, Brooks LJ, Draper KA, et al. Diagnosis and management of childhood obstructive sleep apnea syndrome. *Pediatrics*. 2012;130(3):576–584.
[doi:10.1542/peds.2012-1671](https://doi.org/10.1542/peds.2012-1671)
127. Kaditis AG, Alonso Alvarez ML, Boudewyns A, et al. Obstructive sleep disordered breathing in 2- to 18-year-old children: diagnosis and management. *Eur Respir J*. 2016;47(1):69–94.
[doi:10.1183/13993003.00385-2015](https://doi.org/10.1183/13993003.00385-2015)

128. US Food and Drug Administration. FDA Drug Safety Communication: FDA restricts use of prescription codeine and tramadol medicines in children; recommends against use in breastfeeding women. 20 Apr 2017. [fda.gov](https://www.fda.gov)
129. Hampl SE, Hassink SG, Skinner AC, et al. Clinical Practice Guideline for the Evaluation and Treatment of Children and Adolescents With Obesity. *Pediatrics*. 2023;151(2):e2022060640. [doi:10.1542/peds.2022-060640](https://doi.org/10.1542/peds.2022-060640)
130. Natale S, Bradley J, Nguyen WH, et al. Drug dosing in children with obesity: a narrative updated review. *Ital J Pediatr*. 2022;48(1):168. [doi:10.1186/s13052-022-01361-z](https://doi.org/10.1186/s13052-022-01361-z)
131. Qin KR, Qu LG. Diagnosing with a TWIST: systematic review and meta-analysis of a testicular torsion risk score. *J Urol*. 2022;208(1):62–70. [doi:10.1097/JU.0000000000002496](https://doi.org/10.1097/JU.0000000000002496)
132. Hayashi Y, Kojima Y, Mizuno K, Kohri K. Prepuce: phimosis, paraphimosis, and circumcision. *ScientificWorldJournal*. 2011;11:289–301. [doi:10.1100/tsw.2011.31](https://doi.org/10.1100/tsw.2011.31)
133. Bivalacqua TJ, Allen BK, Brock GB, et al. Acute ischemic priapism: an AUA/SMSNA guideline. *J Urol*. 2021;206(5):1114–1121. [doi:10.1097/JU.0000000000002236](https://doi.org/10.1097/JU.0000000000002236)

134. Evaluation and management of hematuria in children. *Curr Treat Options Pediatr.* 2024;10:127–140. [doi:10.1007/s40746-024-00302-8](https://doi.org/10.1007/s40746-024-00302-8)
135. Greenfield SP, Williot P, Kaplan D. Gross hematuria in children: a ten-year review. *Urology.* 2007;69(1):166–169. [doi:10.1016/j.urology.2006.10.018](https://doi.org/10.1016/j.urology.2006.10.018)
136. Naranje S, Kelly DM, Sawyer JR. A systematic approach to the evaluation of a limping child. *Am Fam Physician.* 2015;92(10):908–916. [PMID 26554284](https://pubmed.ncbi.nlm.nih.gov/26554284/) (no DOI assigned)
137. Bexkens R, Washburn FJ, Eygendaal D, van den Bekerom MPJ, Oh LS. Effectiveness of reduction maneuvers in the treatment of nursemaid's elbow: a systematic review and meta-analysis. *Am J Emerg Med.* 2017;35(1):159–163. [doi:10.1016/j.ajem.2016.10.059](https://doi.org/10.1016/j.ajem.2016.10.059)
138. Perry DC, Metcalfe D, Lane S, Turner S. Childhood obesity and slipped capital femoral epiphysis. *Pediatrics.* 2018;142(5):e20181067. [doi:10.1542/peds.2018-1067](https://doi.org/10.1542/peds.2018-1067)
139. Flaherty EG, Perez-Rossello JM, Levine MA, Hennrikus WL; AAP Committee on Child Abuse and Neglect. Evaluating young children with fractures for child abuse: clinical report. *Pediatrics.* 2025;155(2):e2024070074. [doi:10.1542/peds.2024-070074](https://doi.org/10.1542/peds.2024-070074)

140. Livingston KS, Glotzbecker MP, Shore BJ. Pediatric acute compartment syndrome. *J Am Acad Orthop Surg.* 2017;25(5):358–364. [doi:10.5435/JAAOS-D-15-00655](https://doi.org/10.5435/JAAOS-D-15-00655)
141. Garner MR, Sethuraman SA, Schade MA, Boateng H. Antibiotic prophylaxis in open fractures: evidence, evolving issues, and recommendations. *J Am Acad Orthop Surg.* 2020;28(8):309–315. [doi:10.5435/JAAOS-D-18-00193](https://doi.org/10.5435/JAAOS-D-18-00193)
142. Weaver LK, Hopkins RO, Chan KJ, et al. Hyperbaric oxygen for acute carbon monoxide poisoning. *N Engl J Med.* 2002;347(14):1057–1067. [doi:10.1056/NEJMoa013121](https://doi.org/10.1056/NEJMoa013121)
143. Weaver LK. Carbon monoxide poisoning (reprinted from the 2023 Hyperbaric Indications Manual, 15th edition). *Undersea Hyperb Med.* 2024;51(3):253–276. [PMID 39348519](https://pubmed.ncbi.nlm.nih.gov/39348519/) (no DOI assigned)
144. Yildiz R, et al. How to decide oxygen therapy in childhood carbon monoxide poisoning? *Turk Arch Pediatr.* 2023;58(3):282–288. [doi:10.5152/TurkArchPediatr.2023.22189](https://doi.org/10.5152/TurkArchPediatr.2023.22189)
145. Buckley NA, Juurlink DN, Isbister G, Bennett MH, Lavonas EJ. Hyperbaric oxygen for carbon monoxide poisoning. *Cochrane Database Syst Rev.* 2011;(4):CD002041. [doi:10.1002/14651858.CD002041.pub3](https://doi.org/10.1002/14651858.CD002041.pub3)

146. Walker PF, Buehner MF, Wood LA, et al. Diagnosis and management of inhalation injury: an updated review. Crit Care. 2015;19(1):351. [doi:10.1186/s13054-015-1077-4](https://doi.org/10.1186/s13054-015-1077-4)
147. State of Kuwait. Child Rights Law No. 21 of 2015. Kuwait National Child Protection Program, Ministry of Health. kncpp.moh.gov.kw
148. Kuwait National Child Protection Program (KNCPP), Ministry of Health — Child Protection Hotline 147. kncpp.moh.gov.kw
149. Manterola C, Vial M, Moraga J, Astudillo P. Analgesia in patients with acute abdominal pain. Cochrane Database Syst Rev. 2022;3:CD005660. [Cochrane](https://doi.org/10.1002/CD005660)
150. Poonai N, et al. Opioid analgesia for acute abdominal pain in children: a systematic review and meta-analysis. Acad Emerg Med. 2014;21(11):1183–1192. [doi:10.1111/acem.12509](https://doi.org/10.1111/acem.12509)
151. Podda M, et al. Diagnosis and treatment of acute appendicitis: 2025 edition of the WSES Jerusalem guidelines. JAMA Surg. 2026. [doi:10.1001/jamasurg.2025.6218](https://doi.org/10.1001/jamasurg.2025.6218)
152. Rodriguez L, Ambartsumyan L, Baumgartner K, et al. American Gastroenterological Association–North American Society for Pediatric Gastroenterology, Hepatology and Nutrition Pediatric Functional Constipation Clinical Care Pathway. Clin Gastroenterol Hepatol. 2026. [doi:10.1016/j.cgh.2026.03.003](https://doi.org/10.1016/j.cgh.2026.03.003)

153. Tabbers MM, DiLorenzo C, Berger MY, et al. Evaluation and treatment of functional constipation in infants and children: evidence-based recommendations from ESPGHAN and NASPGHAN. *J Pediatr Gastroenterol Nutr.* 2014;58(2):258–274. [doi:10.1097/MPG.0000000000000266](https://doi.org/10.1097/MPG.0000000000000266)
154. Emapalumab-lzsg injection: US prescribing information. Sobi Inc.; 2025. [DailyMed](#)
155. The Neonatal Inhaled Nitric Oxide Study Group. Inhaled nitric oxide in full-term and nearly full-term infants with hypoxic respiratory failure. *N Engl J Med.* 1997;336(9):597–604. [doi:10.1056/NEJM199702273360901](https://doi.org/10.1056/NEJM199702273360901)
156. Cook DJ, Fuller HD, Guyatt GH, et al. Risk factors for gastrointestinal bleeding in critically ill patients. Canadian Critical Care Trials Group. *N Engl J Med.* 1994;330(6):377–381. [doi:10.1056/NEJM199402103300601](https://doi.org/10.1056/NEJM199402103300601)
157. Canadian Pediatric Society. Management of febrile neutropenia in immunocompetent children and youth. *Pediatr Child Health.* 2023;28(5):324–333. [doi:10.1093/pch/pxac127](https://doi.org/10.1093/pch/pxac127)
158. Barrington K; Canadian Paediatric Society, Fetus and Newborn Committee. Premedication for endotracheal intubation in the newborn infant. *Paediatr Child Health.* 2011;16(3):159–164. [doi:10.1093/pch/16.3.159](https://doi.org/10.1093/pch/16.3.159)

159. Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO Clinical Practice Guideline for Acute Kidney Injury. *Kidney Int Suppl.* 2012;2(1):1–138.
160. Panchal AR, Bartos JA, Cabañas JG, et al. Part 3: Adult Basic and Advanced Life Support: 2020 American Heart Association Guidelines for CPR and Emergency Cardiovascular Care. *Circulation.* 2020;142(16 suppl 2):S366–S468.
[doi:10.1161/CIR.0000000000000916](https://doi.org/10.1161/CIR.0000000000000916)
161. American Academy of Pediatrics. Red Book: 2024–2027 Report of the Committee on Infectious Diseases. 33rd ed. Itasca, IL: American Academy of Pediatrics; 2024.
162. Painter K, Anand S, Philip K. Omphalitis. In: StatPearls. Treasure Island (FL): StatPearls Publishing; 2022. [NBK513338](https://doi.org/10.1161/CIR.0000000000000916)
163. Sawyer RG, Claridge JA, Nathens AB, et al. Trial of short-course antimicrobial therapy for intraabdominal infection. *N Engl J Med.* 2015;372(21):1996–2005.
[doi:10.1056/NEJMoa1411162](https://doi.org/10.1056/NEJMoa1411162)
164. Maitland K, Kiguli S, Opoka RO, et al. (FEAST) Mortality after fluid bolus in African children with severe infection. *N Engl J Med.* 2011;364(26):2483–2495.
[doi:10.1056/NEJMoa1101549](https://doi.org/10.1056/NEJMoa1101549)

165. Hay AD, Costelloe C, Redmond NM, et al. Paracetamol plus ibuprofen for the treatment of fever in children (PITCH): randomized controlled trial. *BMJ*. 2008;337:a1302.
[doi:10.1136/bmj.a1302](https://doi.org/10.1136/bmj.a1302)
166. Bjornson C, Russell K, Vandermeer B, Klassen TP, Johnson DW. Nebulized epinephrine for croup in children. *Cochrane Database Syst Rev*. 2013;(10):CD006619.
[doi:10.1002/14651858.CD006619.pub3](https://doi.org/10.1002/14651858.CD006619.pub3)
167. Hankey PB, Tracy M, Arganbright JM. Emerging findings for the assessment and treatment of pediatric deep neck abscesses. *Curr Opin Otolaryngol Head Neck Surg*. 2023;31(6):407–411.
[doi:10.1097/MOO.0000000000000937](https://doi.org/10.1097/MOO.0000000000000937)
168. Liu Y, Nicotera DJ, Islam AA, Dunsky K, Lieu JEC. Prognostic factors for retropharyngeal abscess in children receiving surgery or antibiotic therapy. *Laryngoscope*. 2024;134(4):1955–1960.
[doi:10.1002/lary.31064](https://doi.org/10.1002/lary.31064)
169. Black AE, Flynn PER, Smith HL, Thomas ML, Wilkinson KA. Development of a guideline for the management of the unanticipated difficult airway in pediatric practice. *Pediatr Anesth*. 2015;25(4):346–362.
[doi:10.1111/pan.12615](https://doi.org/10.1111/pan.12615)

170. Fiadjoe JE, Nishisaki A, Jagannathan N, et al. Airway management complications in children with difficult tracheal intubation from the Pediatric Difficult Intubation (PeDI) registry: a prospective cohort analysis. *Lancet Respir Med*. 2016;4(1):37–48.
[doi:10.1016/S2213-2600\(15\)00508-1](https://doi.org/10.1016/S2213-2600(15)00508-1)
171. Peyton J, Park R, Staffa SJ, et al. Videolaryngoscopy using standard blades or non-standard blades in children in the Pediatric Difficult Intubation Registry. *Br J Anesth*. 2021;126(1):331–339.
[doi:10.1016/j.bja.2020.08.010](https://doi.org/10.1016/j.bja.2020.08.010)
172. Maguire S, Moynihan S, Mann M, Potokar T, Kemp AM. A systematic review of the features that indicate intentional scalds in children. *Burns*. 2008;34(8):1072–1081.
[doi:10.1016/j.burns.2008.02.011](https://doi.org/10.1016/j.burns.2008.02.011)
173. Subcommittee on Febrile Seizures, American Academy of Pediatrics. Neurodiagnostic evaluation of the child with a simple febrile seizure. *Pediatrics*. 2011;127(2):389–394.
[doi:10.1542/peds.2010-3318](https://doi.org/10.1542/peds.2010-3318)
174. Szpilman D, Bierens JJLM, Handley AJ, Orlowski JP. Drowning. *N Engl J Med*. 2012;366(22):2102–2110.
[doi:10.1056/NEJMr1013317](https://doi.org/10.1056/NEJMr1013317)
175. Kuwait PICUs Taskforce. PICU Septic Shock Approach and Management. Version 2.0. Kuwait Ministry of Health; October 2025.

176. Kuwait PICUs Taskforce. PICU Management of Diabetic Ketoacidosis (DKA). Version 2.0. Kuwait Ministry of Health; November 2025.
177. St. Peter SD, Ampofo K, Brogan T, et al. Clinical practice guideline by the Infectious Diseases Society of America and the Pediatric Infectious Diseases Society: 2026 guideline update on the management of community-acquired pneumonia in infants and children older than 3 months of age. *Clin Infect Dis*. 2026. [doi:10.1093/cid/ciag186](https://doi.org/10.1093/cid/ciag186)
178. Moraa I, Sturman N, McGuire TM, van Driel ML. Heliox for croup in children. *Cochrane Database Syst Rev*. 2021;(8):CD006822. doi:10.1002/14651858.CD006822.pub6
179. Rodrigo GJ, Pollack CV, Rodrigo C, Rowe BH. Heliox for non-intubated acute asthma patients. *Cochrane Database Syst Rev*. 2006;(4):CD002884. doi:10.1002/14651858.CD002884.pub2
180. Kim IK, Phrampus E, Venkataraman S, et al. Helium/oxygen-driven albuterol nebulization in the treatment of children with moderate to severe asthma exacerbations: a randomized, controlled trial. *Pediatrics*. 2005;116(5):1127–1133. doi:10.1542/peds.2004-2136
181. Lew A, Morrison JM, Amankwah E, Sochet AA. Heliox for pediatric critical asthma: a multicenter, retrospective, registry-based descriptive study. *J Intensive Care Med*. 2022;37(6):776–783. doi:10.1177/08850666211026550

182. Fouad AF, Abbott PV, Tsilingaridis G, et al. International Association of Dental Traumatology guidelines for the management of traumatic dental injuries: 2. Avulsion of permanent teeth. *Dent Traumatol.* 2020;36(4):331–342. doi:10.1111/edt.12573
183. Creighton J, et al. Diagnosis and management of Stevens–Johnson syndrome and toxic epidermal necrolysis in pediatric patients: a systematic review of clinical guidelines and consensus statements. *Pediatr Dermatol.* 2026. doi:10.1111/pde.70149
184. Rashid R, Goodyear H. Management of paediatric dermatological emergencies. *Paediatr Child Health (Oxford).* 2019;29(2):74–79. [doi:10.1016/j.paed.2018.12.006](https://doi.org/10.1016/j.paed.2018.12.006)
185. McPherson T, Exton LS, Biswas S, et al. British Association of Dermatologists' guidelines for the management of Stevens-Johnson syndrome/toxic epidermal necrolysis in children and young people, 2018. *Br J Dermatol.* 2019;181(1):37–54. doi:10.1111/bjd.17841
186. Heuer L, Zhao Z, et al. S3 guideline: Diagnosis and treatment of epidermal necrolysis (Stevens-Johnson syndrome and toxic epidermal necrolysis) — Part 1: Diagnosis, initial management, and immunomodulating systemic therapy. *JDDG J Dtsch Dermatol Ges.* 2024. doi:10.1111/ddg.15515

187. Zimmermann S, Sekula P, Venhoff M, et al. Systemic immunomodulating therapies for Stevens-Johnson syndrome and toxic epidermal necrolysis: a systematic review and meta-analysis. *JAMA Dermatol.* 2017;153(6):514–522. doi:10.1001/jamadermatol.2016.5668
188. Gray et al. Pediatric staphylococcal scalded skin syndrome: a systematic review of the literature to inform work-up and management. *Pediatr Dermatol.* 2025. doi:10.1111/pde.16029
189. Eczema herpeticum. In: StatPearls. Treasure Island (FL): StatPearls Publishing; 2024. NBK560781
190. Kuyucu et al. Clinical diagnosis and management of drug reaction with eosinophilia and systemic symptoms (DRESS) in children: an EAACI position paper. *Pediatr Allergy Immunol.* 2025. doi:10.1111/pai.70103
191. Stevens DL, Bisno AL, Chambers HF, et al. Practice guidelines for the diagnosis and management of skin and soft tissue infections: 2014 update by the Infectious Diseases Society of America. *Clin Infect Dis.* 2014;59(2):e10–e52. doi:10.1093/cid/ciu444
192. Shah SM, Khanna CL. Ophthalmic emergencies for the clinician. *Mayo Clin Proc.* 2020;95(5):1050–1058. [doi:10.1016/j.mayocp.2020.03.018](https://doi.org/10.1016/j.mayocp.2020.03.018)

193. Children's Hospital of Philadelphia.
Ventricular shunt obstruction/infection
clinical pathway (emergency, ICU, inpatient).
CHOP Clinical Pathways.
194. Characteristics and treatment of pediatric
nasal foreign bodies with button batteries: a
retrospective analysis of 176 cases. *PLoS
One*. 2024;19(8):e0309261.
doi:10.1371/journal.pone.0309261
195. Chen HL, Taylor SA, Lee WS, et al.
Diagnostic approaches for infants with
cholestatic liver diseases: position paper of
the Federation of International Societies of
Pediatric Gastroenterology, Hepatology, and
Nutrition (FISPGHAN). *J Pediatr
Gastroenterol Nutr*. 2025;81(6):1360–1377.
doi:10.1002/jpn3.70207
196. Watson J, McGuire W. Nasal versus oral
route for placing feeding tubes in preterm or
low birth weight infants. *Cochrane Database
Syst Rev*. 2013;(2):CD003952.
doi:10.1002/14651858.CD003952.pub3
197. Sathya P, Fenton TR; Canadian Paediatric
Society, Nutrition and Gastroenterology
Committee. Cow's milk protein allergy in
infants and children. *Paediatr Child Health*.
2024;29(6):382–388.
[doi:10.1093/pch/pxae043](https://doi.org/10.1093/pch/pxae043)
198. American Academy of Pediatric Dentistry.
Useful medications for oral conditions. In:
The Reference Manual of Pediatric Dentistry.
Chicago (IL): American Academy of
Pediatric Dentistry; 2026.

199. Amir J, Harel L, Smetana Z, Varsano I. Treatment of herpes simplex gingivostomatitis with aciclovir in children: a randomized double blind placebo controlled study. *BMJ*. 1997;314(7097):1800–1803. [doi:10.1136/bmj.314.7097.1800](https://doi.org/10.1136/bmj.314.7097.1800)
200. Lin HW, O'Neill A, Cunningham MJ. Ludwig's angina in the pediatric population. *Clin Pediatr (Phila)*. 2009;48(6):583–587. [doi:10.1177/0009922809333095](https://doi.org/10.1177/0009922809333095)
201. Barmayehvar B, Keay AAR, Elsegood R, Parmar A. Paediatric cervical lymphadenopathy: an evidence-based approach for GPs. *Br J Gen Pract*. 2026;76(764):137–140. [doi:10.3399/BJGP.2025.0666](https://doi.org/10.3399/BJGP.2025.0666)
202. From pathology back to the clinic: refining indications for pediatric cervical lymph node biopsy. *Eur J Pediatr*. 2026. [doi:10.1007/s00431-026-07085-0](https://doi.org/10.1007/s00431-026-07085-0)
203. Risk factors for malignancy in pediatric subacute/chronic focal craniocervical lymphadenopathy. *Front Pediatr*. 2024;12:1466116. [doi:10.3389/fped.2024.1466116](https://doi.org/10.3389/fped.2024.1466116)
204. Weinstock MS, Patel NA, Smith LP. Pediatric cervical lymphadenopathy. *Pediatr Rev*. 2018;39(9):433–443. [doi:10.1542/pir.2017-0249](https://doi.org/10.1542/pir.2017-0249)

205. Deosthali A, Donches K, DelVecchio M, Aronoff S. Etiologies of pediatric cervical lymphadenopathy: a systematic review. *Glob Pediatr Health*. 2019;6:2333794X19865440. [doi:10.1177/2333794X19865440](https://doi.org/10.1177/2333794X19865440)
206. Meier JD, Grimmer JF. Evaluation and management of neck masses in children. *Am Fam Physician*. 2014;89(5):353–358. [PMID 24695506](https://pubmed.ncbi.nlm.nih.gov/24695506/) (no DOI assigned)
207. Amin R, Agarwal A, Chiang J, et al. Care of Infants and Children with Tracheostomies: an Official American Thoracic Society Clinical Practice Guideline. *Am J Respir Crit Care Med*. 2025;211(11):2001–2020. [doi:10.1164/rccm.202508-2055ST](https://doi.org/10.1164/rccm.202508-2055ST)
208. Volsko TA, Carroll CL, Bhalla A, et al. AARC Clinical Practice Guideline: Management of Pediatric Patients With Tracheostomy in the Acute Care Setting. *Respir Care*. 2021;66(1):144–155. [doi:10.4187/respcare.08137](https://doi.org/10.4187/respcare.08137)
209. Treatment of pediatric convulsive status epilepticus. *Front Neurol*. 2023;14:1175370. [doi:10.3389/fneur.2023.1175370](https://doi.org/10.3389/fneur.2023.1175370)
210. Manning JC, Pinto NP, Rennick JE, Colville G, Curley MAQ. Conceptualizing Post Intensive Care Syndrome in Children — the PICS-p Framework. *Pediatr Crit Care Med*. 2018;19(4):298–300. [doi:10.1097/PCC.0000000000001476](https://doi.org/10.1097/PCC.0000000000001476)

211. Major P, Lortholary A, Hon J, et al. Zoledronic acid is superior to pamidronate in the treatment of hypercalcemia of malignancy: a pooled analysis of two randomized, controlled clinical trials. *J Clin Oncol*. 2001;19(2):558–567. [doi:10.1200/JCO.2001.19.2.558](https://doi.org/10.1200/JCO.2001.19.2.558)
212. Lee SY, Modzelewski KL, Law AC, Walkey AJ, Pearce EN, Bosch NA. Comparison of propylthiouracil vs methimazole for thyroid storm in critically ill patients. *JAMA Netw Open*. 2023;6(4):e238655. [doi:10.1001/jamanetworkopen.2023.8655](https://doi.org/10.1001/jamanetworkopen.2023.8655)
213. Damari D, et al. Diagnosis and management of reactive infectious mucocutaneous eruption. *J Hosp Med*. 2025. [doi:10.1002/jhm.70099](https://doi.org/10.1002/jhm.70099) (PMID 40619921)
214. Tunkel AR, Hasbun R, Bhimraj A, et al. 2017 Infectious Diseases Society of America's Clinical Practice Guidelines for Healthcare-Associated Ventriculitis and Meningitis. *Clin Infect Dis*. 2017;64(6):e34–e65. [doi:10.1093/cid/ciw861](https://doi.org/10.1093/cid/ciw861)
215. Henter JL, Horne A, Aricó M, et al. HLH-2004: diagnostic and therapeutic guidelines for hemophagocytic lymphohistiocytosis. *Pediatr Blood Cancer*. 2007;48(2):124–131. [doi:10.1002/psc.21039](https://doi.org/10.1002/psc.21039)

216. Ravelli A, Minoia F, Davi S, et al. 2016 classification criteria for macrophage activation syndrome complicating systemic juvenile idiopathic arthritis: a EULAR/ACR/PRINTO collaborative initiative. *Arthritis Rheumatol*. 2016;68(3):566–576. [doi:10.1002/art.39332](https://doi.org/10.1002/art.39332)
217. Ozen S, Marks SD, Brogan P, et al. European consensus-based recommendations for diagnosis and treatment of immunoglobulin A vasculitis — the SHARE initiative. *Rheumatology (Oxford)*. 2019;58(9):1607–1616. [doi:10.1093/rheumatology/kez041](https://doi.org/10.1093/rheumatology/kez041)
218. Henderson LA, Canna SW, Friedman KG, et al. American College of Rheumatology clinical guidance for multisystem inflammatory syndrome in children associated with SARS-CoV-2 and hyperinflammation in pediatric COVID-19: version 3. *Arthritis Rheumatol*. 2022;74(4):e1–e20. [doi:10.1002/art.42062](https://doi.org/10.1002/art.42062)
219. Council of State and Territorial Epidemiologists / CDC. Multisystem inflammatory syndrome in children (MIS-C) associated with SARS-CoV-2 infection: 2023 case definition. Atlanta, GA: CDC; 2023.
220. Asian Pan-Pacific Society for Pediatric Gastroenterology, Hepatology and Nutrition (APSPGHAN). Statement on endoscopic variceal bleeding hemostasis in children. *Pediatr Gastroenterol Hepatol Nutr*. 2026;29(1):1. [doi:10.5223/pghn.2026.29.1.1](https://doi.org/10.5223/pghn.2026.29.1.1)

221. Applegate KE, Anderson JM, Klatte EC. Intestinal malrotation in children: a problem-solving approach to the upper gastrointestinal series. *RadioGraphics*. 2006;26(5):1485–1500. [doi:10.1148/rg.265055167](https://doi.org/10.1148/rg.265055167)
222. Hwang J, Yoon HM, Kim PH, et al. Current diagnosis and image-guided reduction for intussusception in children. *Clin Exp Pediatr*. 2023;66(1):12–21. [doi:10.3345/cep.2021.01816](https://doi.org/10.3345/cep.2021.01816)
223. Hypertrophic pyloric stenosis. In: *Merck Manual (Professional Edition)*. Rahway (NJ): Merck & Co.; 2024.
224. Inguinal hernia in infants and children (incarceration & strangulation). In: *Merck Manual (Professional Edition)*. Rahway (NJ): Merck & Co.; 2024.
225. Gerson R, Malas N, Feuer V, et al. Best practices for evaluation and treatment of agitated children and adolescents (BETA) in the emergency department: consensus statement of the American Association for Emergency Psychiatry. *West J Emerg Med*. 2019;20(2):409–418. [doi:10.5811/westjem.2019.1.41344](https://doi.org/10.5811/westjem.2019.1.41344)
226. Boyer EW, Shannon M. The serotonin syndrome. *N Engl J Med*. 2005;352(11):1112–1120. [doi:10.1056/NEJMr041867](https://doi.org/10.1056/NEJMr041867)
227. Neuroleptic malignant syndrome. In: *StatPearls*. Treasure Island (FL): StatPearls Publishing; 2024. [NBK482282](https://doi.org/10.1007/978-1-4939-9832-2_100000)

228. Rosenberg H, Pollock N, Schiemann A, Bulger T, Stowell K. Malignant hyperthermia: a review. *Orphanet J Rare Dis.* 2015;10:93. [doi:10.1186/s13023-015-0310-1](https://doi.org/10.1186/s13023-015-0310-1)
229. Shain B; American Academy of Pediatrics Committee on Adolescence. Suicide and suicide attempts in adolescents. *Pediatrics.* 2016;138(1):e20161420. [doi:10.1542/peds.2016-1420](https://doi.org/10.1542/peds.2016-1420)
230. Schlapbach LJ, Watson RS, Sorce LR, et al; SCCM Pediatric Sepsis Definition Task Force. International consensus criteria for pediatric sepsis and septic shock. *JAMA.* 2024;331(8):665–674. [doi:10.1001/jama.2024.0179](https://doi.org/10.1001/jama.2024.0179)
231. Law YM, Lal AK, Chen S, et al. Diagnosis and management of myocarditis in children: a scientific statement from the American Heart Association. *Circulation.* 2021;144(6):e123–e135. [doi:10.1161/CIR.0000000000001001](https://doi.org/10.1161/CIR.0000000000001001)
232. Gewitz MH, Baltimore RS, Tani LY, et al. Revision of the Jones criteria for the diagnosis of acute rheumatic fever in the era of Doppler echocardiography: a scientific statement from the American Heart Association. *Circulation.* 2015;131(20):1806–1818. [doi:10.1161/CIR.0000000000000205](https://doi.org/10.1161/CIR.0000000000000205)

233. Gerber MA, Baltimore RS, Eaton CB, et al. Prevention of rheumatic fever and diagnosis and treatment of acute streptococcal pharyngitis: a scientific statement from the American Heart Association. *Circulation*. 2009;119(11):1541–1551.
[doi:10.1161/CIRCULATIONAHA.109.191959](https://doi.org/10.1161/CIRCULATIONAHA.109.191959)
234. Adler Y, Charron P, Imazio M, et al. 2015 ESC guidelines for the diagnosis and management of pericardial diseases. *Eur Heart J*. 2015;36(42):2921–2964.
[doi:10.1093/eurheartj/ehv318](https://doi.org/10.1093/eurheartj/ehv318)
235. Delgado V, Ajmone Marsan N, de Waha S, et al. 2023 ESC guidelines for the management of endocarditis. *Eur Heart J*. 2023;44(39):3948–4042.
[doi:10.1093/eurheartj/ehad193](https://doi.org/10.1093/eurheartj/ehad193)
236. Wilson WR, Gewitz M, Lockhart PB, et al. Prevention of viridans group streptococcal infective endocarditis: a scientific statement from the American Heart Association. *Circulation*. 2021;143(20):e963–e978.
[doi:10.1161/CIR.0000000000000969](https://doi.org/10.1161/CIR.0000000000000969)
237. Li JS, Sexton DJ, Mick N, et al. Proposed modifications to the Duke criteria for the diagnosis of infective endocarditis. *Clin Infect Dis*. 2000;30(4):633–638.
[doi:10.1086/313753](https://doi.org/10.1086/313753)
238. Fowler VG, Durack DT, Selton-Suty C, et al. The 2023 Duke-ISCVID criteria for infective endocarditis: updating the modified Duke criteria. *Clin Infect Dis*. 2023;77(4):518–526.
[doi:10.1093/cid/ciad271](https://doi.org/10.1093/cid/ciad271)

239. Singh Y, Mikrou P. Use of prostaglandins in duct-dependent congenital heart conditions. *Arch Dis Child Educ Pract Ed*. 2018;103(3):137–140. [doi:10.1136/archdischild-2017-313654](https://doi.org/10.1136/archdischild-2017-313654)
240. NHS Greater Glasgow & Clyde & PIER Network. Acute management of the cyanosed infant with a blocked systemic-to-pulmonary (modified Blalock–Taussig / Sano) shunt. Paediatric clinical guideline.
241. National Institute for Health and Care Excellence. Fever in under 5s: assessment and initial management (NG143). London: NICE; 2021. [nice.org.uk/guidance/ng143](https://www.nice.org.uk/guidance/ng143)
242. Sullivan JE, Farrar HC; American Academy of Pediatrics, Section on Clinical Pharmacology and Therapeutics, Committee on Drugs. Fever and antipyretic use in children. *Pediatrics*. 2011;127(3):580–587. [doi:10.1542/peds.2010-3852](https://doi.org/10.1542/peds.2010-3852)
243. American College of Obstetricians and Gynecologists. Committee Opinion No. 785: Screening and management of bleeding disorders in adolescents with heavy menstrual bleeding. *Obstet Gynecol*. 2019;134(3):e71–e83. [doi:10.1097/AOG.0000000000003411](https://doi.org/10.1097/AOG.0000000000003411)
244. Extracorporeal Life Support Organization (ELSO). ELSO guidelines for pediatric respiratory and cardiac failure. Ann Arbor (MI): ELSO; 2021. [elseo.org](https://www.elseo.org)

245. British Thoracic Society Air Travel Working Party. Managing passengers with stable respiratory disease planning air travel: BTS recommendations. *Thorax*. 2011;66(Suppl 1):i1–i30. [doi:10.1136/thoraxjnl-2011-200295](https://doi.org/10.1136/thoraxjnl-2011-200295)
246. State of Kuwait, Ministry of Health. Aeromedical evacuation & transport policy (A-ADM-014). Kuwait: MOH; 2025 (effective Oct 2025; review 2027).
247. Bornstein SR, Allolio B, Arlt W, et al. Diagnosis and treatment of primary adrenal insufficiency: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab*. 2016;101(2):364–389. [doi:10.1210/jc.2015-1710](https://doi.org/10.1210/jc.2015-1710)
248. Lehnhardt A, Kemper MJ. Pathogenesis, diagnosis and management of hyperkalemia. *Pediatr Nephrol*. 2011;26(3):377–384. [doi:10.1007/s00467-010-1699-3](https://doi.org/10.1007/s00467-010-1699-3)
249. Davies JH, Shaw NJ. Investigation and management of hypercalcaemia in children. *Arch Dis Child*. 2012;97(6):533–538. [doi:10.1136/archdischild-2011-301284](https://doi.org/10.1136/archdischild-2011-301284)
250. Ross DS, Burch HB, Cooper DS, et al. 2016 American Thyroid Association guidelines for diagnosis and management of hyperthyroidism and other causes of thyrotoxicosis. *Thyroid*. 2016;26(10):1343–1421. [doi:10.1089/thy.2016.0229](https://doi.org/10.1089/thy.2016.0229)

251. Häberle J, Burlina A, Chakrapani A, et al. Suggested guidelines for the diagnosis and management of urea cycle disorders: first revision. *J Inherit Metab Dis*. 2019;42(6):1192–1230. [doi:10.1002/jimd.12100](https://doi.org/10.1002/jimd.12100)
252. Assa A, Aloï M, Van Biervliet S, et al. Management of paediatric ulcerative colitis, part 2: Acute severe colitis—An updated evidence-based consensus guideline from the European Society of Paediatric Gastroenterology, Hepatology and Nutrition and the European Crohn's and Colitis Organization. *J Pediatr Gastroenterol Nutr*. 2025;81(3):816–851. [doi:10.1002/jpn3.70096](https://doi.org/10.1002/jpn3.70096) (supersedes Turner et al. 2018, JPGN 67:292–310)
253. Neunert C, Terrell DR, Arnold DM, et al. American Society of Hematology 2019 guidelines for immune thrombocytopenia. *Blood Adv*. 2019;3(23):3829–3866. [doi:10.1182/bloodadvances.2019000966](https://doi.org/10.1182/bloodadvances.2019000966)
254. Stephanos K, Dubbs SB. Pediatric Hematologic and Oncologic Emergencies. *Emerg Med Clin North Am*. 2021;39(3):555–571. [doi:10.1016/j.emc.2021.04.007](https://doi.org/10.1016/j.emc.2021.04.007)
255. Puopolo KM, Benitz WE, Zaoutis TE; Committee on Fetus and Newborn, Committee on Infectious Diseases. Management of Neonates Born at ≥ 35 0/7 Weeks' Gestation With Suspected or Proven Early-Onset Bacterial Sepsis. *Pediatrics*. 2018;142(6):e20182894. [doi:10.1542/peds.2018-2894](https://doi.org/10.1542/peds.2018-2894)

256. Hand I, Noble L, Abrams SA; Committee on Fetus and Newborn. Vitamin K and the Newborn Infant. *Pediatrics*. 2022;149(3):e2021056036.
[doi:10.1542/peds.2021-056036](https://doi.org/10.1542/peds.2021-056036)
257. Young LW, Ounpraseuth ST, Merhar SL, et al. Eat, Sleep, Console Approach or Usual Care for Neonatal Opioid Withdrawal. *N Engl J Med*. 2023;388(25):2326–2337.
[doi:10.1056/NEJMoa2214470](https://doi.org/10.1056/NEJMoa2214470)
258. Schmidt B, Roberts RS, Davis P, et al; Caffeine for Apnea of Prematurity Trial Group. Caffeine therapy for apnea of prematurity. *N Engl J Med*. 2006;354(20):2112–2121.
[doi:10.1056/NEJMoa054065](https://doi.org/10.1056/NEJMoa054065)
259. Rybak MJ, Le J, Lodise TP, et al. Executive Summary: Therapeutic Monitoring of Vancomycin for Serious Methicillin-Resistant *Staphylococcus aureus* Infections: A Revised Consensus Guideline and Review (ASHP/IDSA/PIDS/SIDP). *J Pediatric Infect Dis Soc*. 2020;9(3):281–284.
[doi:10.1093/jpids/piaa057](https://doi.org/10.1093/jpids/piaa057)
260. Squires JE, Alonso EM, Ibrahim SH, et al. North American Society for Pediatric Gastroenterology, Hepatology, and Nutrition Position Paper on the Diagnosis and Management of Pediatric Acute Liver Failure. *J Pediatr Gastroenterol Nutr*. 2022;74(1):138–158.
[doi:10.1097/MPG.0000000000003268](https://doi.org/10.1097/MPG.0000000000003268)

261. Romano C, Oliva S, Martellosi S, et al. Pediatric gastrointestinal bleeding: Perspectives from the Italian Society of Pediatric Gastroenterology. *World J Gastroenterol*. 2017;23(8):1328–1337. [doi:10.3748/wjg.v23.i8.1328](https://doi.org/10.3748/wjg.v23.i8.1328)
262. National Institute for Health and Care Excellence (NICE). Constipation in children and young people: diagnosis and management. Clinical guideline [CG99]. London: NICE; 2010 (updated 2017). nice.org.uk/guidance/cg99
263. Wilne S, Collier J, Kennedy C, Koller K, Grundy R, Walker D. Presentation of childhood CNS tumours: a systematic review and meta-analysis. *Lancet Oncol*. 2007;8(8):685–695. [doi:10.1016/S1470-2045\(07\)70207-3](https://doi.org/10.1016/S1470-2045(07)70207-3)
264. van Doorn PA, Van den Bergh PYK, Hadden RDM, et al. European Academy of Neurology/Peripheral Nerve Society Guideline on diagnosis and treatment of Guillain-Barré syndrome. *Eur J Neurol*. 2023;30(12):3646–3674. [doi:10.1111/ene.16073](https://doi.org/10.1111/ene.16073)
265. Birnkrant DJ, Bushby K, Bann CM, et al. Diagnosis and management of Duchenne muscular dystrophy, part 2: respiratory, cardiac, bone health, and orthopaedic management. *Lancet Neurol*. 2018;17(4):347–361. [doi:10.1016/S1474-4422\(18\)30025-5](https://doi.org/10.1016/S1474-4422(18)30025-5)

266. Venkatesan A, Tunkel AR, Bloch KC, et al; International Encephalitis Consortium. Case definitions, diagnostic algorithms, and priorities in encephalitis: consensus statement. *Clin Infect Dis*. 2013;57(8):1114–1128. [doi:10.1093/cid/cit458](https://doi.org/10.1093/cid/cit458)
267. Pierce MC, Kaczor K, Lorenz DJ, et al. Validation of a Clinical Decision Rule to Predict Abuse in Young Children Based on Bruising Characteristics (TEN-4-FACESp). *JAMA Netw Open*. 2021;4(4):e215832. [doi:10.1001/jamanetworkopen.2021.5832](https://doi.org/10.1001/jamanetworkopen.2021.5832)
268. Rossaint R, Afshari A, Bouillon B, et al. The European guideline on management of major bleeding and coagulopathy following trauma: sixth edition. *Crit Care*. 2023;27(1):80. [doi:10.1186/s13054-023-04327-7](https://doi.org/10.1186/s13054-023-04327-7)
269. Holcomb JB, Tilley BC, Baraniuk S, et al. Transfusion of plasma, platelets, and red blood cells in a 1:1:1 vs a 1:1:2 ratio and mortality in patients with severe trauma: the PROPPR randomized clinical trial. *JAMA*. 2015;313(5):471–482. [doi:10.1001/jama.2015.12](https://doi.org/10.1001/jama.2015.12)
270. CRASH-2 trial collaborators; Shakur H, Roberts I, et al. Effects of tranexamic acid on death, vascular occlusive events, and blood transfusion in trauma patients with significant haemorrhage (CRASH-2): a randomised, placebo-controlled trial. *Lancet*. 2010;376(9734):23–32. [doi:10.1016/S0140-6736\(10\)60835-5](https://doi.org/10.1016/S0140-6736(10)60835-5)

271. Fakhouri F, Zuber J, Frémeaux-Bacchi V, Loirat C. Haemolytic uraemic syndrome. *Lancet*. 2017;390(10095):681–696. [doi:10.1016/S0140-6736\(17\)30062-4](https://doi.org/10.1016/S0140-6736(17)30062-4)
272. Mannix R, Tan ML, Wright R, Baskin M. Acute pediatric rhabdomyolysis: causes and rates of renal failure. *Pediatrics*. 2006;118(5):2119–2125. [doi:10.1542/peds.2006-1352](https://doi.org/10.1542/peds.2006-1352)
273. Maurer M, Magerl M, Betschel S, et al. The international WAO/EAACI guideline for the management of hereditary angioedema—the 2021 revision and update. *Allergy*. 2022;77(7):1961–1990. [doi:10.1111/all.15214](https://doi.org/10.1111/all.15214)
274. Contini S, Scarpignato C. Caustic injury of the upper gastrointestinal tract: a comprehensive review. *World J Gastroenterol*. 2013;19(25):3918–3930. [doi:10.3748/wjg.v19.i25.3918](https://doi.org/10.3748/wjg.v19.i25.3918)
275. ISBI Practice Guidelines Committee. ISBI Practice Guidelines for Burn Care. *Burns*. 2016;42(5):953–1021. [doi:10.1016/j.burns.2016.05.013](https://doi.org/10.1016/j.burns.2016.05.013)
276. Epstein Y, Yanovich R. Heatstroke. *N Engl J Med*. 2019;380(25):2449–2459. [doi:10.1056/NEJMr1810762](https://doi.org/10.1056/NEJMr1810762)
277. Bawaskar HS, Bawaskar PH. Efficacy and safety of scorpion antivenom plus prazosin compared with prazosin alone for venomous scorpion (*Mesobuthus tamulus*) sting: randomised open label clinical trial. *BMJ*. 2011;342:c7136. [doi:10.1136/bmj.c7136](https://doi.org/10.1136/bmj.c7136)

278. World Health Organization, Regional Office for South-East Asia. Guidelines for the Management of Snakebites. 2nd ed. New Delhi: WHO-SEARO; 2016. who.int
279. World Health Organization. Rabies vaccines: WHO position paper, April 2018 – Recommendations. *Wkly Epidemiol Rec.* 2018;93(16):201–220. who.int
280. Candy DCA, Edwards D, Geraint M. Treatment of faecal impaction with polyethylene glycol plus electrolytes (PEG+E) followed by a double-blind comparison of PEG+E versus lactulose as maintenance therapy. *J Pediatr Gastroenterol Nutr.* 2006;43(1):65–70. [doi:10.1097/01.mpg.0000228097.58960.e6](https://doi.org/10.1097/01.mpg.0000228097.58960.e6)
281. Bekkali NLH, Hoekman DR, Liem O, et al. Polyethylene glycol 3350 with electrolytes versus polyethylene glycol 4000 for constipation: a randomized, controlled trial. *J Pediatr Gastroenterol Nutr.* 2018;66(1):10–15. [doi:10.1097/MPG.0000000000001726](https://doi.org/10.1097/MPG.0000000000001726)
282. Mínguez M, López Higuera A, Júdez J. Use of polyethylene glycol in functional constipation and fecal impaction. *Rev Esp Enferm Dig.* 2016;108(12):790–806. [doi:10.17235/reed.2016.4571/2016](https://doi.org/10.17235/reed.2016.4571/2016)
283. Gralnek IM, Stanley AJ, Morris AJ, et al. Endoscopic diagnosis and management of nonvariceal upper gastrointestinal hemorrhage (NVUGIH): European Society of Gastrointestinal Endoscopy (ESGE) Guideline – Update 2021. *Endoscopy.* 2021;53(3):300–332. [doi:10.1055/a-1369-5274](https://doi.org/10.1055/a-1369-5274)

284. Rockey DC, Ahn C, de Melo SW Jr. Randomized pragmatic trial of nasogastric tube placement in patients with upper gastrointestinal tract bleeding. *J Investig Med*. 2017;65(4):759–764. [doi:10.1136/jim-2016-000375](https://doi.org/10.1136/jim-2016-000375)
285. DeBaun MR, Jordan LC, King AA, et al. American Society of Hematology 2020 guidelines for sickle cell disease: prevention, diagnosis, and treatment of cerebrovascular disease in children and adults. *Blood Adv*. 2020;4(8):1554–1588. [doi:10.1182/bloodadvances.2019001142](https://doi.org/10.1182/bloodadvances.2019001142)
286. Alghounaim M, Alonaizi E, Hussain E, Burhamah M. Guideline for the Management of Fever and Neutropenia in Children with Cancer or Bone Marrow Failure Syndromes. Version 4. Kuwait: NBK Children's Hospital; 2019.
287. Sanz MA, Fenaux P, Tallman MS, et al. Management of acute promyelocytic leukemia: updated recommendations from an expert panel of the European LeukemiaNet. *Blood*. 2019;133(15):1630–1643. [doi:10.1182/blood-2019-01-894980](https://doi.org/10.1182/blood-2019-01-894980)
288. Tan A, Nolan JA. Anesthesia for children with anterior mediastinal masses. *Paediatr Anaesth*. 2022;32(1):4–9. [doi:10.1111/pan.14319](https://doi.org/10.1111/pan.14319)
289. Kelly LE, Ohlsson A, Shah PS. Sildenafil for pulmonary hypertension in neonates. *Cochrane Database Syst Rev*. 2017;8:CD005494. [doi:10.1002/14651858.CD005494.pub4](https://doi.org/10.1002/14651858.CD005494.pub4)

290. Tourneux P, Rakza T, Bouissou A, Krim G, Storme L. Pulmonary circulatory effects of norepinephrine in newborn infants with persistent pulmonary hypertension. *J Pediatr*. 2008;153(3):345–349. [doi:10.1016/j.jpeds.2008.03.007](https://doi.org/10.1016/j.jpeds.2008.03.007)
291. Lawn ND, Fletcher DD, Henderson RD, Wolter TD, Wijdicks EF. Anticipating mechanical ventilation in Guillain-Barré syndrome. *Arch Neurol*. 2001;58(6):893–898. [doi:10.1001/archneur.58.6.893](https://doi.org/10.1001/archneur.58.6.893)
292. Gralnek IM, Morris J, Laursen SB, et al. Endoscopic diagnosis and management of peptic ulcer bleeding: European Society of Gastrointestinal Endoscopy (ESGE) Guideline – Update 2026. *Endoscopy*. 2026;58(8):899–924. [doi:10.1055/a-2863-8314](https://doi.org/10.1055/a-2863-8314)
293. Venter C, Meyer R, Groetch M, et al. World Allergy Organization (WAO) Diagnosis and Rationale for Action against Cow's Milk Allergy (DRACMA) guidelines update – XVI – Nutritional management of cow's milk allergy. *World Allergy Organ J*. 2024;17(8):100931. [doi:10.1016/j.waojou.2024.100931](https://doi.org/10.1016/j.waojou.2024.100931)
294. Bivalacqua TJ, Allen BK, Brock GB, et al. The diagnosis and management of recurrent ischemic priapism, priapism in sickle cell patients, and non-ischemic priapism: an AUA/SMSNA guideline. *J Urol*. 2022;208(1):43–52. [doi:10.1097/JU.0000000000002767](https://doi.org/10.1097/JU.0000000000002767)

295. Jone PN, Tremoulet A, Choueiter N, et al; American Heart Association. Update on diagnosis and management of Kawasaki disease: a scientific statement from the American Heart Association. *Circulation*. 2024;150(23):e481–e500.
[doi:10.1161/CIR.0000000000001295](https://doi.org/10.1161/CIR.0000000000001295)
296. Steering Committee on Quality Improvement and Management, Subcommittee on Febrile Seizures; American Academy of Pediatrics. Febrile seizures: clinical practice guideline for the long-term management of the child with simple febrile seizures. *Pediatrics*. 2008;121(6):1281–1286.
[doi:10.1542/peds.2008-0939](https://doi.org/10.1542/peds.2008-0939)
297. National Institute for Health and Care Excellence. Epilepsies in children, young people and adults (NG217). London: NICE; 2022, updated January 2025.
nice.org.uk/guidance/ng217
298. British Orthopaedic Association. BOAST: Supracondylar fractures of the humerus in children. London: BOA; updated October 2020. boa.ac.uk/resource/boast-11-pdf.html
299. British Orthopaedic Association. BOAST: Diagnosis and management of arterial injuries associated with musculoskeletal trauma. London: BOA; revised June 2026.
[boa.ac.uk \(BOAST arterial injuries\)](https://boa.ac.uk/BOAST%20arterial%20injuries)
300. Benedick A, Lopas L, Daley E, Jang Y. Traumatic hip dislocation: pediatric and adult evaluation and management. *J Am Acad Orthop Surg*. 2024;32(14):637–646.
[doi:10.5435/JAAOS-D-23-01013](https://doi.org/10.5435/JAAOS-D-23-01013)

301. Kenawey M, Howard A, Tile M.
Pediatric-specific lesions in unstable pelvic ring fractures of skeletally immature children: impact on diagnosis and management decisions. *J Am Acad Orthop Surg.* 2024;32(14):e683–e694.
[doi:10.5435/JAAOS-D-23-01249](https://doi.org/10.5435/JAAOS-D-23-01249)
302. Kwon BK, Tetreault LA, Martin AR, et al. A clinical practice guideline for the management of patients with acute spinal cord injury: recommendations on hemodynamic management. *Global Spine J.* 2024;14(3 Suppl):187S–211S.
[doi:10.1177/21925682231202348](https://doi.org/10.1177/21925682231202348)
303. Fehlings MG, Vaccaro A, Wilson JR, et al. Early versus delayed decompression for traumatic cervical spinal cord injury: results of the Surgical Timing in Acute Spinal Cord Injury Study (STASCIS). *PLoS One.* 2012;7(2):e32037.
[doi:10.1371/journal.pone.0032037](https://doi.org/10.1371/journal.pone.0032037)
304. Fehlings MG, Wilson JR, Tetreault LA, et al. A clinical practice guideline for the management of patients with acute spinal cord injury: recommendations on the use of methylprednisolone sodium succinate. *Global Spine J.* 2017;7(3 Suppl):203S–211S.
[doi:10.1177/2192568217703085](https://doi.org/10.1177/2192568217703085)

305. Ng E, Altit G, Joynt C, Radziminski N, Narvey M; Canadian Paediatric Society, Fetus and Newborn Committee. Guidelines for detection and management of hyperbilirubinemia in term and late preterm newborns (≥ 35 weeks gestational age). *Paediatr Child Health*. 2026;31(5):496–510. [doi:10.1093/pch/pxaf034](https://doi.org/10.1093/pch/pxaf034)
306. Muniraman H, Sardesai T, Sardesai S. Disorders of the umbilical cord. *Pediatr Rev*. 2018;39(7):332–341. [doi:10.1542/pir.2017-0202](https://doi.org/10.1542/pir.2017-0202)
307. Inarejos Clemente EJ, Navarro OM, Navallas Irujo M, et al. Omphalomesenteric duct anomalies in children: a multimodality overview. *RadioGraphics*. 2021;41(7):2090–2110. [doi:10.1148/rg.2021210048](https://doi.org/10.1148/rg.2021210048)
308. Naiditch JA, Radhakrishnan J, Chin AC. Current diagnosis and management of urachal remnants. *J Pediatr Surg*. 2013;48(10):2148–2152. [doi:10.1016/j.jpedsurg.2013.02.069](https://doi.org/10.1016/j.jpedsurg.2013.02.069)
309. Solomon-Cohen E, Lapidoth M, Snast I, et al. Cutaneous presentations of omphalomesenteric duct remnant: a systematic review of the literature. *J Am Acad Dermatol*. 2019;81(5):1120–1126. [doi:10.1016/j.jaad.2019.02.033](https://doi.org/10.1016/j.jaad.2019.02.033)

310. American Academy of Pediatrics Section on Orthopaedics, Committee on Pediatric Emergency Medicine, Section on Critical Care, Section on Surgery, Section on Transport Medicine, Committee on Pediatric Emergency Medicine; American College of Surgeons Committee on Trauma. Management of pediatric trauma (policy statement). *Pediatrics*. 2016;138(2):e20161569. [doi:10.1542/peds.2016-1569](https://doi.org/10.1542/peds.2016-1569)
311. Suttipongkaset P, Chaikittisilpa N, Vavilala MS, et al. Blood pressure thresholds and mortality in pediatric traumatic brain injury. *Pediatrics*. 2018;142(2):e20180594. [doi:10.1542/peds.2018-0594](https://doi.org/10.1542/peds.2018-0594)
312. Fawaz R, Baumann U, Ekong U, et al. Guideline for the evaluation of cholestatic jaundice in infants: joint recommendations of the North American Society for Pediatric Gastroenterology, Hepatology, and Nutrition and the European Society for Pediatric Gastroenterology, Hepatology, and Nutrition. *J Pediatr Gastroenterol Nutr*. 2017;64(1):154–168. [doi:10.1097/MPG.0000000000001334](https://doi.org/10.1097/MPG.0000000000001334)
313. Turner D, Otley AR, Mack D, et al. Development, validation, and evaluation of a pediatric ulcerative colitis activity index: a prospective multicenter study. *Gastroenterology*. 2007;133(2):423–432. [doi:10.1053/j.gastro.2007.05.029](https://doi.org/10.1053/j.gastro.2007.05.029)

314. Cho J, Kogosov A, Scheinfeld L, et al. What to do after it's swallowed? Management strategies after superabsorbent polymer bead ingestion: a scoping review. *J Pediatr Surg*. 2026;61(6):162957.
[doi:10.1016/j.jpedsurg.2026.162957](https://doi.org/10.1016/j.jpedsurg.2026.162957)
315. Expert Panel on Pediatric Imaging; Meyers ML, Moore MM, Baker JB, et al. ACR Appropriateness Criteria® Ingested or Aspirated Foreign Body—Child. *J Am Coll Radiol*. 2026;23(4):732–747.
[doi:10.1016/j.jacr.2025.12.013](https://doi.org/10.1016/j.jacr.2025.12.013)
316. Expert Panel on Pediatric Imaging; Mirsky DM, Bardo DME, Iyer RS, et al. ACR Appropriateness Criteria® Suspected Physical Abuse—Child: Update 2025. *J Am Coll Radiol*. 2026;23(8):1613–1646.
[doi:10.1016/j.jacr.2026.02.017](https://doi.org/10.1016/j.jacr.2026.02.017)
317. The Royal College of Radiologists, Society and College of Radiographers. The radiological investigation of suspected physical abuse in children. London: RCR; 2017 (revised 2018). [rcr.ac.uk](https://www.rcr.ac.uk)
318. Glenn K, Nickerson E, Bennett CV, et al. Head computed tomography in suspected physical abuse: time to rethink? *Arch Dis Child*. 2021;106(5):461–466.
[doi:10.1136/archdischild-2020-320192](https://doi.org/10.1136/archdischild-2020-320192)
319. Rubin DM, Christian CW, Bilaniuk LT, Zazyczny KA, Durbin DR. Occult head injury in high-risk abused children. *Pediatrics*. 2003;111(6 Pt 1):1382–1386.
[doi:10.1542/peds.111.6.1382](https://doi.org/10.1542/peds.111.6.1382)

320. Christian CW, Levin AV; AAP Council on Child Abuse and Neglect, Section on Ophthalmology; American Association of Certified Orthoptists; American Association for Pediatric Ophthalmology and Strabismus; American Academy of Ophthalmology. The eye examination in the evaluation of child abuse. *Pediatrics*. 2018;142(2):e20181411. [doi:10.1542/peds.2018-1411](https://doi.org/10.1542/peds.2018-1411)
321. Anderst J, Carpenter SL, Abshire TC, Killough E; AAP Section on Hematology/Oncology, Council on Child Abuse and Neglect. Evaluation for bleeding disorders in suspected child abuse. *Pediatrics*. 2022;150(4):e2022059276. [doi:10.1542/peds.2022-059276](https://doi.org/10.1542/peds.2022-059276)
322. Lindberg D, Makoroff K, Harper N, et al. Utility of hepatic transaminases to recognize abuse in children. *Pediatrics*. 2009;124(2):509–516. [doi:10.1542/peds.2008-2348](https://doi.org/10.1542/peds.2008-2348)
323. Loirat C, Fakhouri F, Ariceta G, et al.; HUS International. An international consensus approach to the management of atypical hemolytic uremic syndrome in children. *Pediatr Nephrol*. 2016;31(1):15–39. [doi:10.1007/s00467-015-3076-8](https://doi.org/10.1007/s00467-015-3076-8)
324. European Association of Urology. EAU Guidelines on Paediatric Urology — Acute scrotum. Edn. presented at the EAU Annual Congress, London 2026. Arnheim: EAU Guidelines Office; 2026. ISBN 978-94-92671-32-5. uroweb.org

325. Garnier A, Brochard K, Kwon T, et al. Efficacy and safety of eculizumab in pediatric patients affected by Shiga toxin-related hemolytic and uremic syndrome: a randomized, placebo-controlled trial. *J Am Soc Nephrol*. 2023;34(9):1561–1573. [doi:10.1681/ASN.0000000000000182](https://doi.org/10.1681/ASN.0000000000000182)
326. Centers for Disease Control and Prevention. Clinical guidance for managing meningococcal disease risk in patients receiving complement inhibitor therapy. Atlanta: CDC. [cdc.gov](https://www.cdc.gov)
327. McNamara LA, Topaz N, Wang X, Hariri S, Fox L, MacNeil JR. High risk for invasive meningococcal disease among patients receiving eculizumab (Soliris) despite receipt of meningococcal vaccine. *MMWR Morb Mortal Wkly Rep*. 2017;66(27):734–737. [doi:10.15585/mmwr.mm6627e1](https://doi.org/10.15585/mmwr.mm6627e1)
328. Kidney Disease: Improving Global Outcomes (KDIGO) Glomerular Diseases Work Group. KDIGO 2021 Clinical Practice Guideline for the Management of Glomerular Diseases. *Kidney Int*. 2021;100(4S):S1–S276. [doi:10.1016/j.kint.2021.05.021](https://doi.org/10.1016/j.kint.2021.05.021)
329. Abugrain K, McCulloch MI, Muloiwa R, Luyckx VA, Buys H. A 6-year review of acute post-streptococcal glomerulonephritis at a public children's hospital in Cape Town, South Africa. *Pediatr Nephrol*. 2024;39(6):1809–1816. [doi:10.1007/s00467-023-06247-8](https://doi.org/10.1007/s00467-023-06247-8)

330. Kuwait Pediatric Council. Pediatric Antibiotic Policy (Ministry of Health, Kuwait) — local formulary guidance; edition and date to be confirmed by the Council.
331. Dribin TE, Sampson HA, Camargo CA Jr, et al. Anaphylaxis diagnosis: Consensus Report of the Global Allergy and Asthma European Network (GA²LEN). *J Allergy Clin Immunol*. 2025;156(2):406–417.e6.
[doi:10.1016/j.jaci.2025.01.021](https://doi.org/10.1016/j.jaci.2025.01.021)
332. National Institute for Health and Care Excellence. Anaphylaxis: assessment and referral after emergency treatment. NICE guideline NG258. London: NICE; May 2026.
nice.org.uk/guidance/ng258
333. Cheng A; Canadian Paediatric Society, Acute Care Committee. Emergency treatment of anaphylaxis in infants and children. *Paediatr Child Health*. 2011;16(1):35–40. [doi:10.1093/pch/16.1.35](https://doi.org/10.1093/pch/16.1.35)
334. Stróżyk A, Ruszczyński M, Horvath A, et al. World Allergy Organization (WAO) Diagnosis and Rationale for Action against Cow's Milk Allergy (DRACMA) guidelines update – IV – A quality appraisal with the AGREE II instrument. *World Allergy Organ J*. 2022;15(2):100613.
[doi:10.1016/j.waojou.2021.100613](https://doi.org/10.1016/j.waojou.2021.100613)
335. Biggs CM, Haddad E, Issekutz TB, Roifman CM, Turvey SE. Newborn screening for severe combined immunodeficiency: a primer for clinicians. *CMAJ*. 2017;189(50):E1551–E1557.
[doi:10.1503/cmaj.170561](https://doi.org/10.1503/cmaj.170561)

336. Box 6.2 Clinical pearls: therapeutic principles — treatment of infants with SCID. Textbook chapter supplied by the section reviewer (Dr. Maysoon Al Rushood) — title, edition and chapter to be confirmed.

Kuwait Pediatric Council

Neonatal & pediatric acute care for pediatricians and primary care physicians.

- Issued June 2026 · review June 2028.
- Compiled from current international guidelines; newer evidence takes precedence before renewal.
- Verify all doses locally and follow Ministry of Health protocols.
- Self-contained single file — works offline once saved; print to PDF to distribute; add to the home screen for an app-like experience.